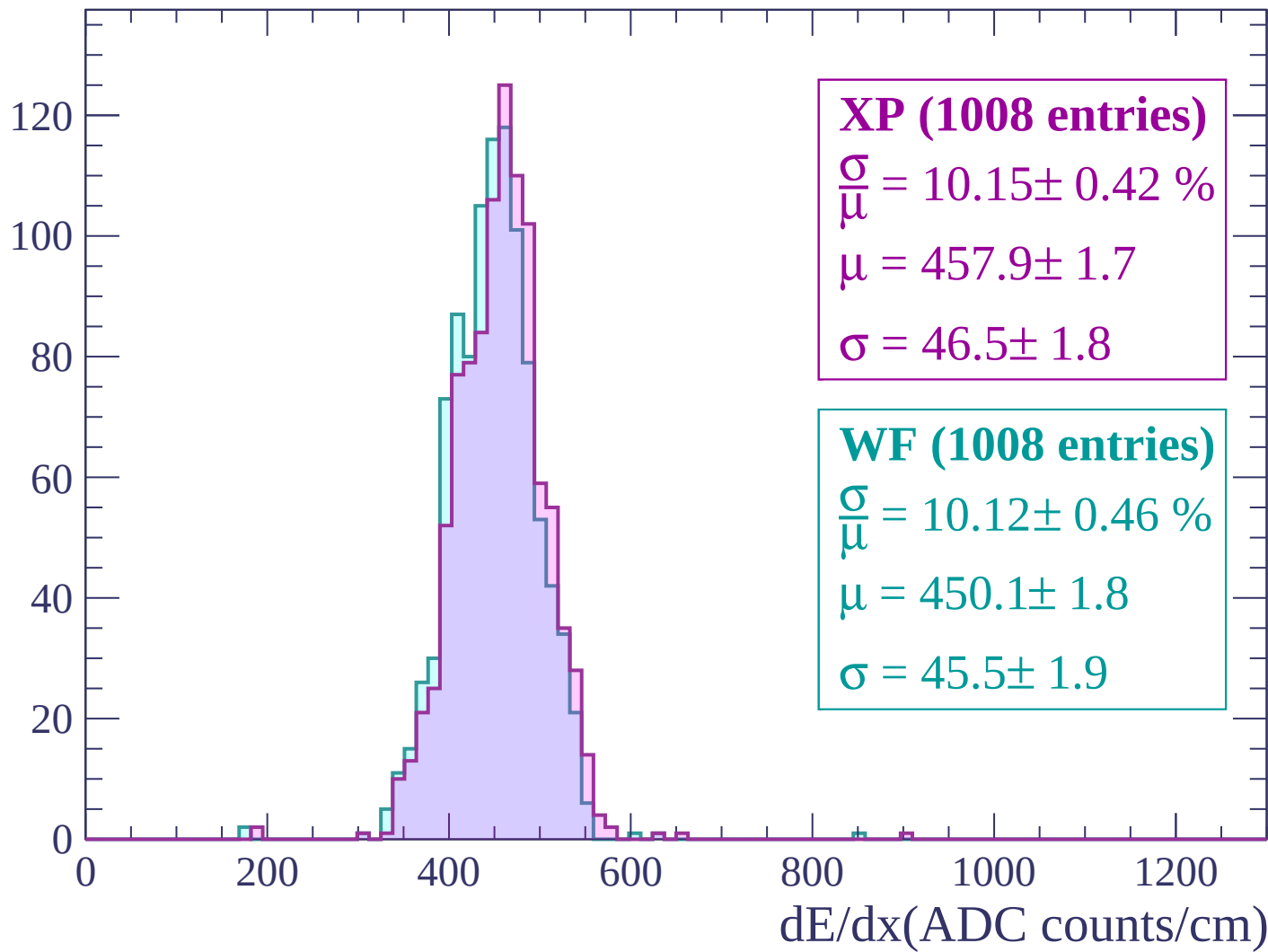
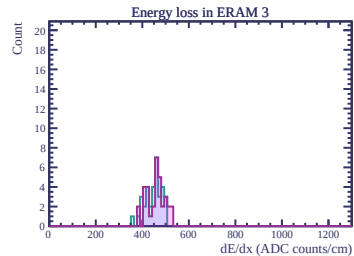
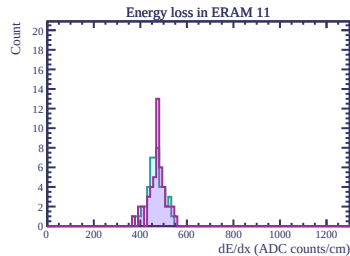
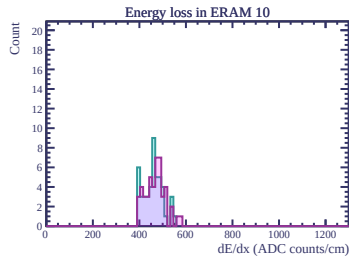
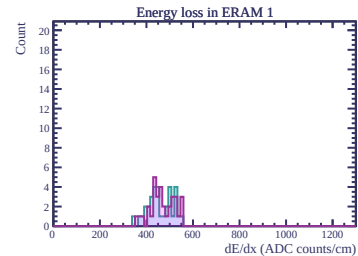
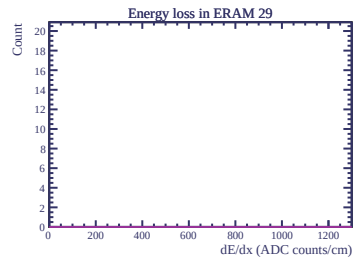
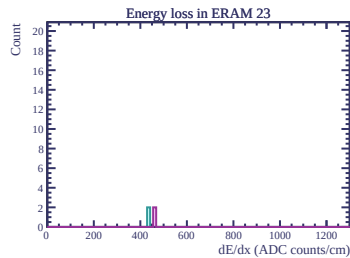
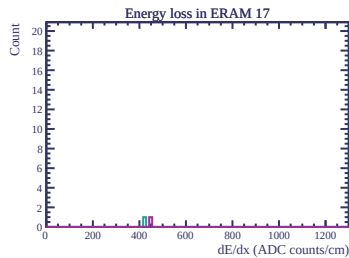
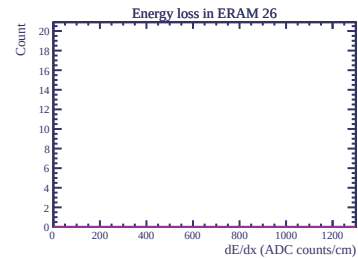
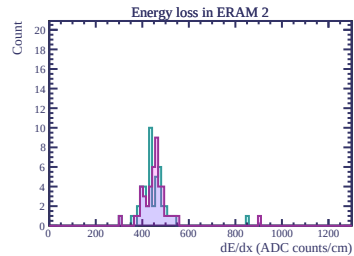
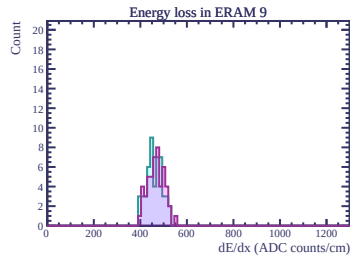
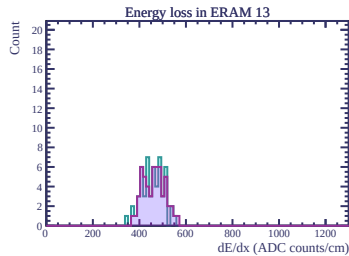
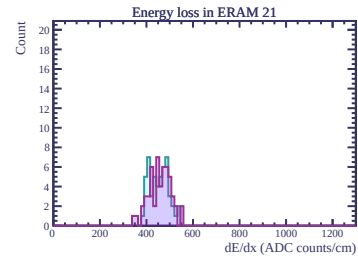
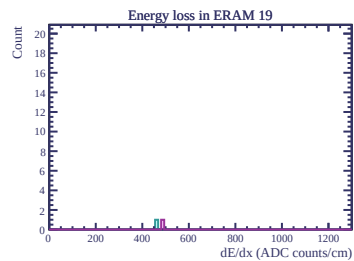
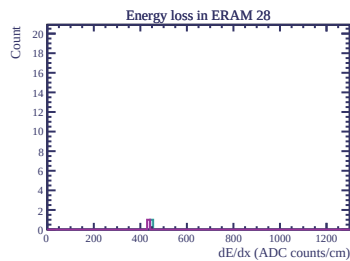
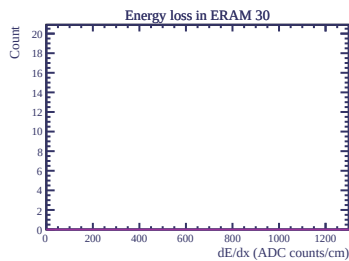
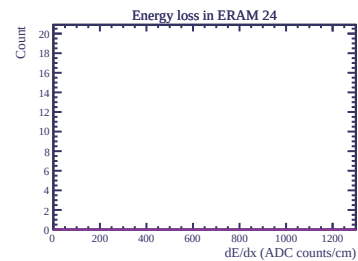
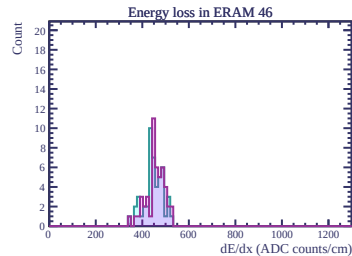
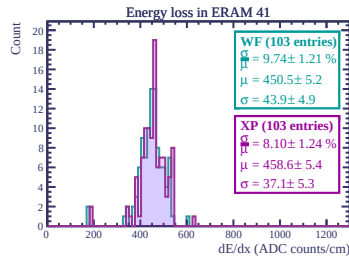
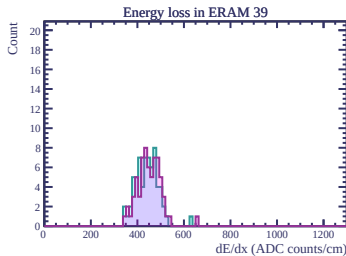
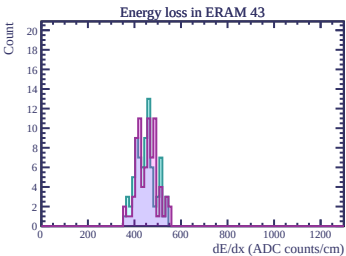
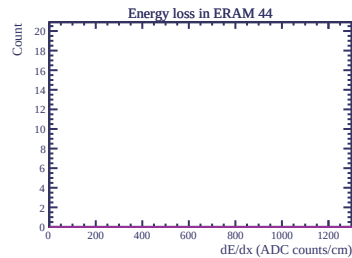
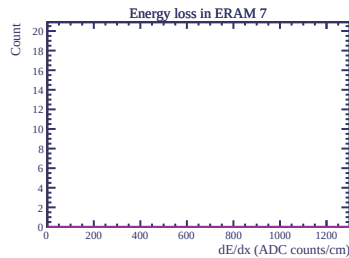
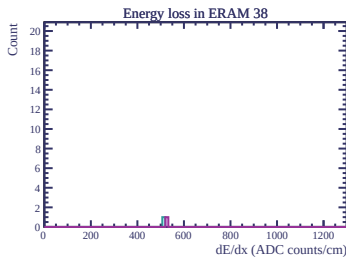
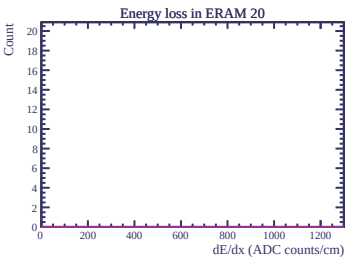
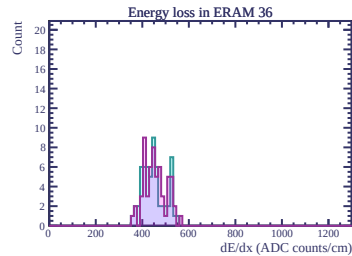
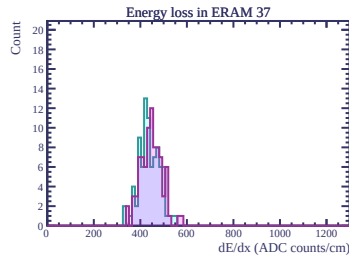
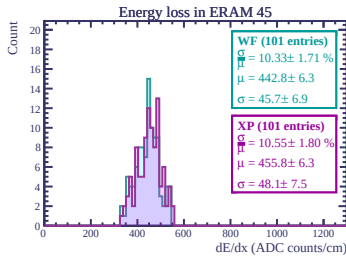
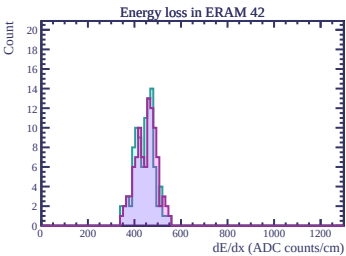
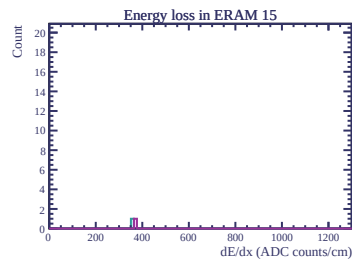
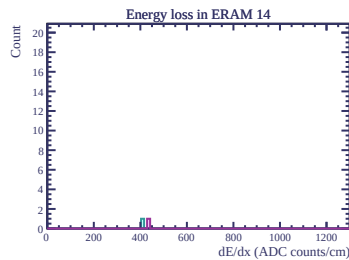
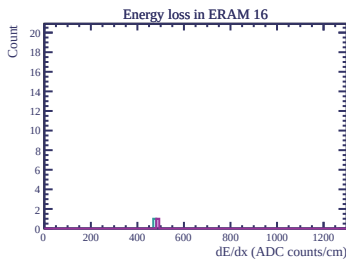
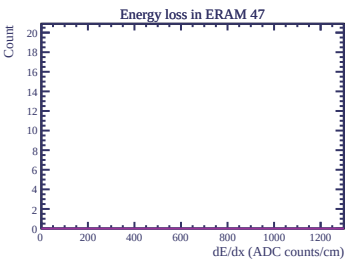
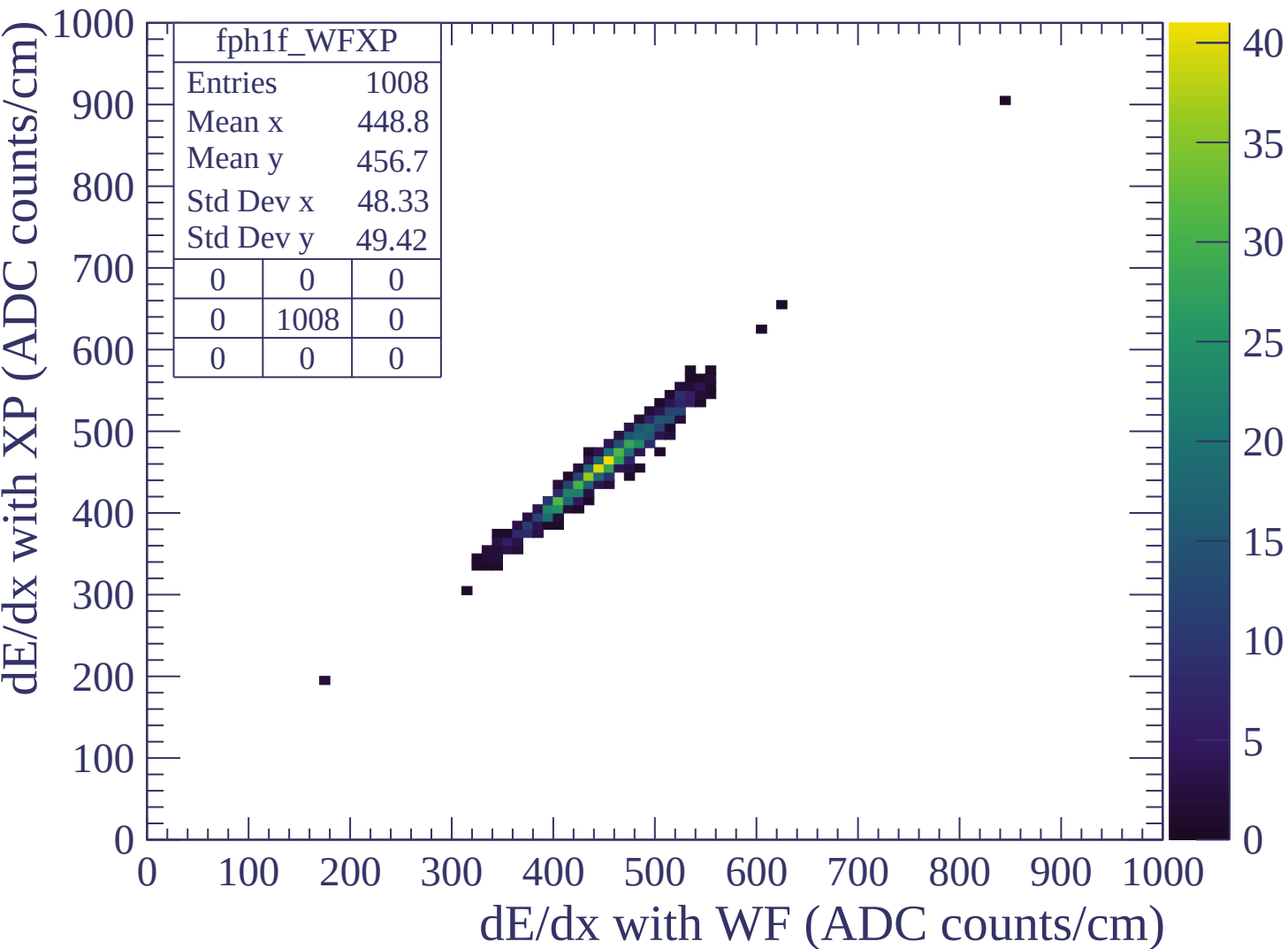


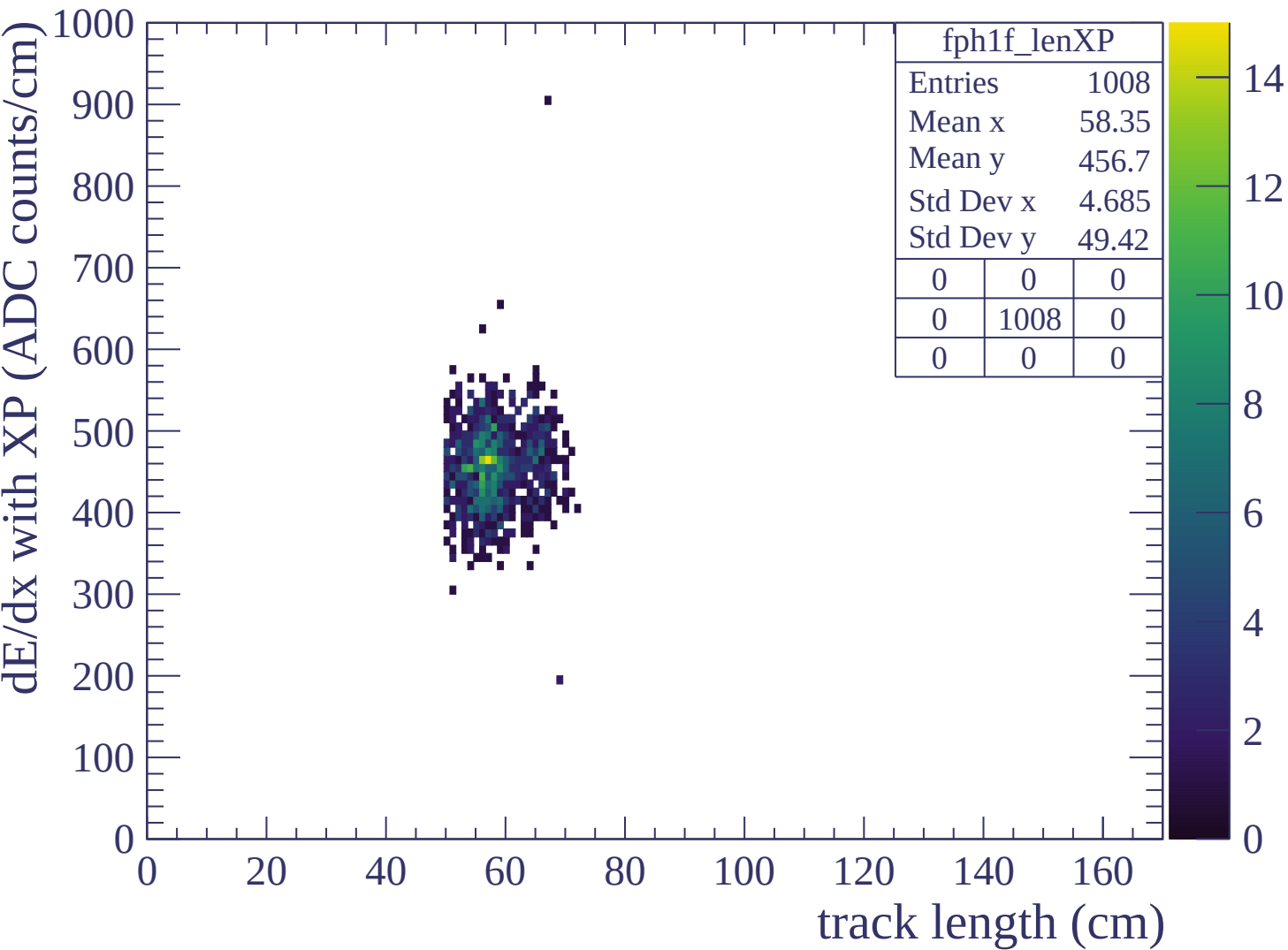
Count

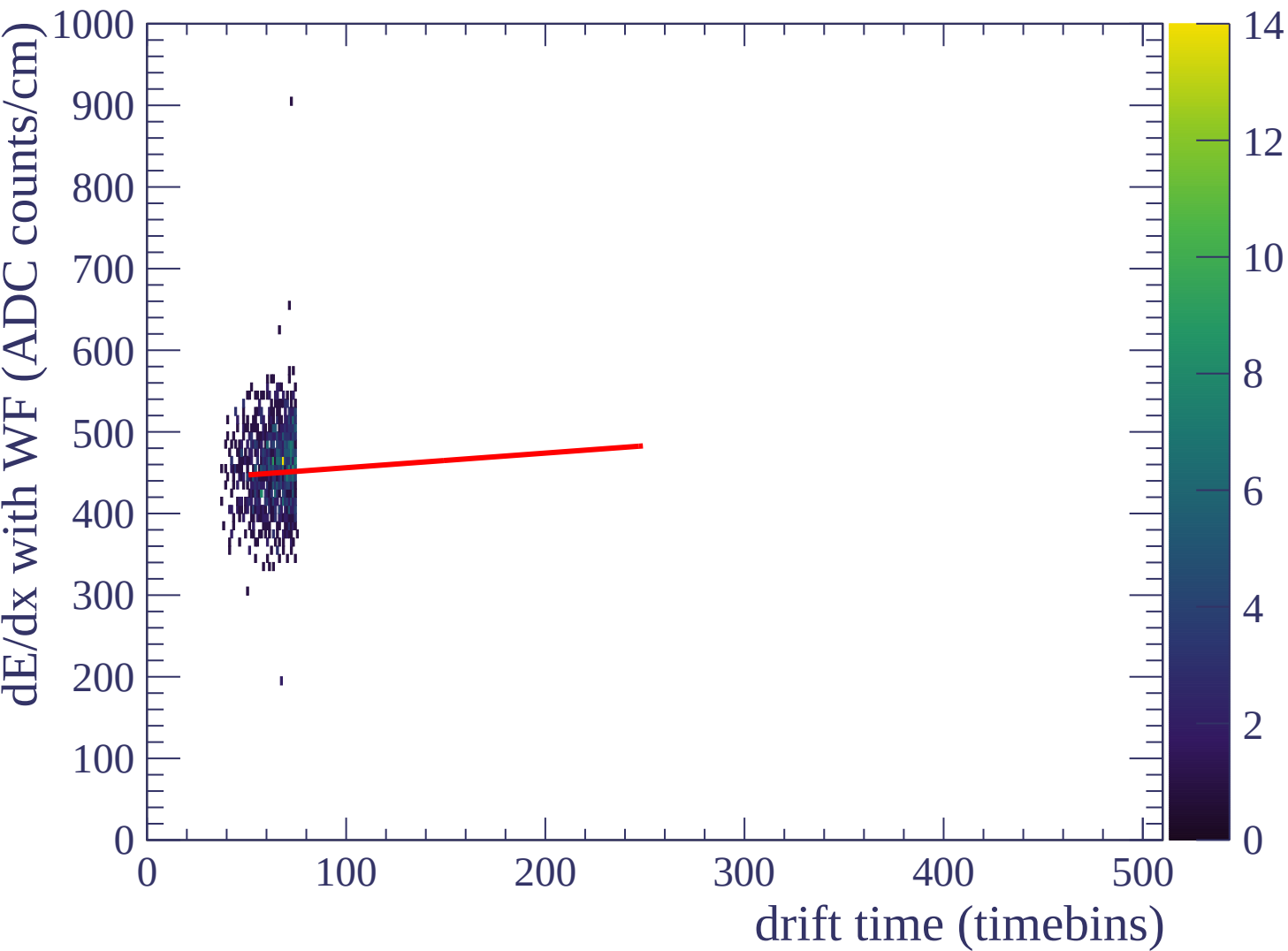


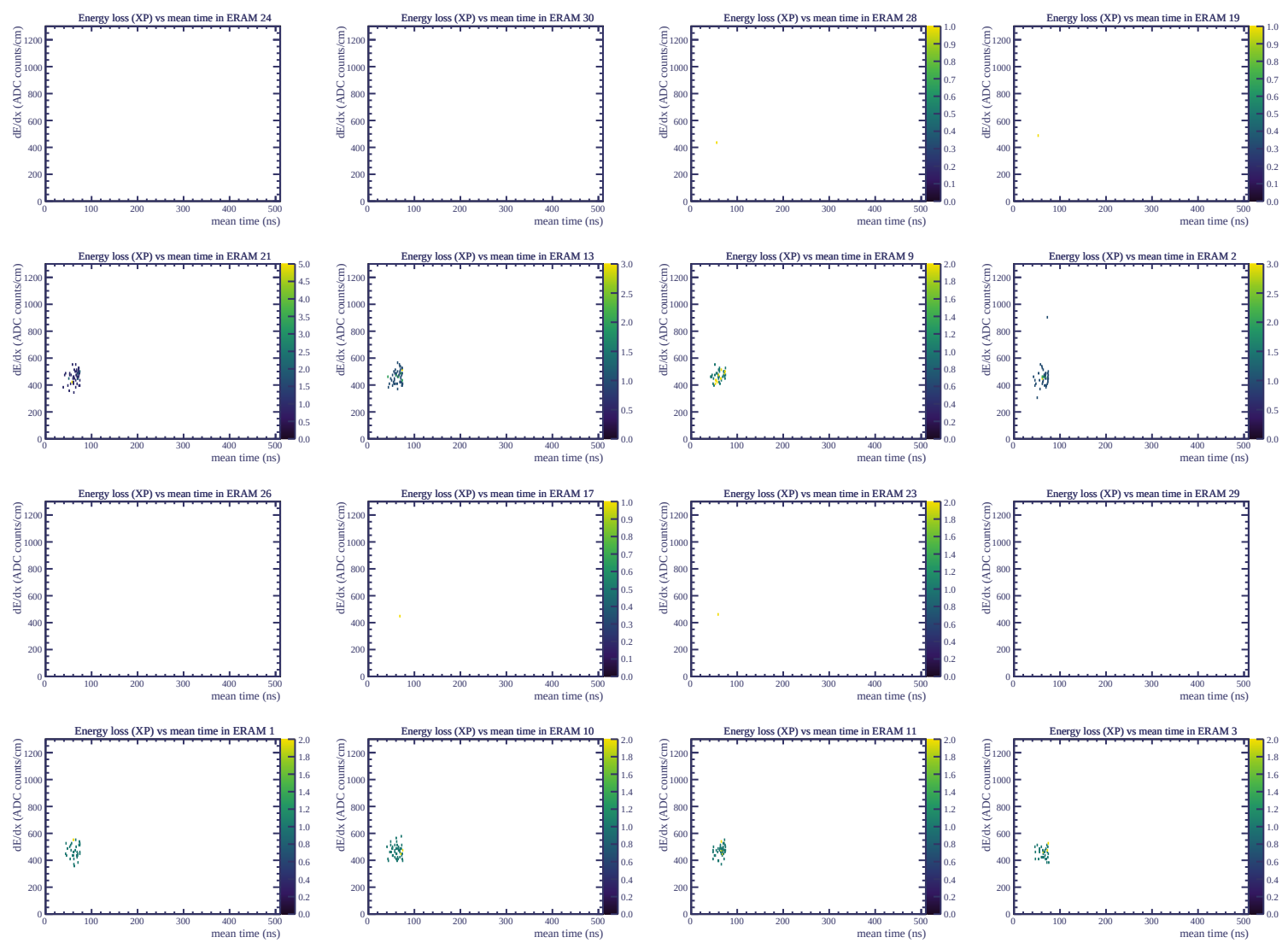


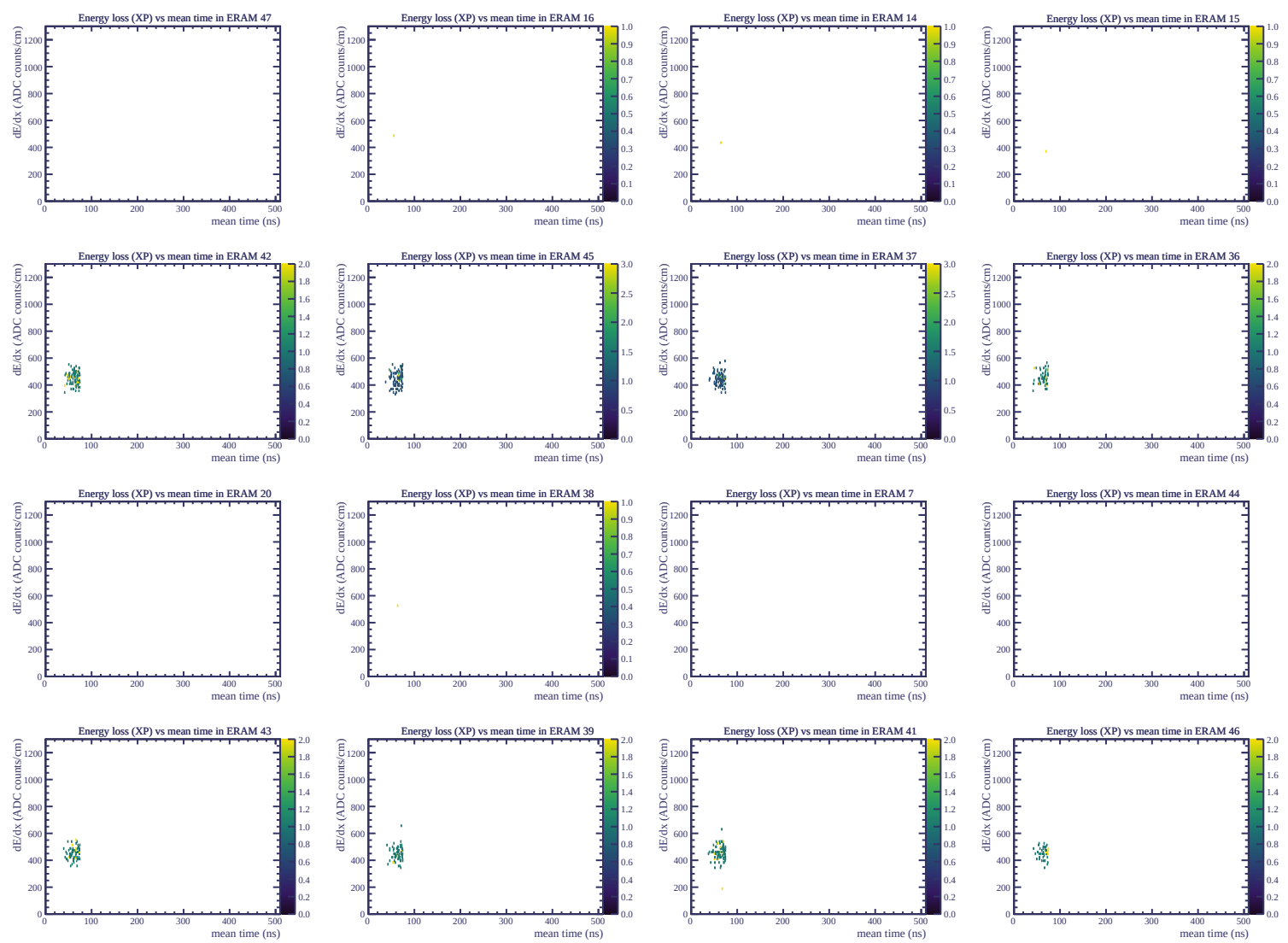












◆ Waveforms sum

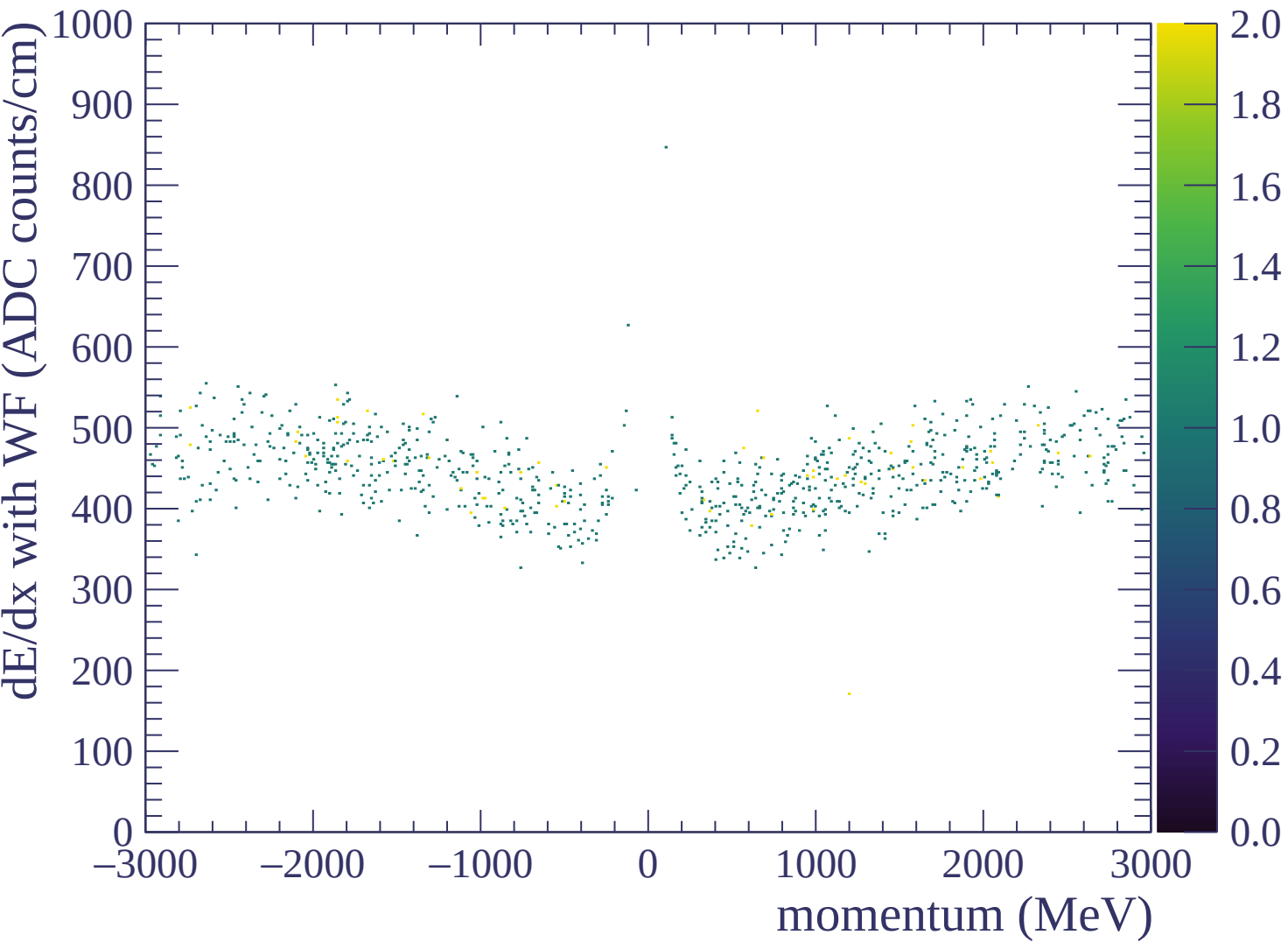
✕ Crossed pads

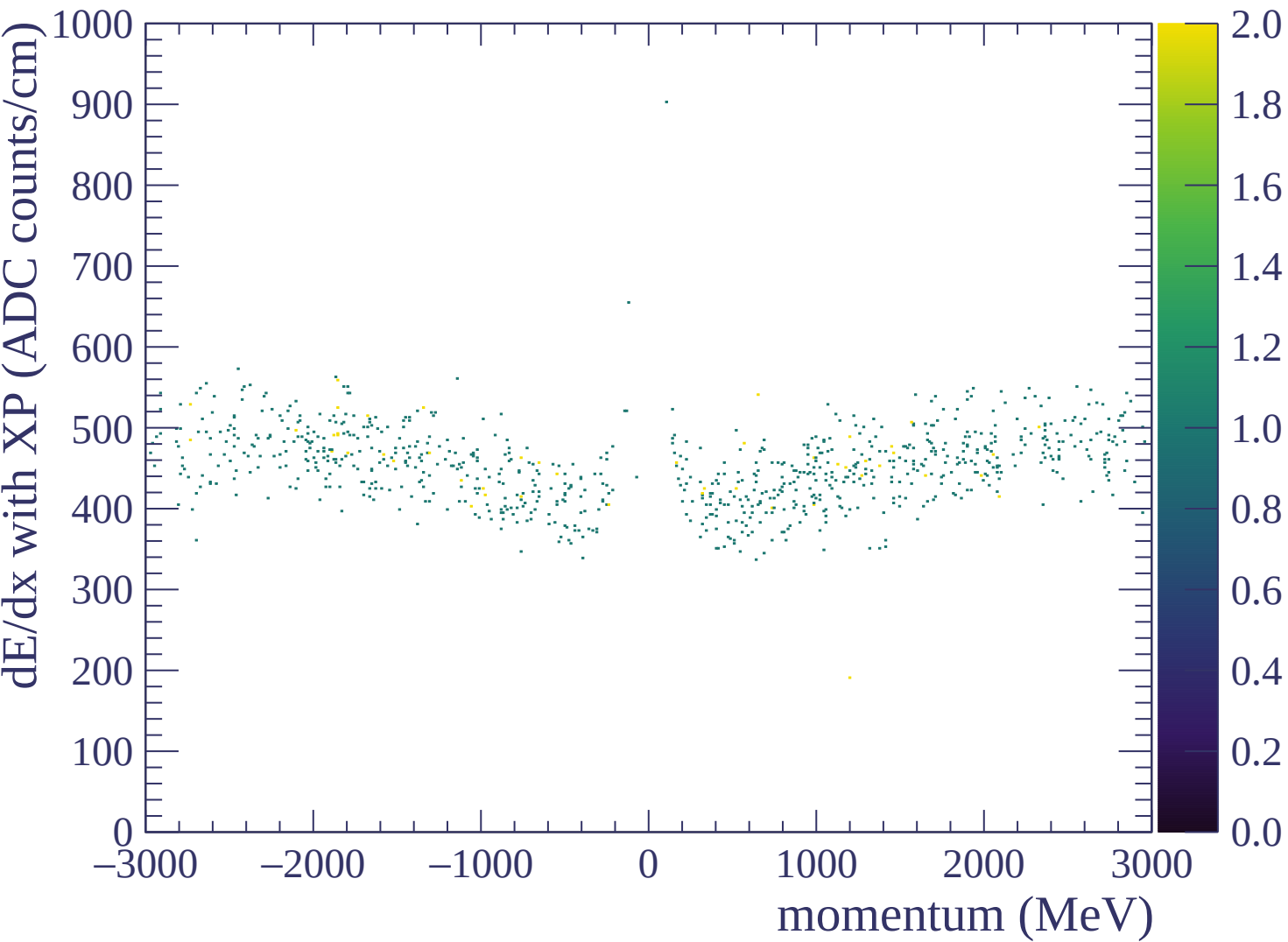
◆ Waveforms sum

✕ Crossed pads

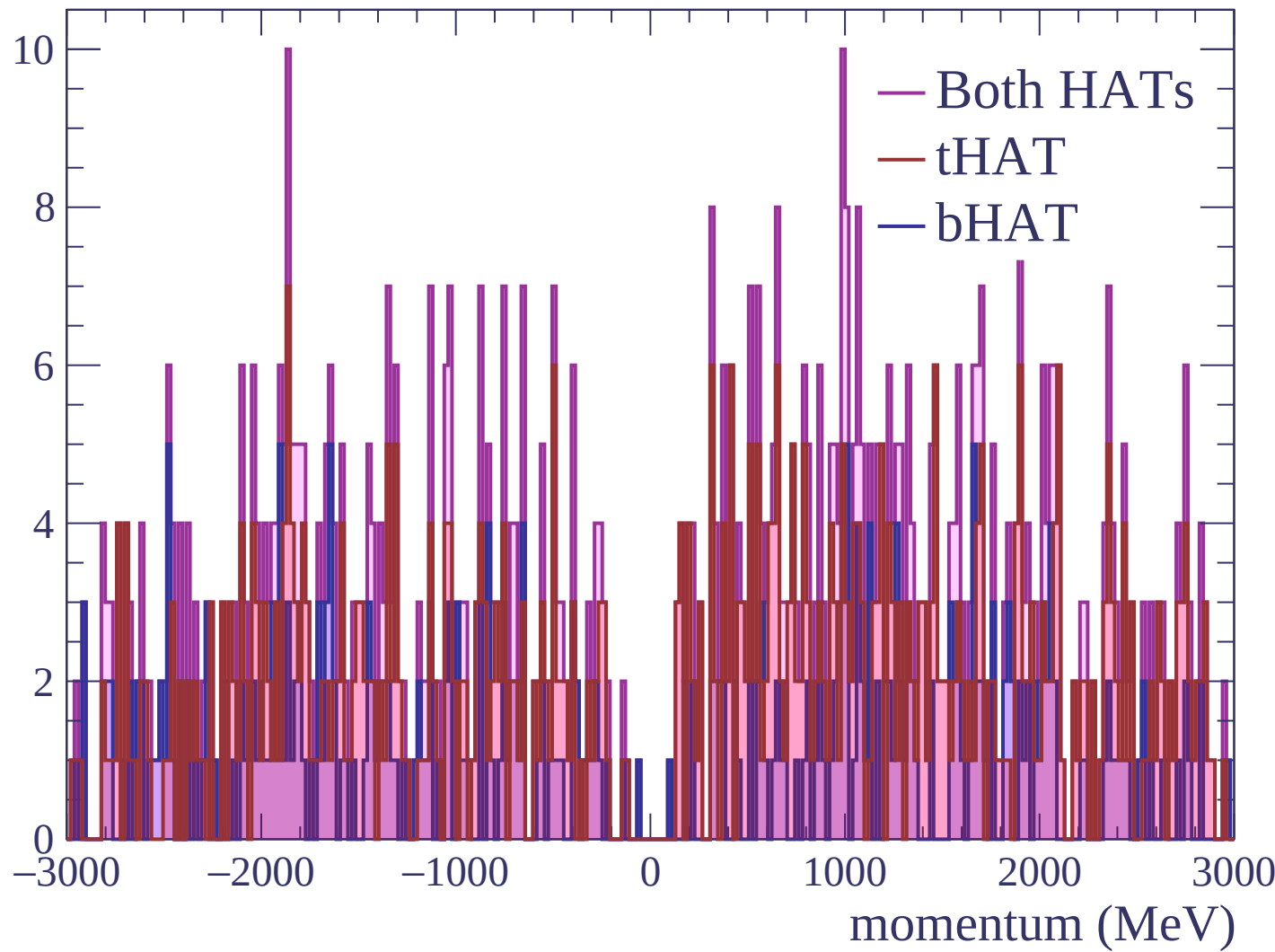
◆ Waveforms sum

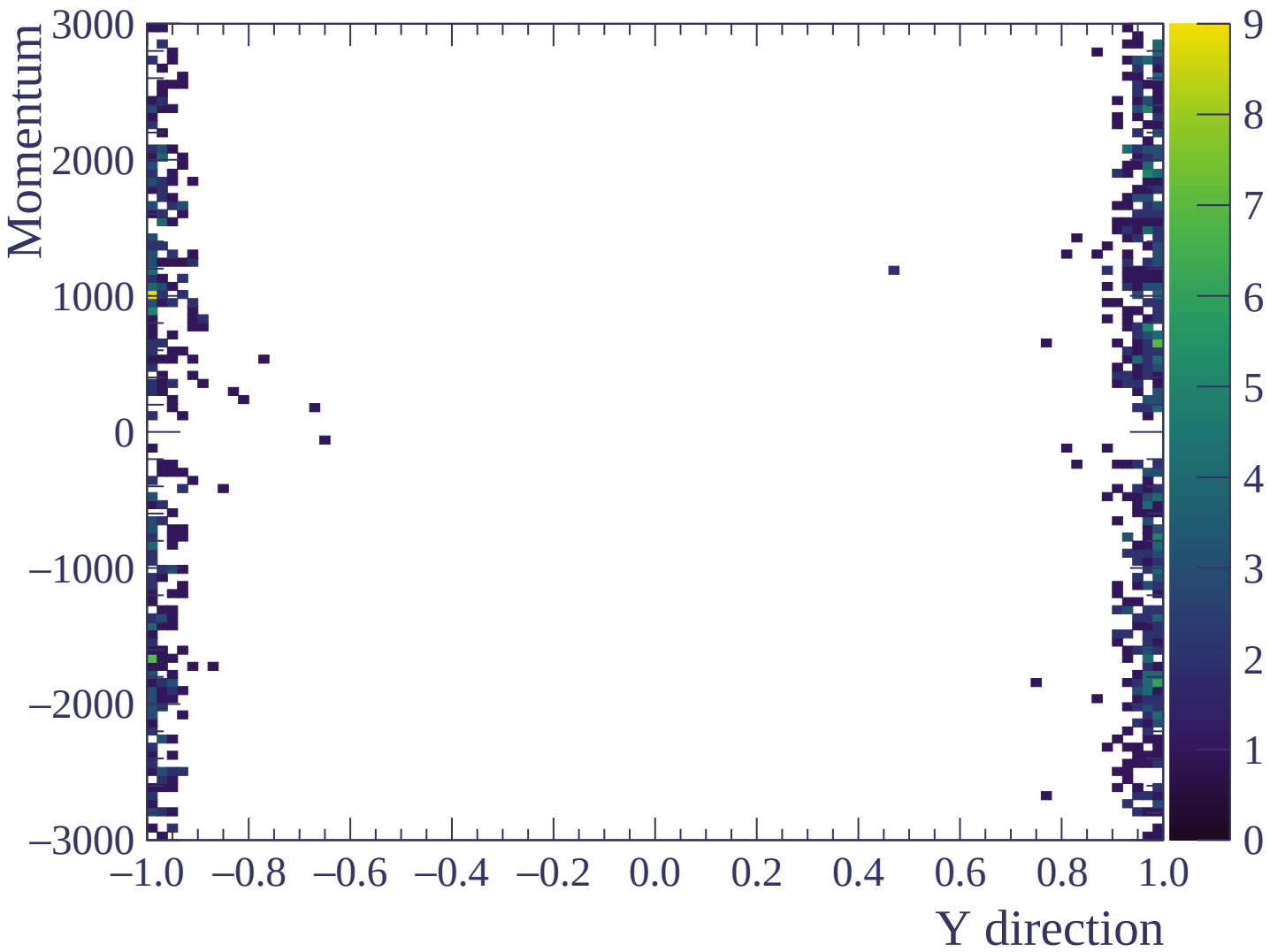
✖ Crossed pads

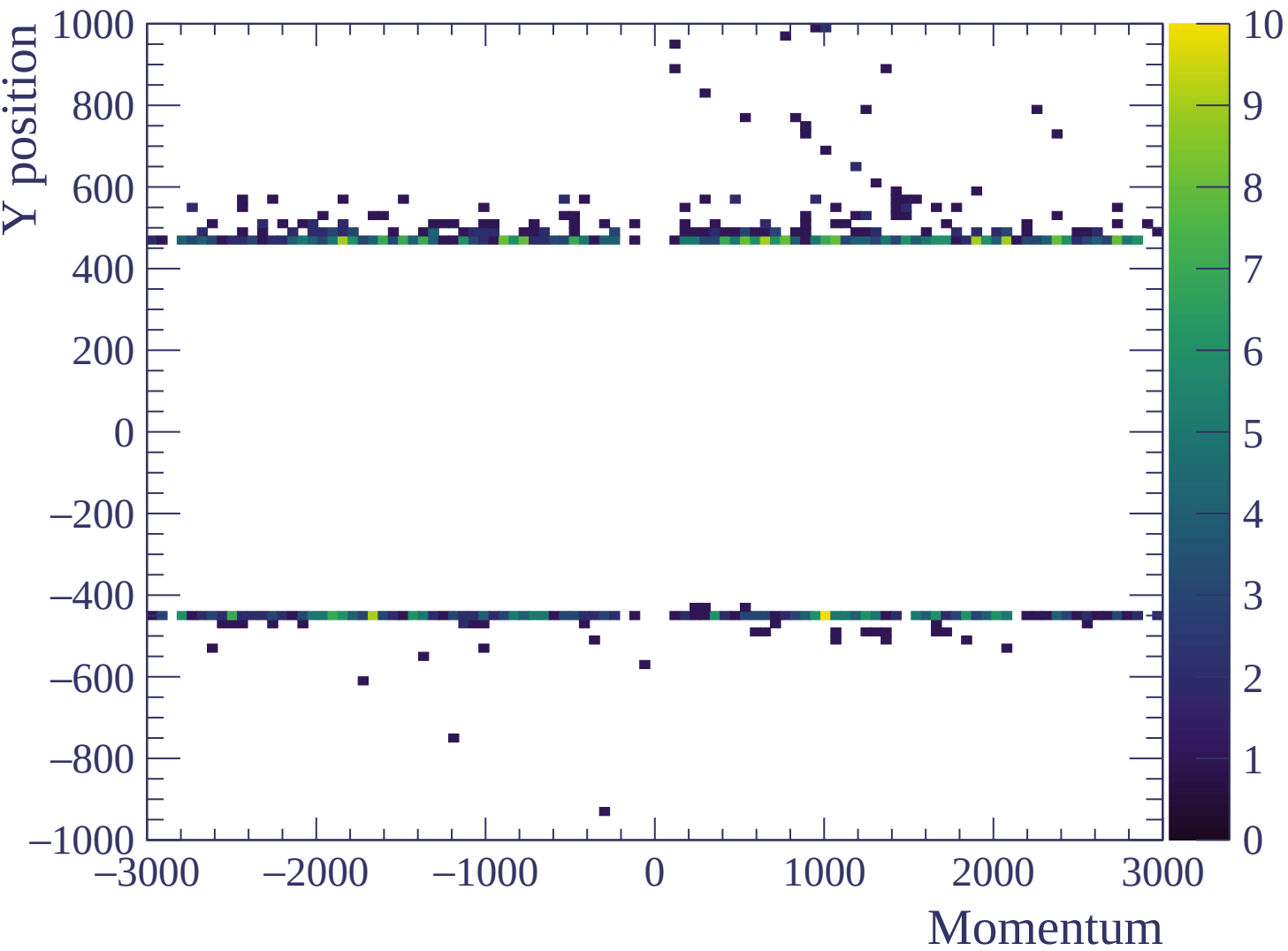


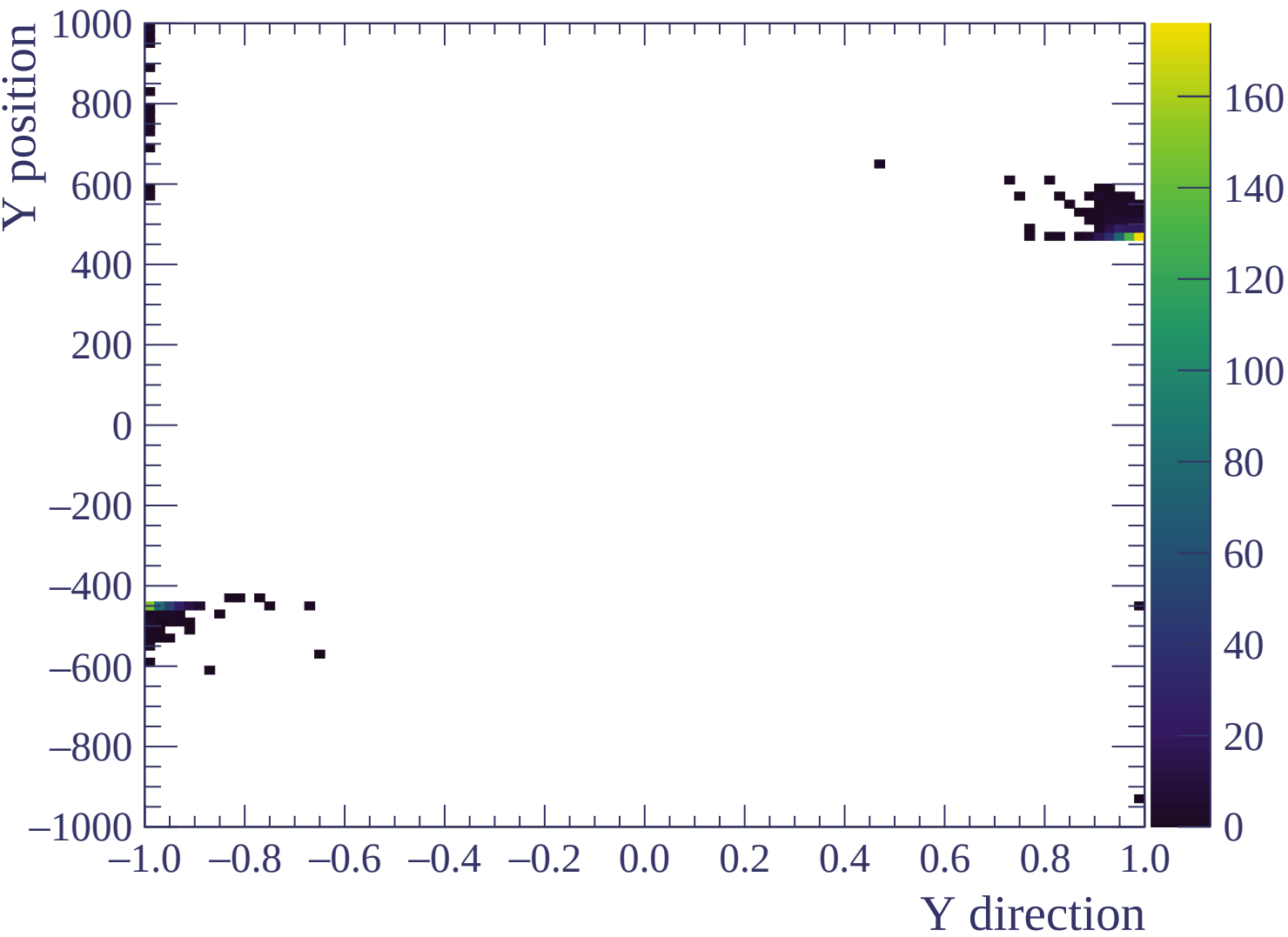


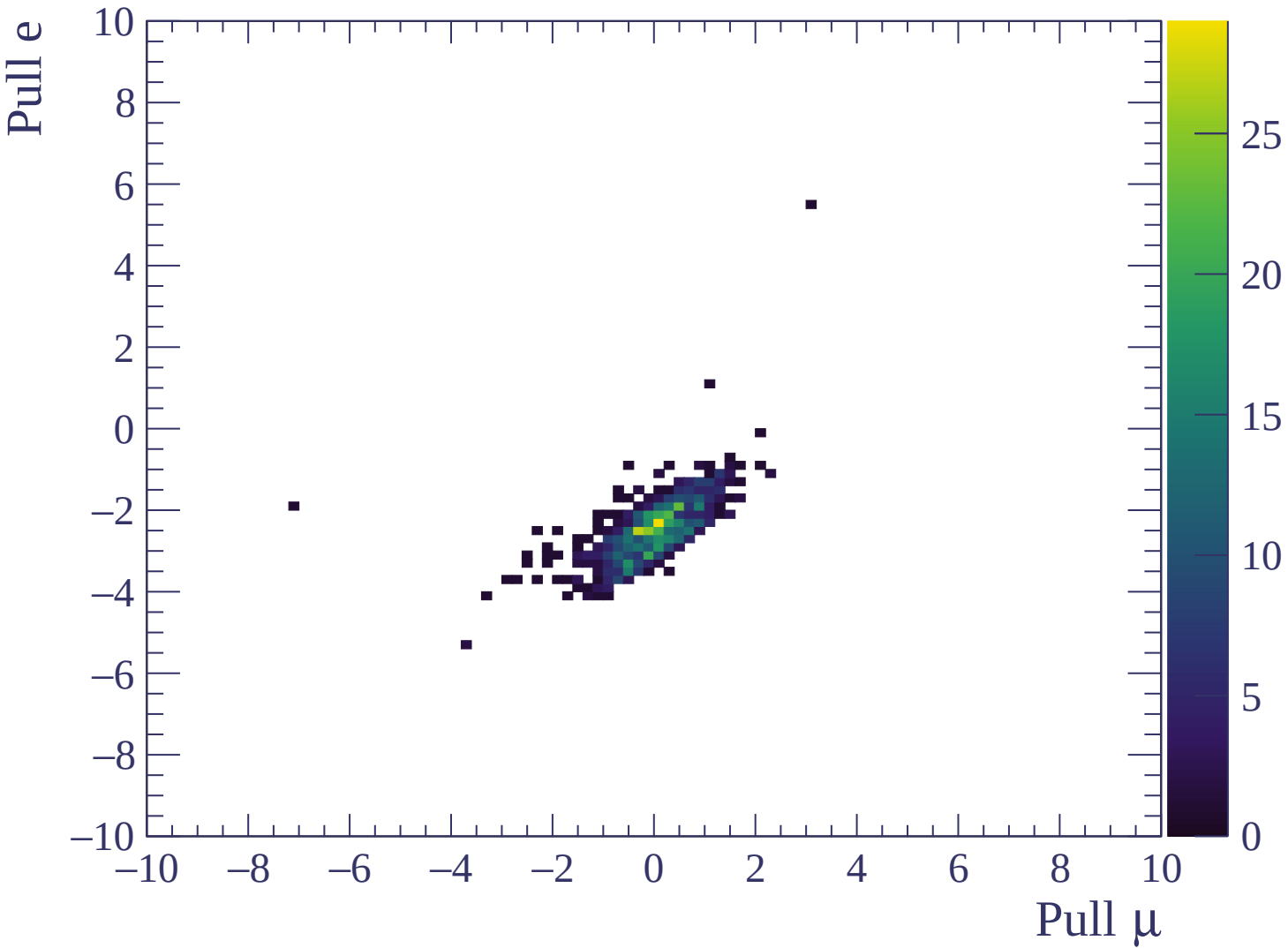
Count











Energy loss | $-3000 < p < -2940$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

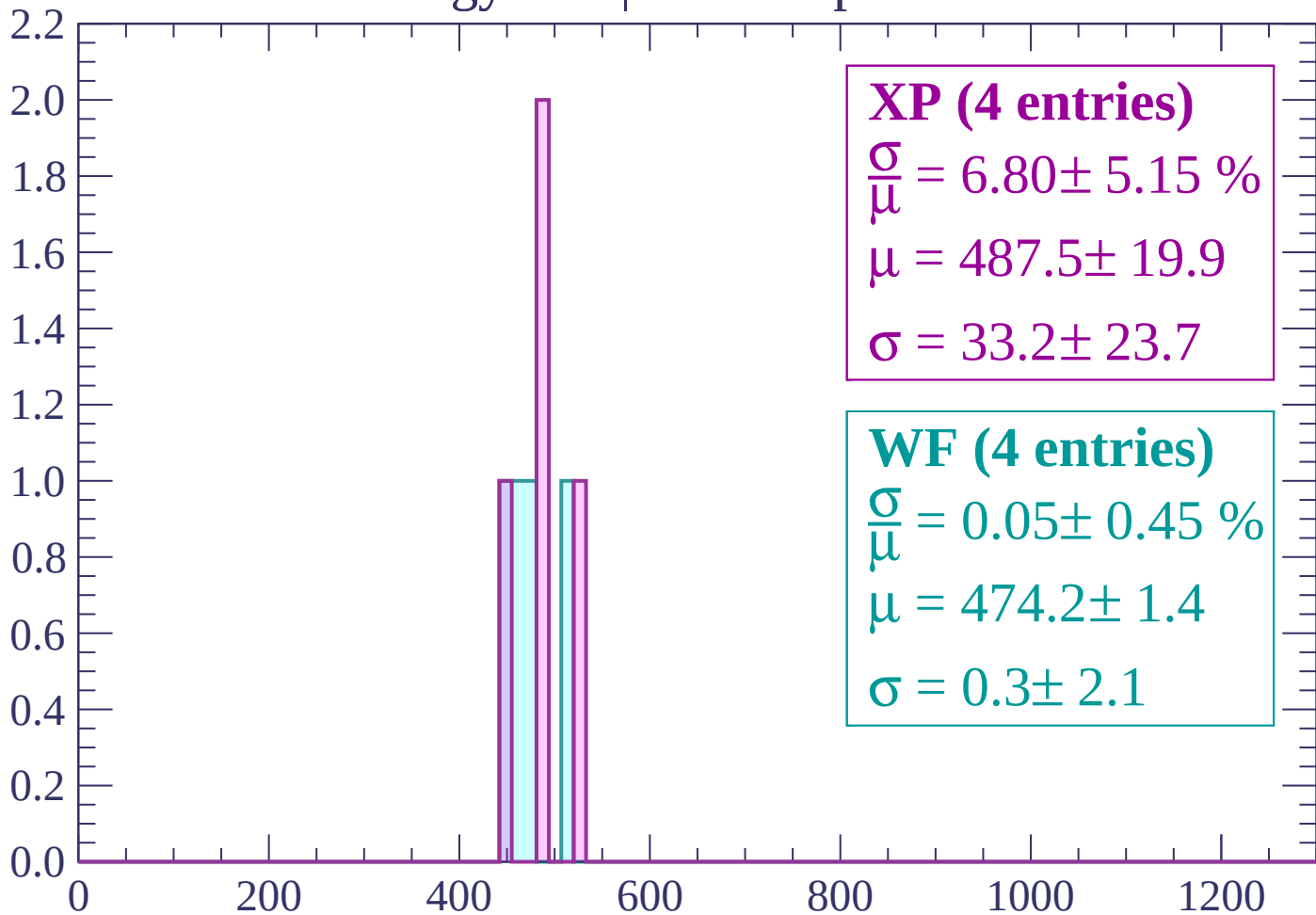
$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2940 < p < -2880$

Count



XP (4 entries)

$\frac{\sigma}{\mu} = 6.80 \pm 5.15 \%$

$\mu = 487.5 \pm 19.9$

$\sigma = 33.2 \pm 23.7$

WF (4 entries)

$\frac{\sigma}{\mu} = 0.05 \pm 0.45 \%$

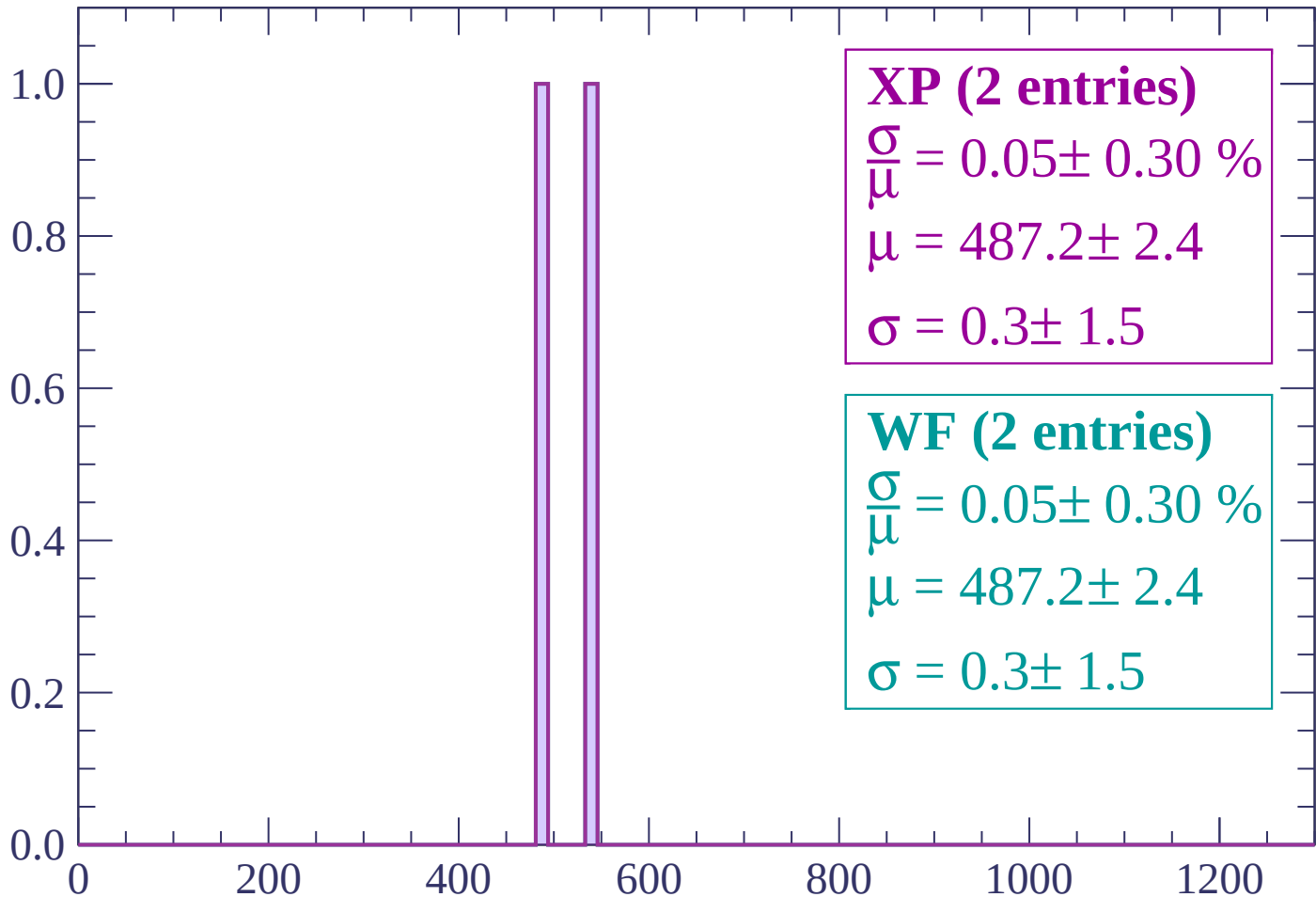
$\mu = 474.2 \pm 1.4$

$\sigma = 0.3 \pm 2.1$

dE/dx (ADC counts/cm)

Energy loss | $-2880 < p < -2820$

Count



XP (2 entries)

$\frac{\sigma}{\mu} = 0.05 \pm 0.30 \%$

$\mu = 487.2 \pm 2.4$

$\sigma = 0.3 \pm 1.5$

WF (2 entries)

$\frac{\sigma}{\mu} = 0.05 \pm 0.30 \%$

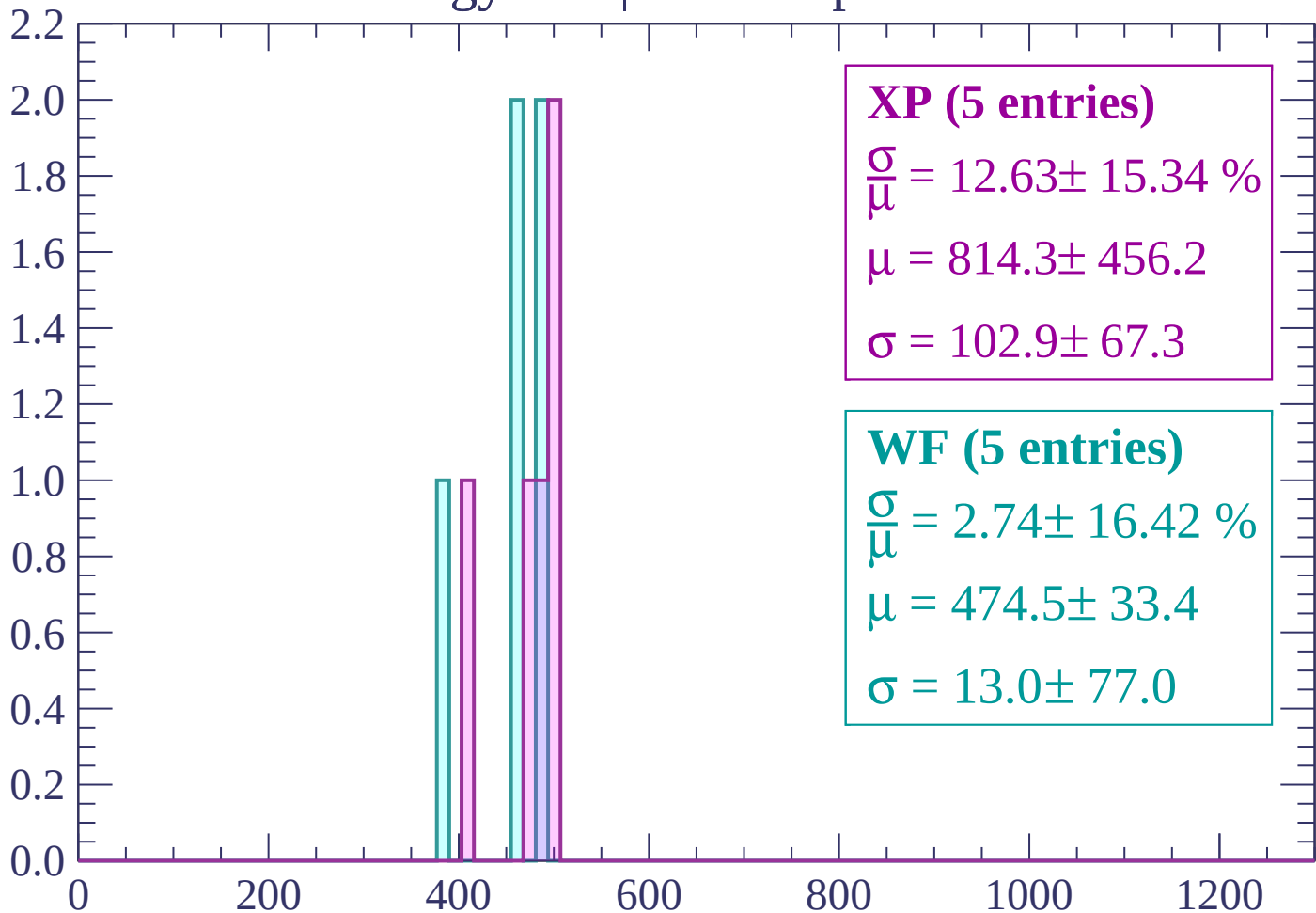
$\mu = 487.2 \pm 2.4$

$\sigma = 0.3 \pm 1.5$

dE/dx (ADC counts/cm)

Energy loss | -2820 < p < -2760

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 12.63 \pm 15.34 \%$$

$$\mu = 814.3 \pm 456.2$$

$$\sigma = 102.9 \pm 67.3$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 2.74 \pm 16.42 \%$$

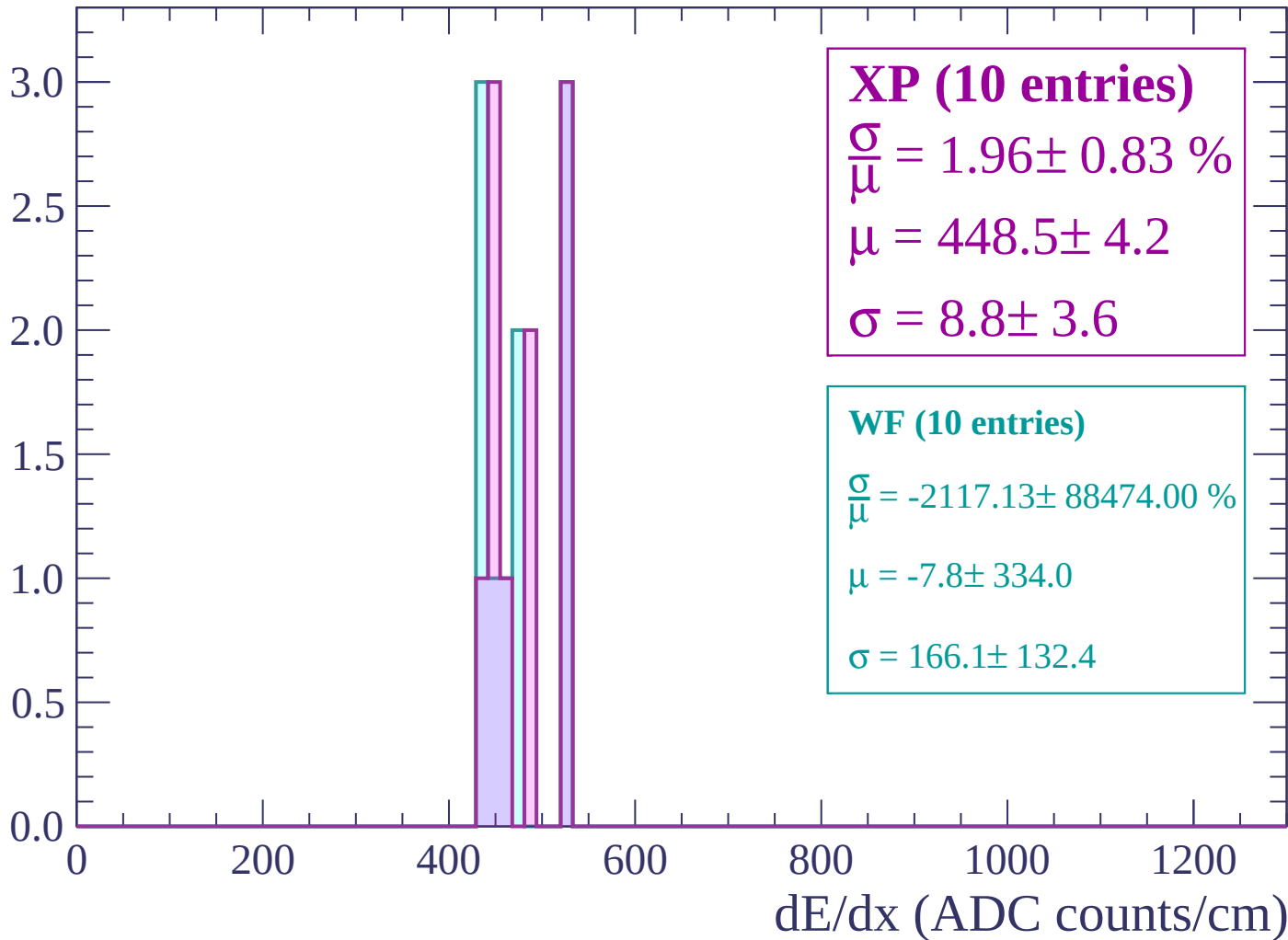
$$\mu = 474.5 \pm 33.4$$

$$\sigma = 13.0 \pm 77.0$$

dE/dx (ADC counts/cm)

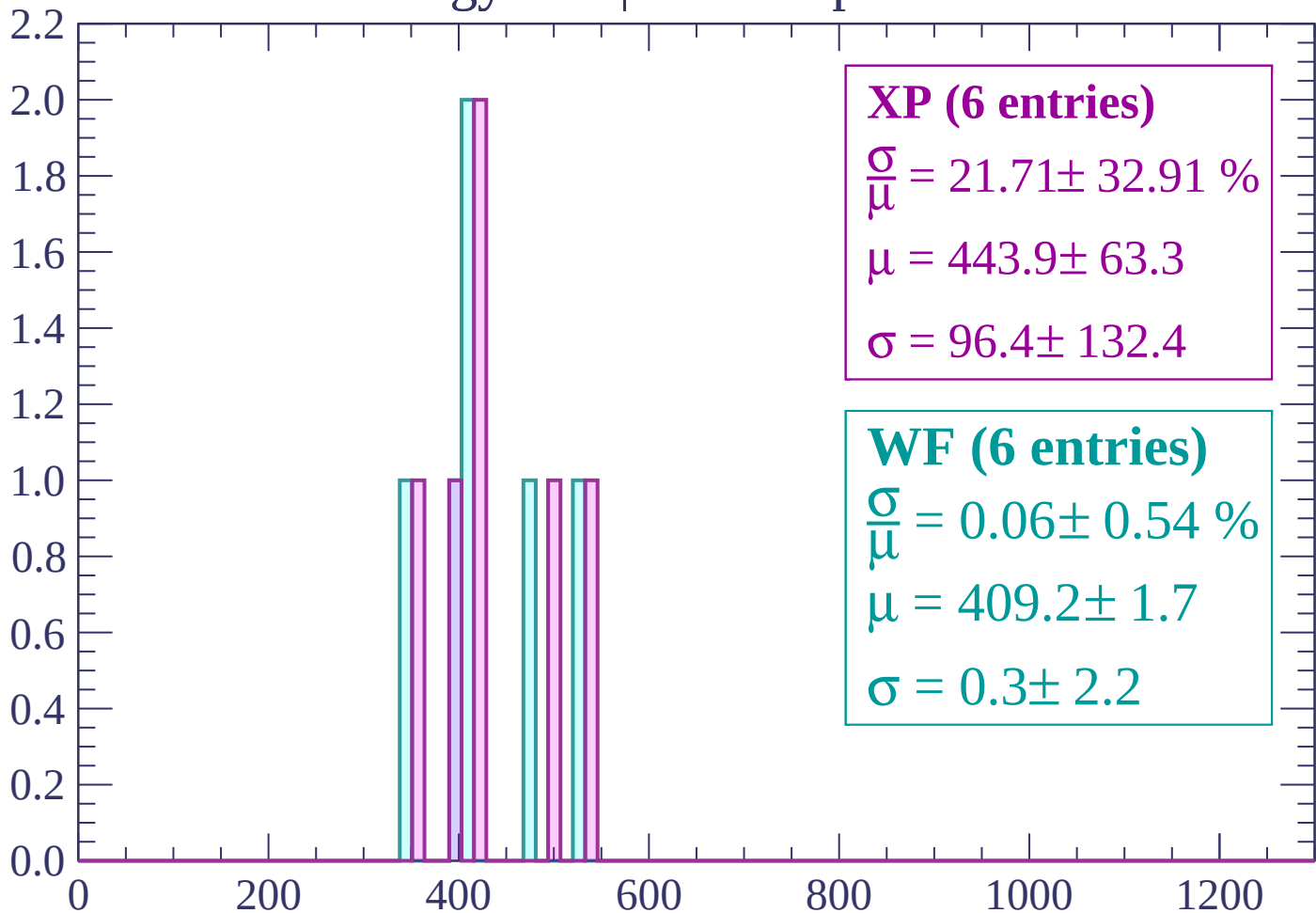
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (6 entries)

$$\frac{\sigma}{\mu} = 21.71 \pm 32.91 \%$$

$$\mu = 443.9 \pm 63.3$$

$$\sigma = 96.4 \pm 132.4$$

WF (6 entries)

$$\frac{\sigma}{\mu} = 0.06 \pm 0.54 \%$$

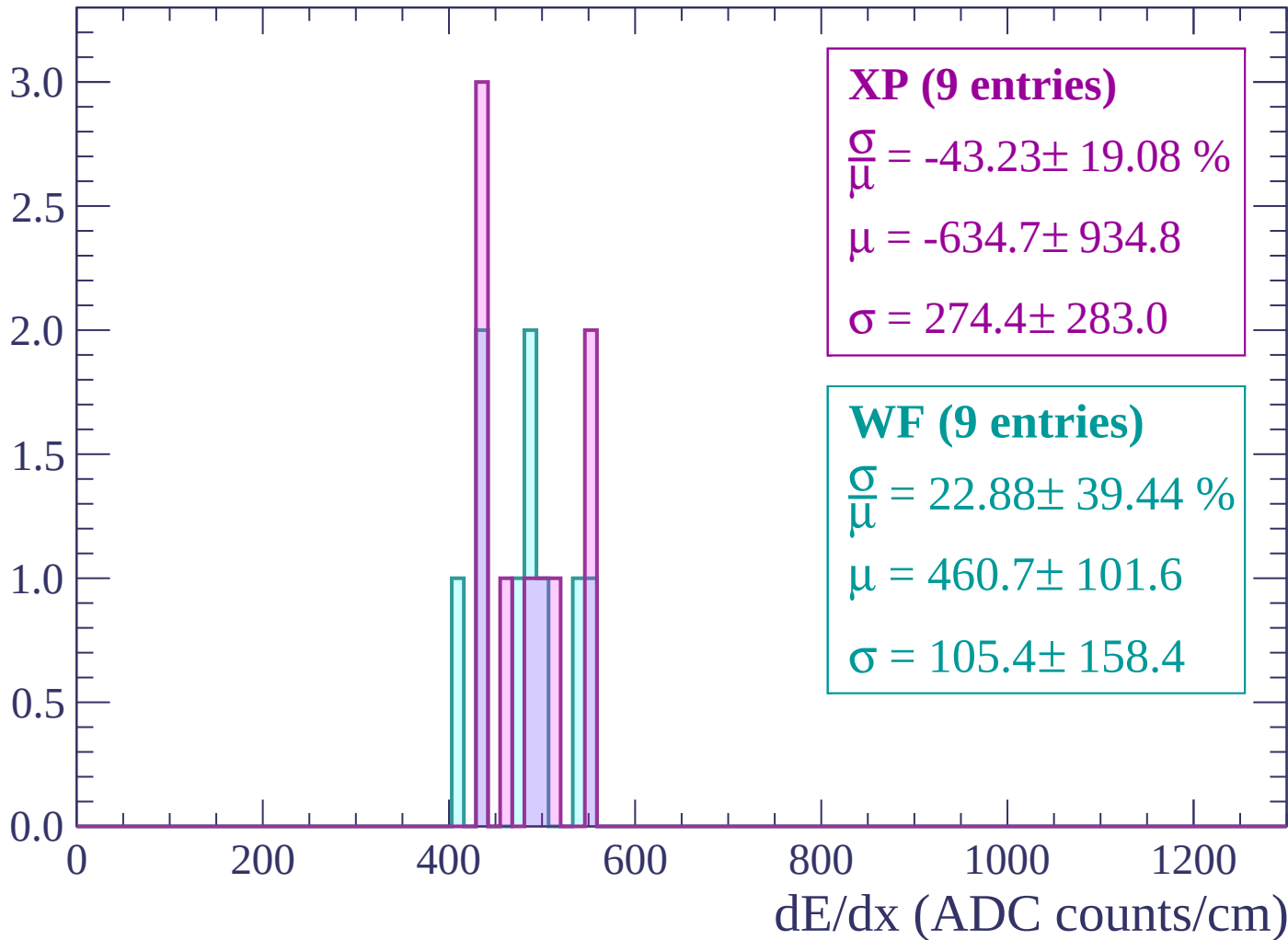
$$\mu = 409.2 \pm 1.7$$

$$\sigma = 0.3 \pm 2.2$$

dE/dx (ADC counts/cm)

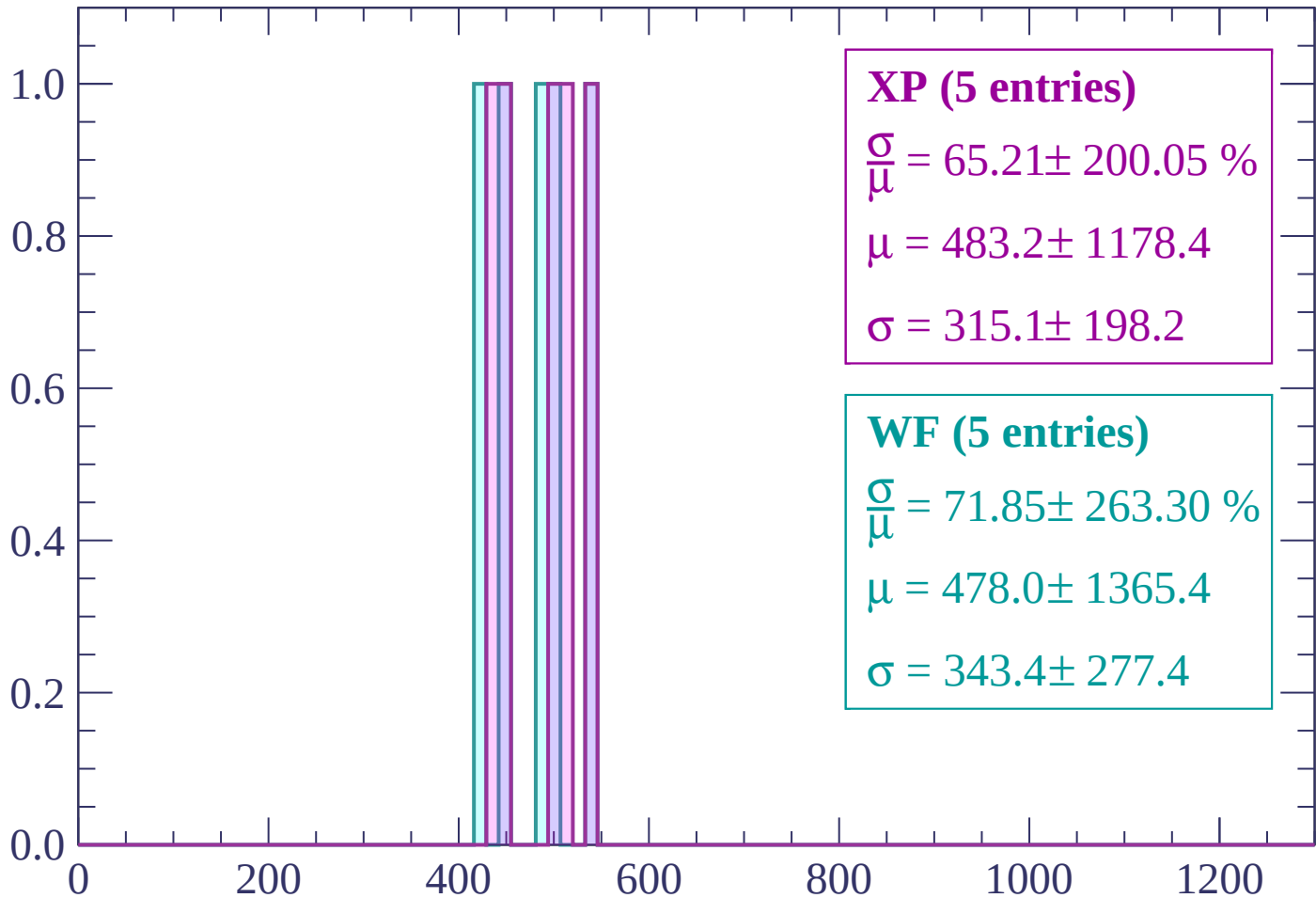
Energy loss | -2640 < p < -2580

Count



Energy loss | -2580 < p < -2520

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 65.21 \pm 200.05 \%$$

$$\mu = 483.2 \pm 1178.4$$

$$\sigma = 315.1 \pm 198.2$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 71.85 \pm 263.30 \%$$

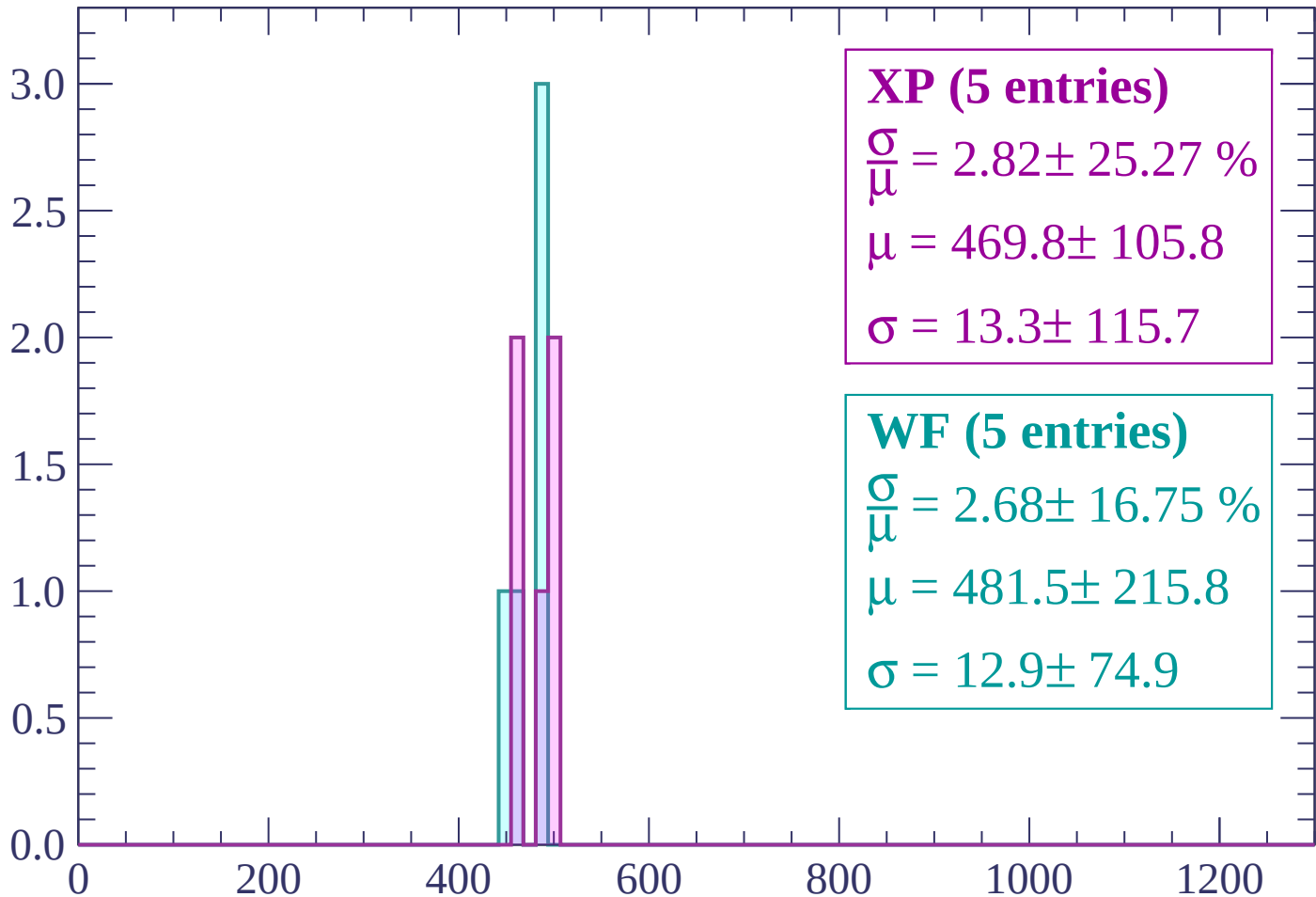
$$\mu = 478.0 \pm 1365.4$$

$$\sigma = 343.4 \pm 277.4$$

dE/dx (ADC counts/cm)

Energy loss | -2520 < p < -2460

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 2.82 \pm 25.27 \%$$

$$\mu = 469.8 \pm 105.8$$

$$\sigma = 13.3 \pm 115.7$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 2.68 \pm 16.75 \%$$

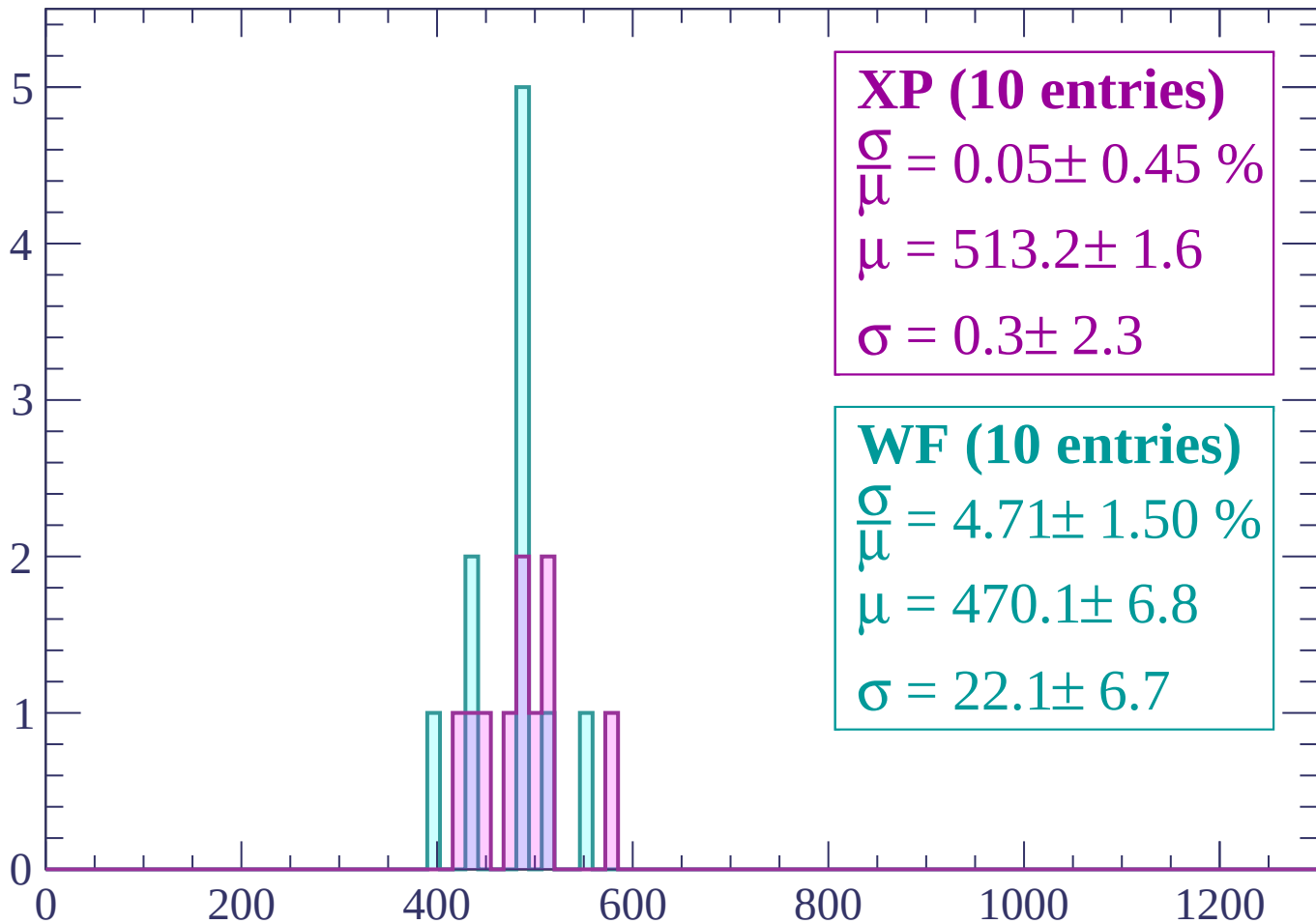
$$\mu = 481.5 \pm 215.8$$

$$\sigma = 12.9 \pm 74.9$$

dE/dx (ADC counts/cm)

Energy loss | $-2460 < p < -2400$

Count



XP (10 entries)

$\frac{\sigma}{\mu} = 0.05 \pm 0.45 \%$

$\mu = 513.2 \pm 1.6$

$\sigma = 0.3 \pm 2.3$

WF (10 entries)

$\frac{\sigma}{\mu} = 4.71 \pm 1.50 \%$

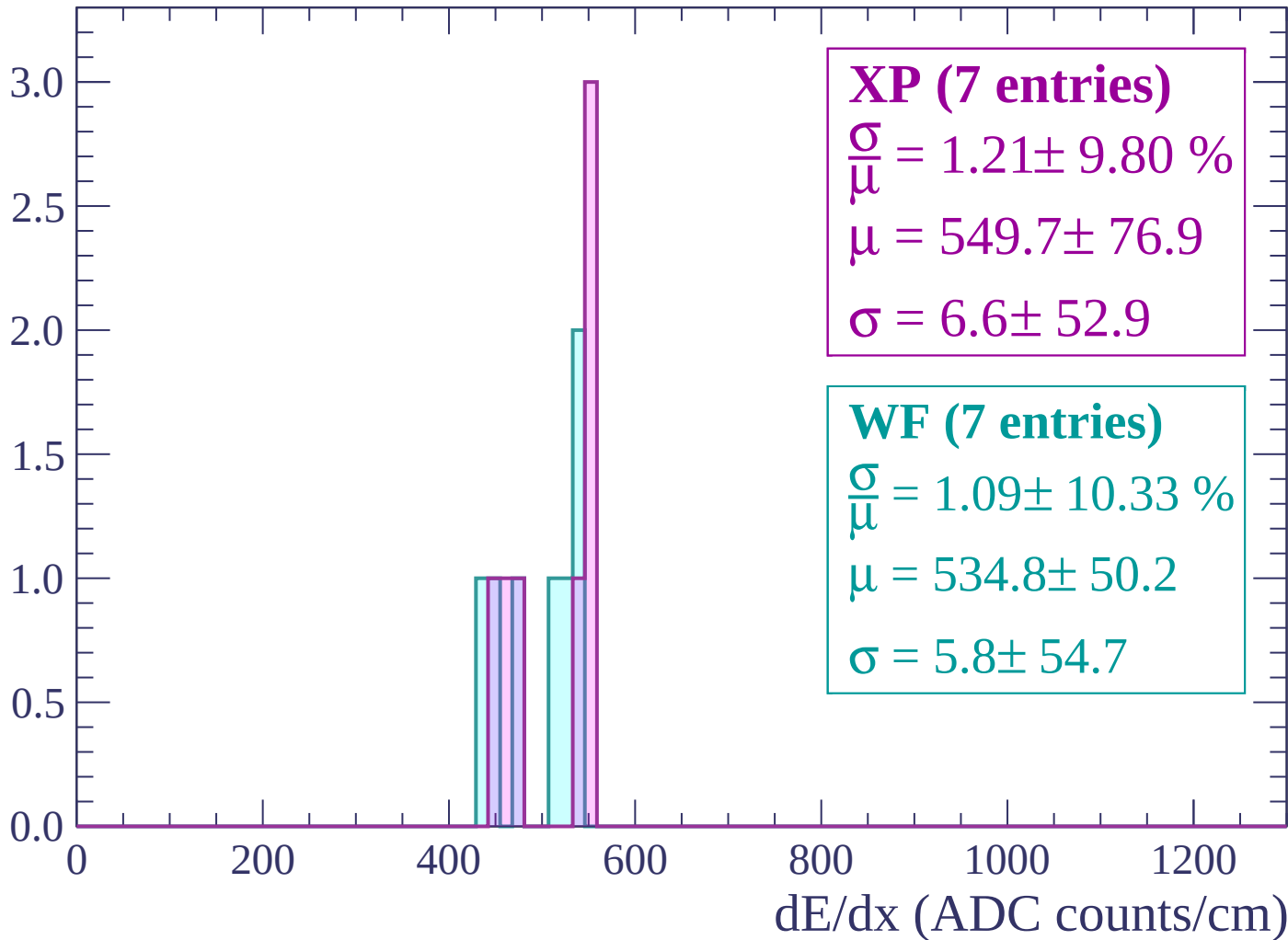
$\mu = 470.1 \pm 6.8$

$\sigma = 22.1 \pm 6.7$

dE/dx (ADC counts/cm)

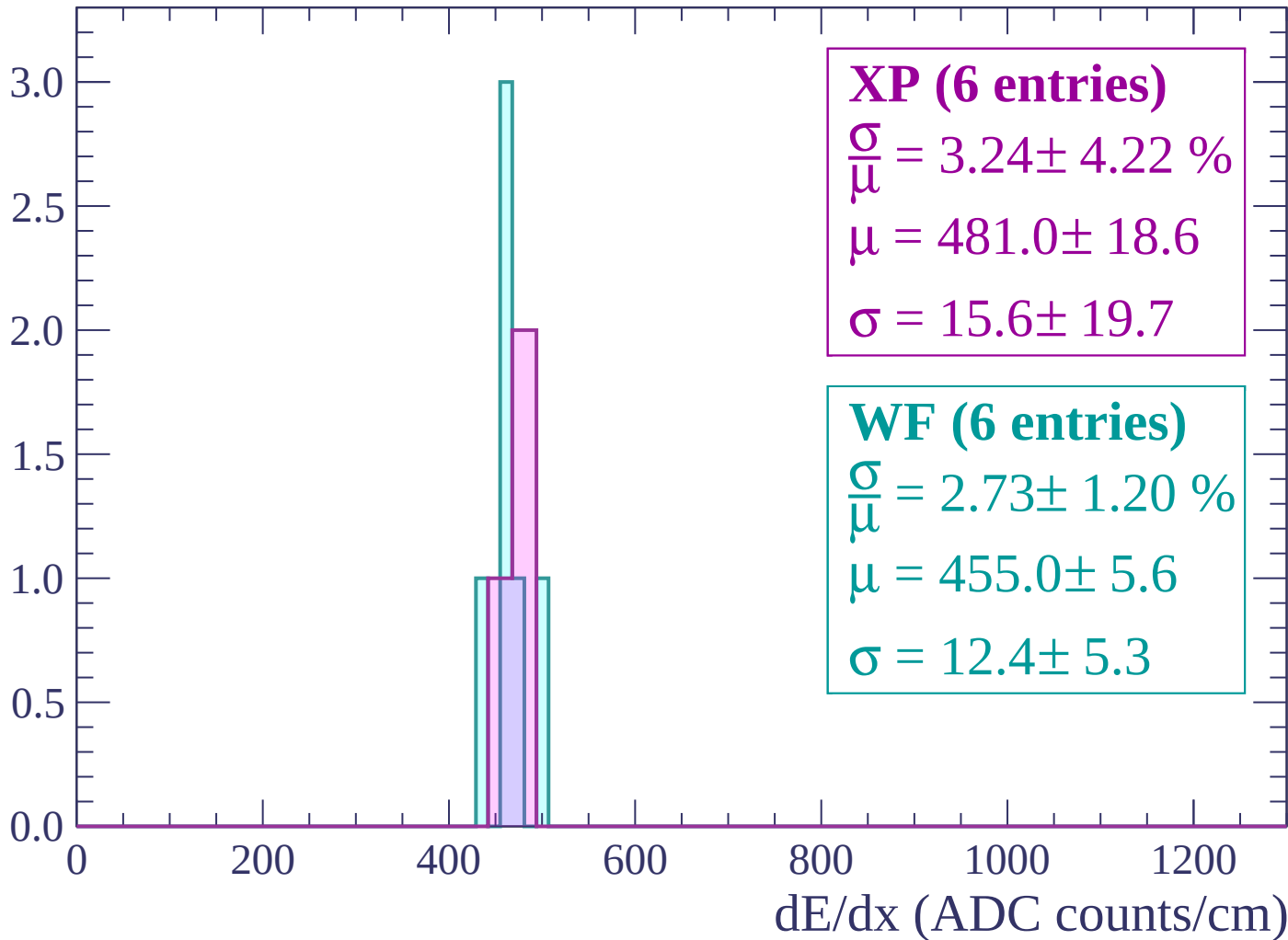
Energy loss | -2400 < p < -2340

Count



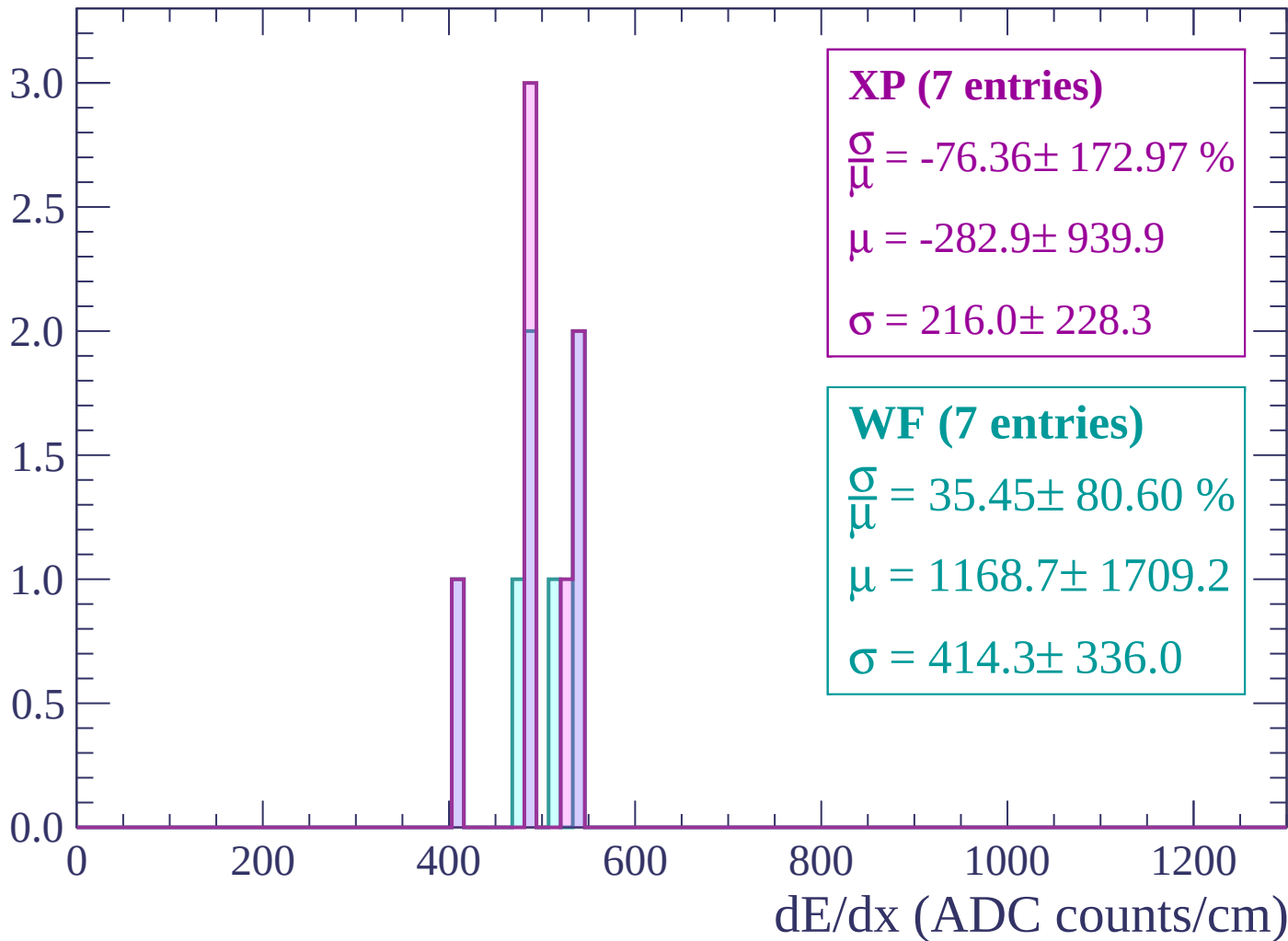
Energy loss | $-2340 < p < -2280$

Count



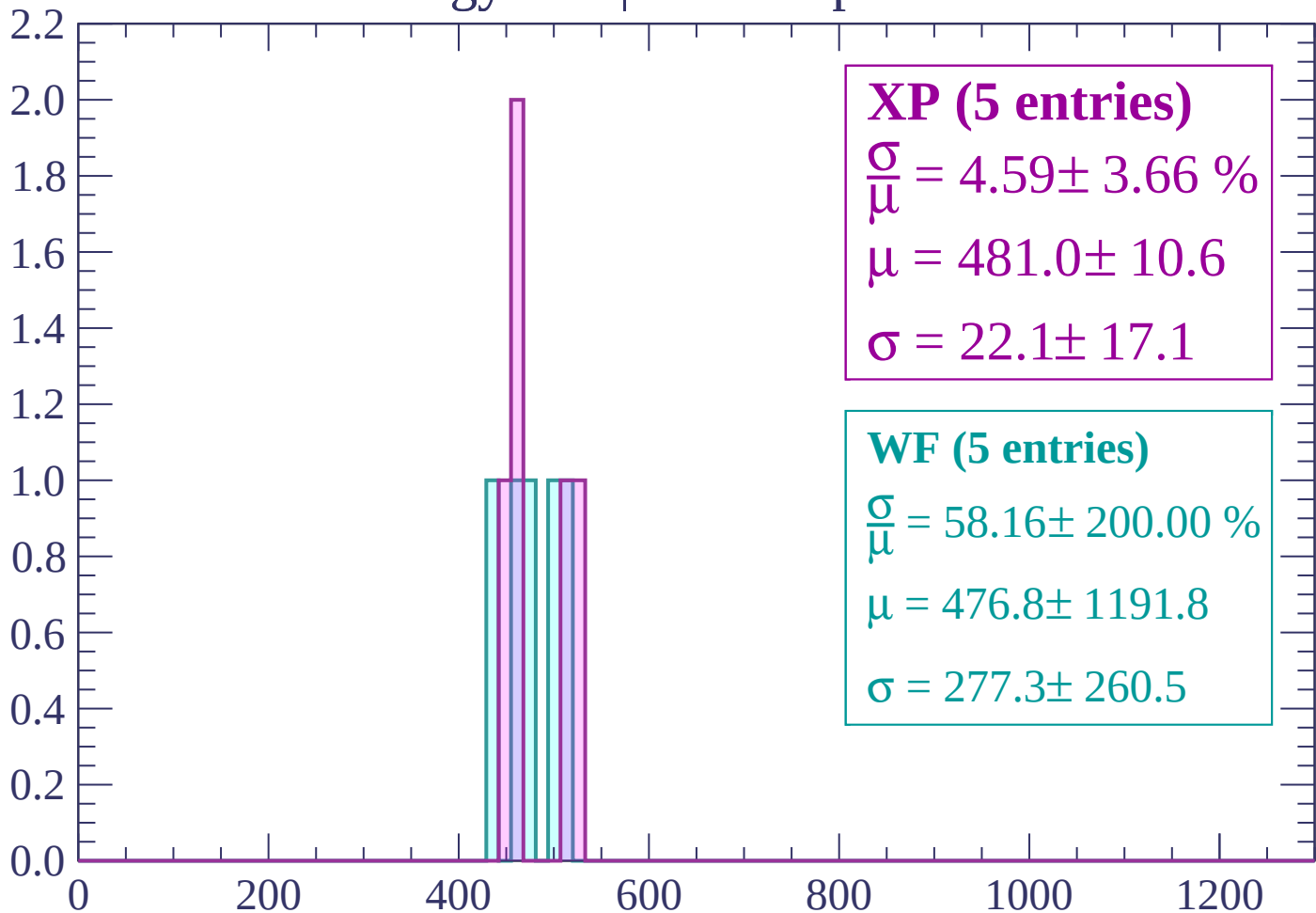
Energy loss | -2280 < p < -2220

Count



Energy loss | -2220 < p < -2160

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 4.59 \pm 3.66 \%$$

$$\mu = 481.0 \pm 10.6$$

$$\sigma = 22.1 \pm 17.1$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 58.16 \pm 200.00 \%$$

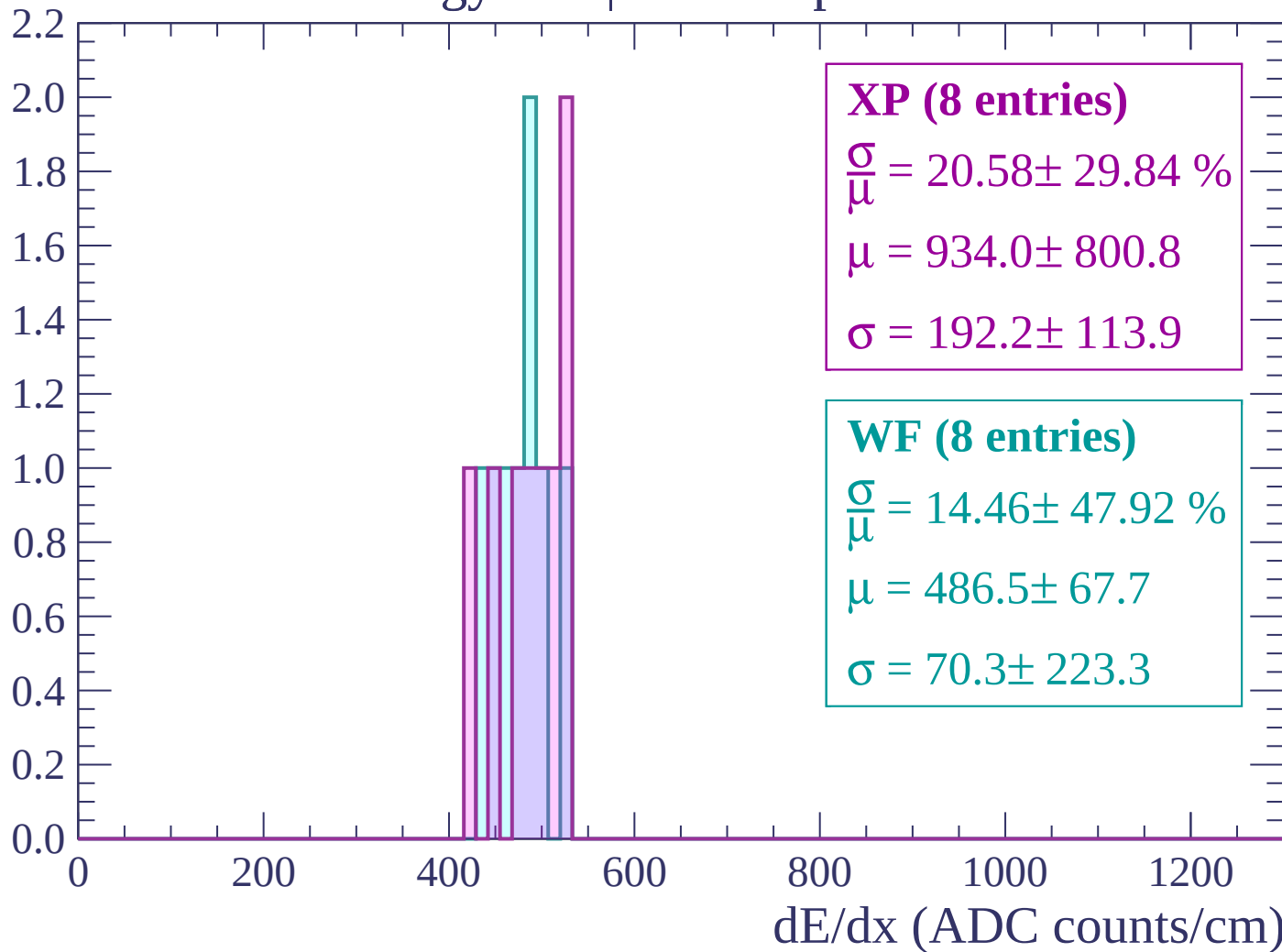
$$\mu = 476.8 \pm 1191.8$$

$$\sigma = 277.3 \pm 260.5$$

dE/dx (ADC counts/cm)

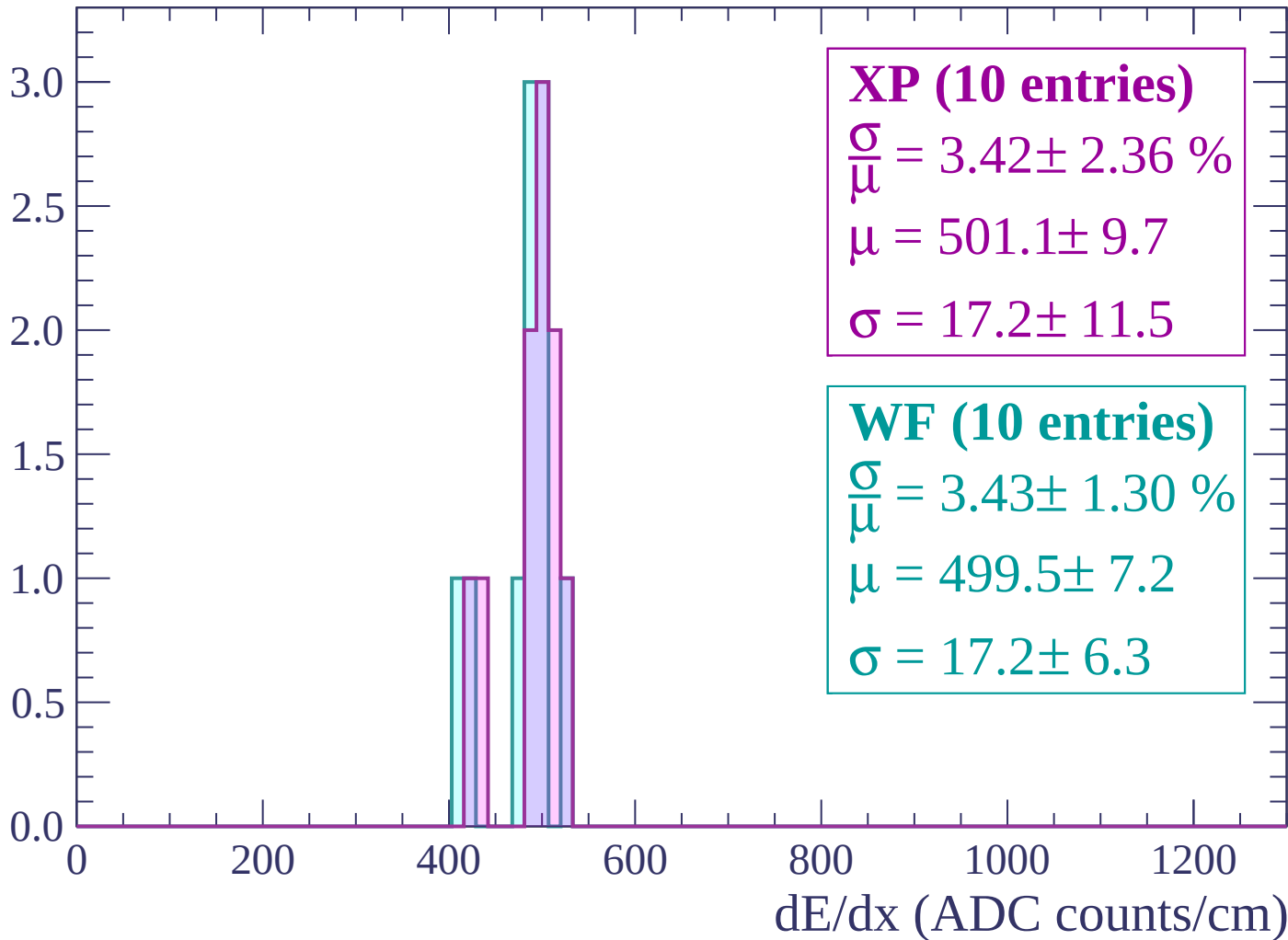
Energy loss | -2160 < p < -2100

Count



Energy loss | -2100 < p < -2040

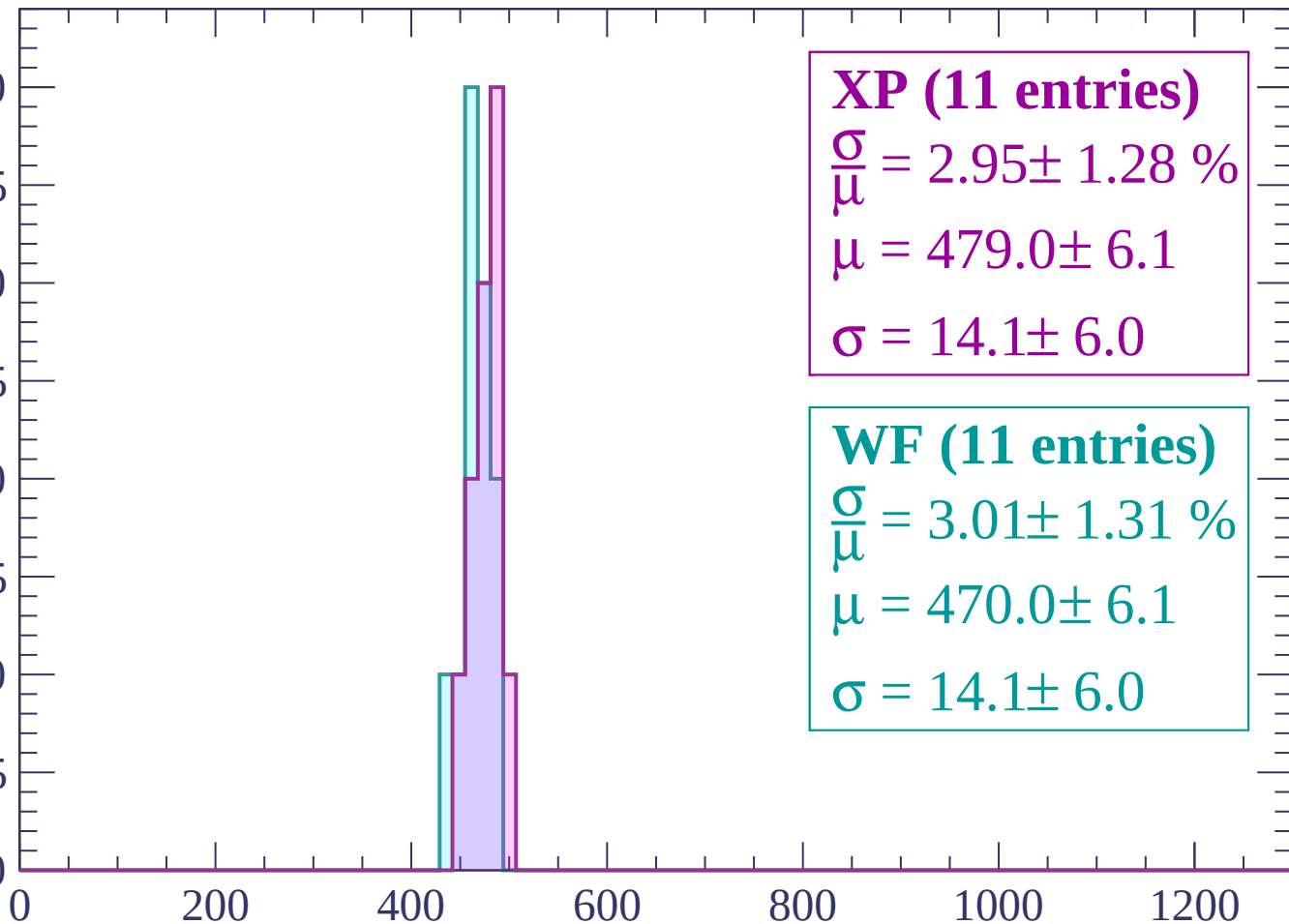
Count



Energy loss | -2040 < p < -1980

Count

4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0



XP (11 entries)

$\frac{\sigma}{\mu} = 2.95 \pm 1.28 \%$

$\mu = 479.0 \pm 6.1$

$\sigma = 14.1 \pm 6.0$

WF (11 entries)

$\frac{\sigma}{\mu} = 3.01 \pm 1.31 \%$

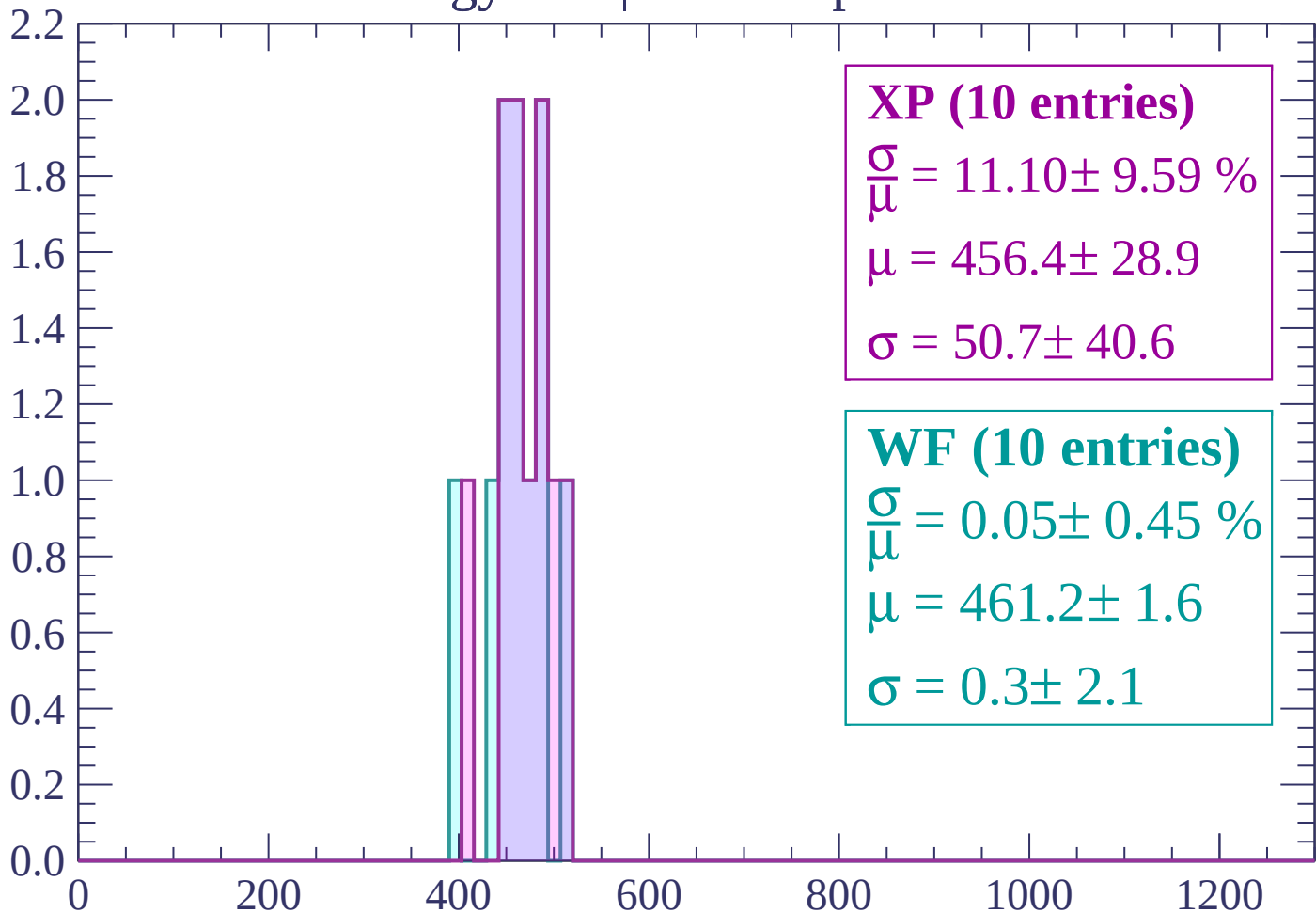
$\mu = 470.0 \pm 6.1$

$\sigma = 14.1 \pm 6.0$

dE/dx (ADC counts/cm)

Energy loss | -1980 < p < -1920

Count



XP (10 entries)

$$\frac{\sigma}{\mu} = 11.10 \pm 9.59 \%$$

$$\mu = 456.4 \pm 28.9$$

$$\sigma = 50.7 \pm 40.6$$

WF (10 entries)

$$\frac{\sigma}{\mu} = 0.05 \pm 0.45 \%$$

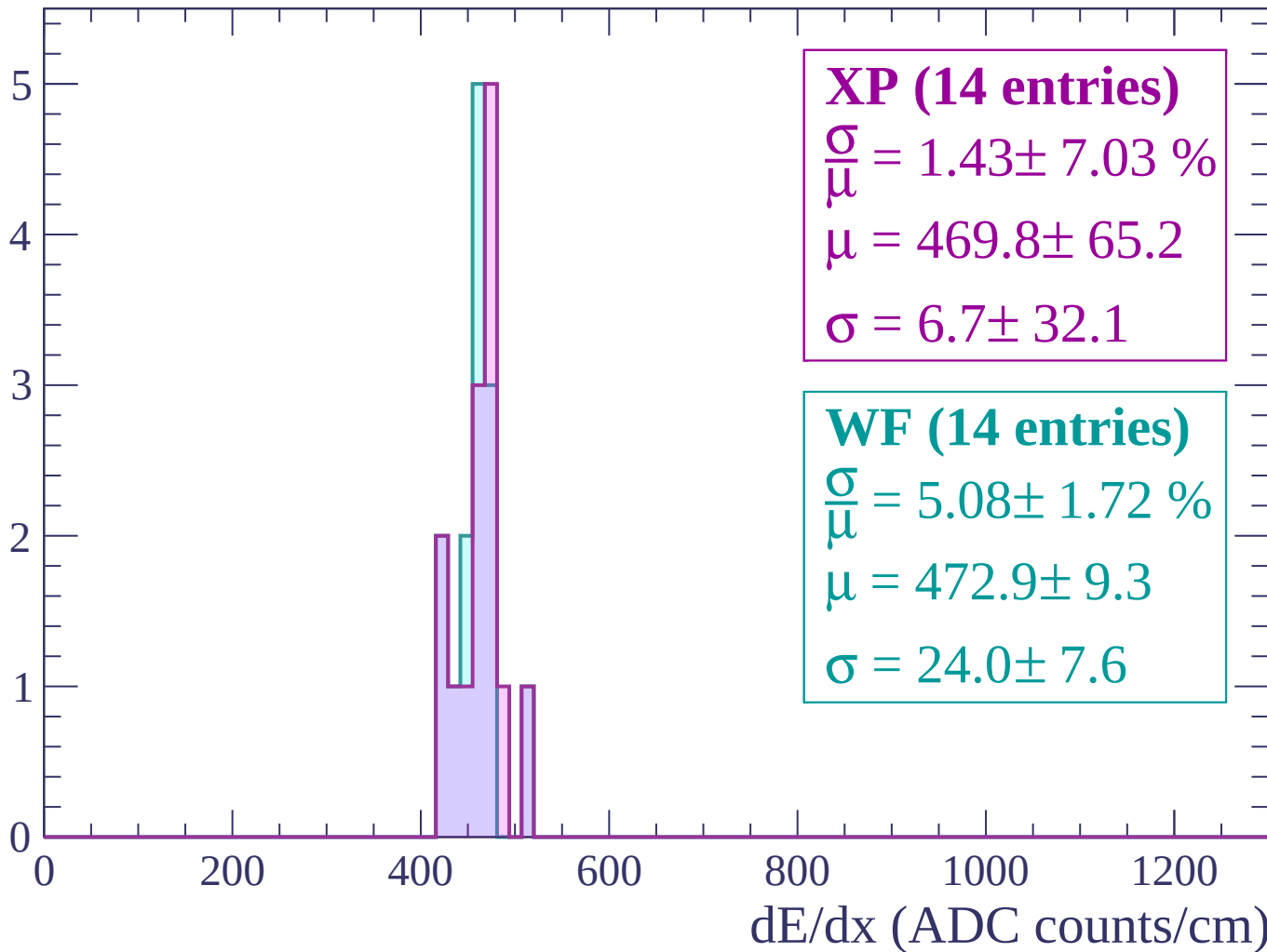
$$\mu = 461.2 \pm 1.6$$

$$\sigma = 0.3 \pm 2.1$$

dE/dx (ADC counts/cm)

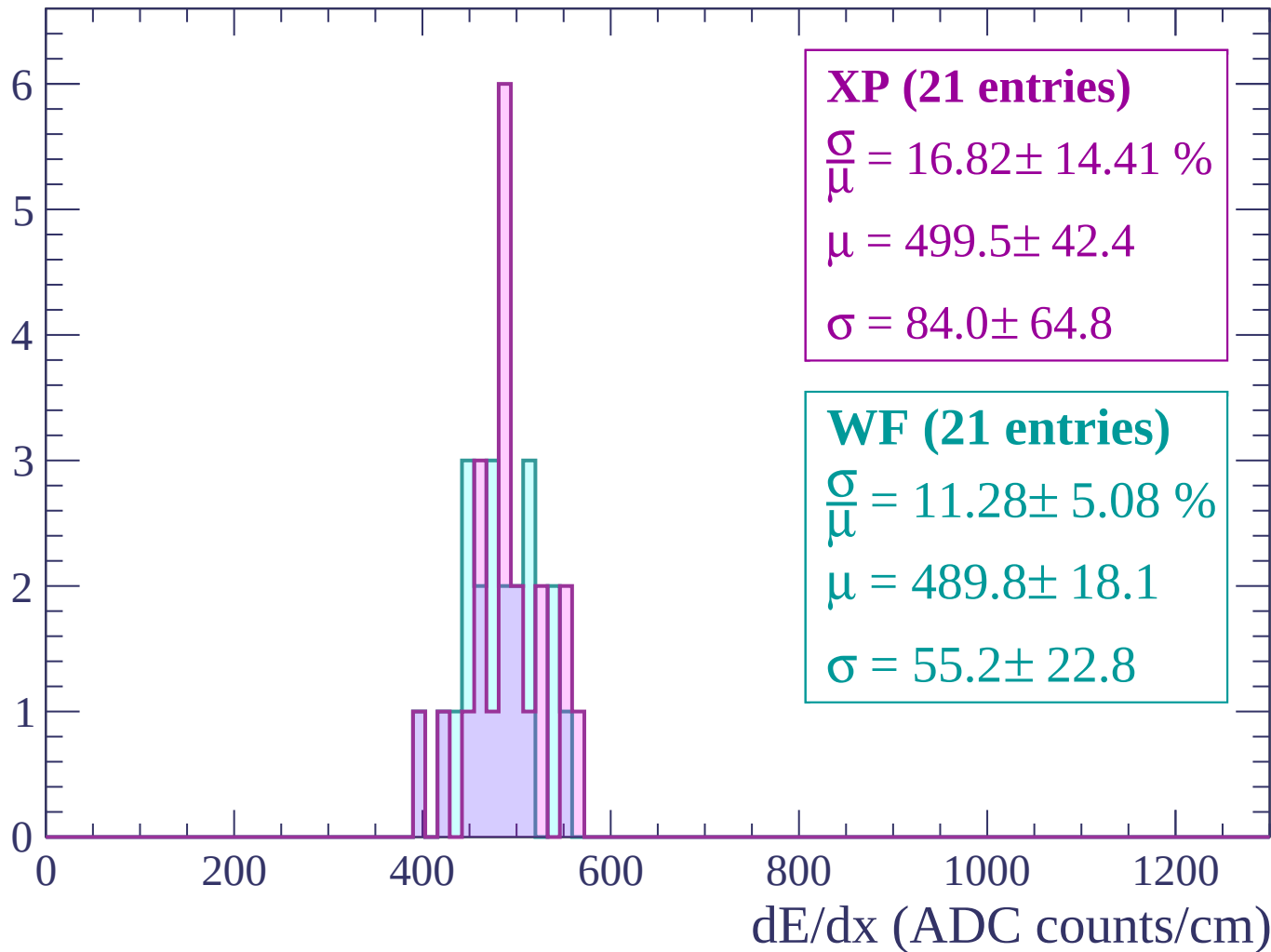
Energy loss | $-1920 < p < -1860$

Count



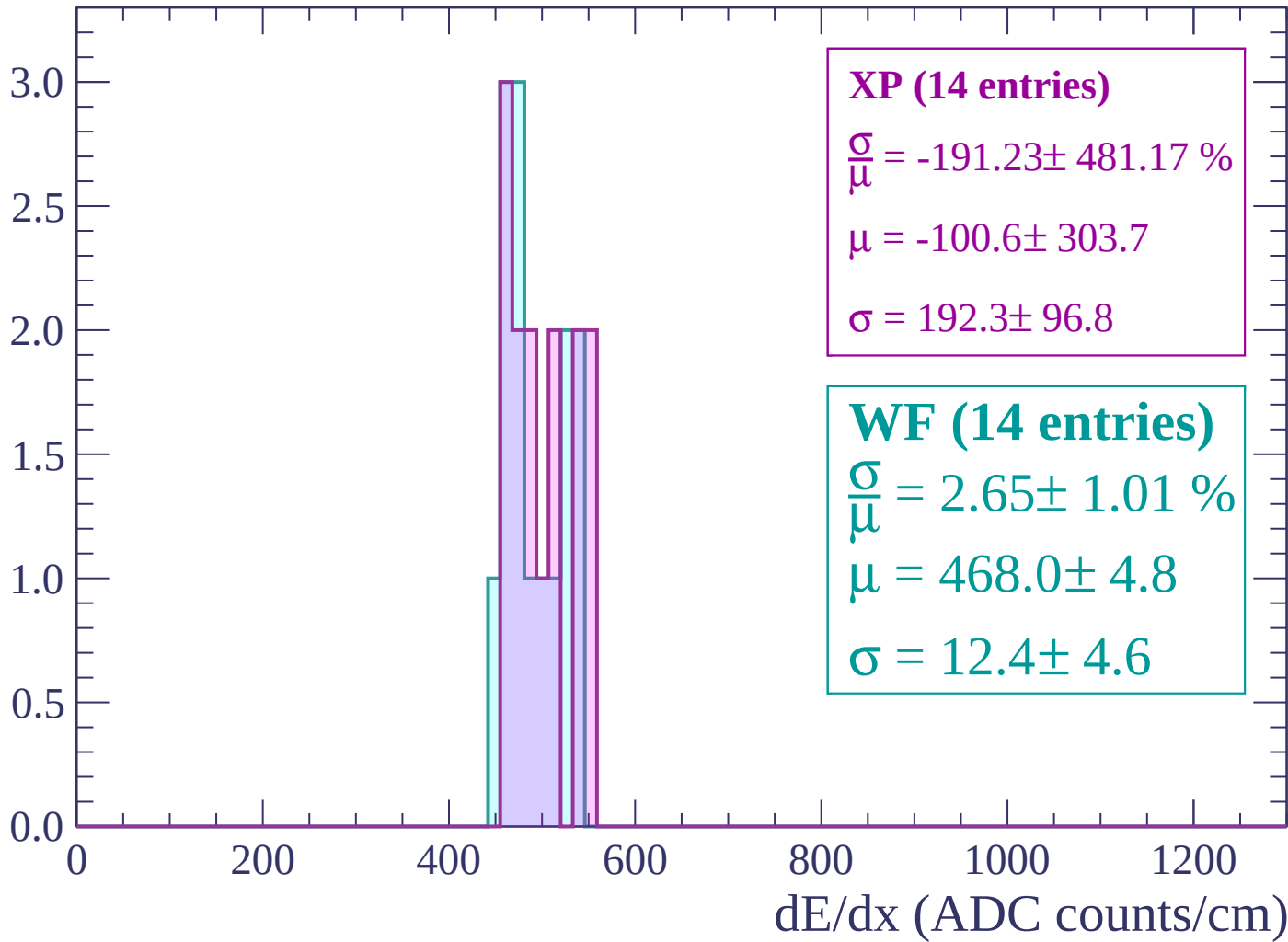
Energy loss | -1860 < p < -1800

Count



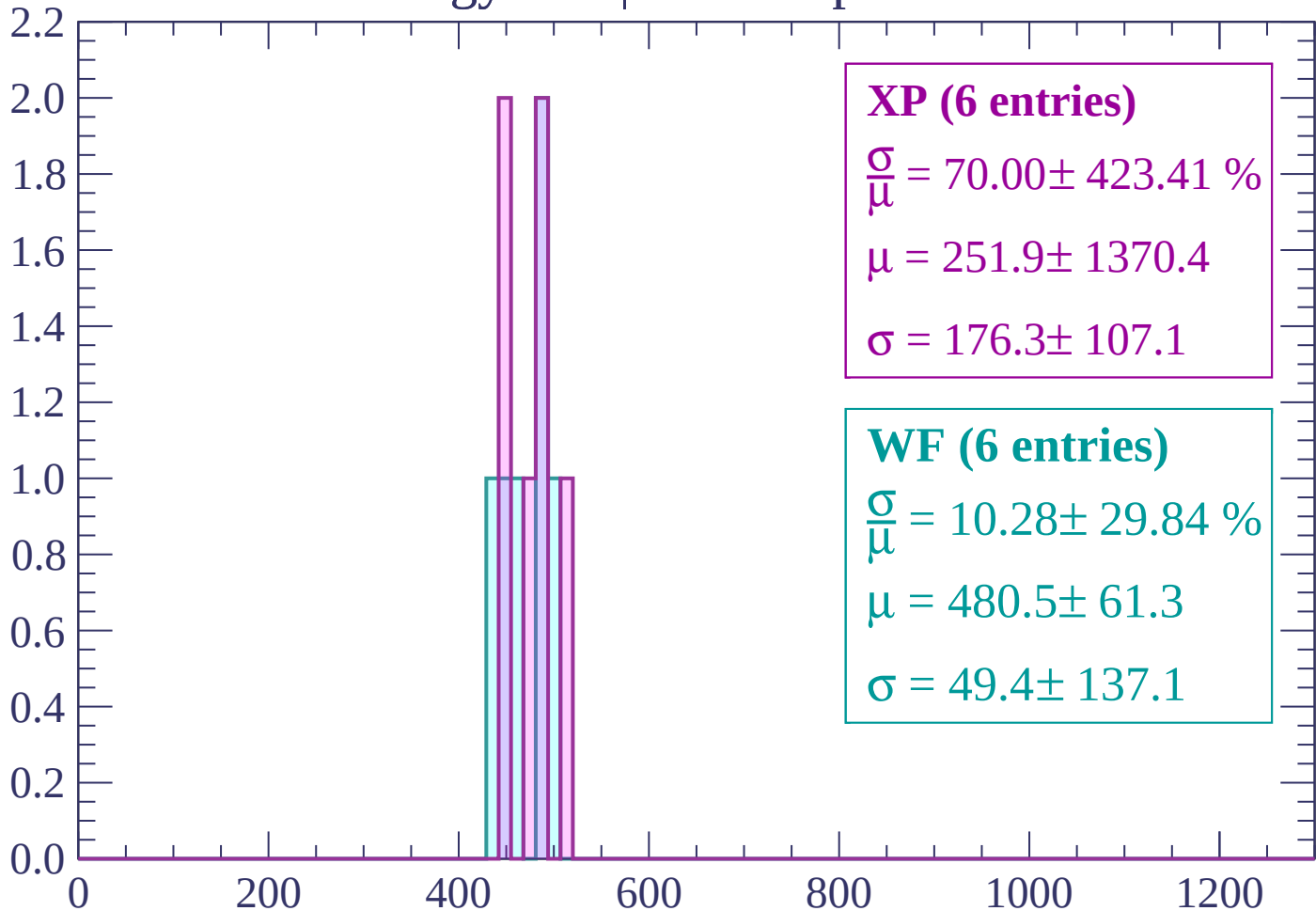
Energy loss | $-1800 < p < -1740$

Count



Energy loss | -1740 < p < -1680

Count



XP (6 entries)

$\frac{\sigma}{\mu} = 70.00 \pm 423.41 \%$

$\mu = 251.9 \pm 1370.4$

$\sigma = 176.3 \pm 107.1$

WF (6 entries)

$\frac{\sigma}{\mu} = 10.28 \pm 29.84 \%$

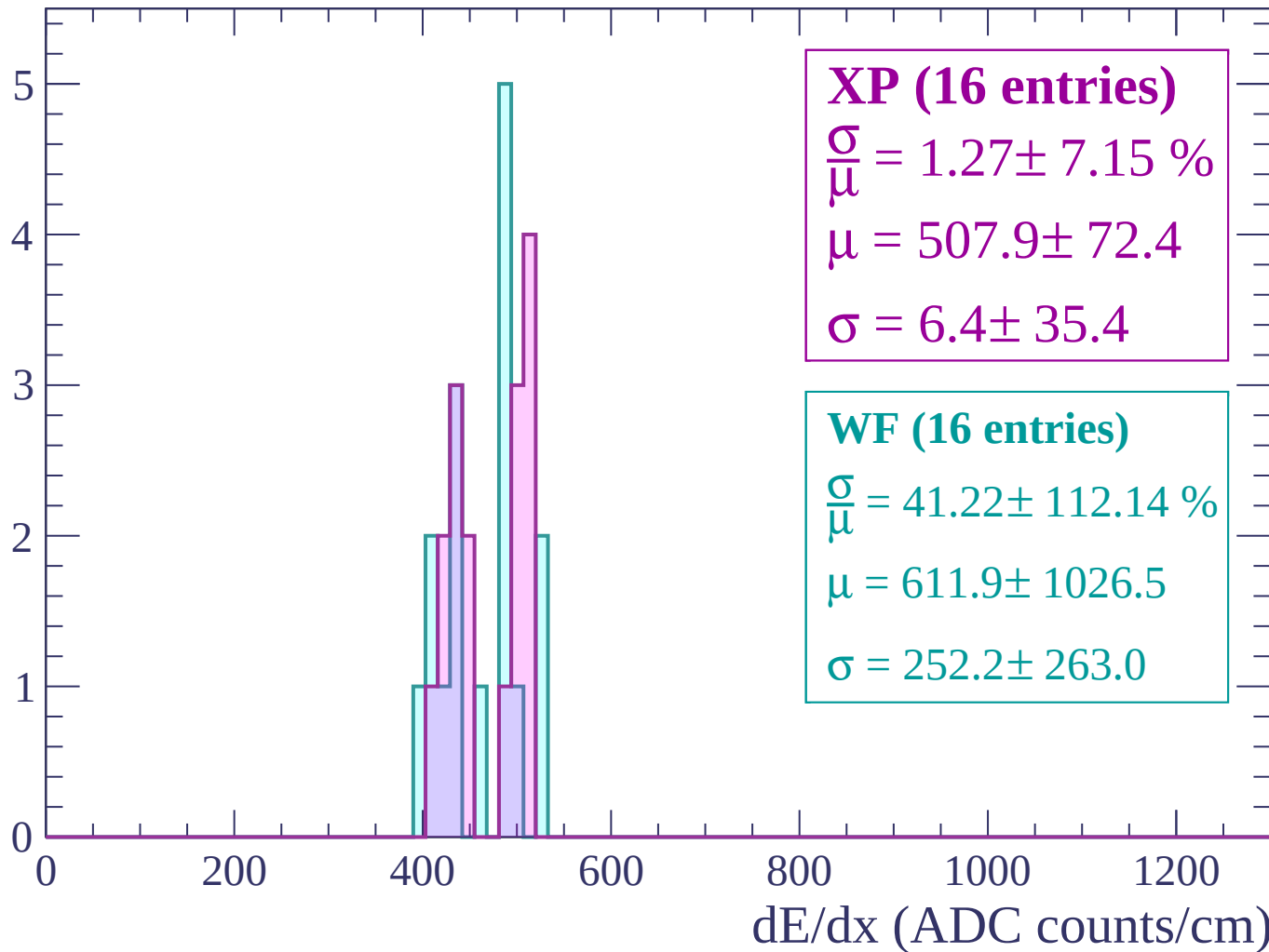
$\mu = 480.5 \pm 61.3$

$\sigma = 49.4 \pm 137.1$

dE/dx (ADC counts/cm)

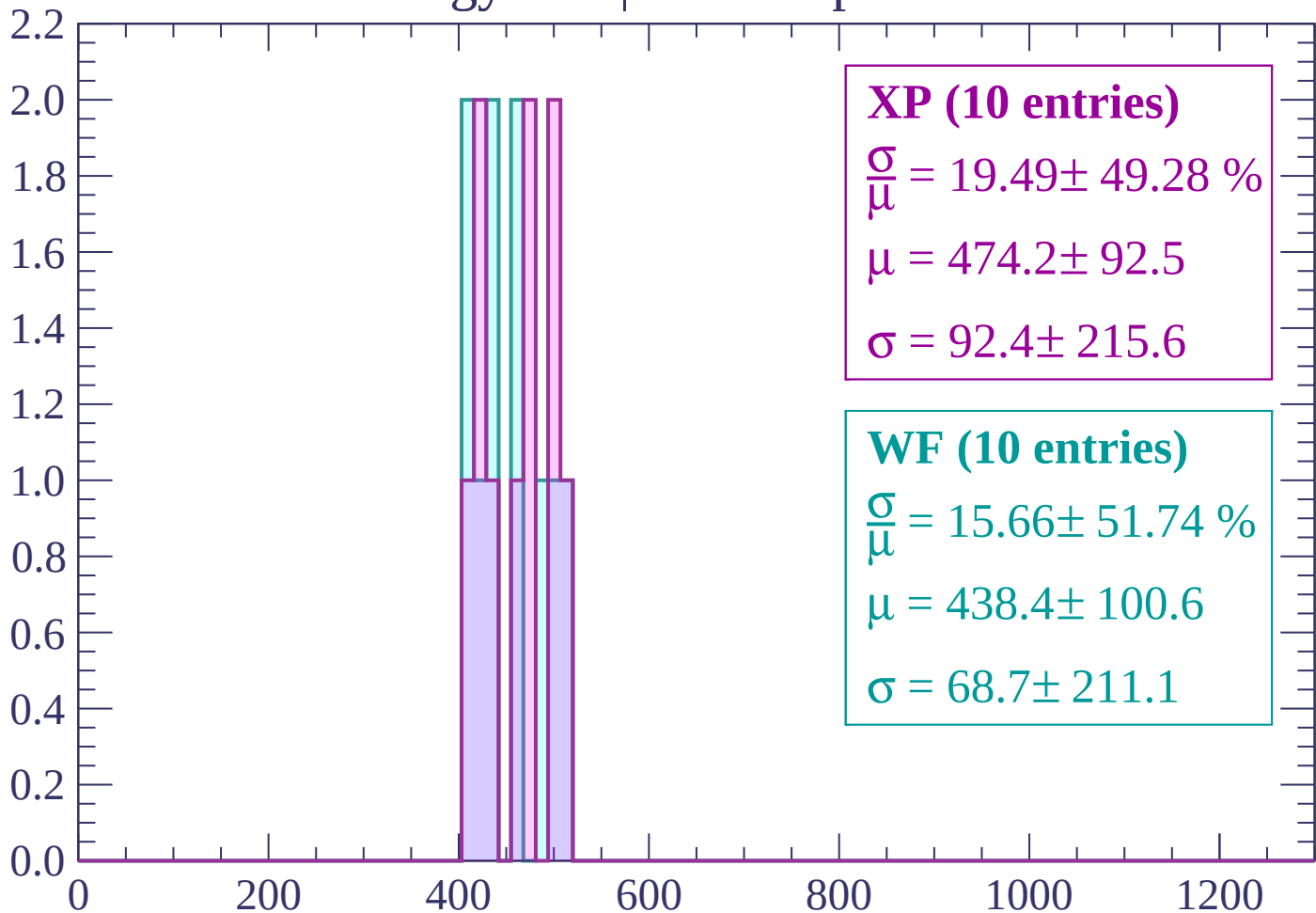
Energy loss | $-1680 < p < -1620$

Count



Energy loss | -1620 < p < -1560

Count



XP (10 entries)

$$\frac{\sigma}{\mu} = 19.49 \pm 49.28 \%$$

$$\mu = 474.2 \pm 92.5$$

$$\sigma = 92.4 \pm 215.6$$

WF (10 entries)

$$\frac{\sigma}{\mu} = 15.66 \pm 51.74 \%$$

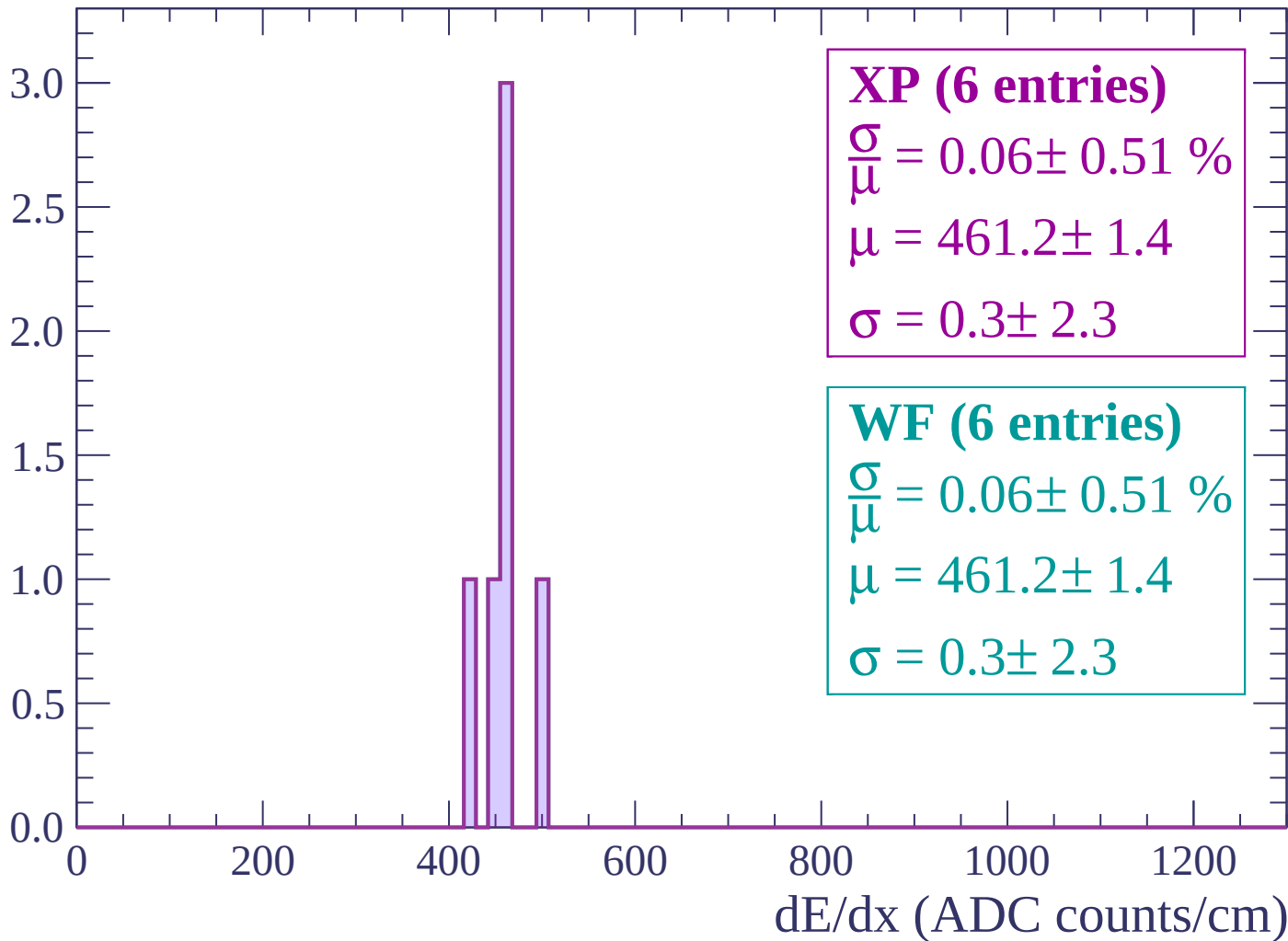
$$\mu = 438.4 \pm 100.6$$

$$\sigma = 68.7 \pm 211.1$$

dE/dx (ADC counts/cm)

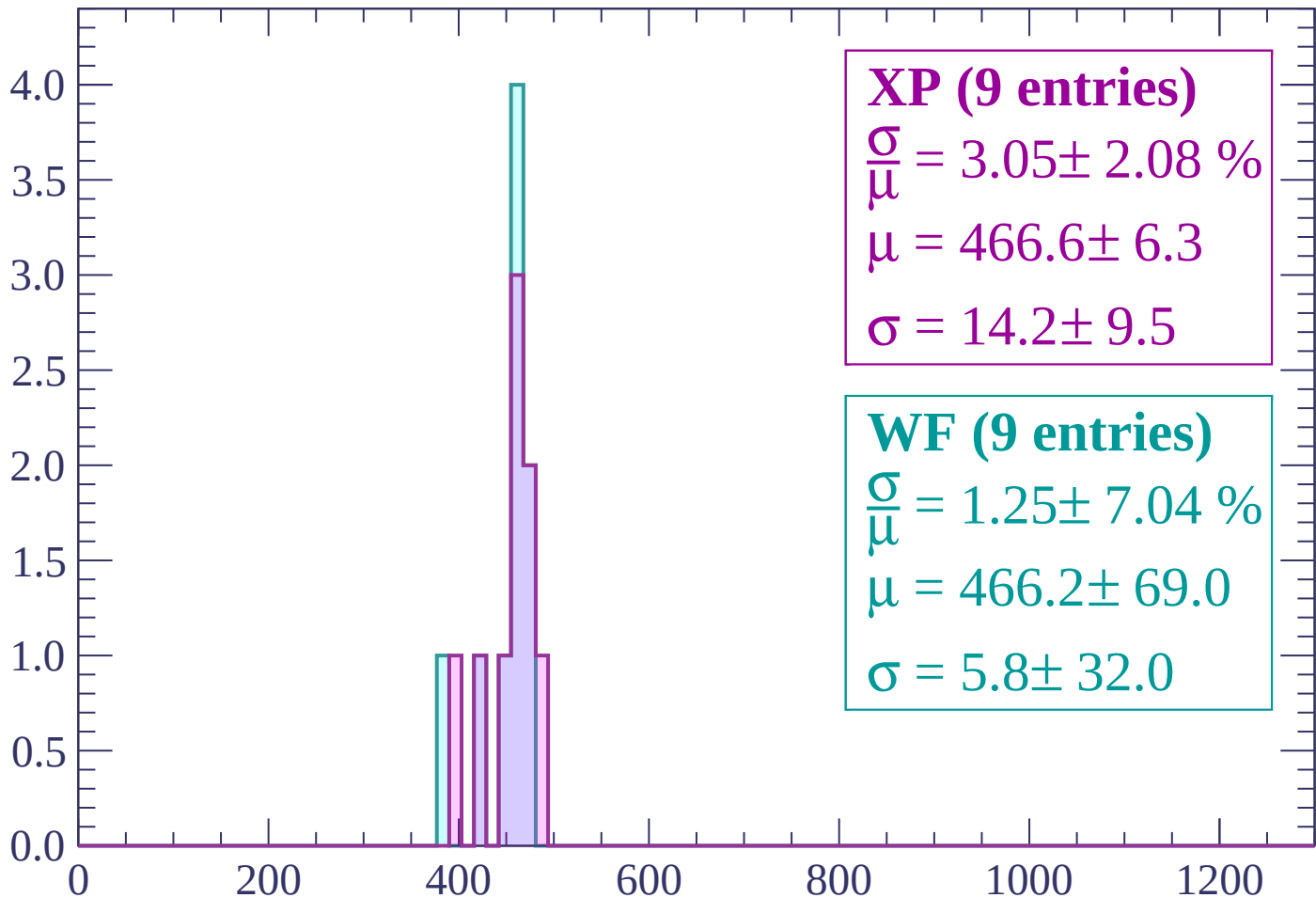
Energy loss | -1560 < p < -1500

Count



Energy loss | -1500 < p < -1440

Count



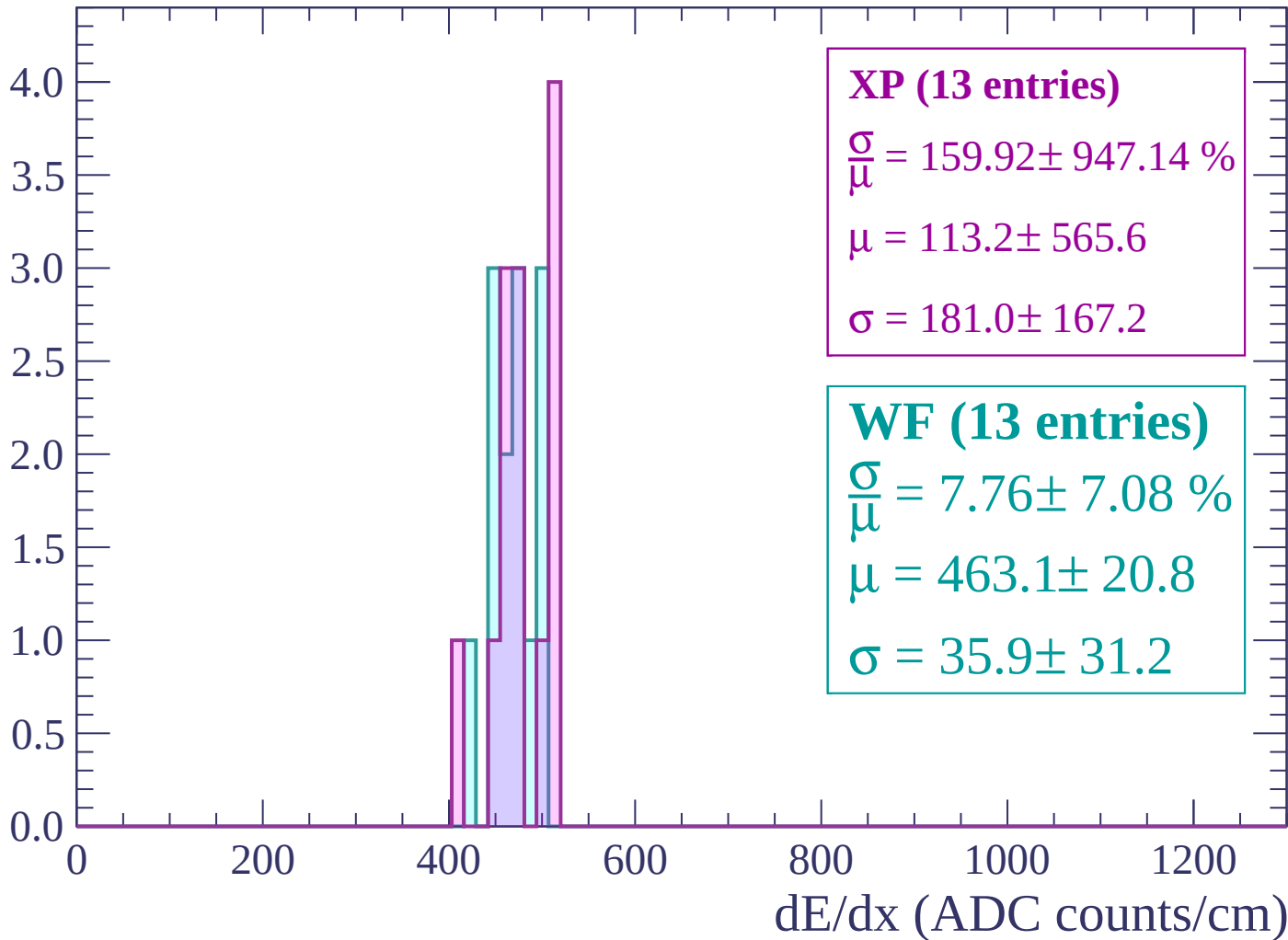
XP (9 entries)
 $\frac{\sigma}{\mu} = 3.05 \pm 2.08 \%$
 $\mu = 466.6 \pm 6.3$
 $\sigma = 14.2 \pm 9.5$

WF (9 entries)
 $\frac{\sigma}{\mu} = 1.25 \pm 7.04 \%$
 $\mu = 466.2 \pm 69.0$
 $\sigma = 5.8 \pm 32.0$

dE/dx (ADC counts/cm)

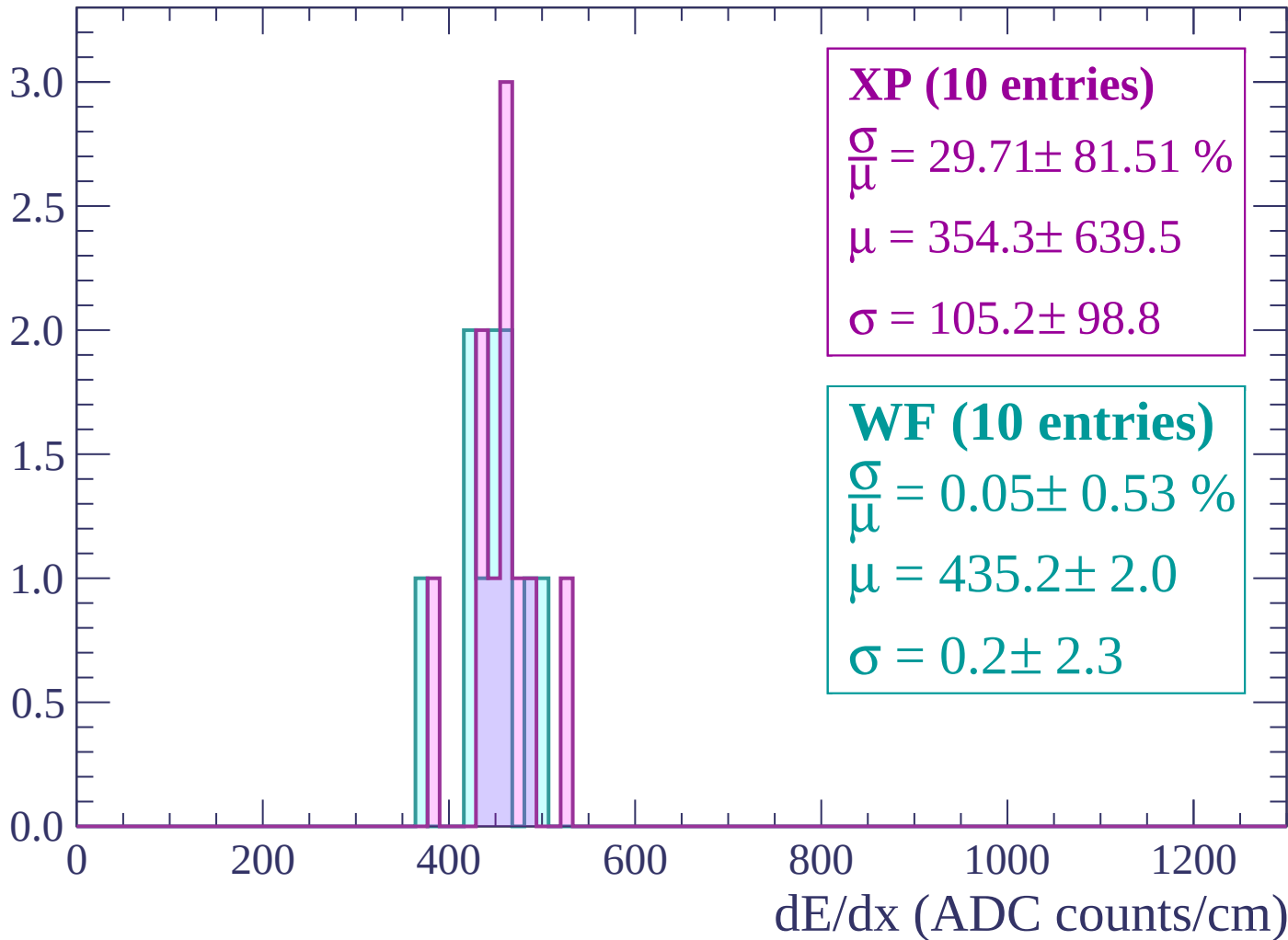
Energy loss | -1440 < p < -1380

Count



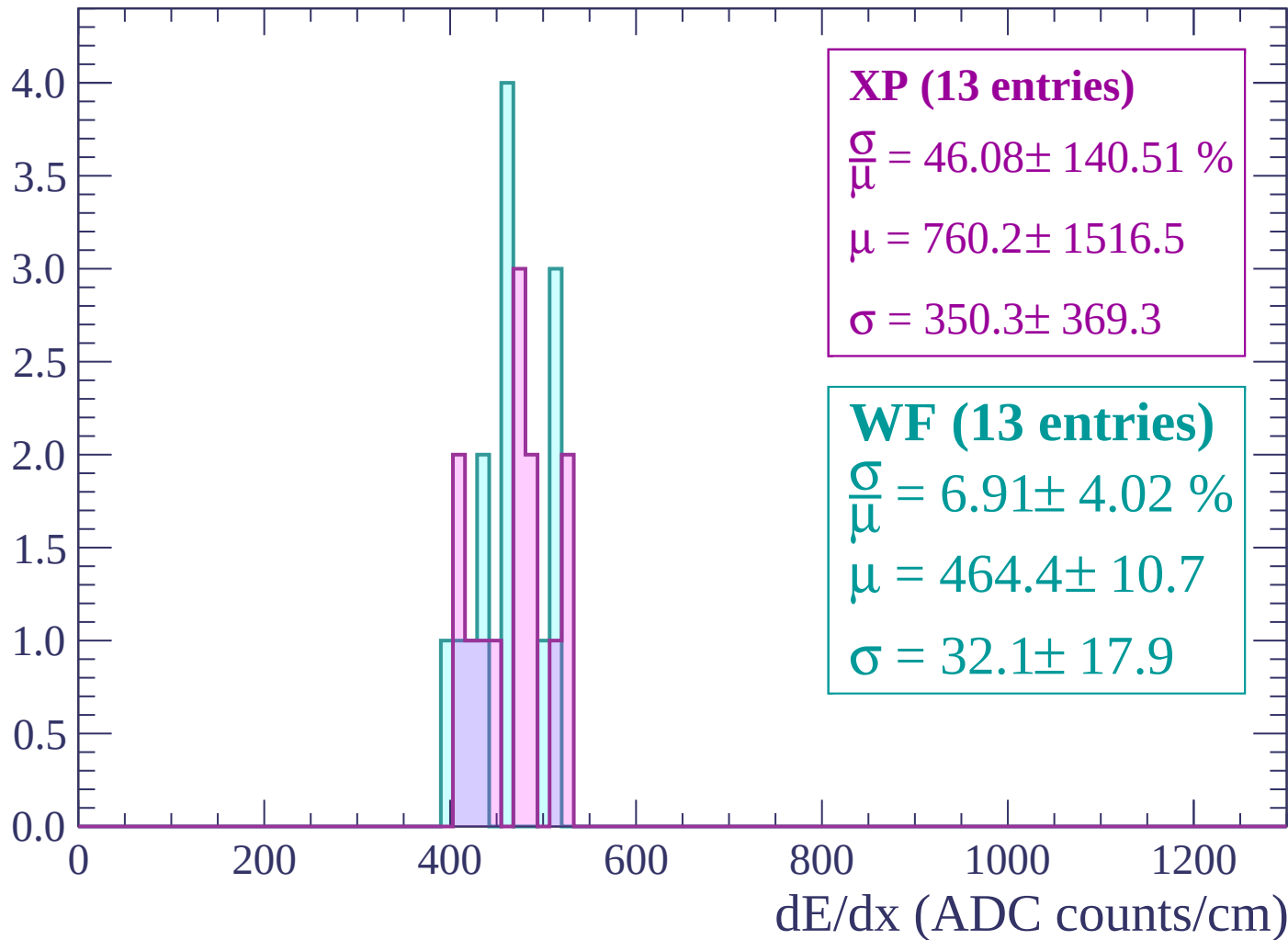
Energy loss | -1380 < p < -1320

Count



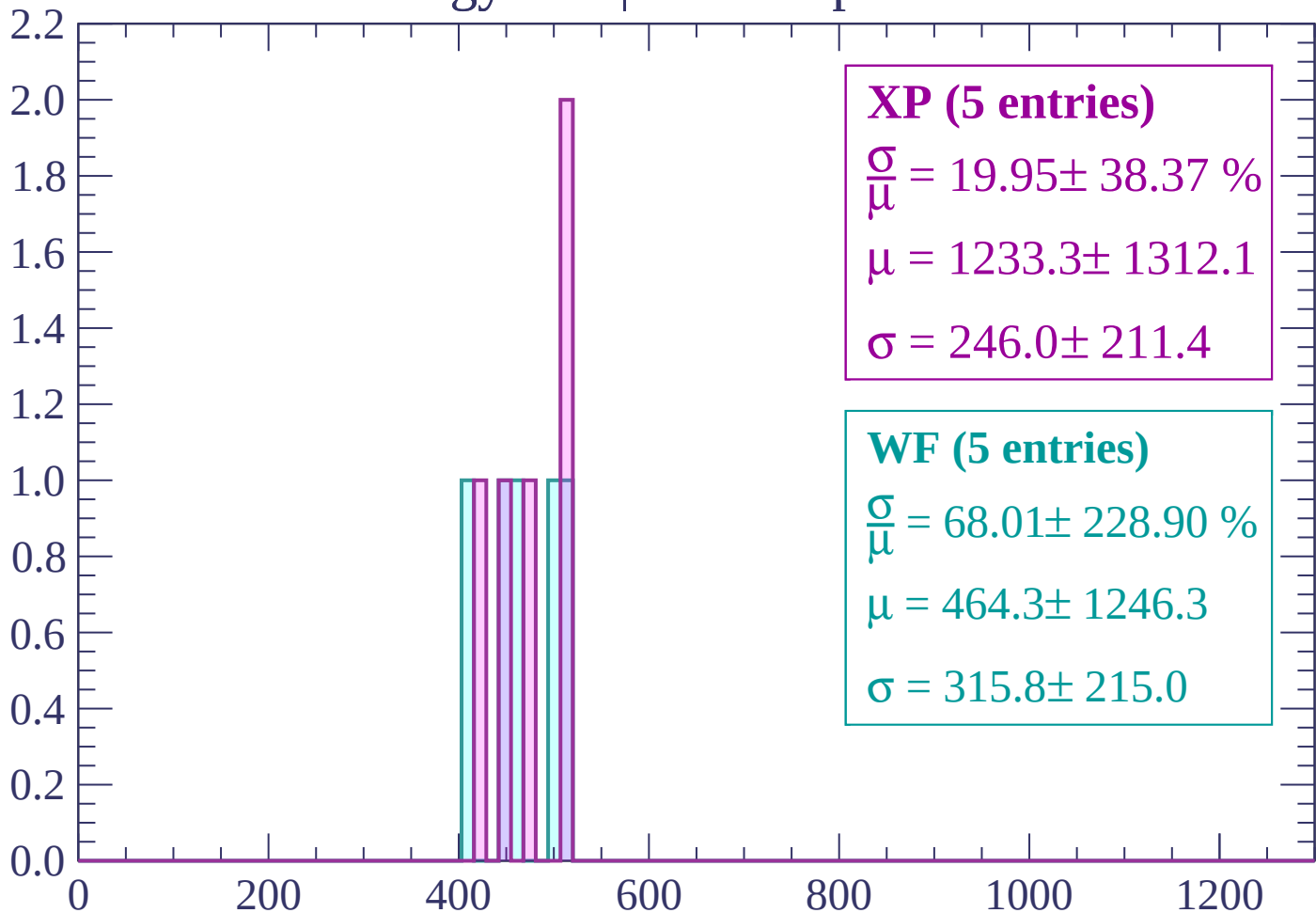
Energy loss | -1320 < p < -1260

Count



Energy loss | -1260 < p < -1200

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 19.95 \pm 38.37 \%$$

$$\mu = 1233.3 \pm 1312.1$$

$$\sigma = 246.0 \pm 211.4$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 68.01 \pm 228.90 \%$$

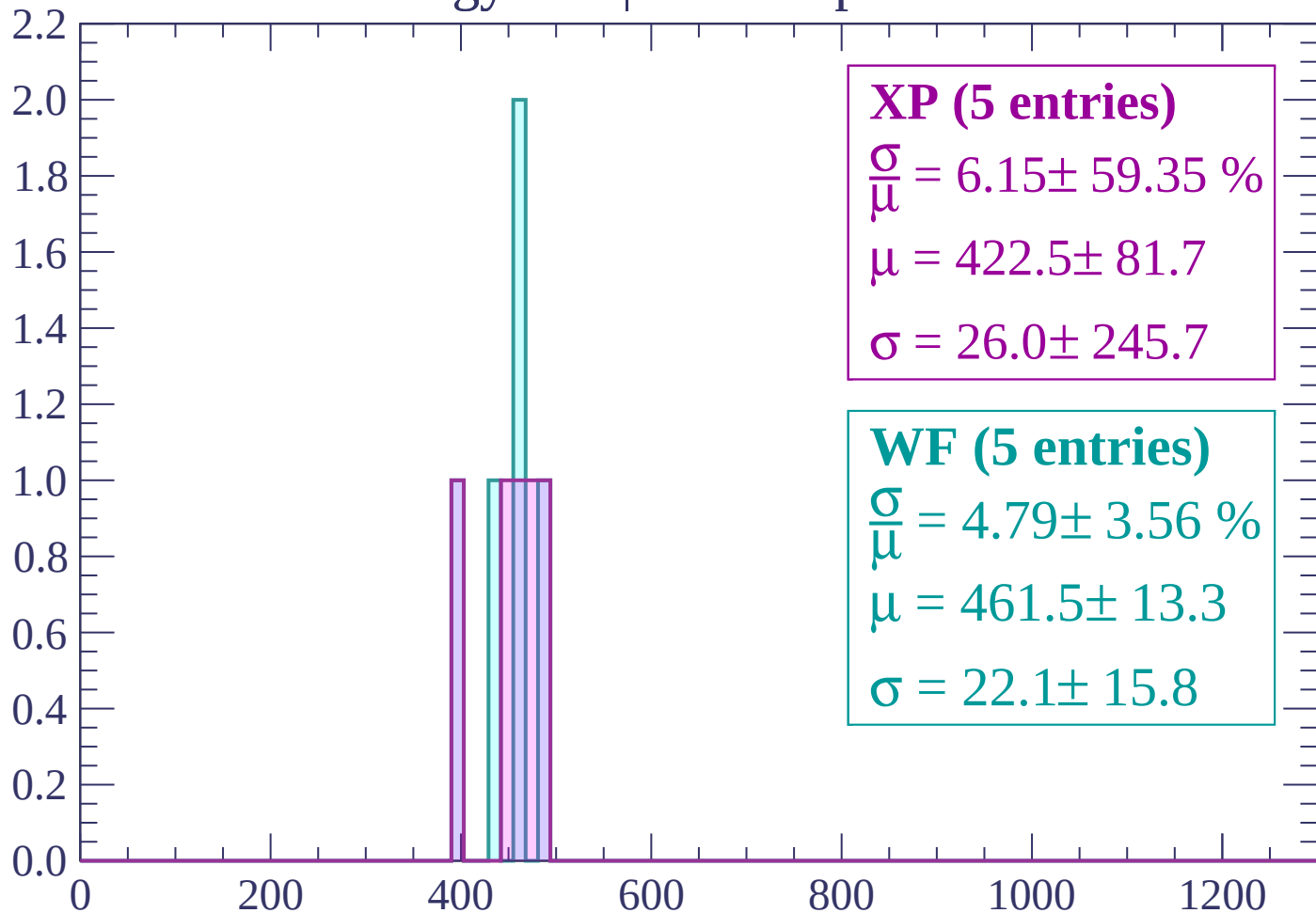
$$\mu = 464.3 \pm 1246.3$$

$$\sigma = 315.8 \pm 215.0$$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 6.15 \pm 59.35 \%$$

$$\mu = 422.5 \pm 81.7$$

$$\sigma = 26.0 \pm 245.7$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 4.79 \pm 3.56 \%$$

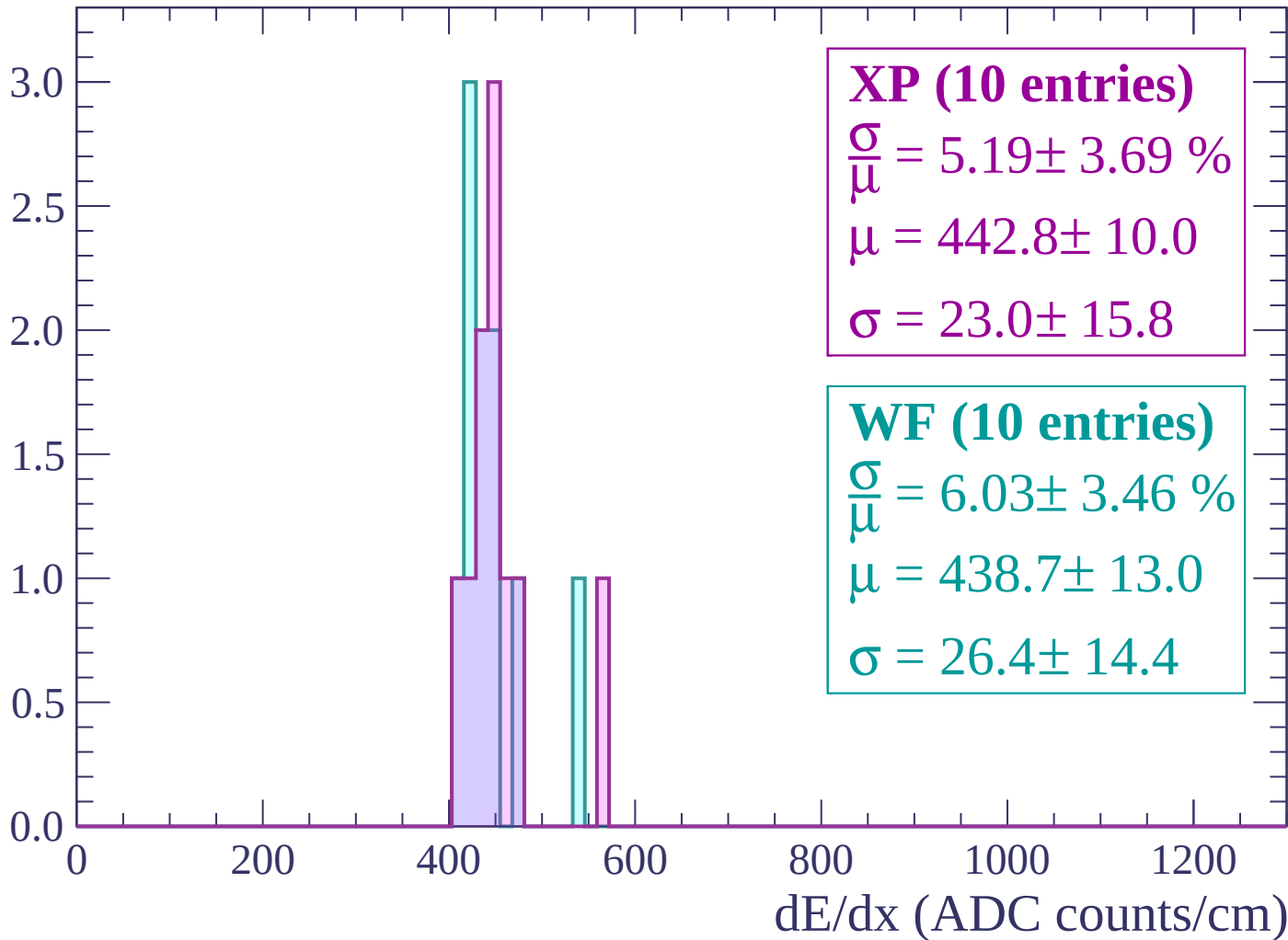
$$\mu = 461.5 \pm 13.3$$

$$\sigma = 22.1 \pm 15.8$$

dE/dx (ADC counts/cm)

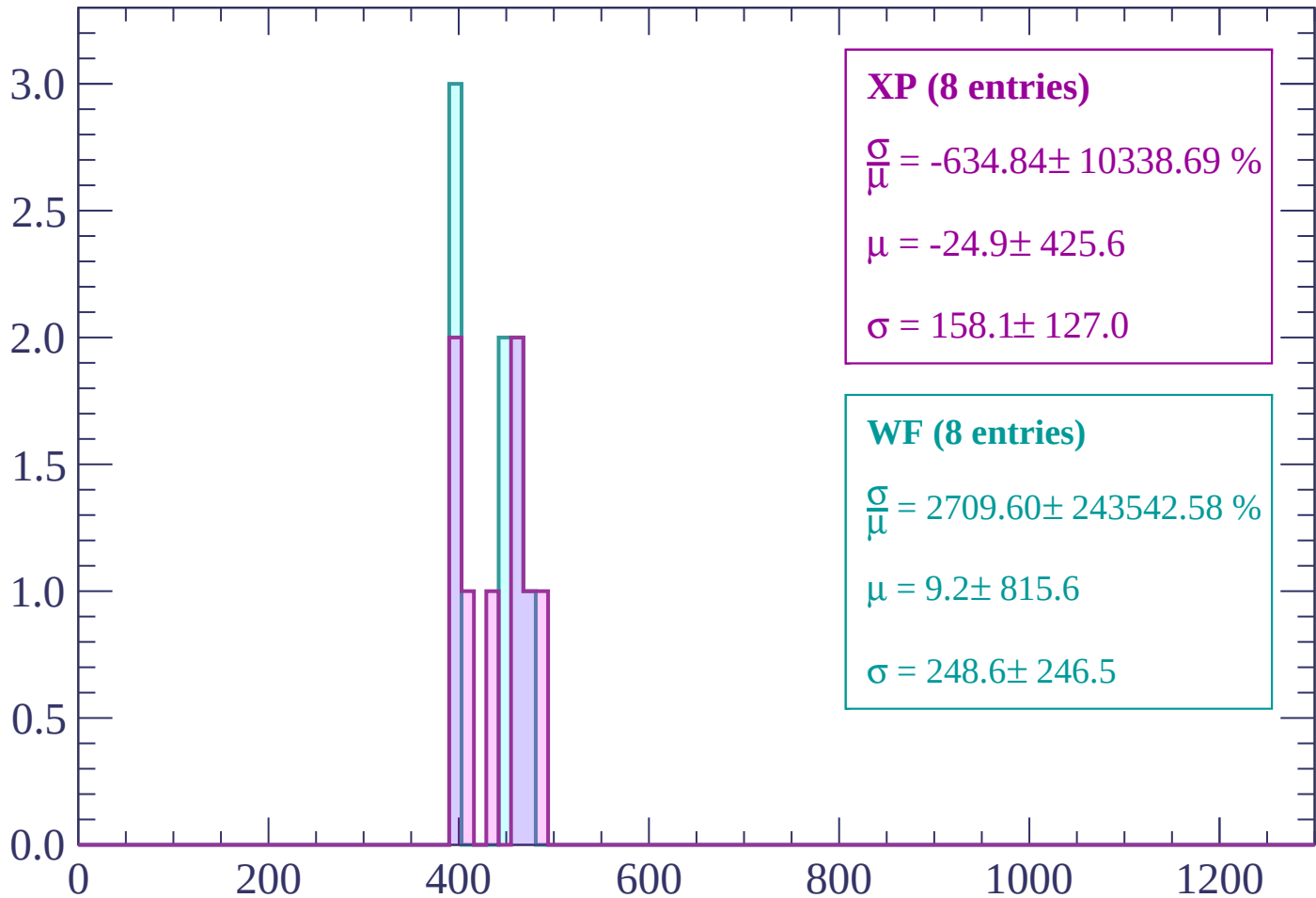
Energy loss | -1140 < p < -1080

Count



Energy loss | -1080 < p < -1020

Count



XP (8 entries)

$\frac{\sigma}{\mu} = -634.84 \pm 10338.69 \%$

$\mu = -24.9 \pm 425.6$

$\sigma = 158.1 \pm 127.0$

WF (8 entries)

$\frac{\sigma}{\mu} = 2709.60 \pm 243542.58 \%$

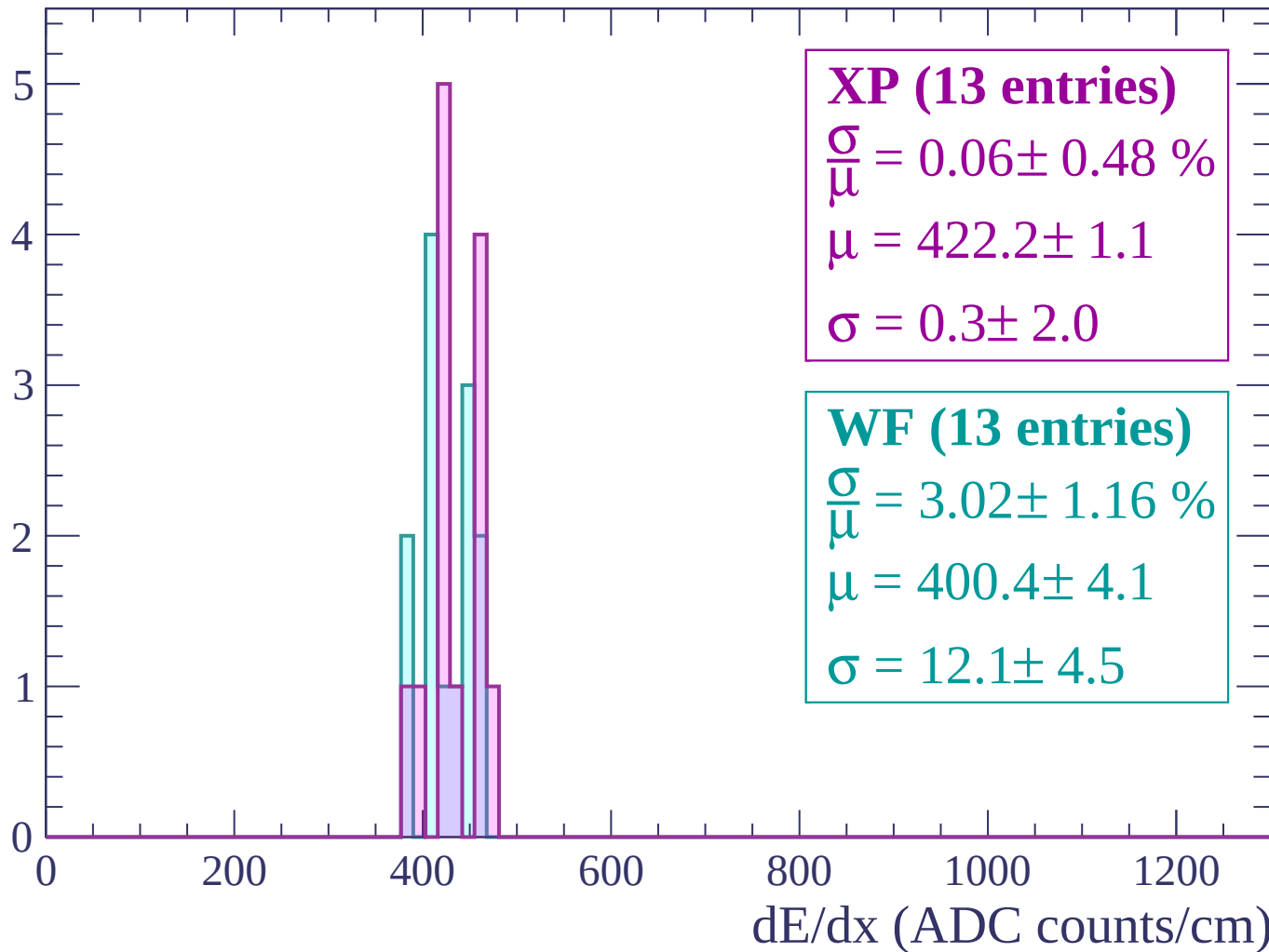
$\mu = 9.2 \pm 815.6$

$\sigma = 248.6 \pm 246.5$

dE/dx (ADC counts/cm)

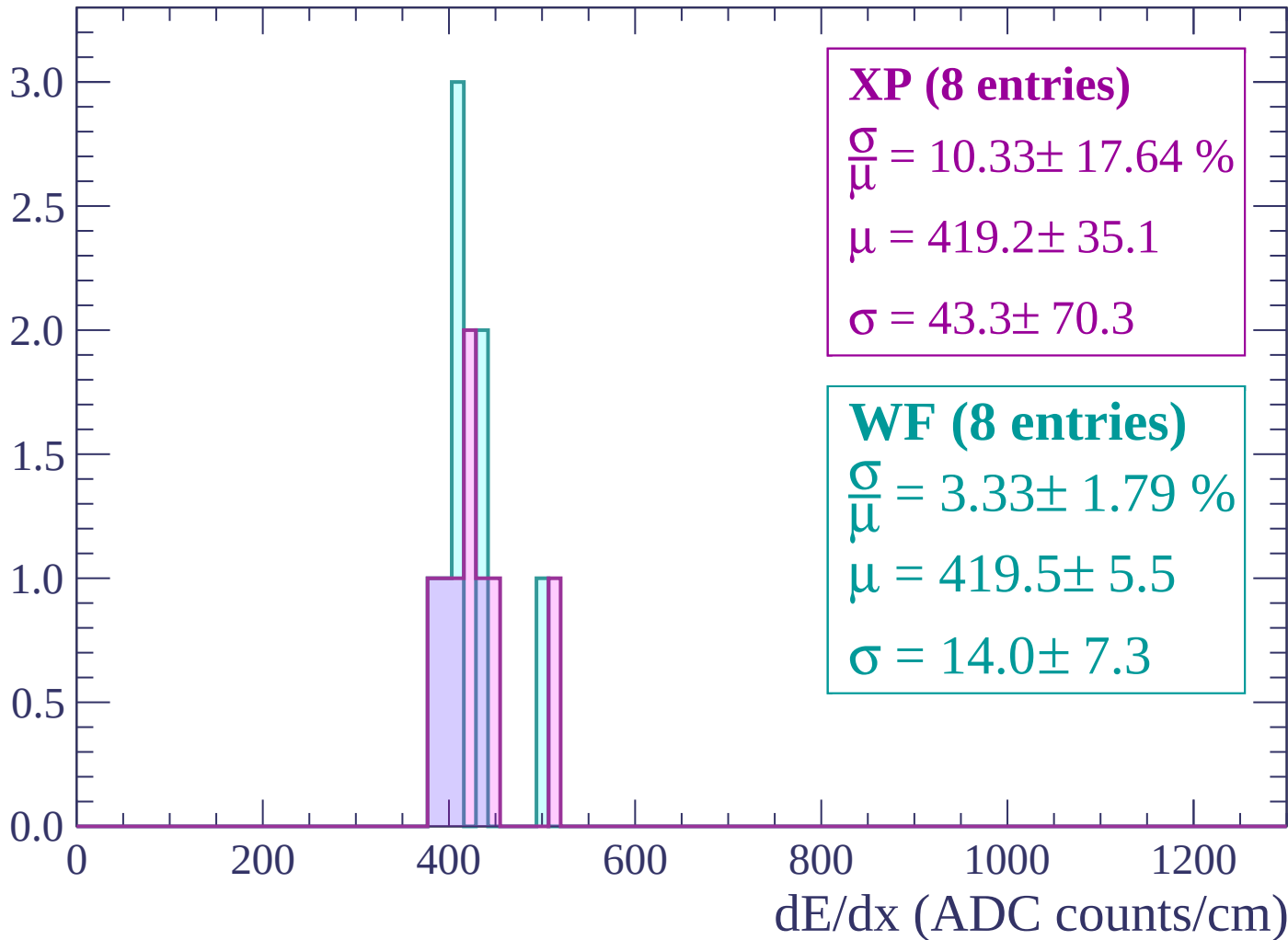
Energy loss | $-1020 < p < -960$

Count



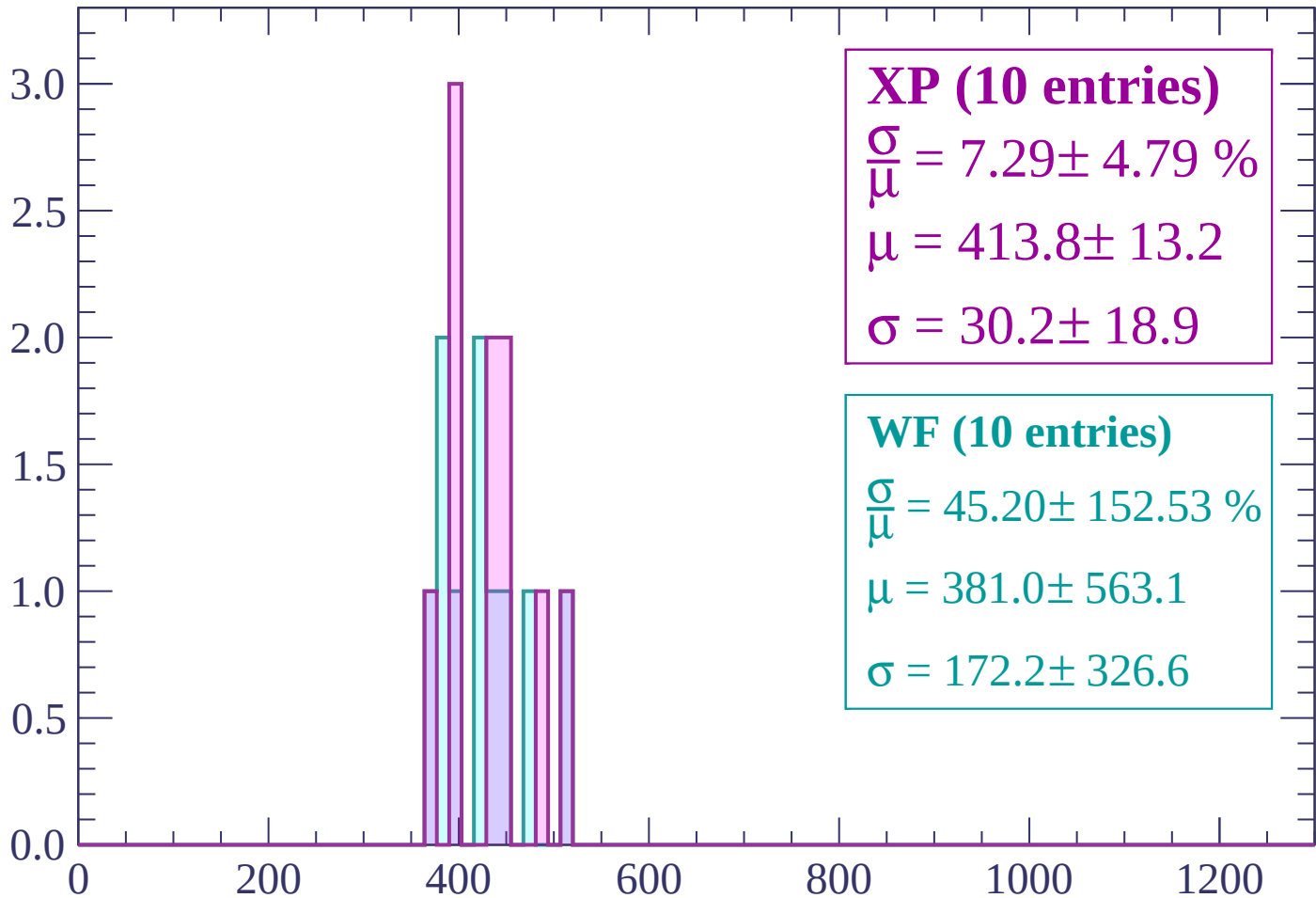
Energy loss | -960 < p < -900

Count



Energy loss | $-900 < p < -840$

Count



XP (10 entries)

$\frac{\sigma}{\mu} = 7.29 \pm 4.79 \%$

$\mu = 413.8 \pm 13.2$

$\sigma = 30.2 \pm 18.9$

WF (10 entries)

$\frac{\sigma}{\mu} = 45.20 \pm 152.53 \%$

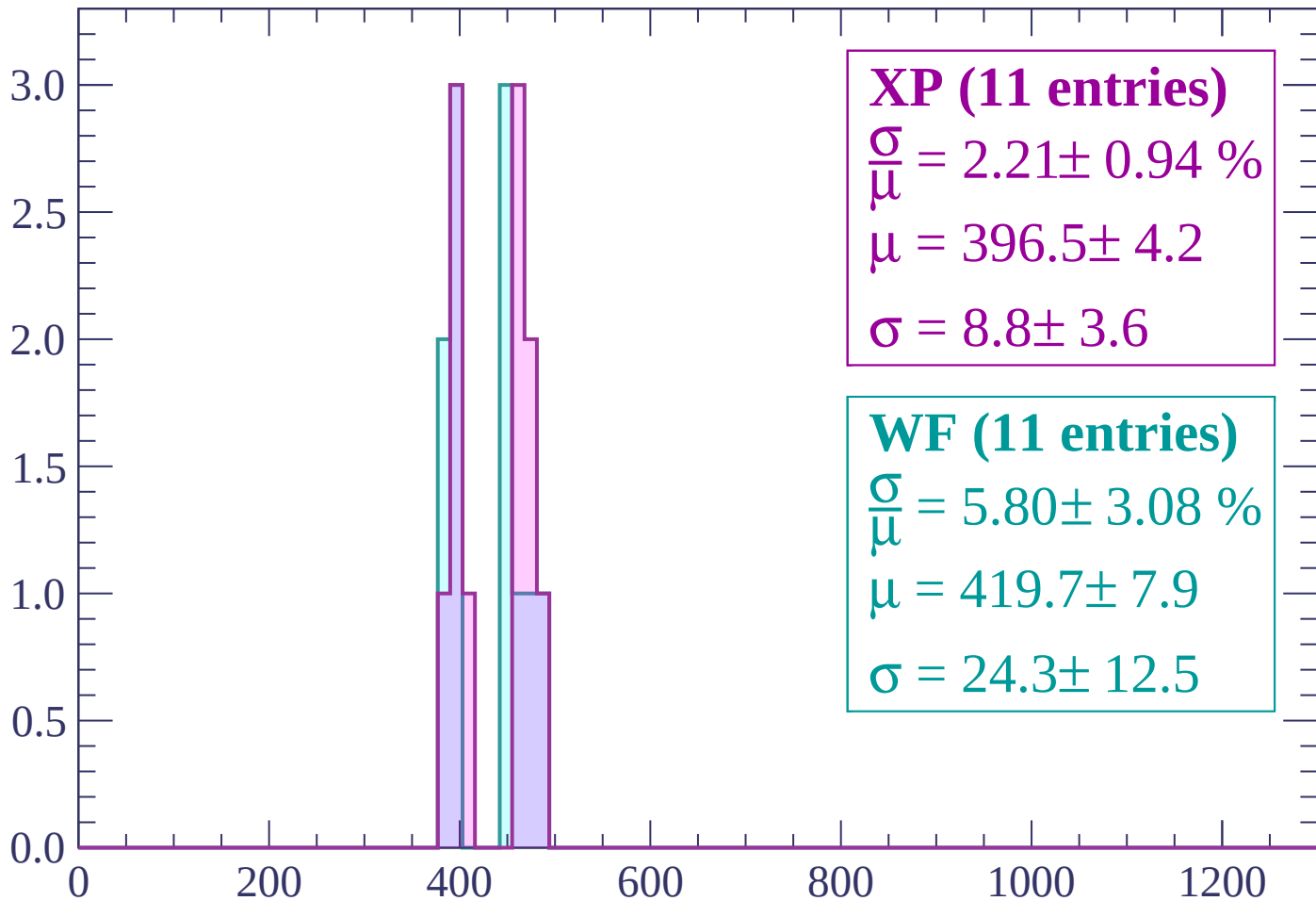
$\mu = 381.0 \pm 563.1$

$\sigma = 172.2 \pm 326.6$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (11 entries)

$\frac{\sigma}{\mu} = 2.21 \pm 0.94 \%$

$\mu = 396.5 \pm 4.2$

$\sigma = 8.8 \pm 3.6$

WF (11 entries)

$\frac{\sigma}{\mu} = 5.80 \pm 3.08 \%$

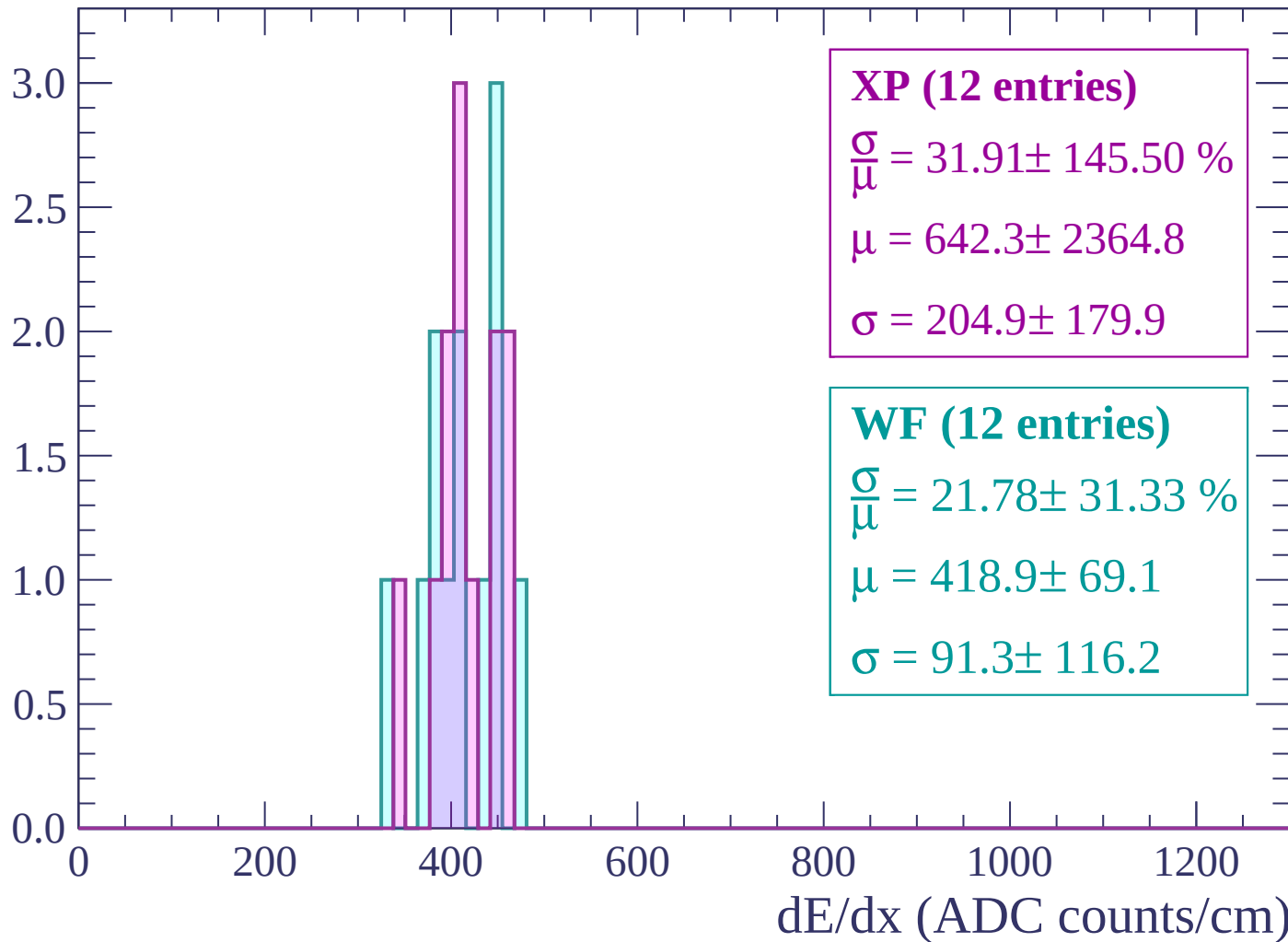
$\mu = 419.7 \pm 7.9$

$\sigma = 24.3 \pm 12.5$

dE/dx (ADC counts/cm)

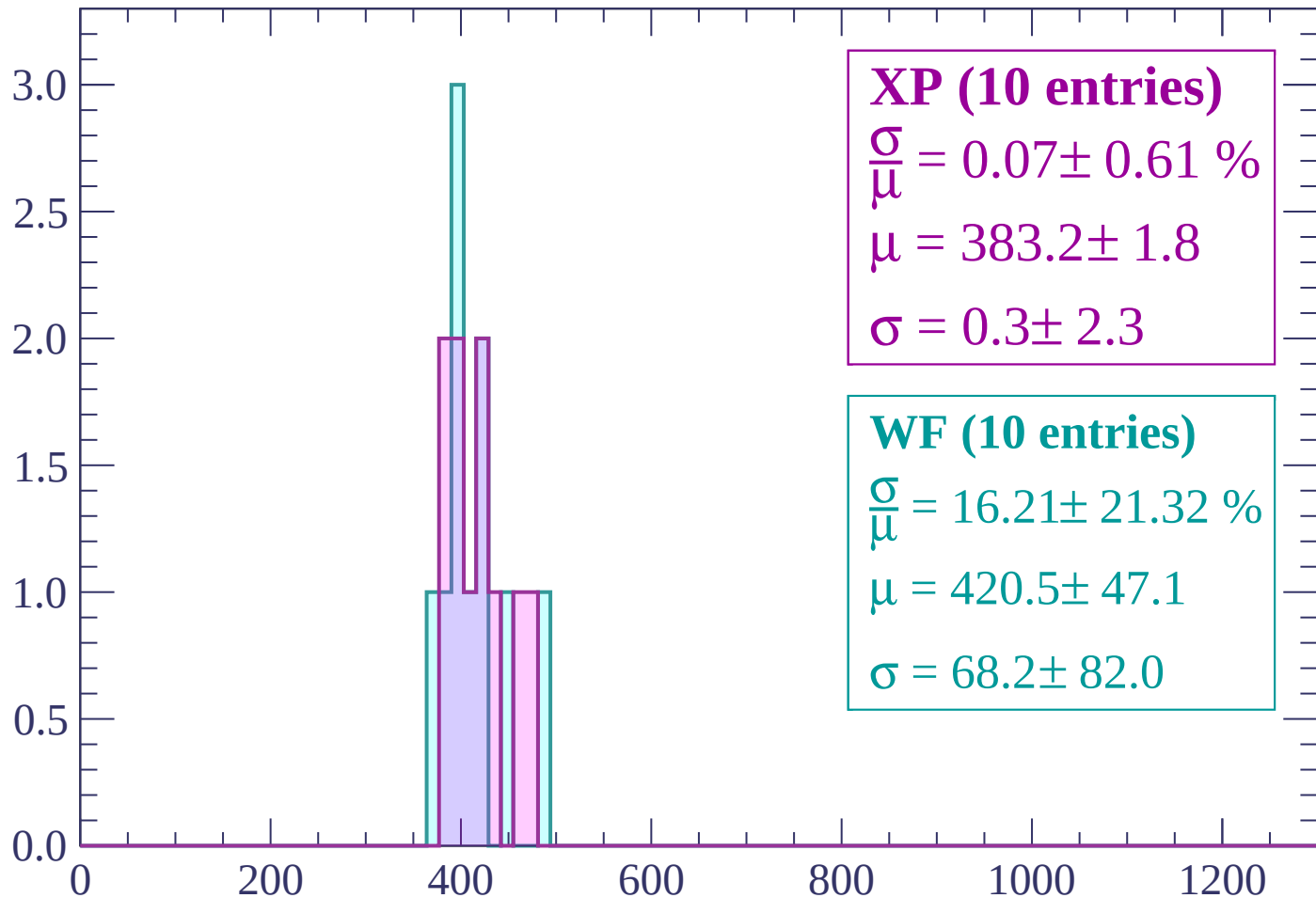
Energy loss | $-780 < p < -720$

Count



Energy loss | $-720 < p < -660$

Count



XP (10 entries)

$$\frac{\sigma}{\mu} = 0.07 \pm 0.61 \%$$

$$\mu = 383.2 \pm 1.8$$

$$\sigma = 0.3 \pm 2.3$$

WF (10 entries)

$$\frac{\sigma}{\mu} = 16.21 \pm 21.32 \%$$

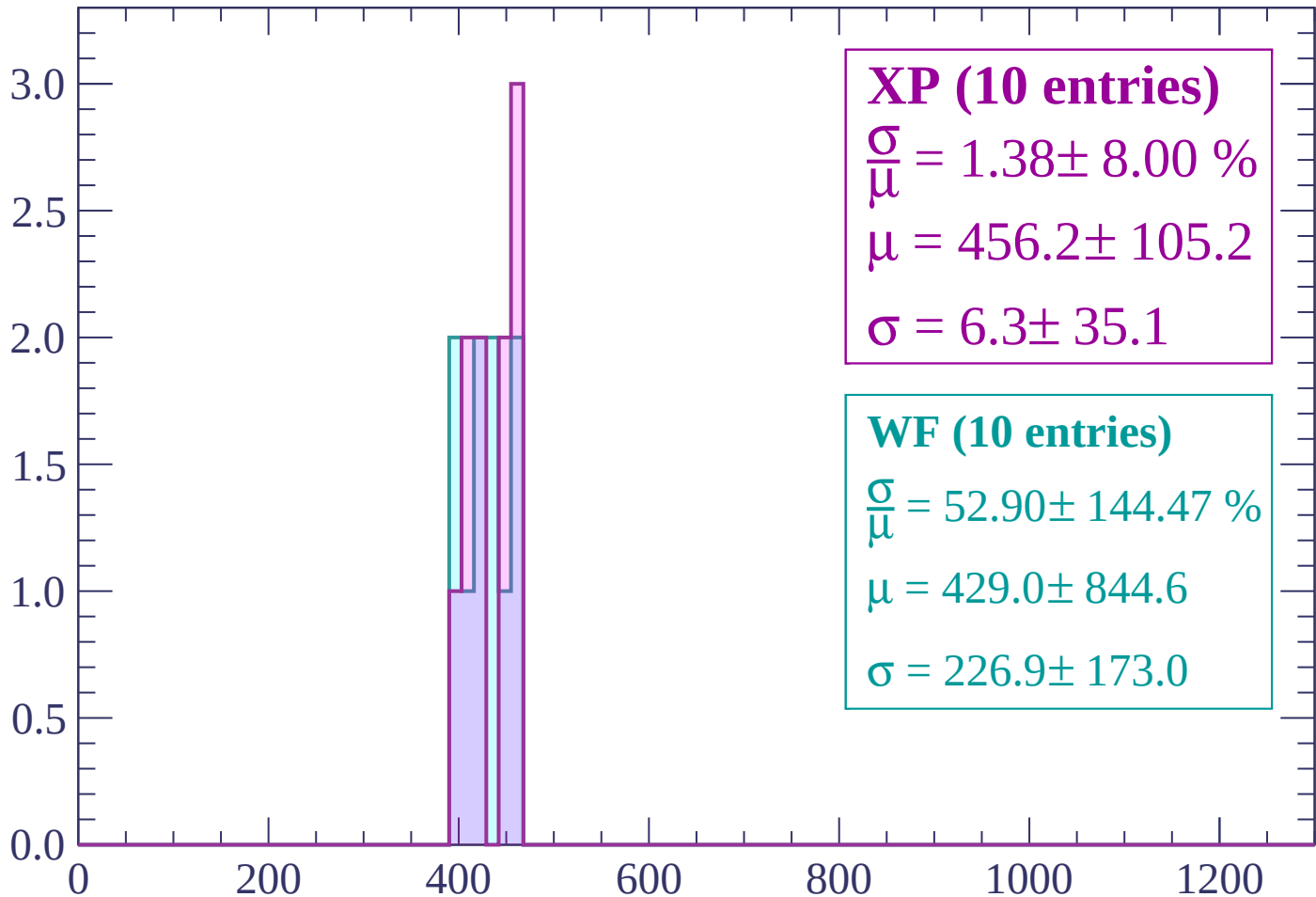
$$\mu = 420.5 \pm 47.1$$

$$\sigma = 68.2 \pm 82.0$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (10 entries)

$\frac{\sigma}{\mu} = 1.38 \pm 8.00 \%$

$\mu = 456.2 \pm 105.2$

$\sigma = 6.3 \pm 35.1$

WF (10 entries)

$\frac{\sigma}{\mu} = 52.90 \pm 144.47 \%$

$\mu = 429.0 \pm 844.6$

$\sigma = 226.9 \pm 173.0$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (3 entries)

$$\frac{\sigma}{\bar{\mu}} = 1.67 \pm 9.28 \%$$

$$\mu = 390.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.7$$

WF (3 entries)

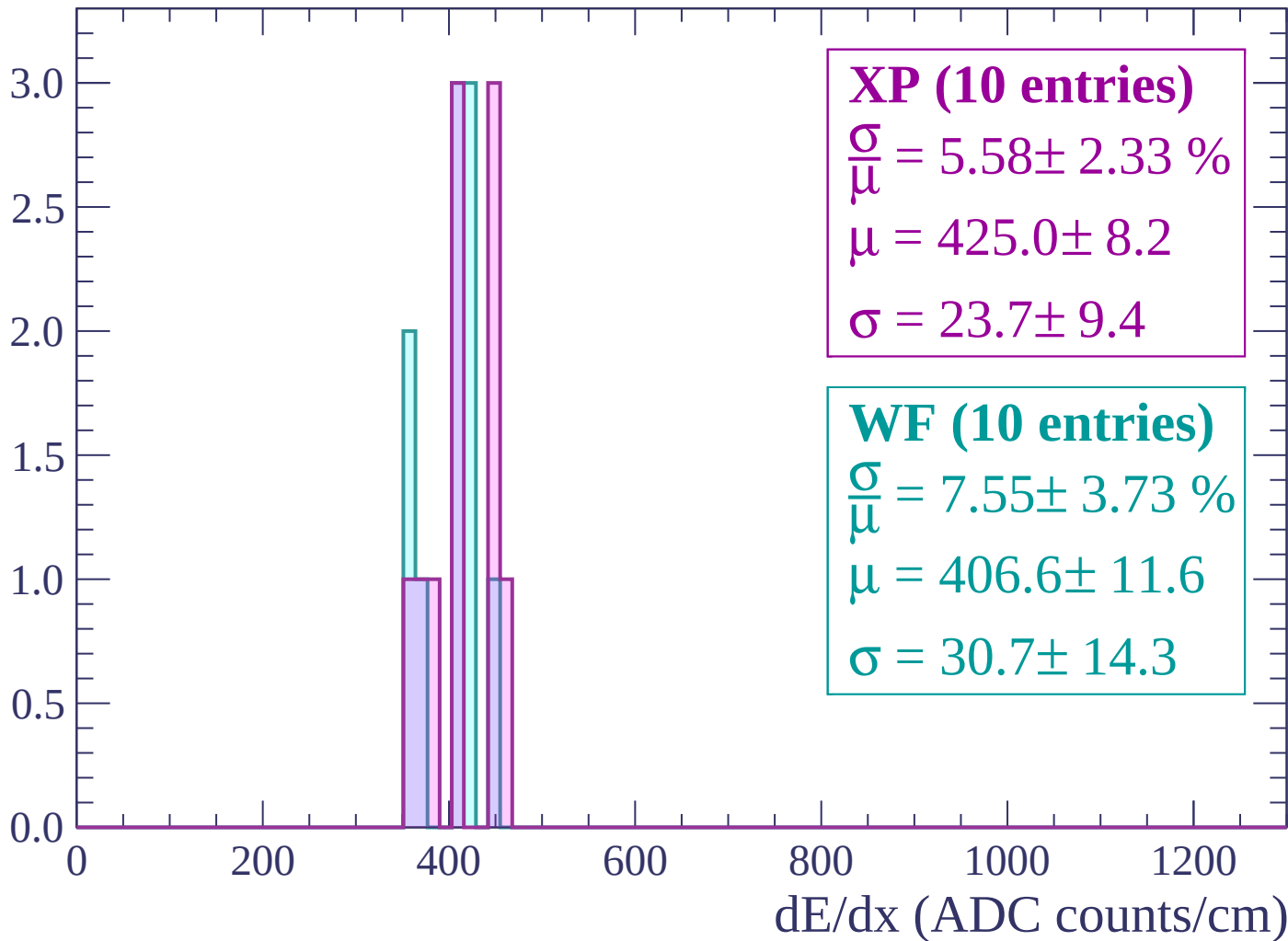
$$\frac{\sigma}{\bar{\mu}} = 1.72 \pm 9.61 \%$$

$$\mu = 377.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.7$$

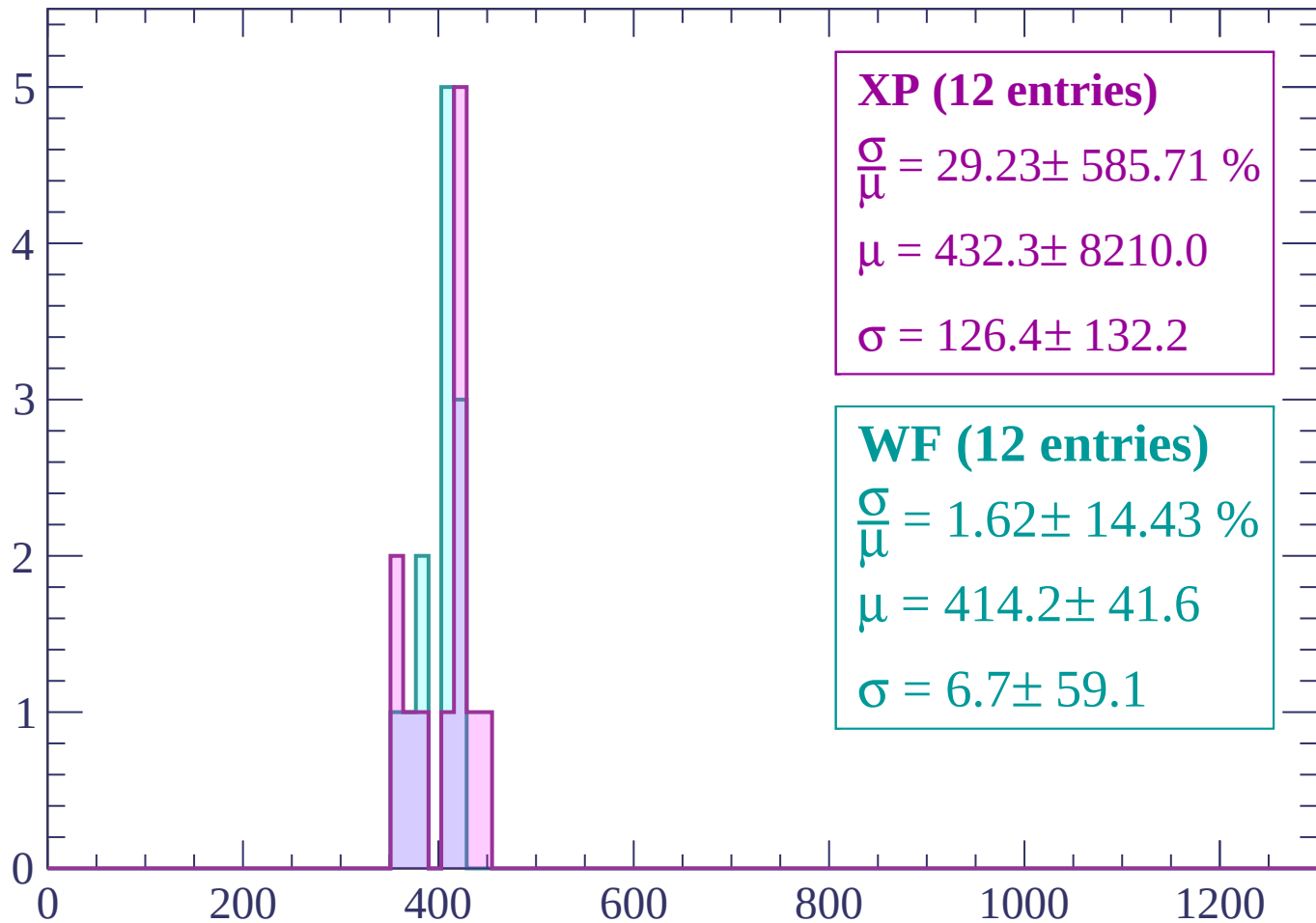
Energy loss | $-540 < p < -480$

Count



Energy loss | $-480 < p < -420$

Count



XP (12 entries)

$\frac{\sigma}{\mu} = 29.23 \pm 585.71 \%$

$\mu = 432.3 \pm 8210.0$

$\sigma = 126.4 \pm 132.2$

WF (12 entries)

$\frac{\sigma}{\mu} = 1.62 \pm 14.43 \%$

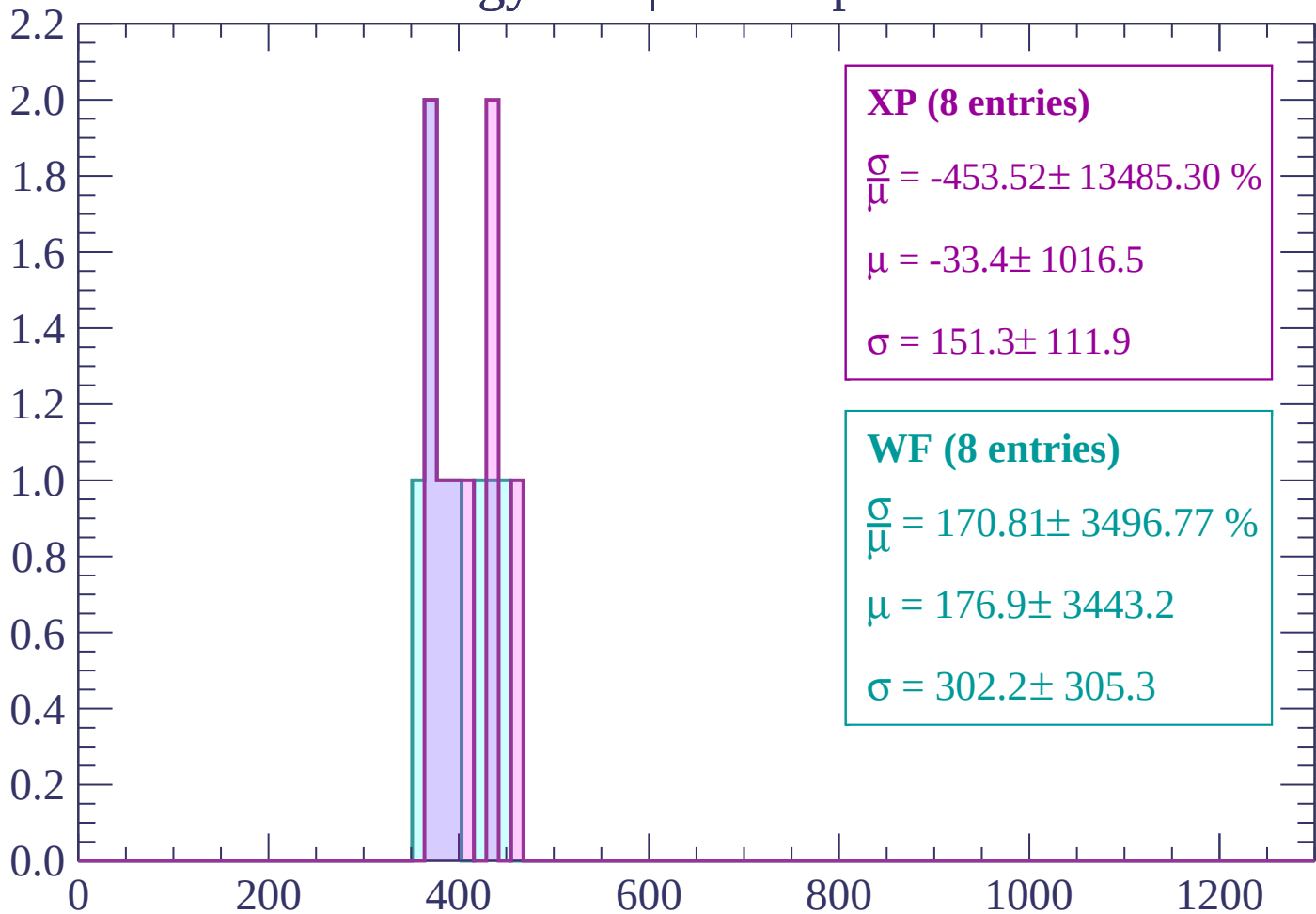
$\mu = 414.2 \pm 41.6$

$\sigma = 6.7 \pm 59.1$

dE/dx (ADC counts/cm)

Energy loss | -420 < p < -360

Count



XP (8 entries)

$\frac{\sigma}{\mu} = -453.52 \pm 13485.30 \%$

$\mu = -33.4 \pm 1016.5$

$\sigma = 151.3 \pm 111.9$

WF (8 entries)

$\frac{\sigma}{\mu} = 170.81 \pm 3496.77 \%$

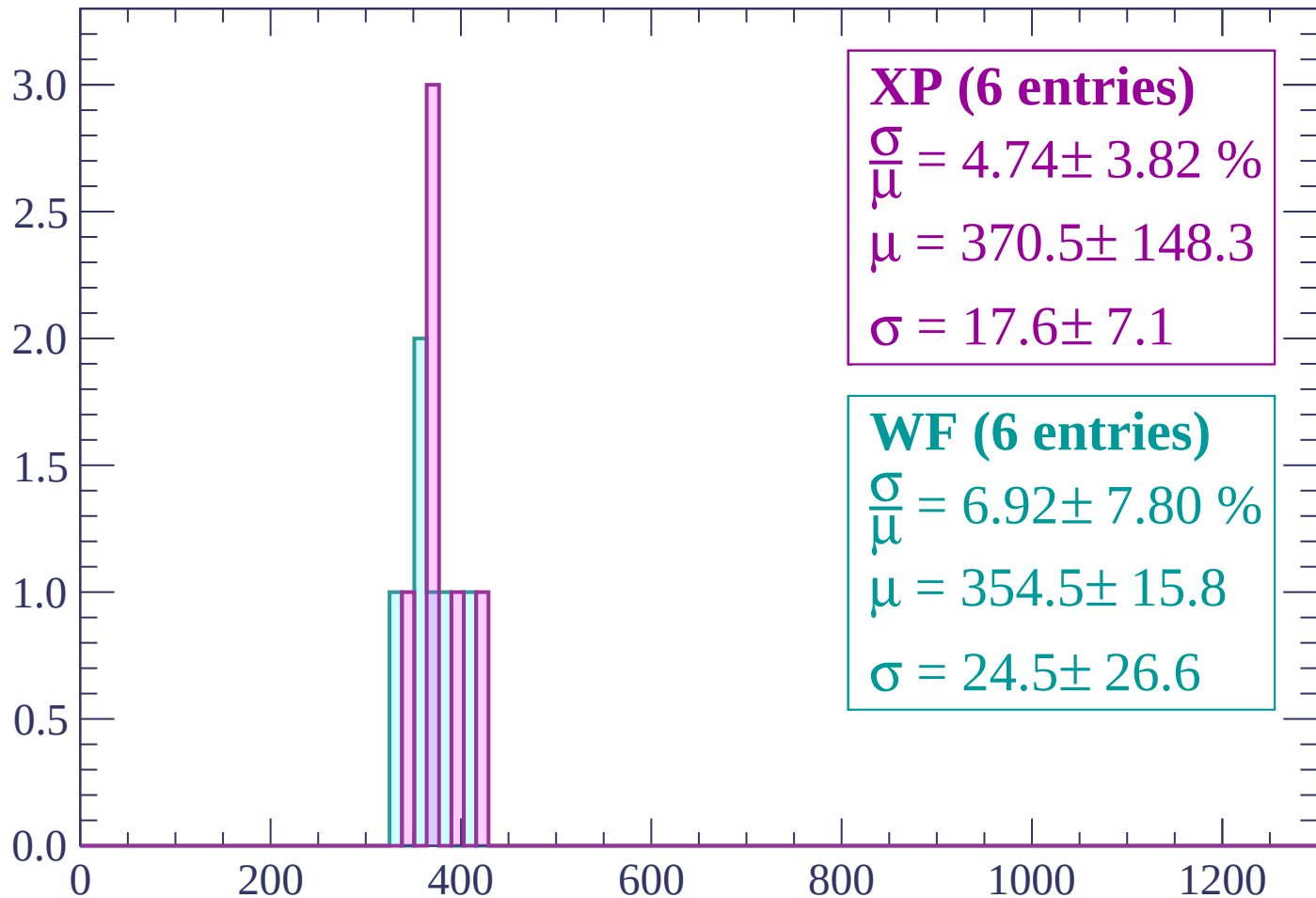
$\mu = 176.9 \pm 3443.2$

$\sigma = 302.2 \pm 305.3$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP (6 entries)

$$\frac{\sigma}{\mu} = 4.74 \pm 3.82 \%$$

$$\mu = 370.5 \pm 148.3$$

$$\sigma = 17.6 \pm 7.1$$

WF (6 entries)

$$\frac{\sigma}{\mu} = 6.92 \pm 7.80 \%$$

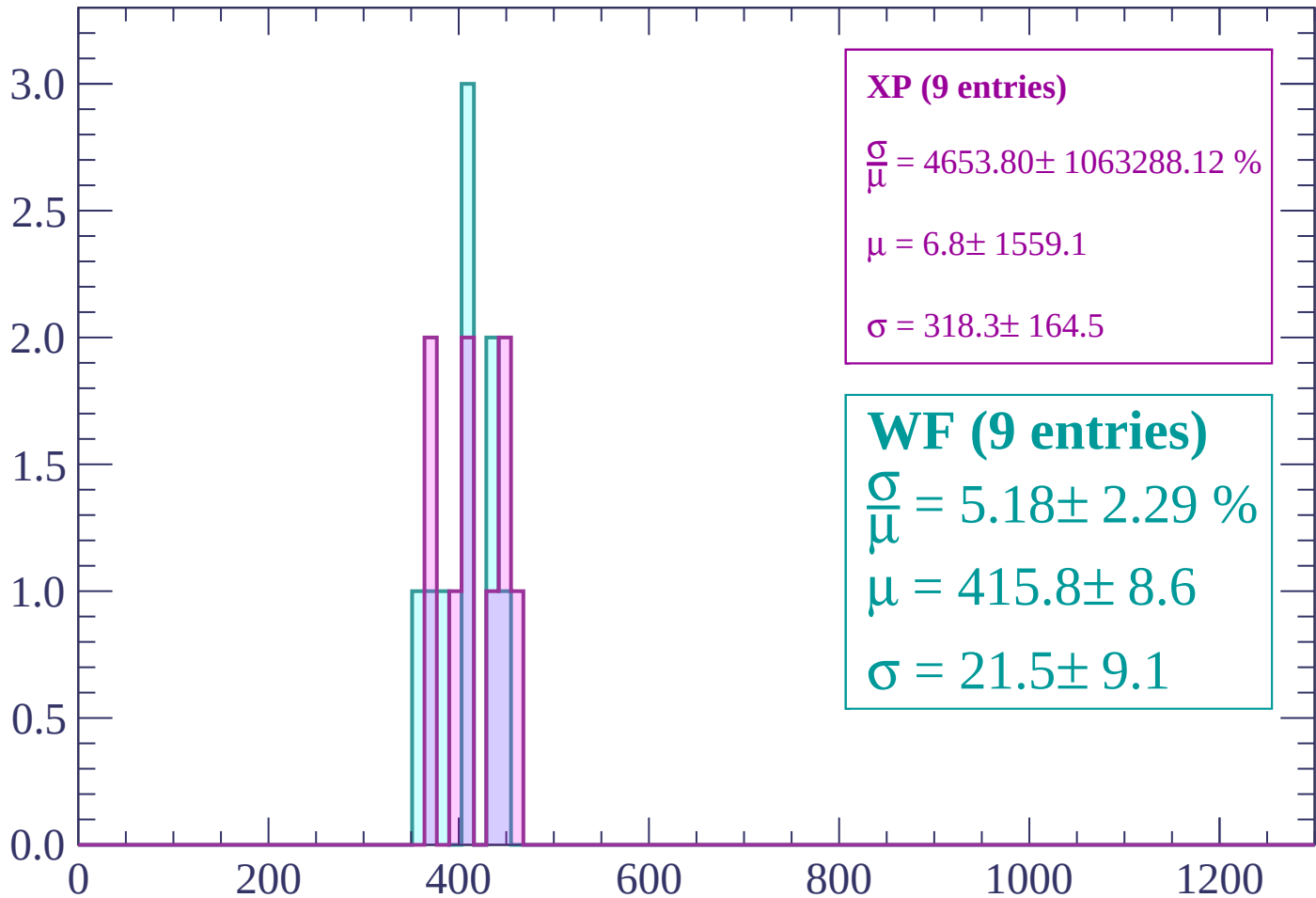
$$\mu = 354.5 \pm 15.8$$

$$\sigma = 24.5 \pm 26.6$$

dE/dx (ADC counts/cm)

Energy loss | -300 < p < -240

Count



XP (9 entries)

$\frac{\sigma}{\mu} = 4653.80 \pm 1063288.12 \%$

$\mu = 6.8 \pm 1559.1$

$\sigma = 318.3 \pm 164.5$

WF (9 entries)

$\frac{\sigma}{\mu} = 5.18 \pm 2.29 \%$

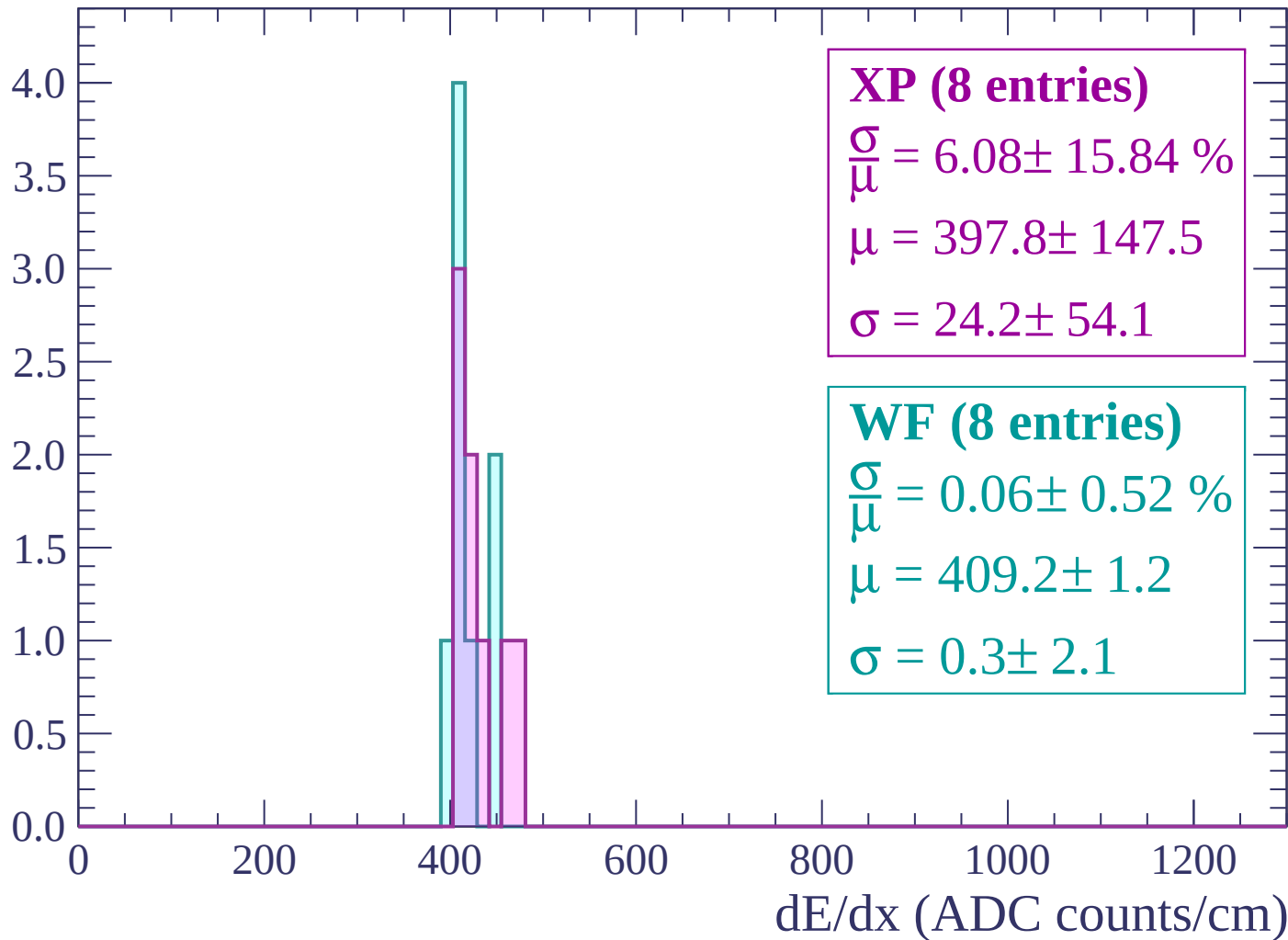
$\mu = 415.8 \pm 8.6$

$\sigma = 21.5 \pm 9.1$

dE/dx (ADC counts/cm)

Energy loss | -240 < p < -180

Count



Energy loss | -180 < p < -120

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

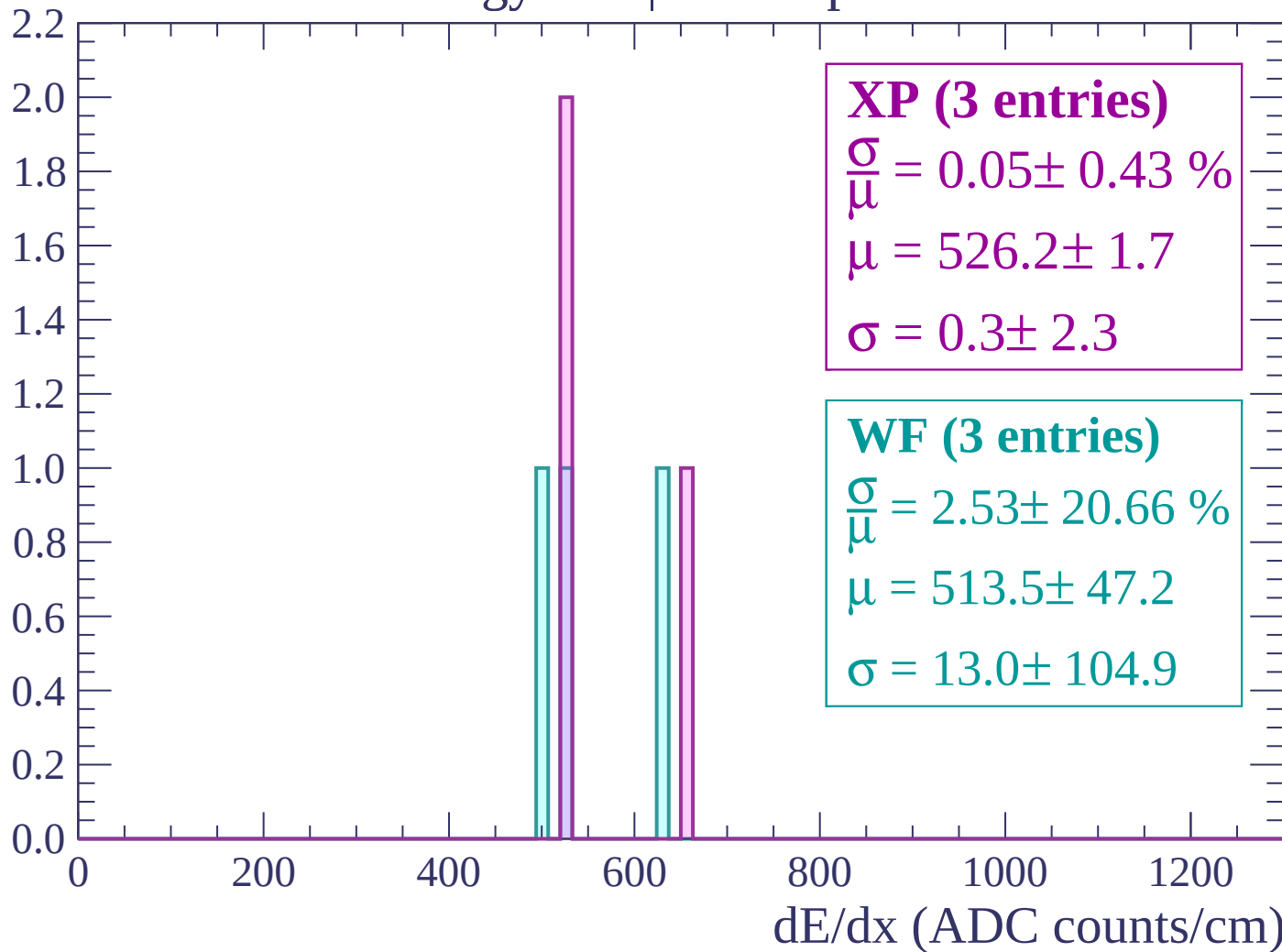
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

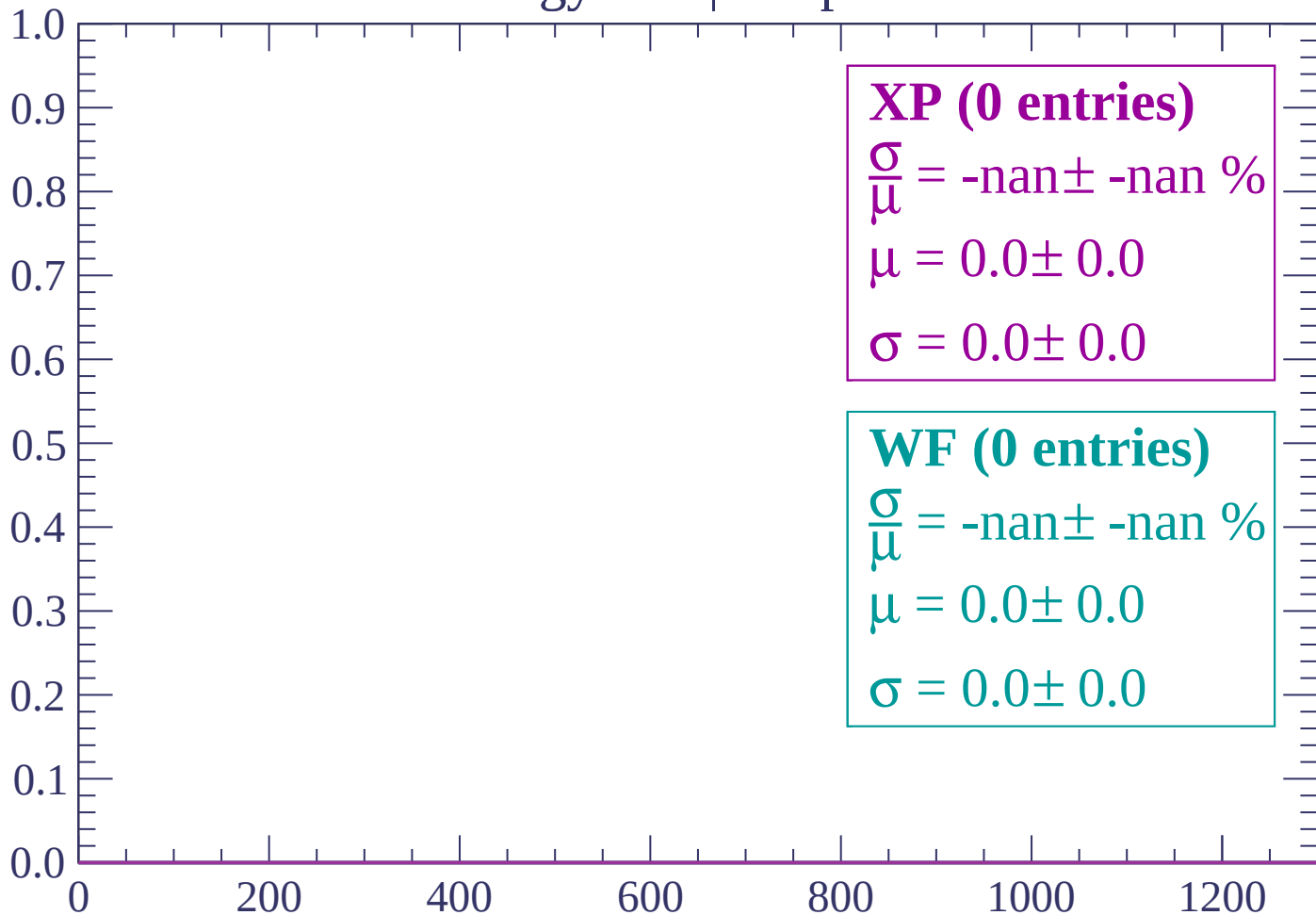
$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $0 < p < 60$

Count



dE/dx (ADC counts/cm)

Energy loss | $60 < p < 120$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

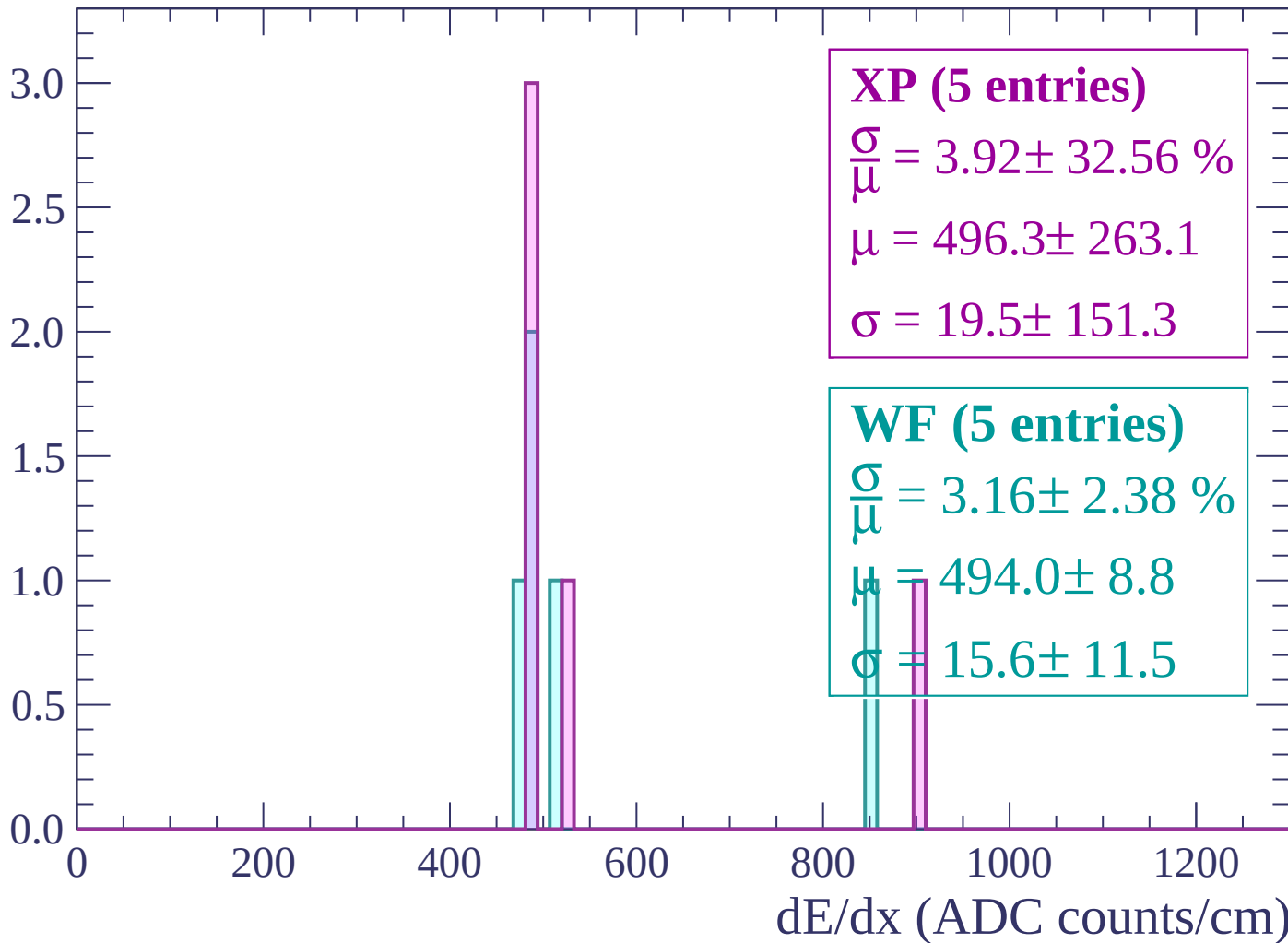
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

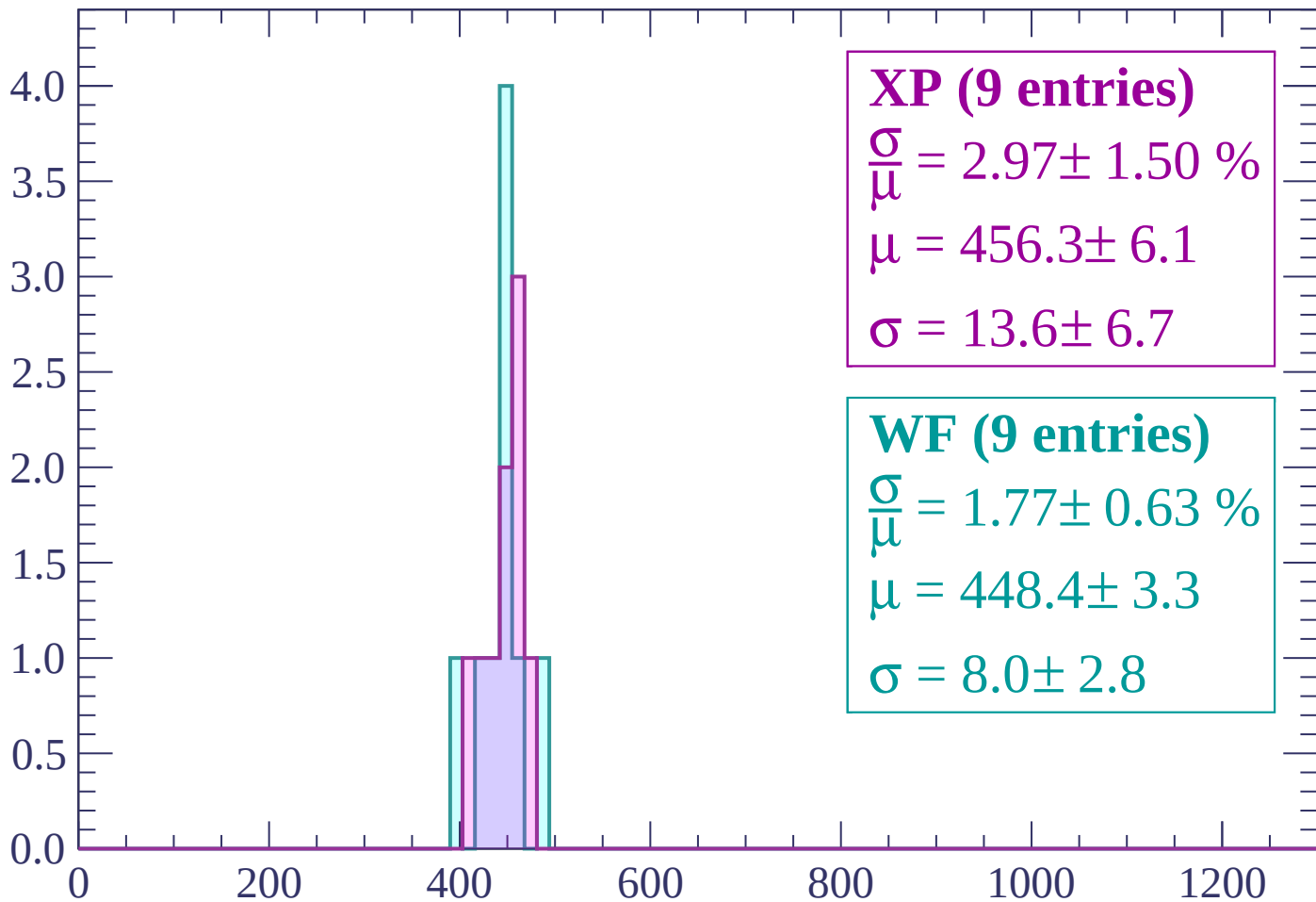
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (9 entries)

$\frac{\sigma}{\mu} = 2.97 \pm 1.50 \%$

$\mu = 456.3 \pm 6.1$

$\sigma = 13.6 \pm 6.7$

WF (9 entries)

$\frac{\sigma}{\mu} = 1.77 \pm 0.63 \%$

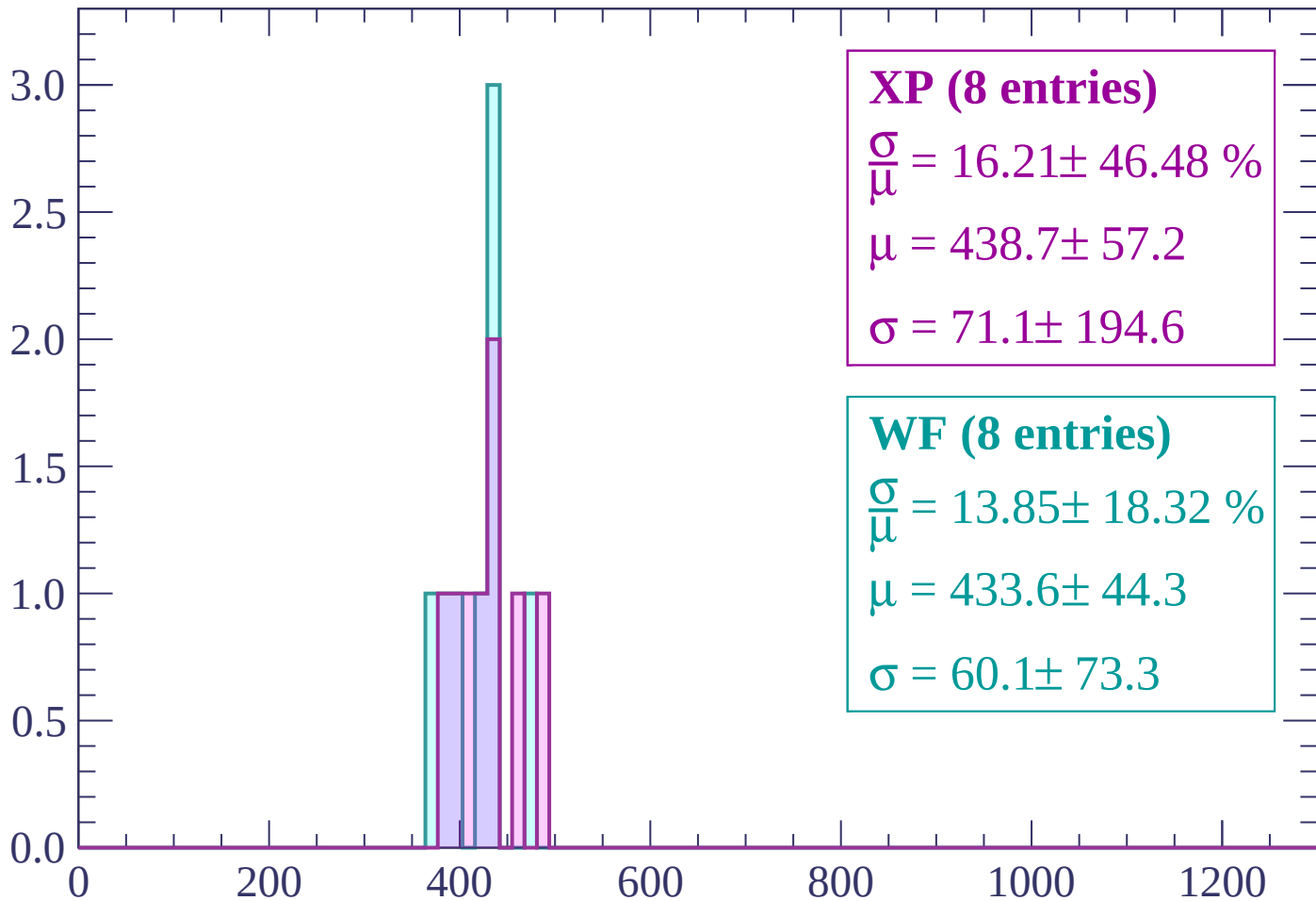
$\mu = 448.4 \pm 3.3$

$\sigma = 8.0 \pm 2.8$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (8 entries)

$$\frac{\sigma}{\mu} = 16.21 \pm 46.48 \%$$

$$\mu = 438.7 \pm 57.2$$

$$\sigma = 71.1 \pm 194.6$$

WF (8 entries)

$$\frac{\sigma}{\mu} = 13.85 \pm 18.32 \%$$

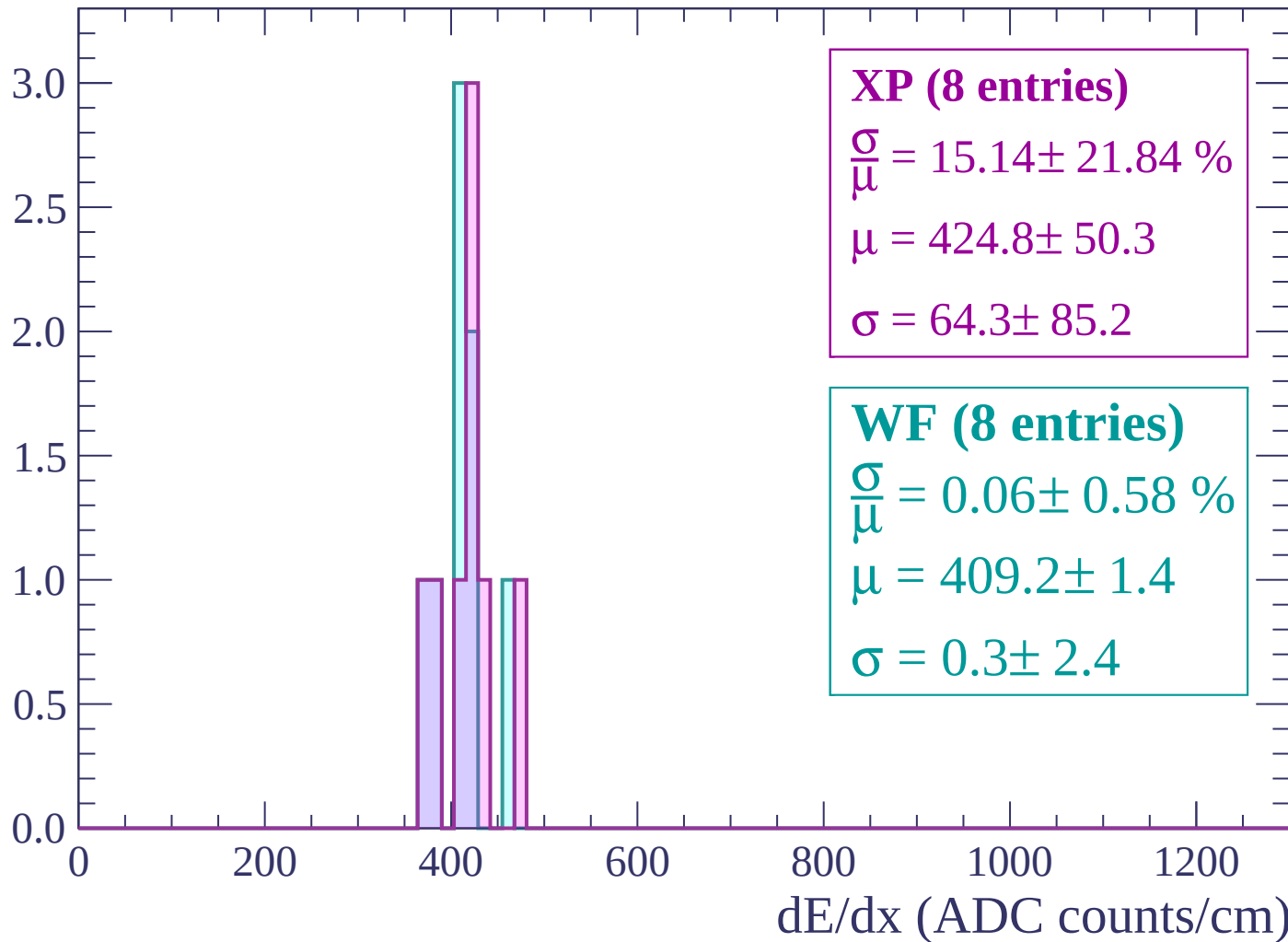
$$\mu = 433.6 \pm 44.3$$

$$\sigma = 60.1 \pm 73.3$$

dE/dx (ADC counts/cm)

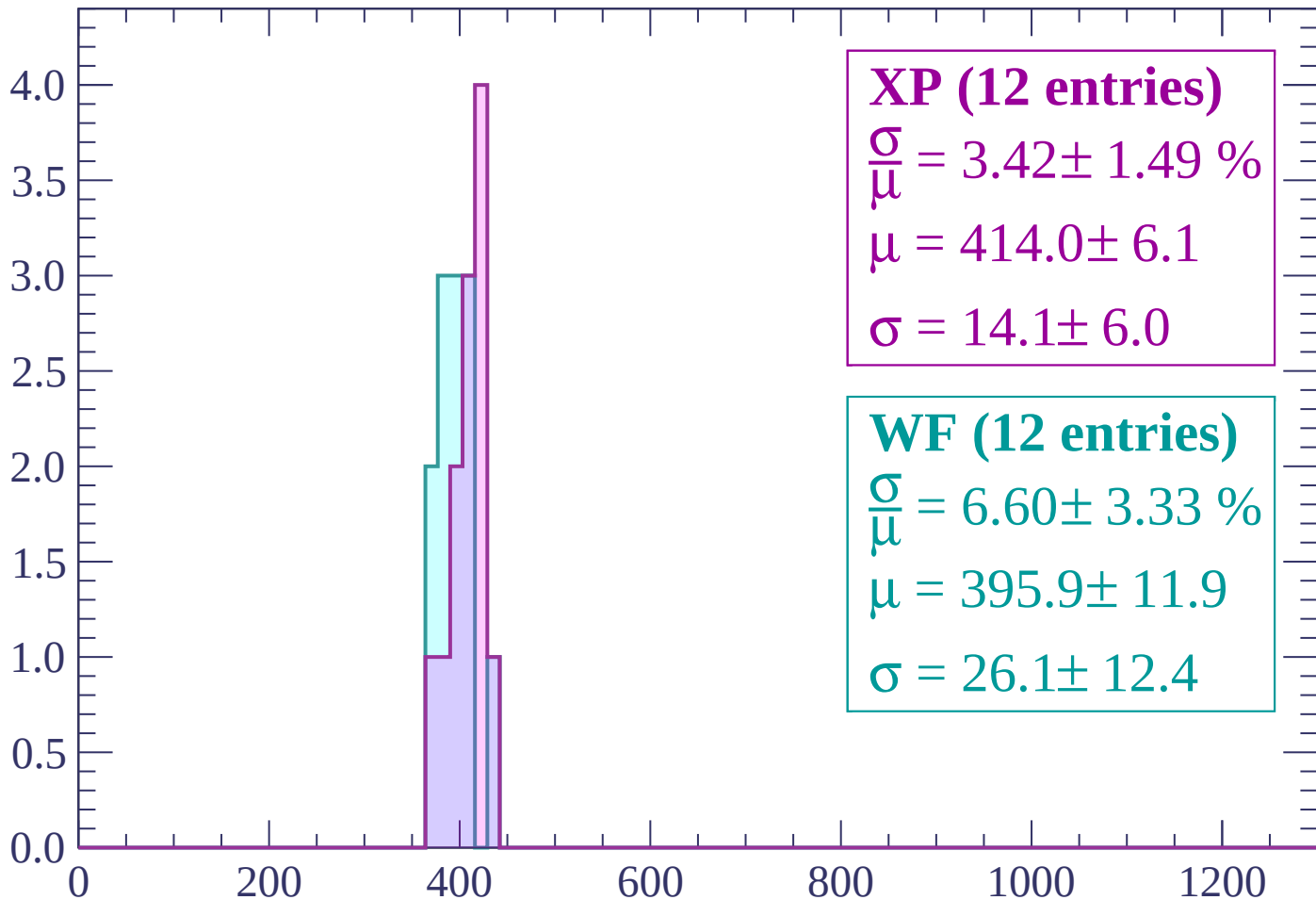
Energy loss | $300 < p < 360$

Count



Energy loss | $360 < p < 420$

Count



XP (12 entries)

$\frac{\sigma}{\mu} = 3.42 \pm 1.49 \%$

$\mu = 414.0 \pm 6.1$

$\sigma = 14.1 \pm 6.0$

WF (12 entries)

$\frac{\sigma}{\mu} = 6.60 \pm 3.33 \%$

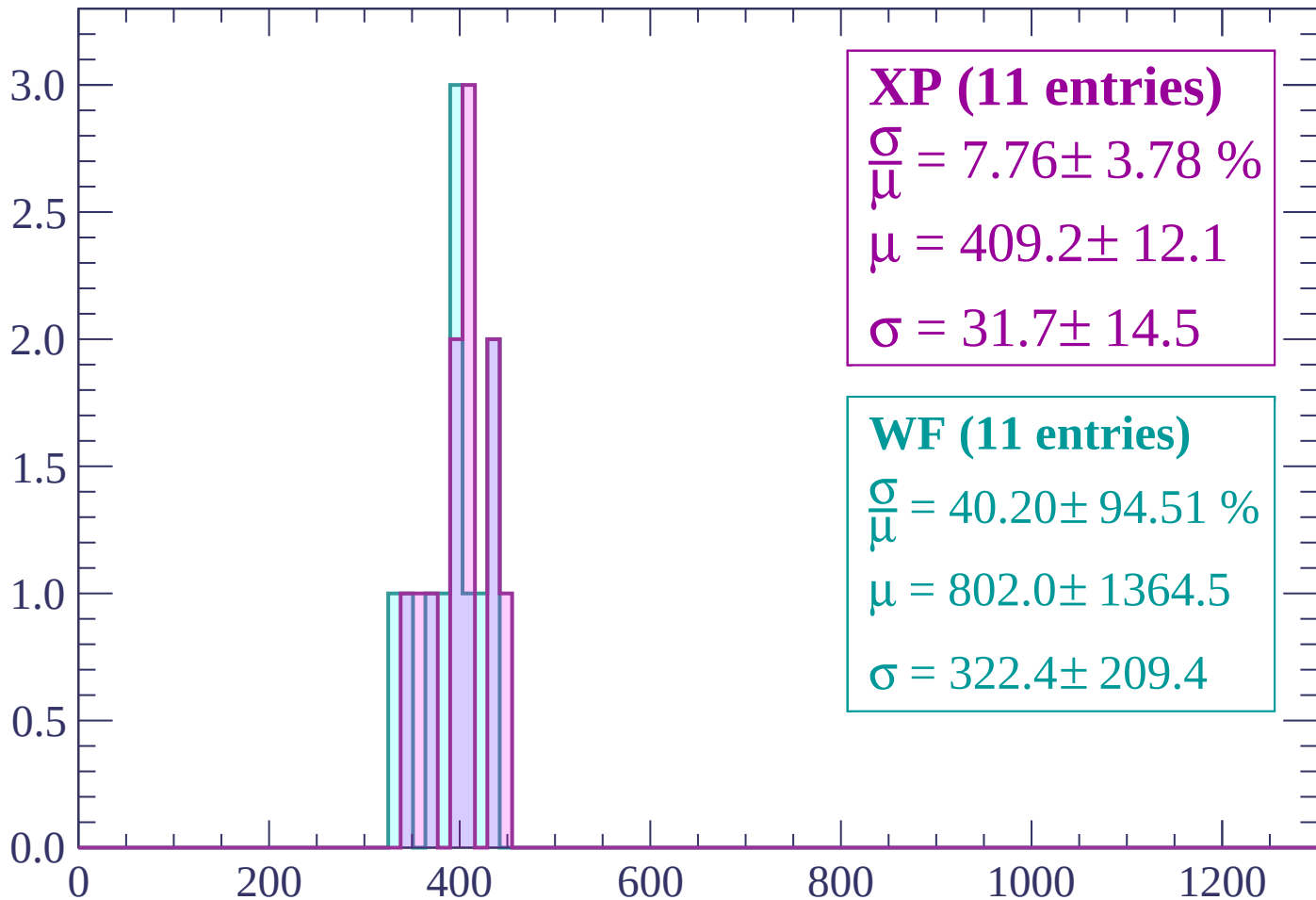
$\mu = 395.9 \pm 11.9$

$\sigma = 26.1 \pm 12.4$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (11 entries)

$\frac{\sigma}{\mu} = 7.76 \pm 3.78 \%$

$\mu = 409.2 \pm 12.1$

$\sigma = 31.7 \pm 14.5$

WF (11 entries)

$\frac{\sigma}{\mu} = 40.20 \pm 94.51 \%$

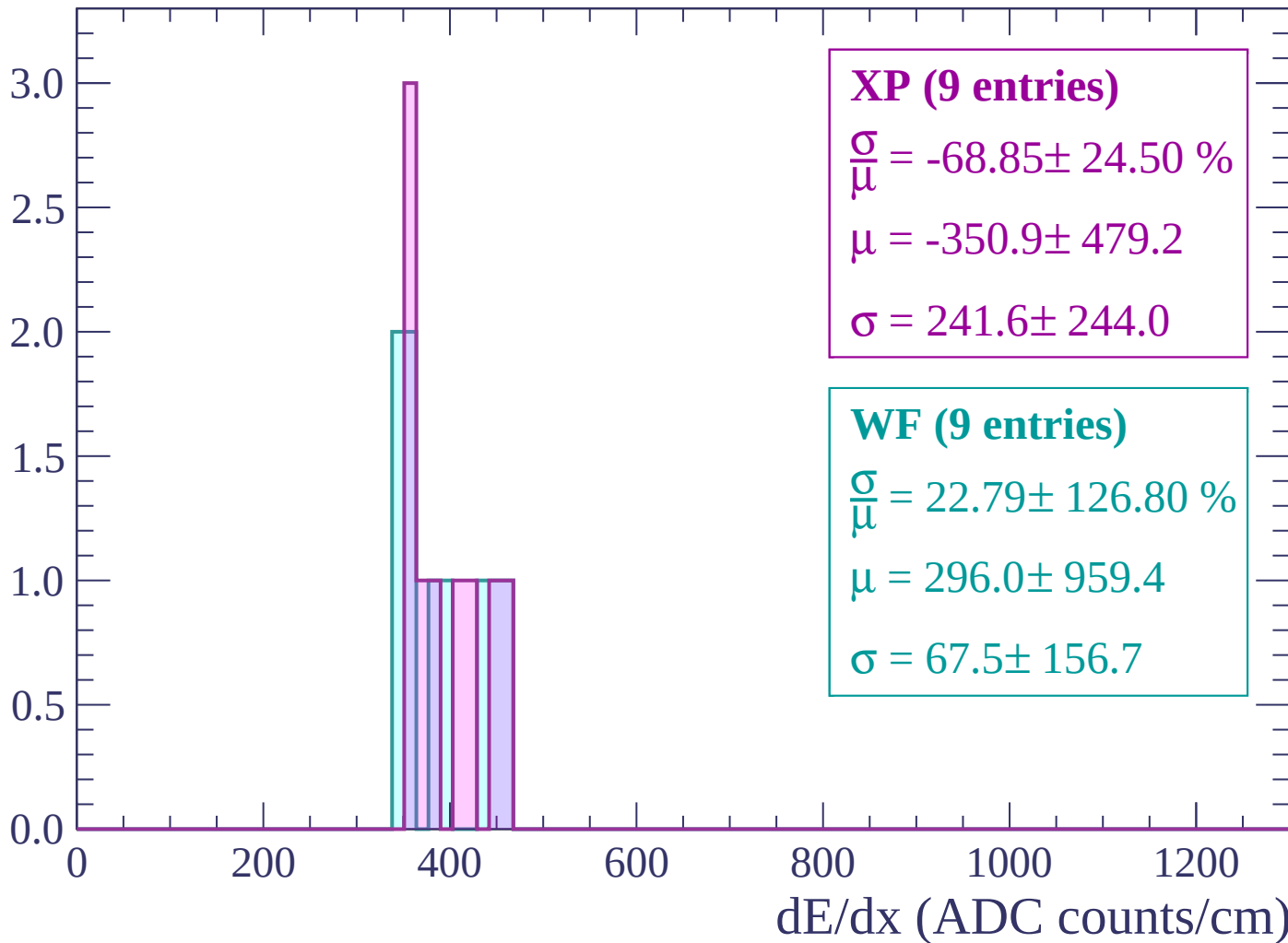
$\mu = 802.0 \pm 1364.5$

$\sigma = 322.4 \pm 209.4$

dE/dx (ADC counts/cm)

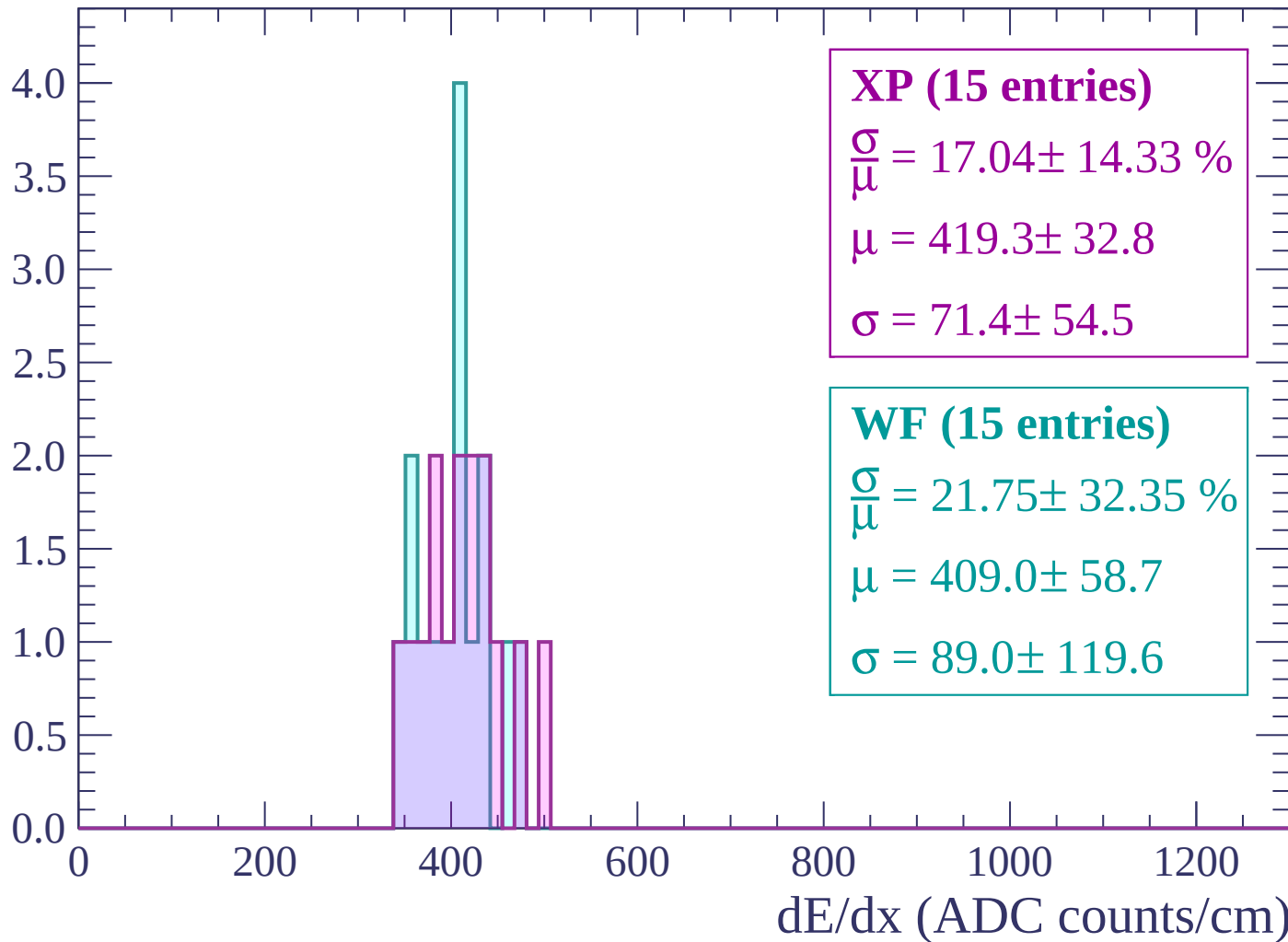
Energy loss | $480 < p < 540$

Count



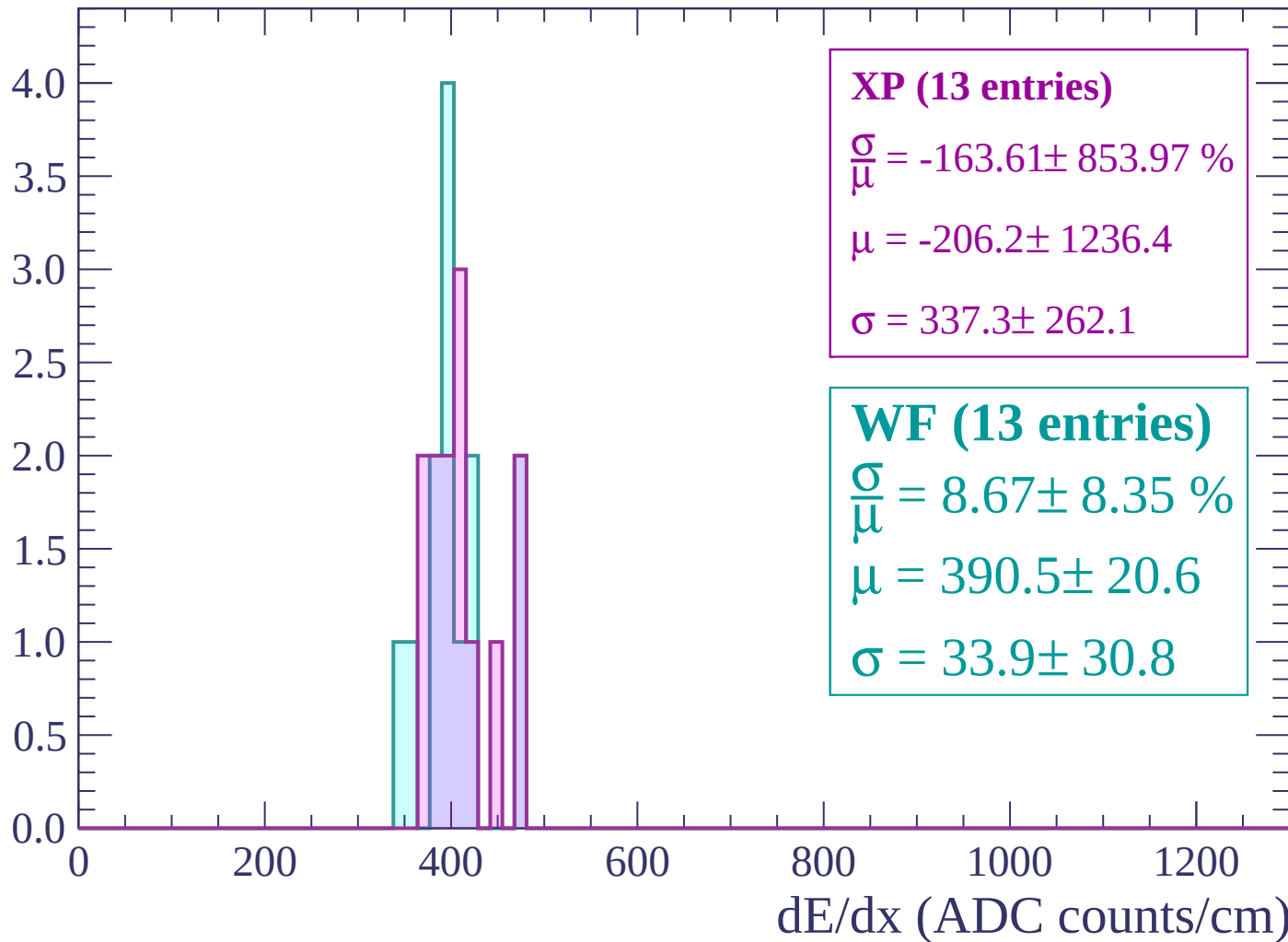
Energy loss | $540 < p < 600$

Count



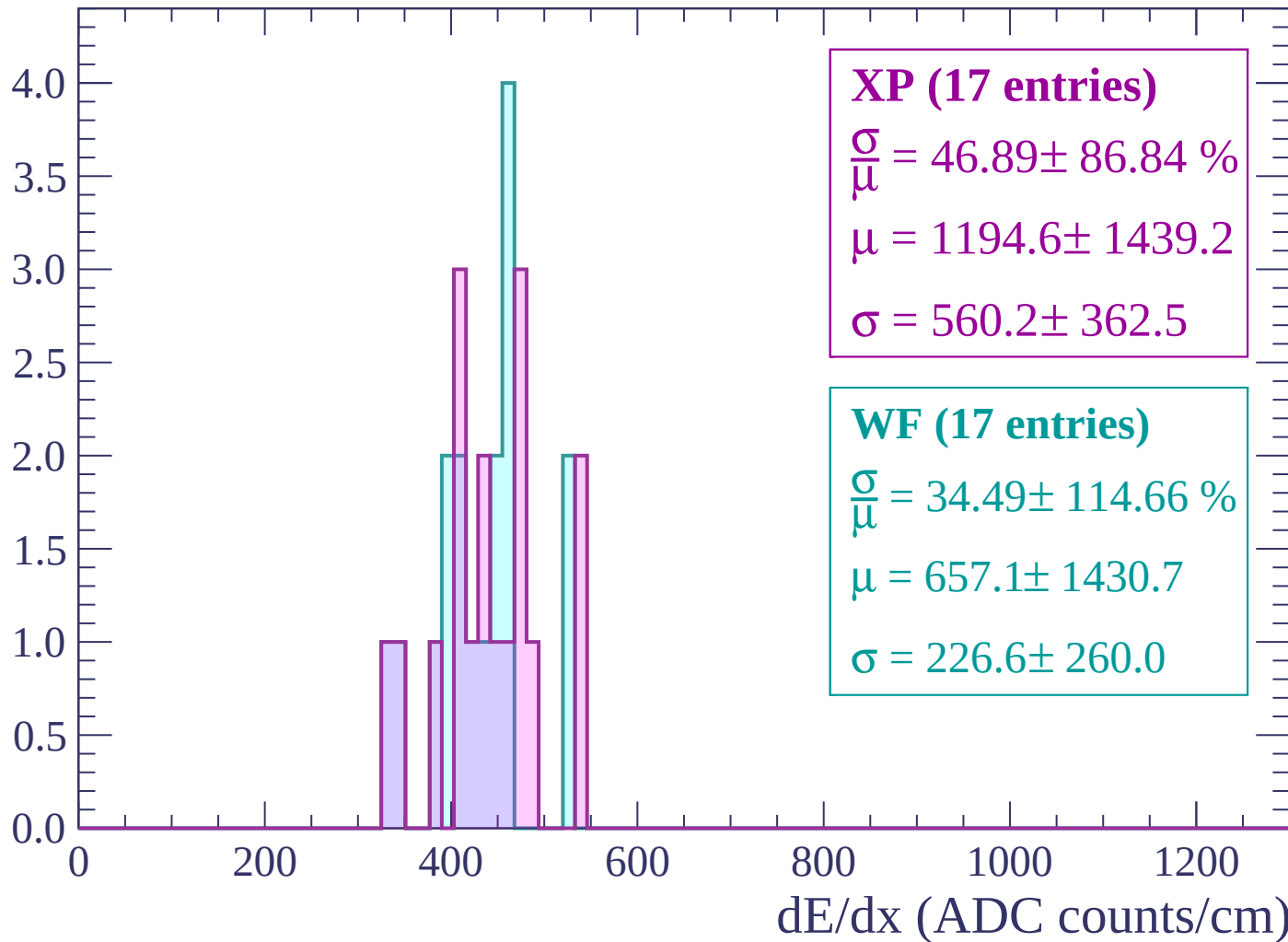
Energy loss | $600 < p < 660$

Count



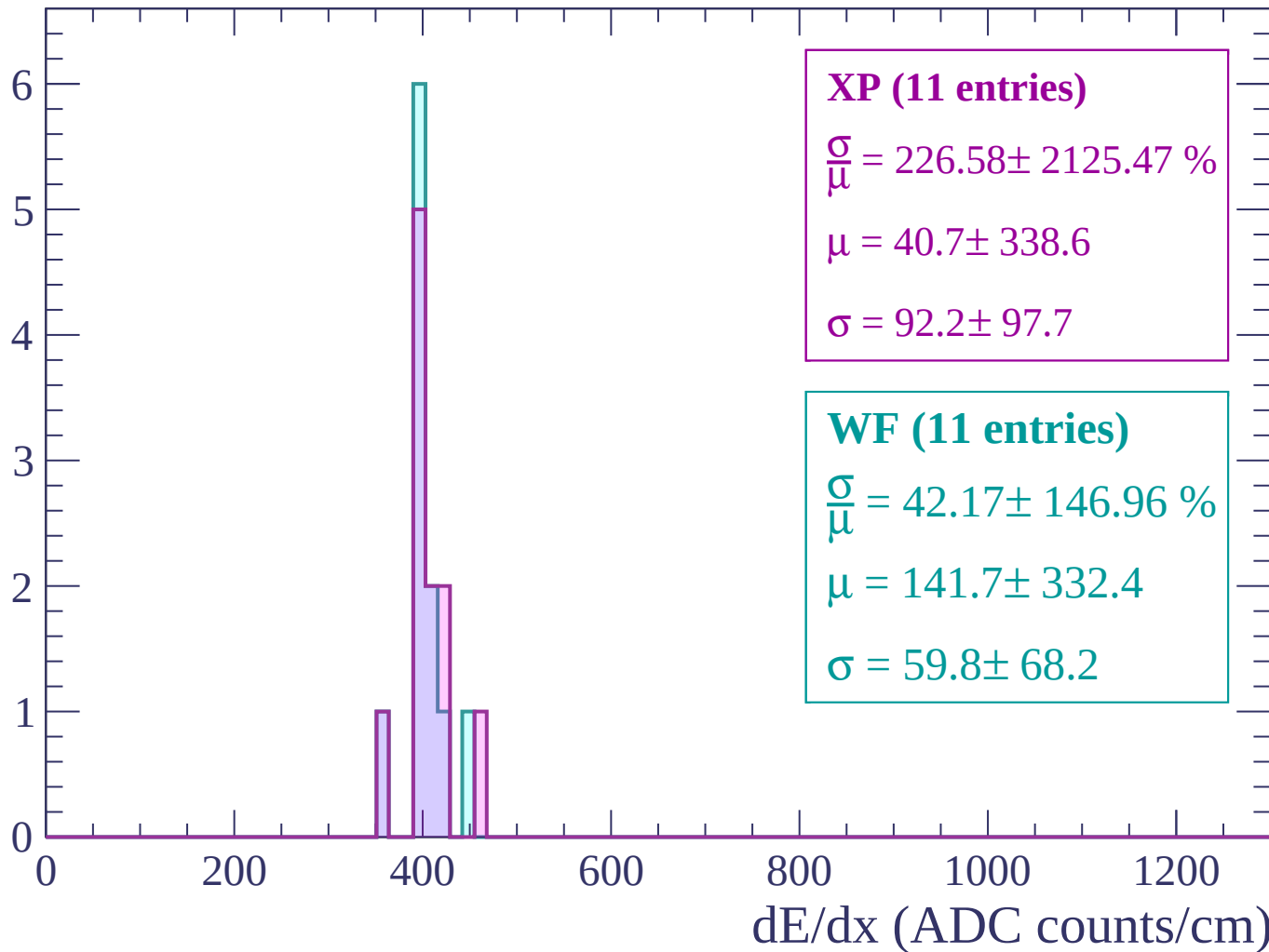
Energy loss | $660 < p < 720$

Count



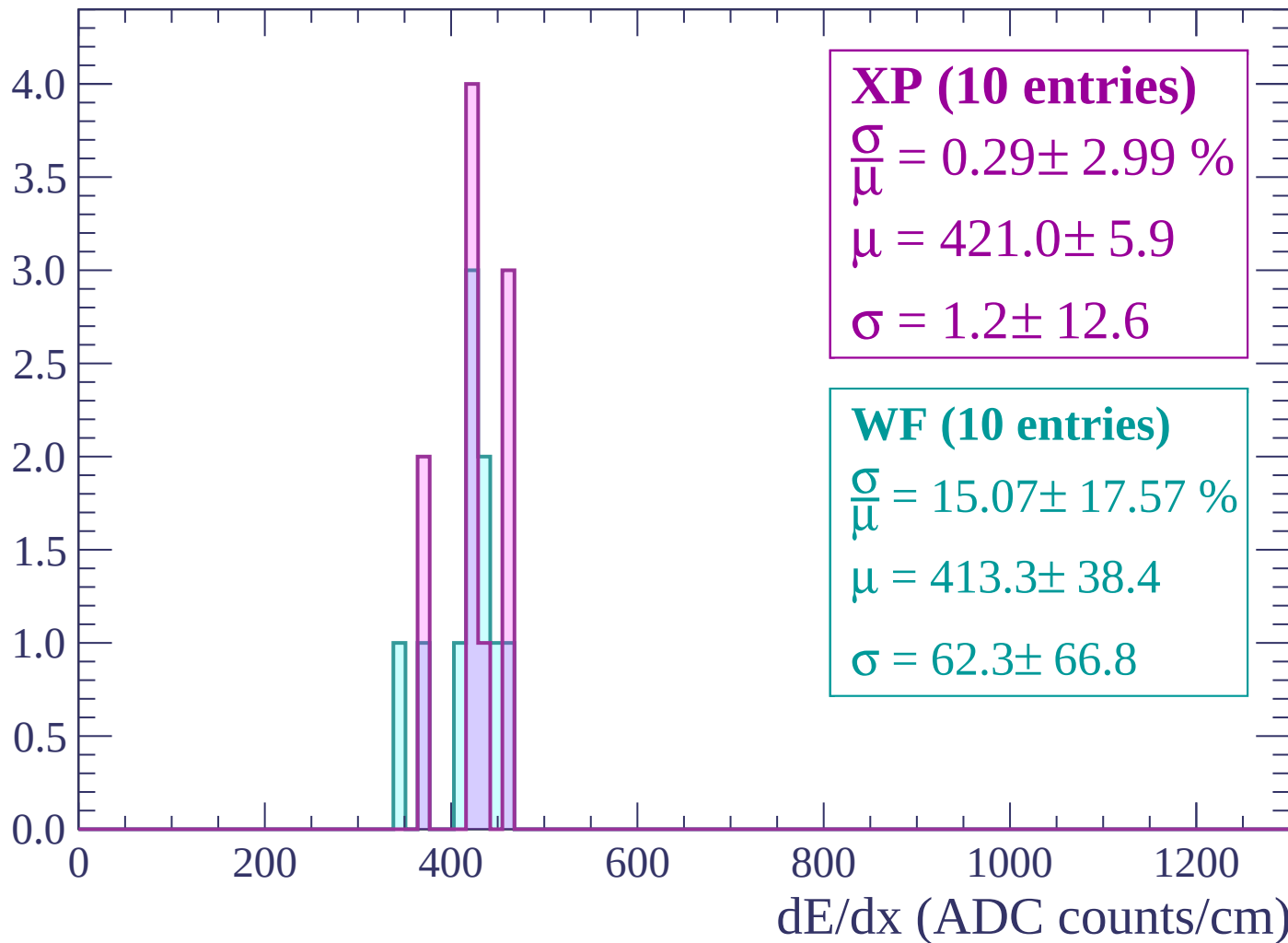
Energy loss | $720 < p < 780$

Count



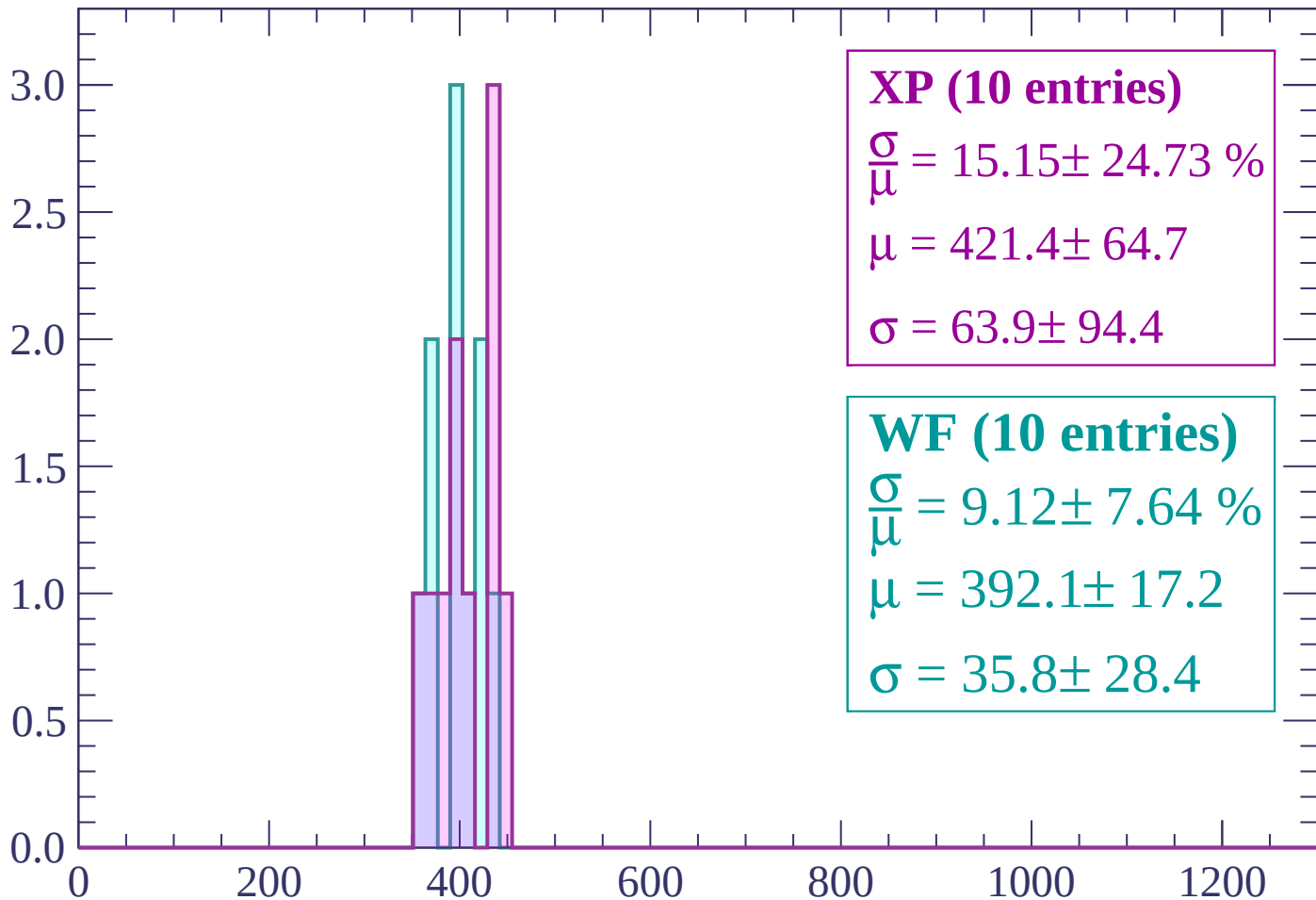
Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count



XP (10 entries)

$$\frac{\sigma}{\mu} = 15.15 \pm 24.73 \%$$

$$\mu = 421.4 \pm 64.7$$

$$\sigma = 63.9 \pm 94.4$$

WF (10 entries)

$$\frac{\sigma}{\mu} = 9.12 \pm 7.64 \%$$

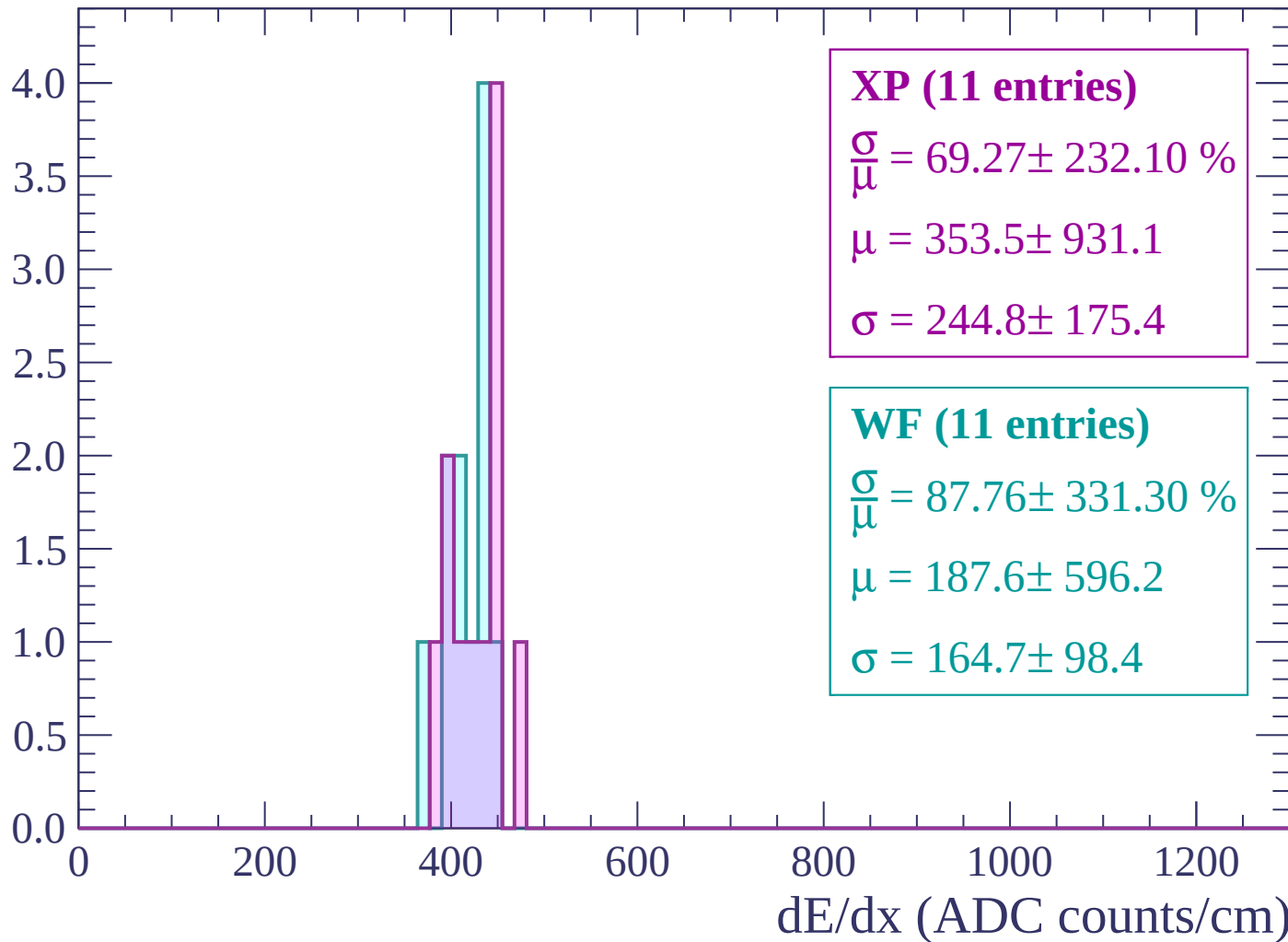
$$\mu = 392.1 \pm 17.2$$

$$\sigma = 35.8 \pm 28.4$$

dE/dx (ADC counts/cm)

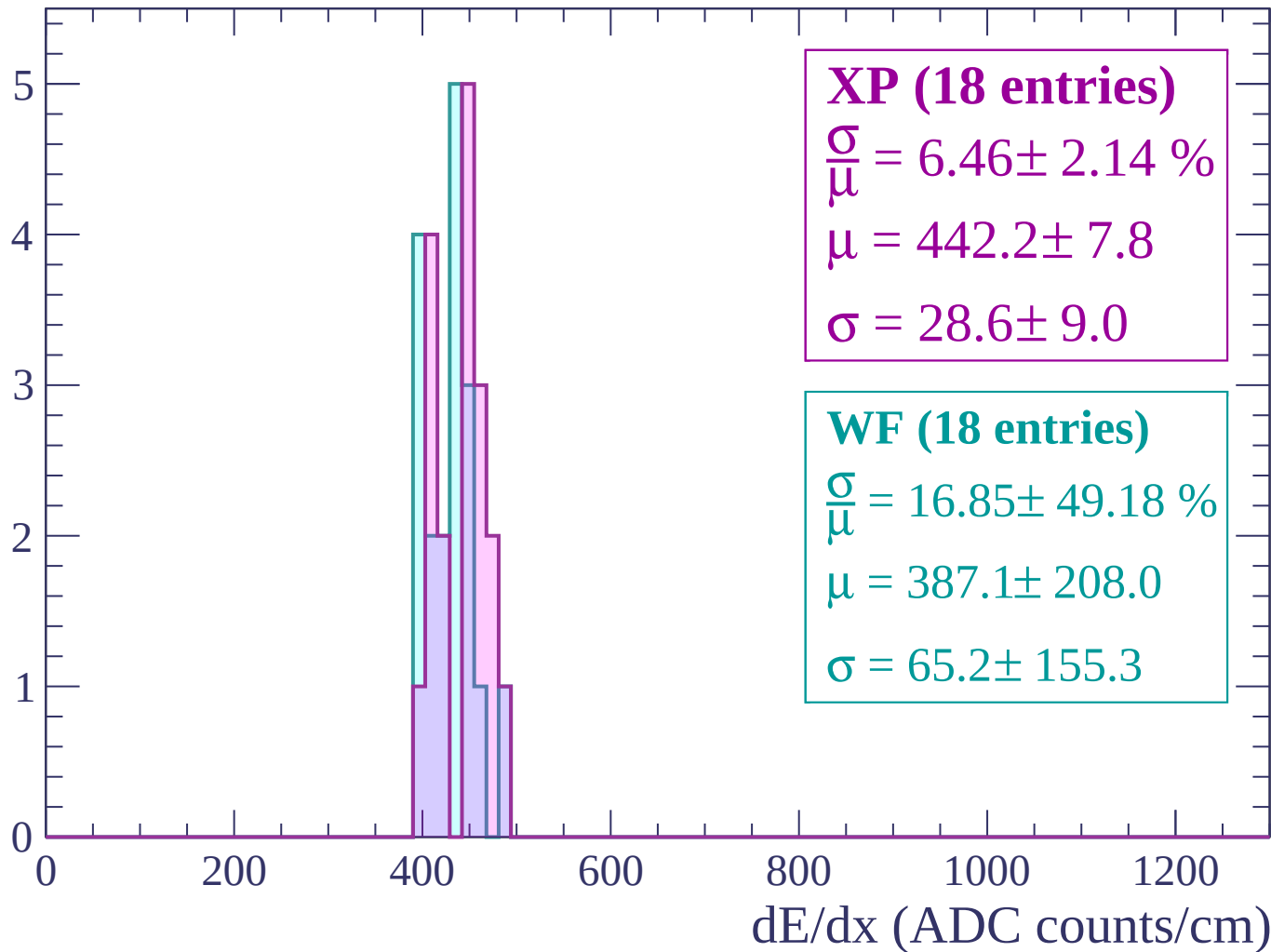
Energy loss | $900 < p < 960$

Count



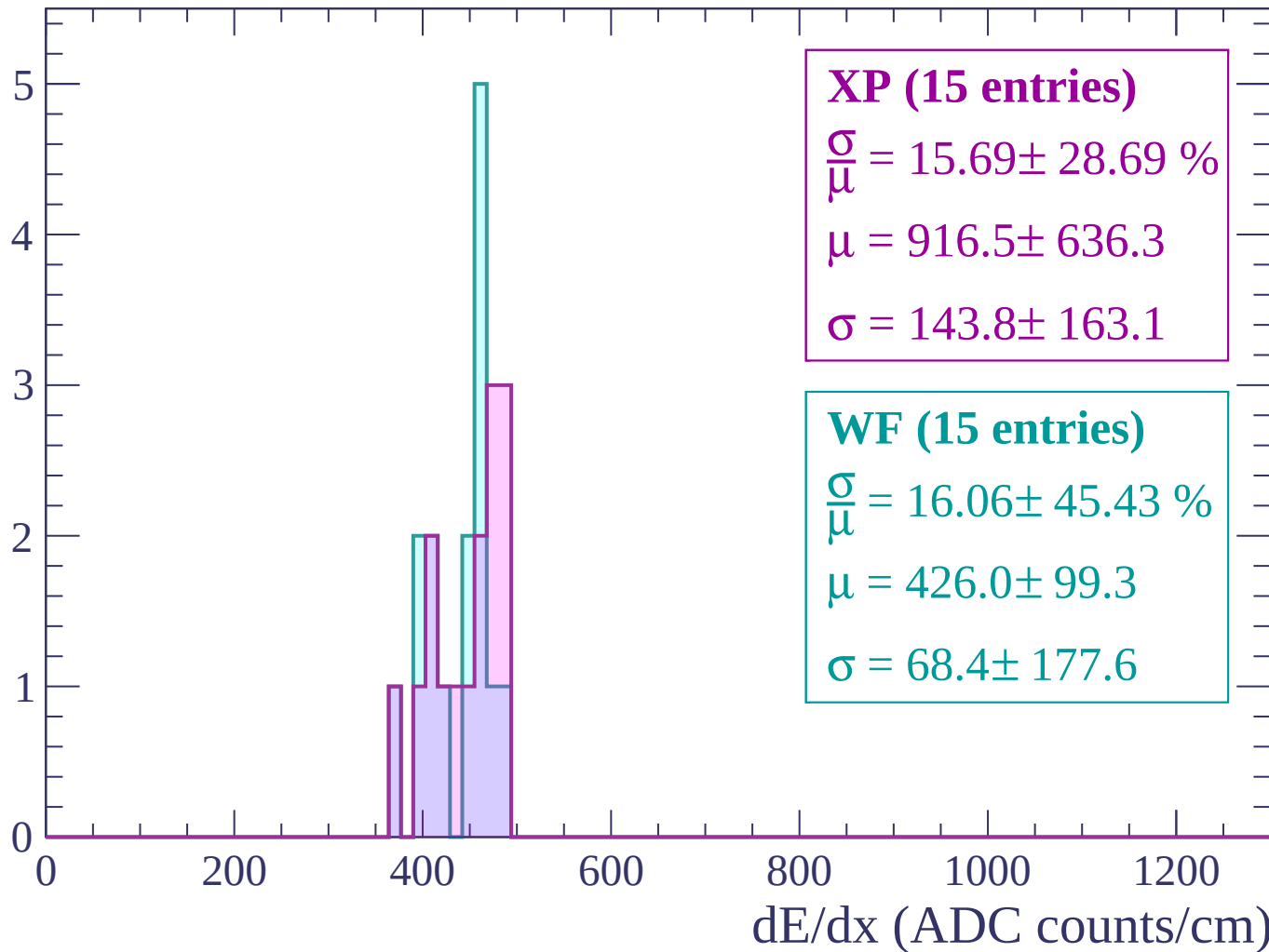
Energy loss | $960 < p < 1020$

Count



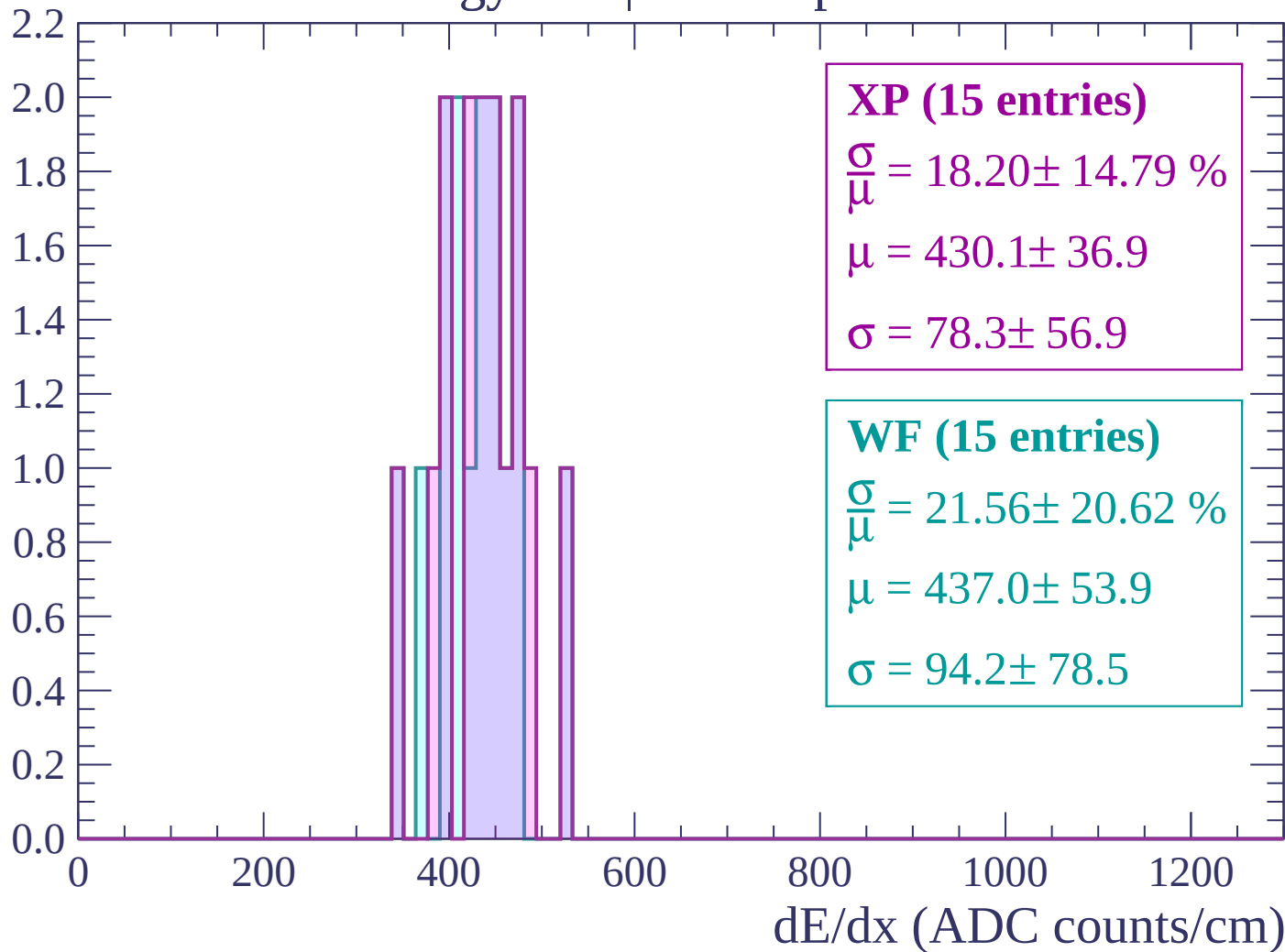
Energy loss | $1020 < p < 1080$

Count



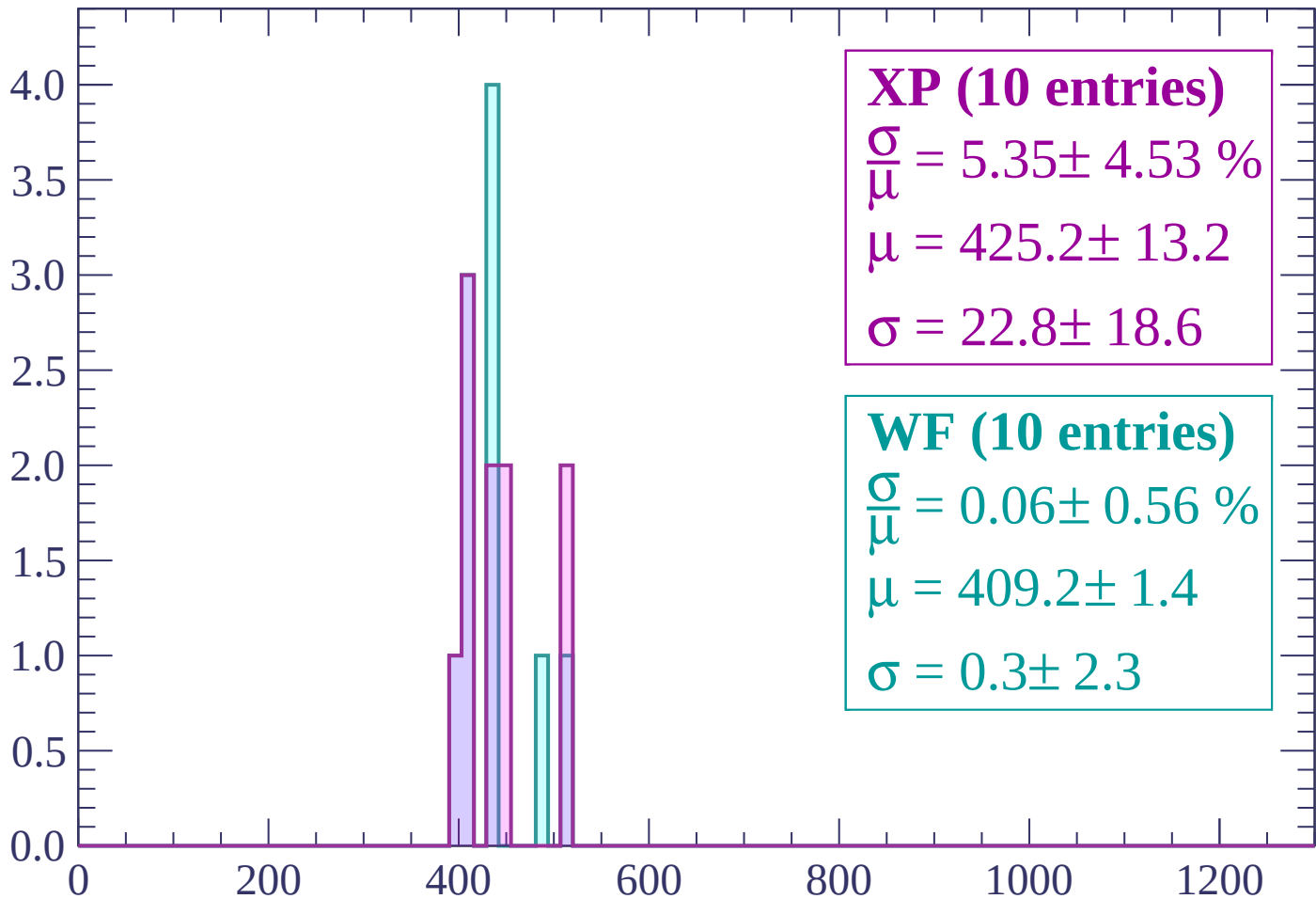
Energy loss | $1080 < p < 1140$

Count



Energy loss | $1140 < p < 1200$

Count



XP (10 entries)

$$\frac{\sigma}{\mu} = 5.35 \pm 4.53 \%$$

$$\mu = 425.2 \pm 13.2$$

$$\sigma = 22.8 \pm 18.6$$

WF (10 entries)

$$\frac{\sigma}{\mu} = 0.06 \pm 0.56 \%$$

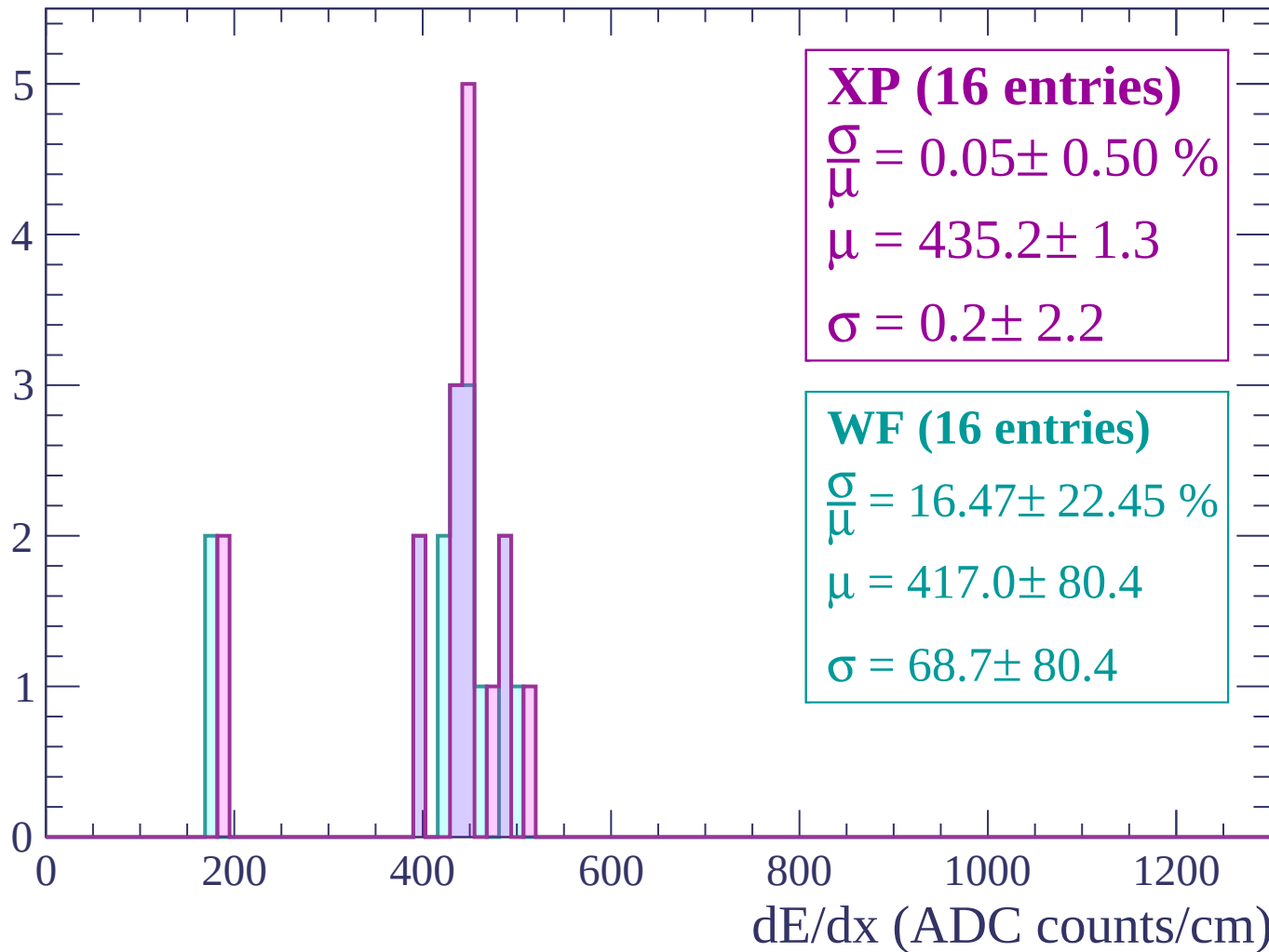
$$\mu = 409.2 \pm 1.4$$

$$\sigma = 0.3 \pm 2.3$$

dE/dx (ADC counts/cm)

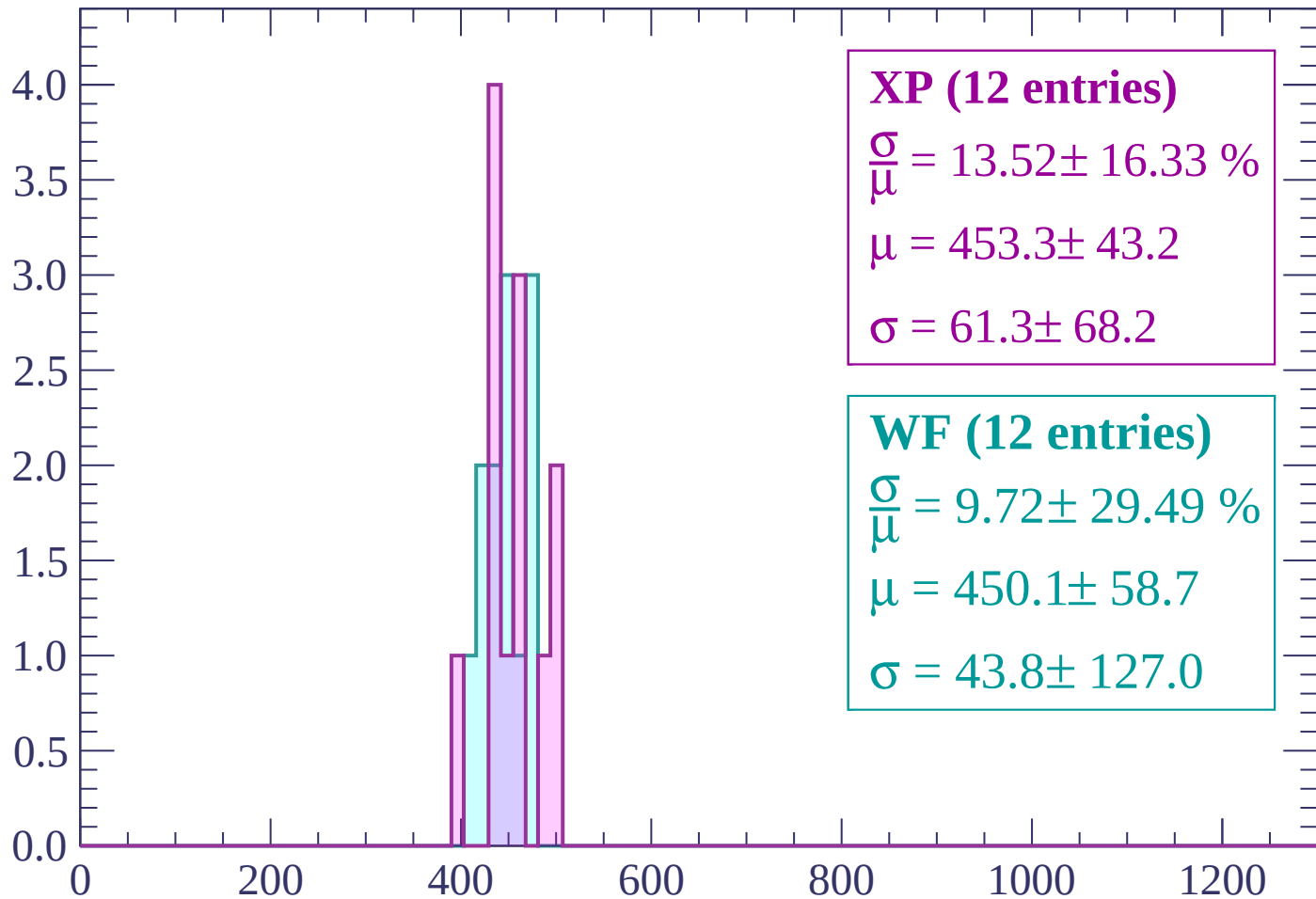
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP (12 entries)

$\frac{\sigma}{\mu} = 13.52 \pm 16.33 \%$

$\mu = 453.3 \pm 43.2$

$\sigma = 61.3 \pm 68.2$

WF (12 entries)

$\frac{\sigma}{\mu} = 9.72 \pm 29.49 \%$

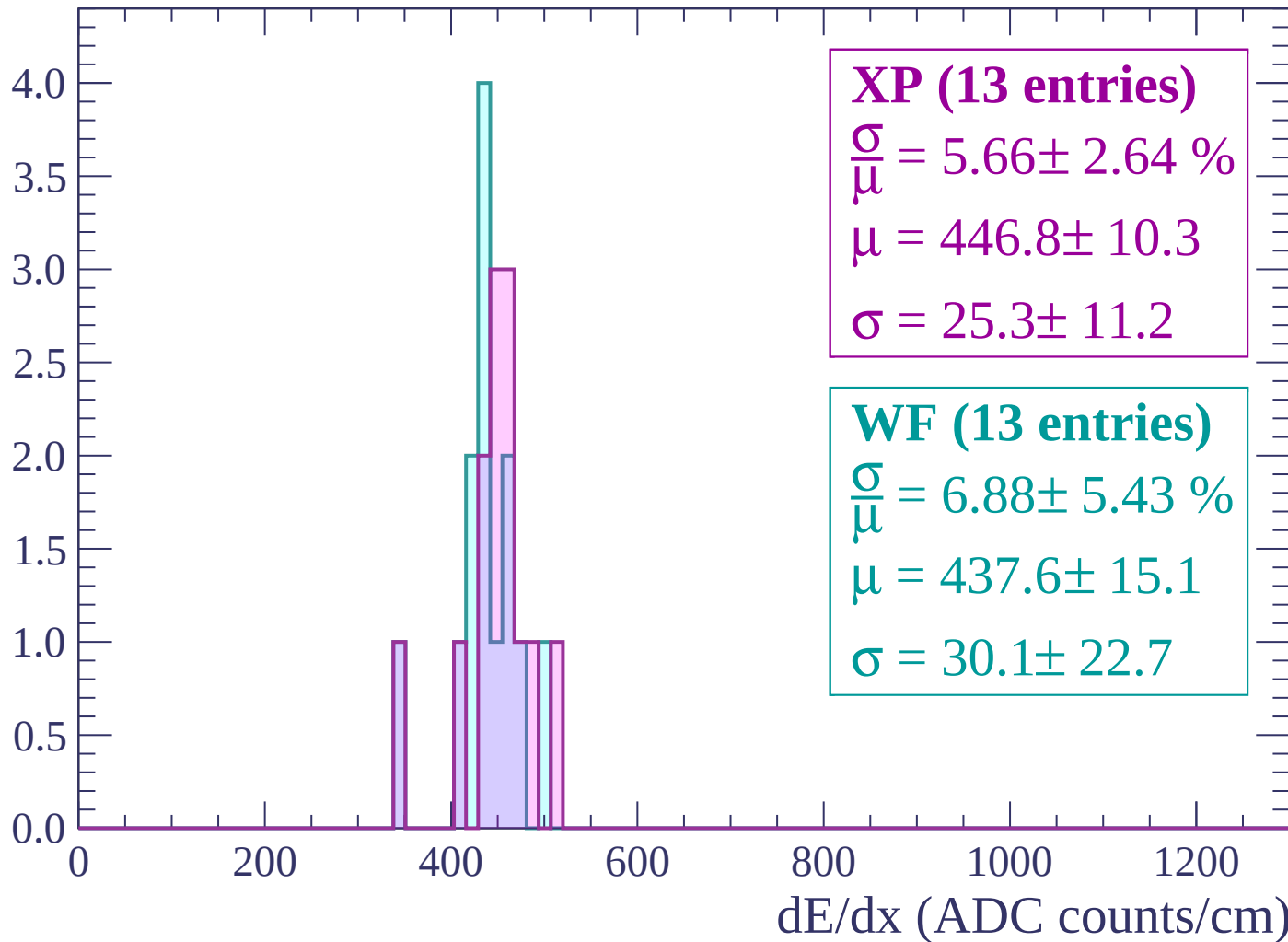
$\mu = 450.1 \pm 58.7$

$\sigma = 43.8 \pm 127.0$

dE/dx (ADC counts/cm)

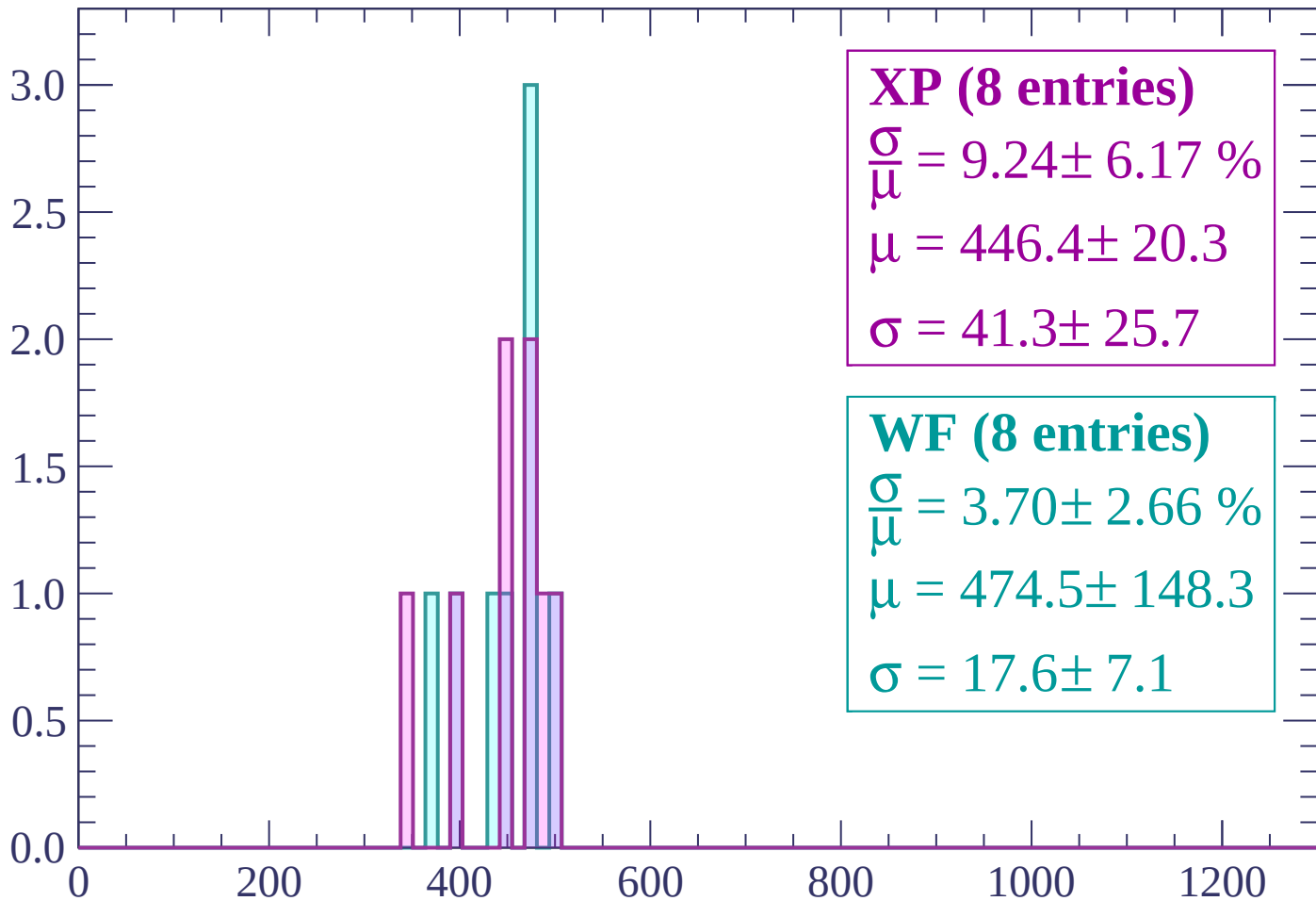
Energy loss | $1320 < p < 1380$

Count



Energy loss | $1380 < p < 1440$

Count



XP (8 entries)

$$\frac{\sigma}{\mu} = 9.24 \pm 6.17 \%$$

$$\mu = 446.4 \pm 20.3$$

$$\sigma = 41.3 \pm 25.7$$

WF (8 entries)

$$\frac{\sigma}{\mu} = 3.70 \pm 2.66 \%$$

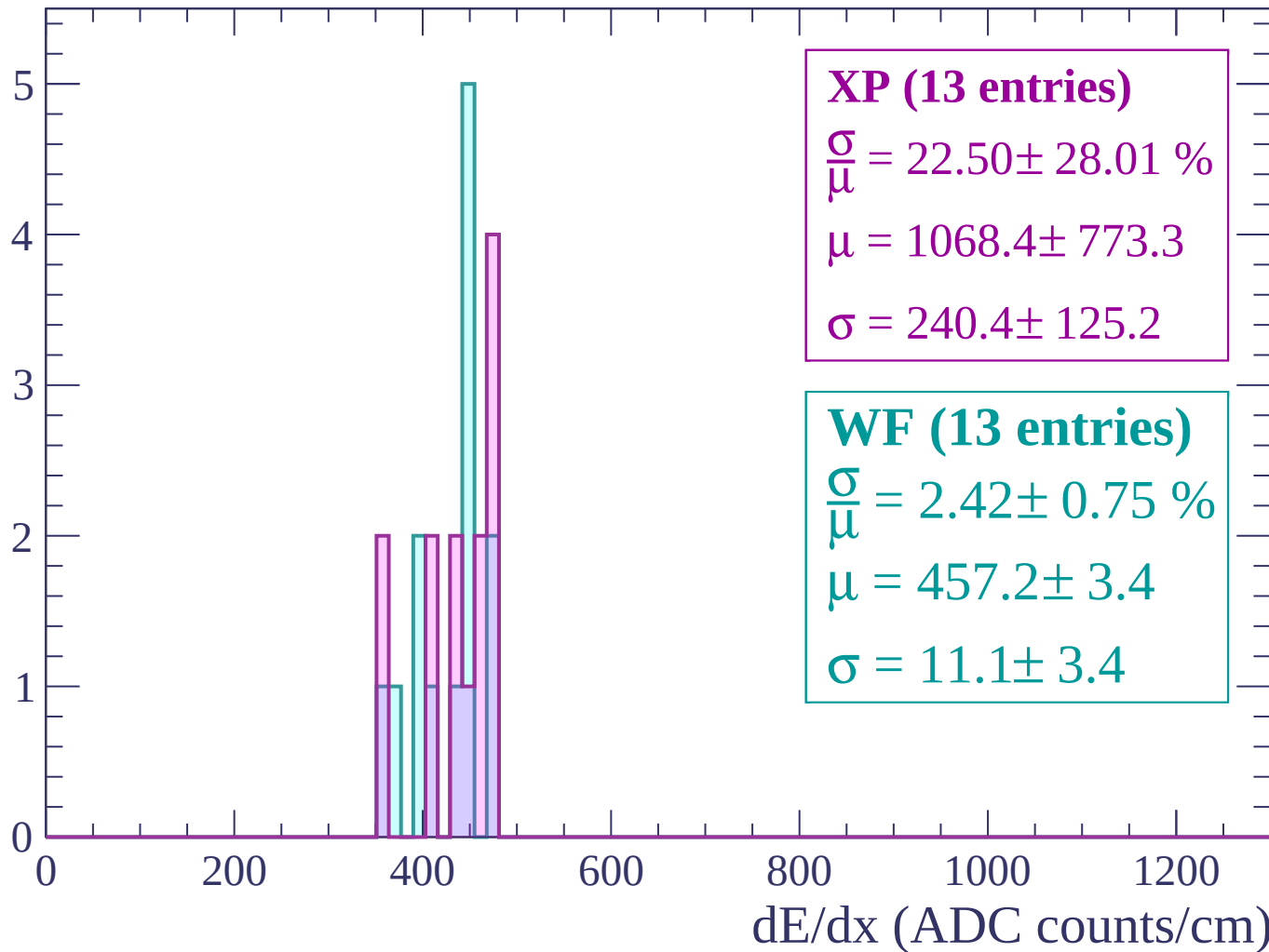
$$\mu = 474.5 \pm 148.3$$

$$\sigma = 17.6 \pm 7.1$$

dE/dx (ADC counts/cm)

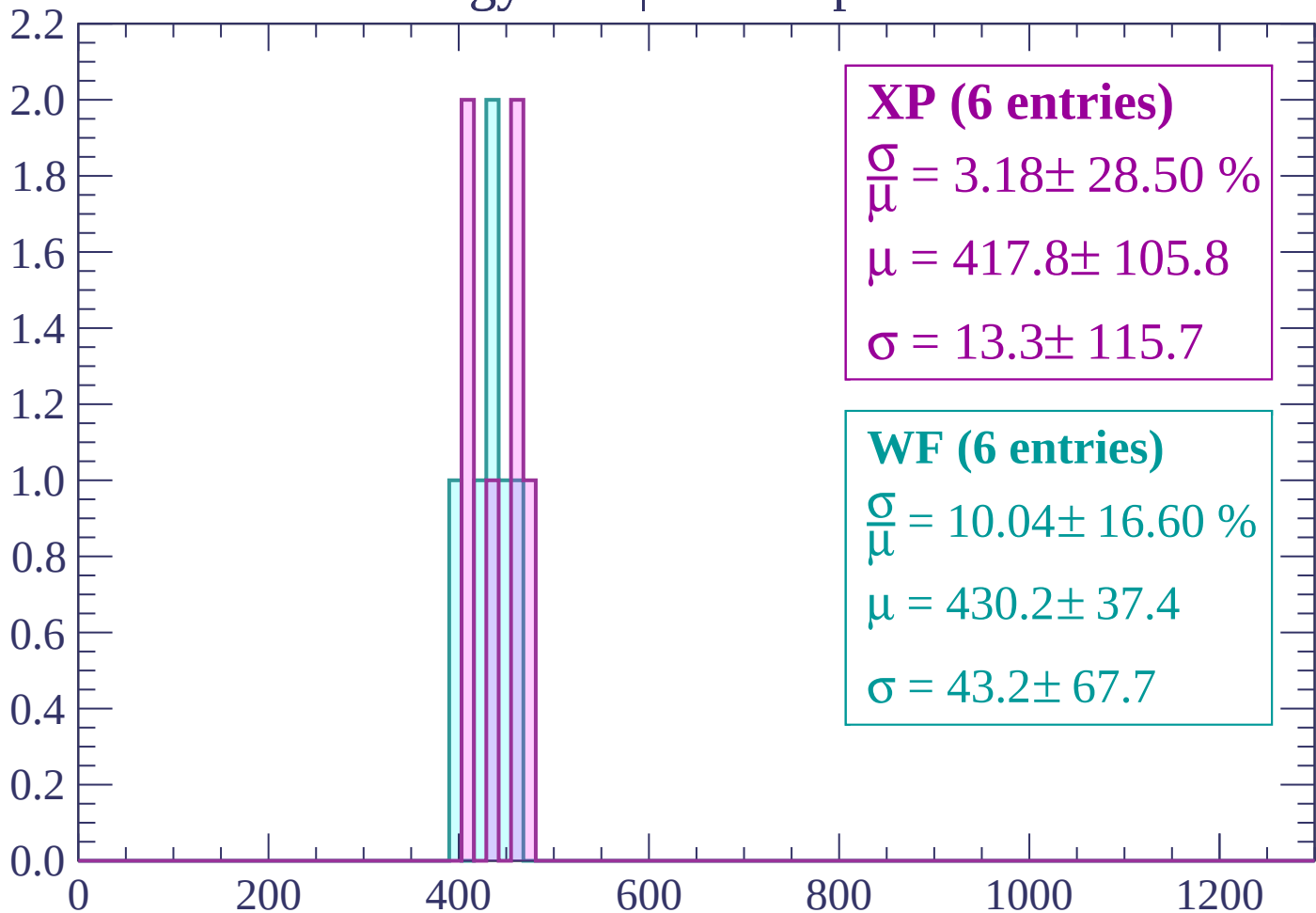
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP (6 entries)

$$\frac{\sigma}{\mu} = 3.18 \pm 28.50 \%$$

$$\mu = 417.8 \pm 105.8$$

$$\sigma = 13.3 \pm 115.7$$

WF (6 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 16.60 \%$$

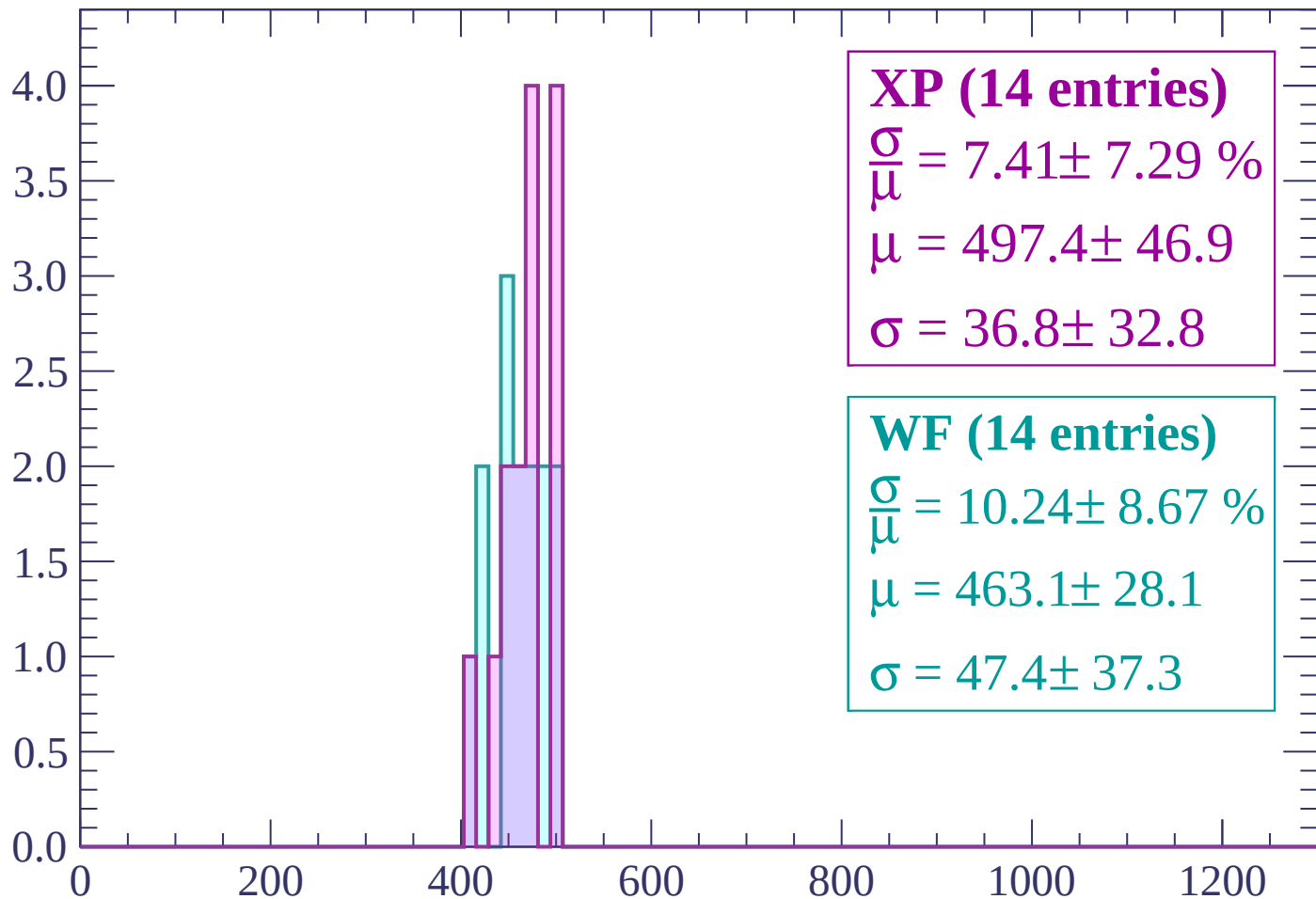
$$\mu = 430.2 \pm 37.4$$

$$\sigma = 43.2 \pm 67.7$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (14 entries)

$\frac{\sigma}{\mu} = 7.41 \pm 7.29 \%$

$\mu = 497.4 \pm 46.9$

$\sigma = 36.8 \pm 32.8$

WF (14 entries)

$\frac{\sigma}{\mu} = 10.24 \pm 8.67 \%$

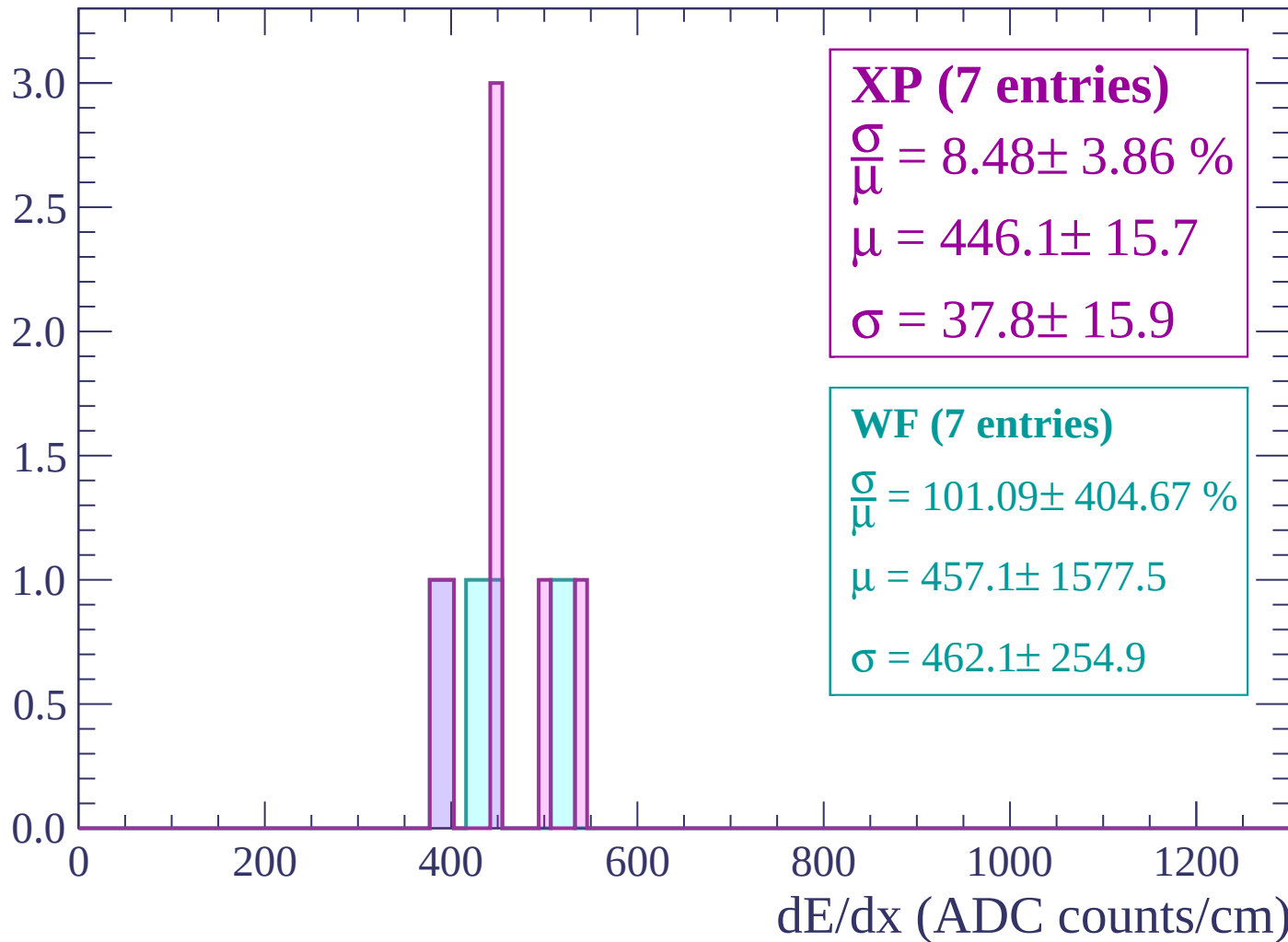
$\mu = 463.1 \pm 28.1$

$\sigma = 47.4 \pm 37.3$

dE/dx (ADC counts/cm)

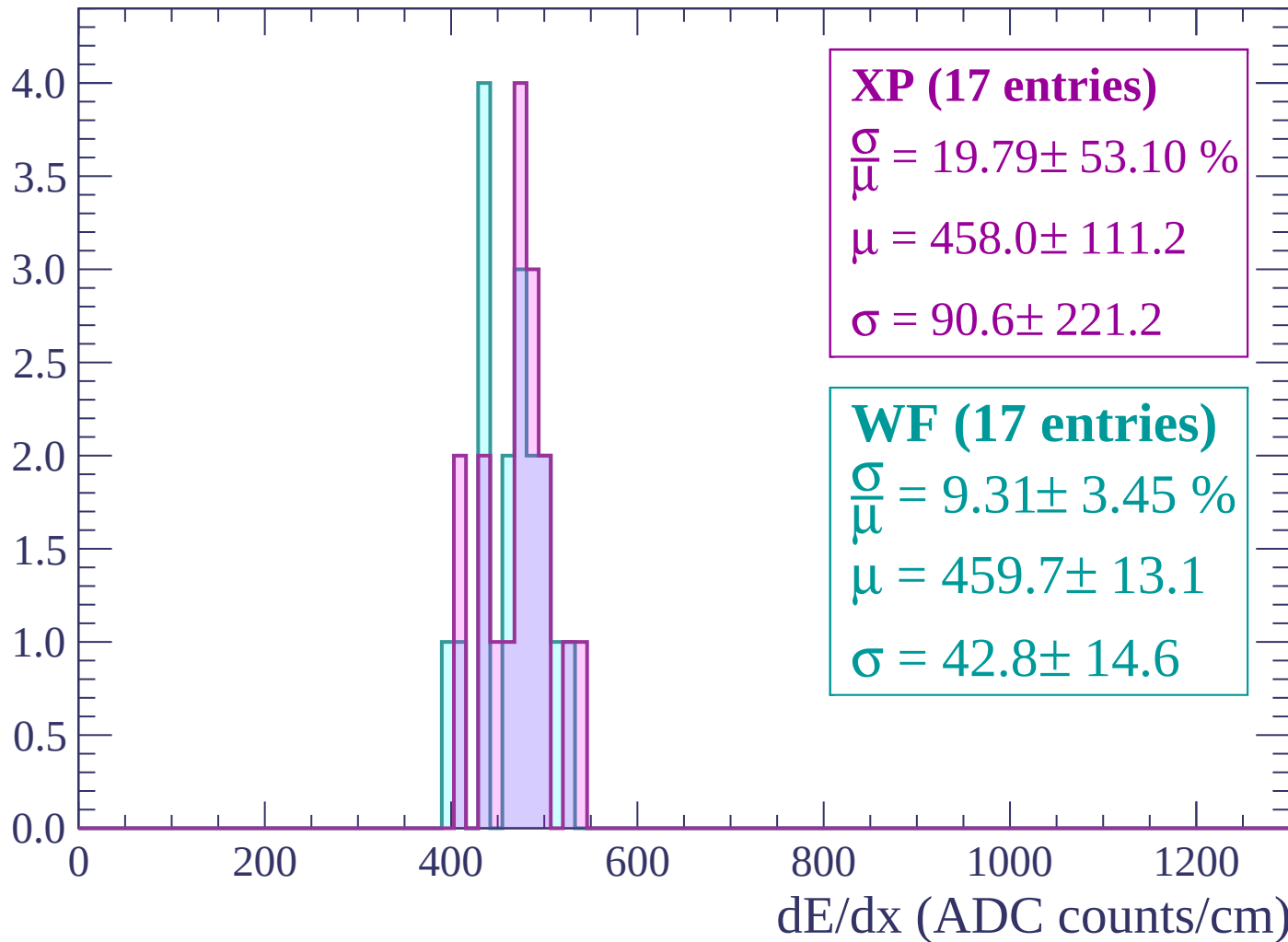
Energy loss | $1620 < p < 1680$

Count



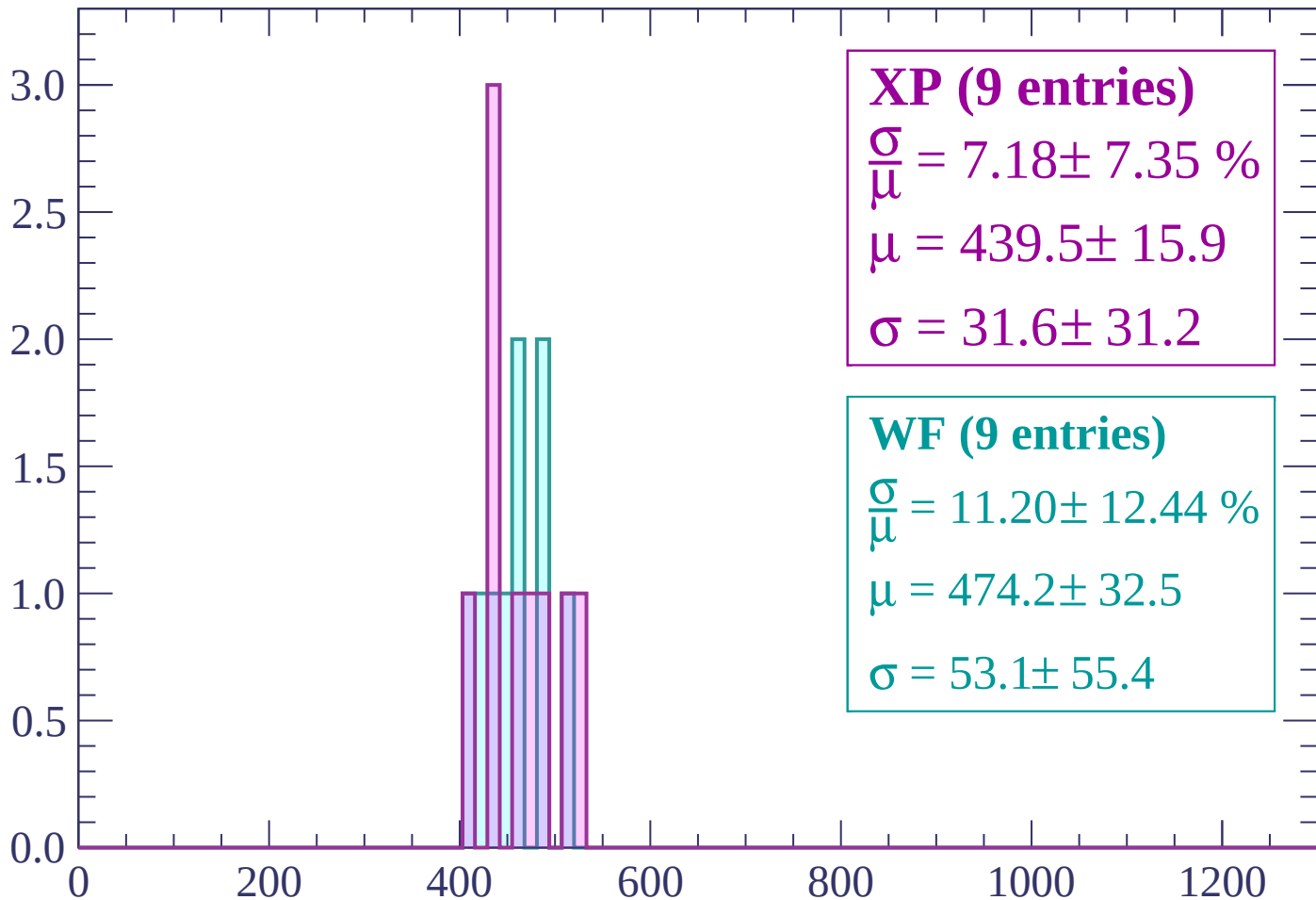
Energy loss | $1680 < p < 1740$

Count



Energy loss | $1740 < p < 1800$

Count



XP (9 entries)

$$\frac{\sigma}{\mu} = 7.18 \pm 7.35 \%$$

$$\mu = 439.5 \pm 15.9$$

$$\sigma = 31.6 \pm 31.2$$

WF (9 entries)

$$\frac{\sigma}{\mu} = 11.20 \pm 12.44 \%$$

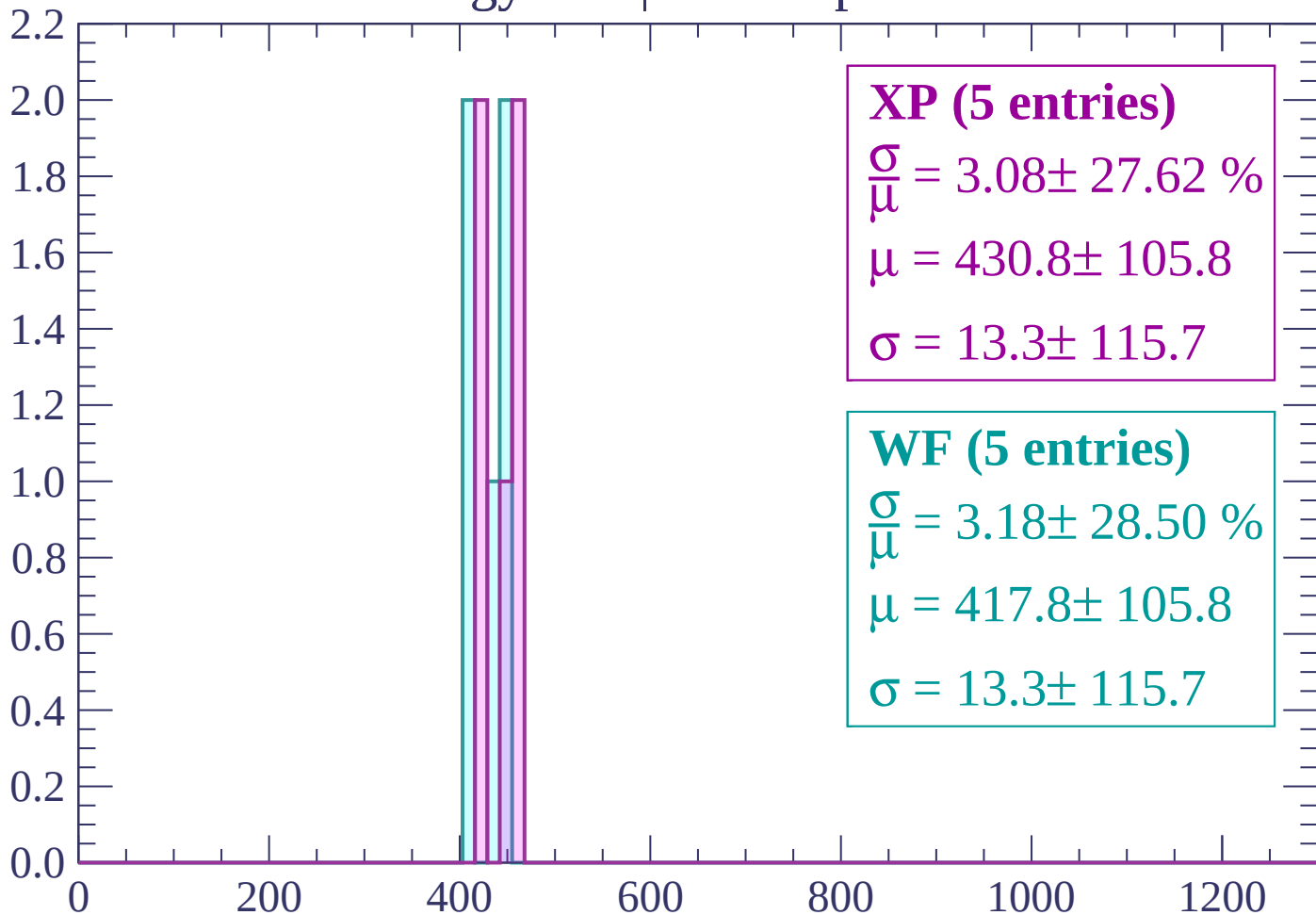
$$\mu = 474.2 \pm 32.5$$

$$\sigma = 53.1 \pm 55.4$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 3.08 \pm 27.62 \%$$

$$\mu = 430.8 \pm 105.8$$

$$\sigma = 13.3 \pm 115.7$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 3.18 \pm 28.50 \%$$

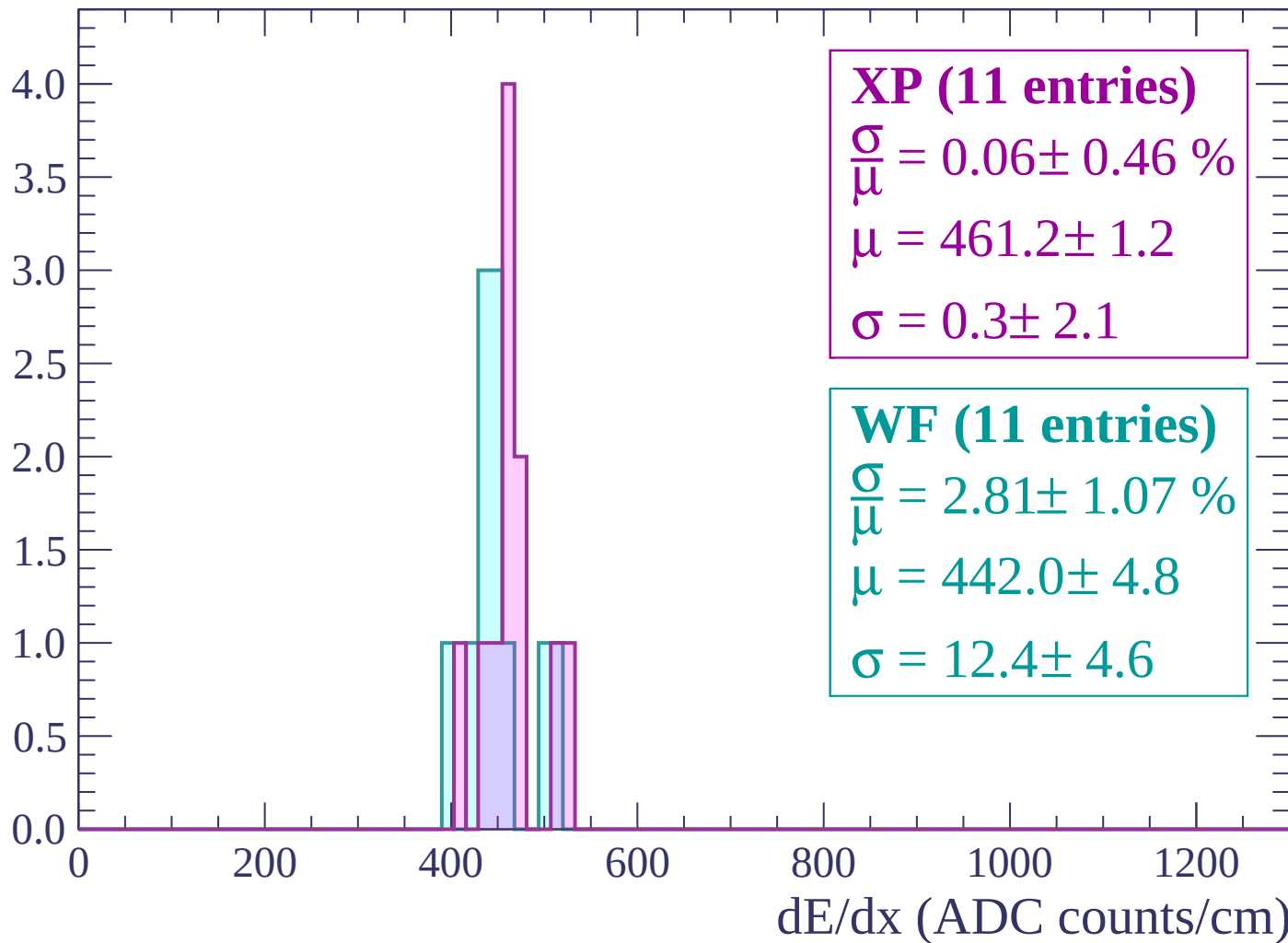
$$\mu = 417.8 \pm 105.8$$

$$\sigma = 13.3 \pm 115.7$$

dE/dx (ADC counts/cm)

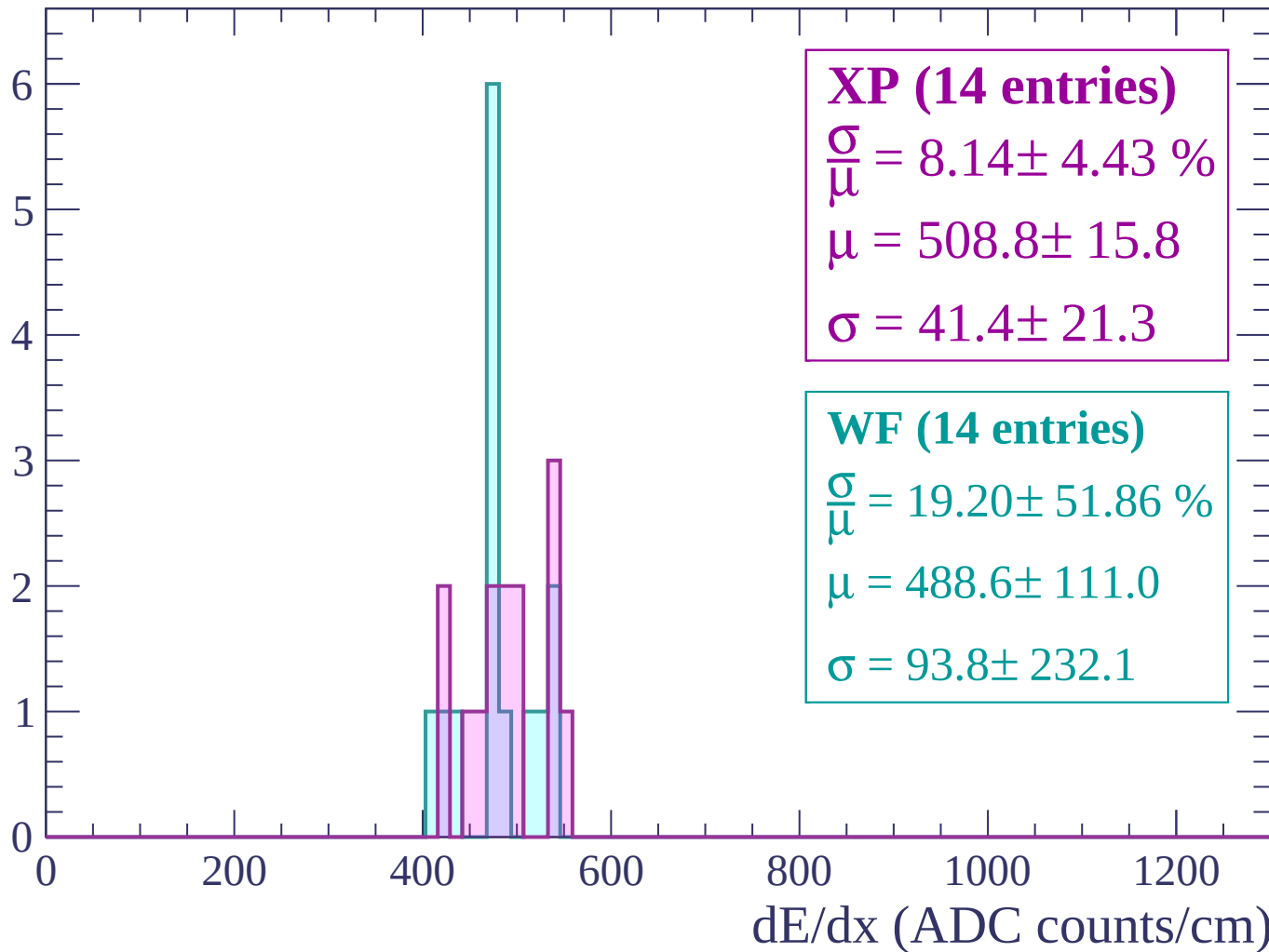
Energy loss | $1860 < p < 1920$

Count



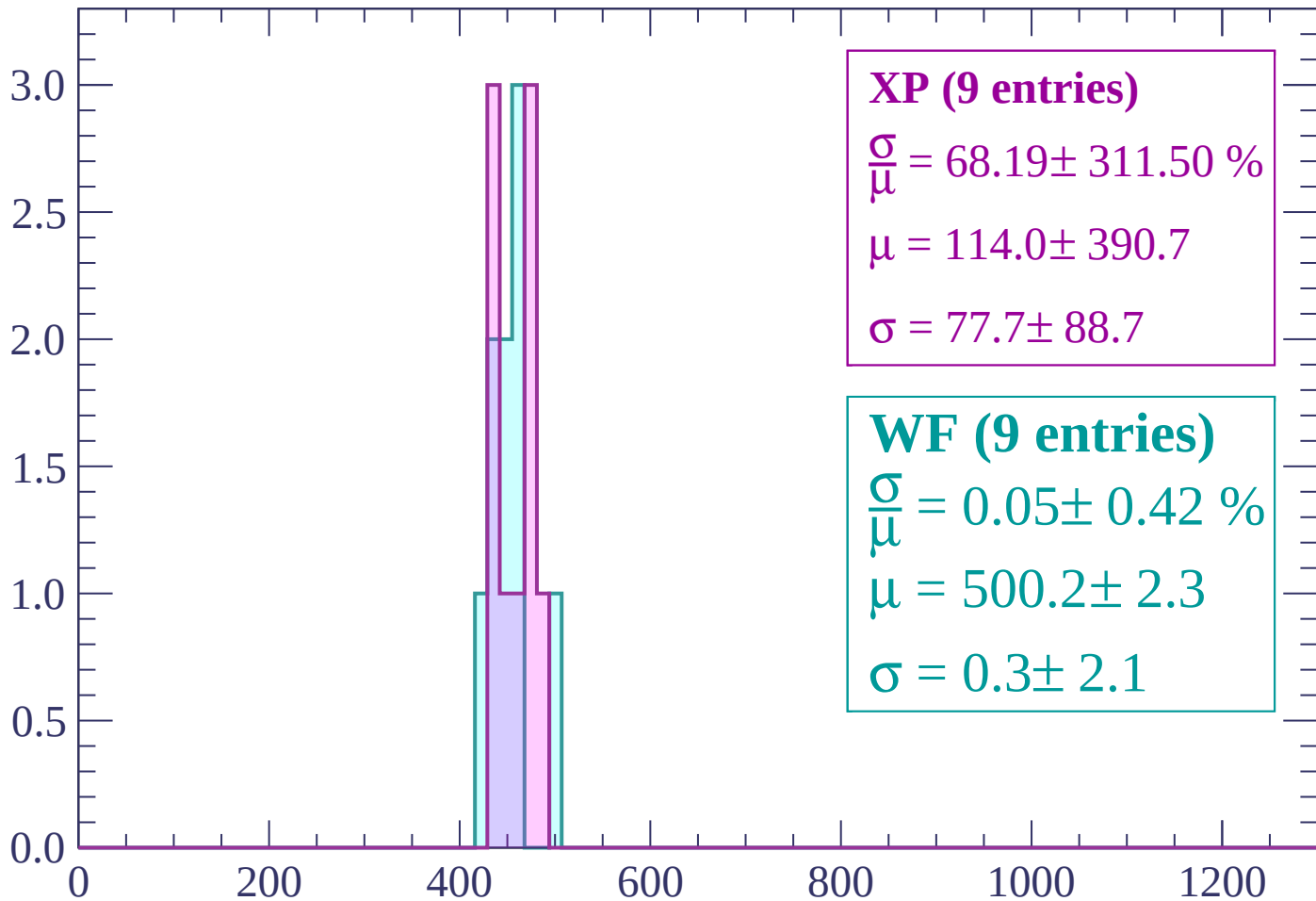
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP (9 entries)

$\frac{\sigma}{\mu} = 68.19 \pm 311.50 \%$

$\mu = 114.0 \pm 390.7$

$\sigma = 77.7 \pm 88.7$

WF (9 entries)

$\frac{\sigma}{\mu} = 0.05 \pm 0.42 \%$

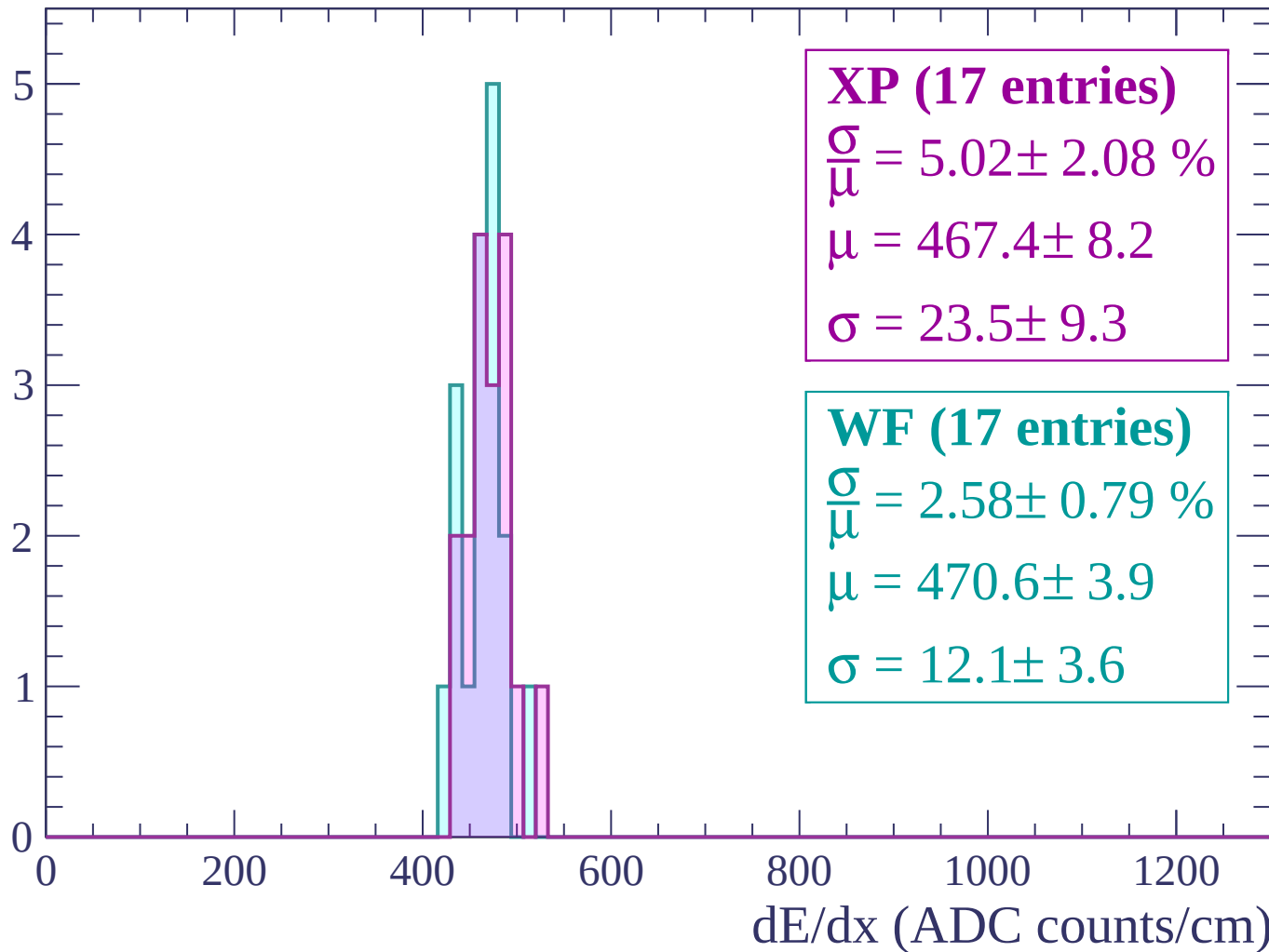
$\mu = 500.2 \pm 2.3$

$\sigma = 0.3 \pm 2.1$

dE/dx (ADC counts/cm)

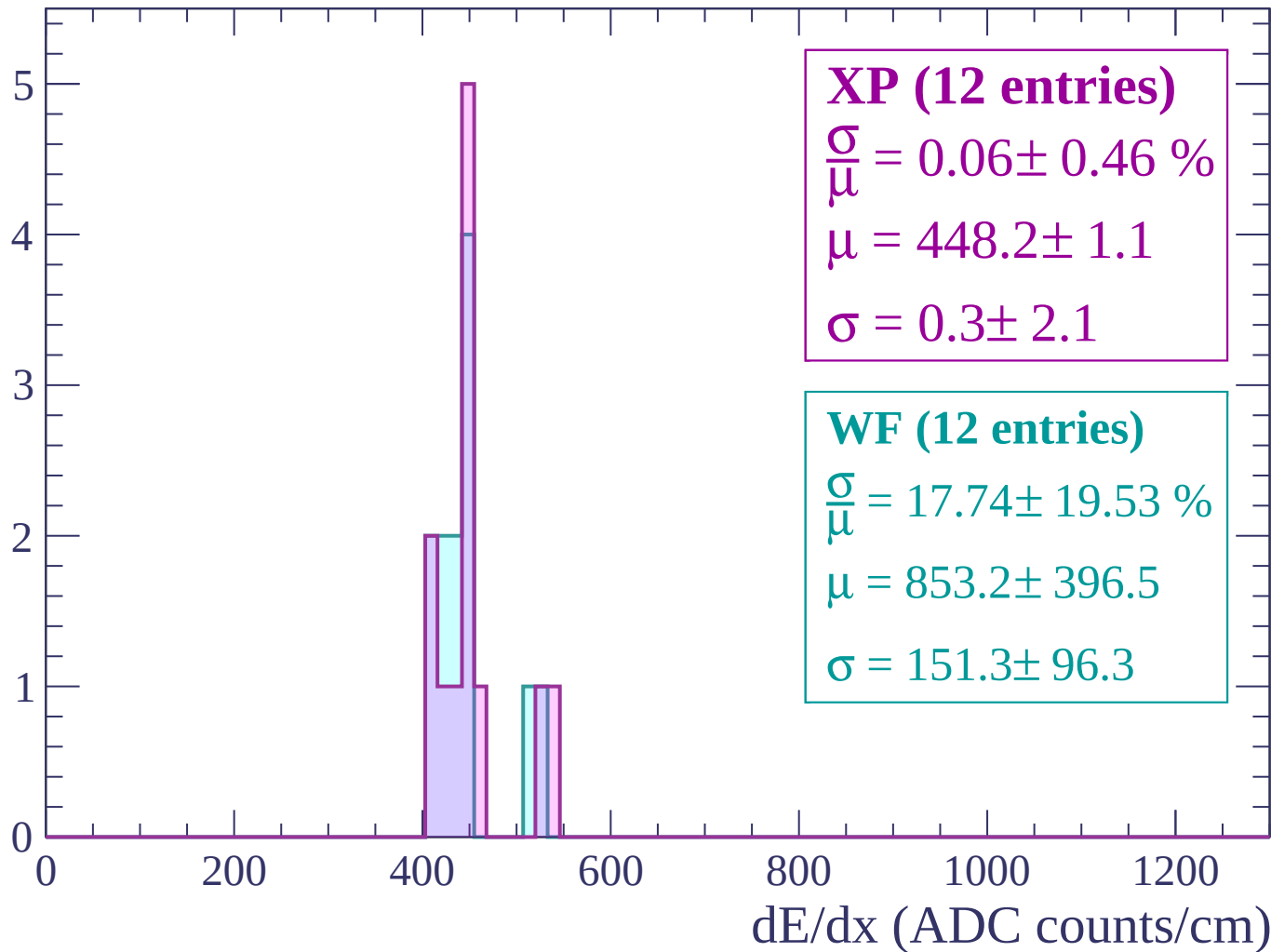
Energy loss | $2040 < p < 2100$

Count



Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

XP (2 entries)

$\frac{\sigma}{\mu} = 0.06 \pm 0.52 \%$

$\mu = 448.2 \pm 2.5$

$\sigma = 0.3 \pm 2.3$

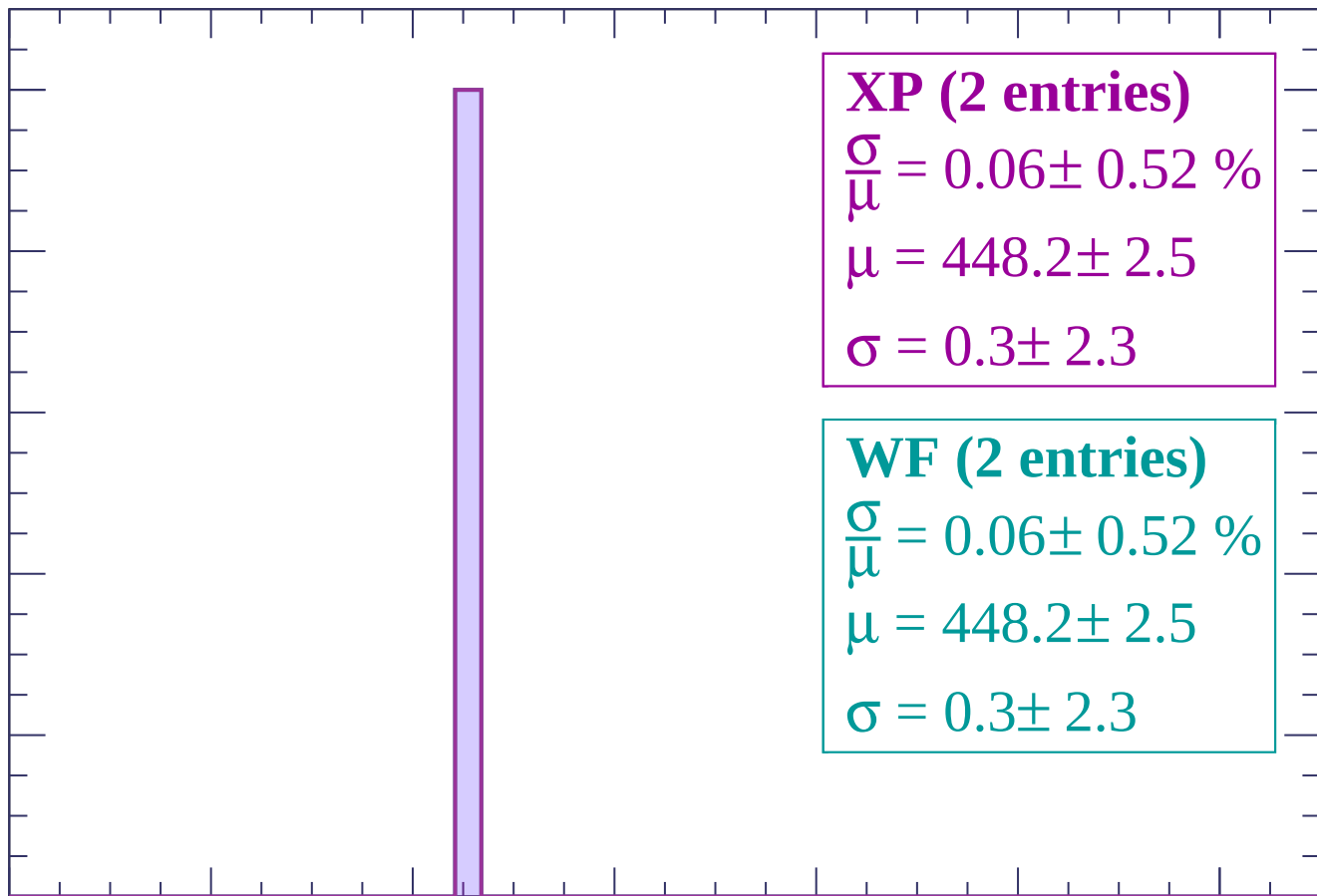
WF (2 entries)

$\frac{\sigma}{\mu} = 0.06 \pm 0.52 \%$

$\mu = 448.2 \pm 2.5$

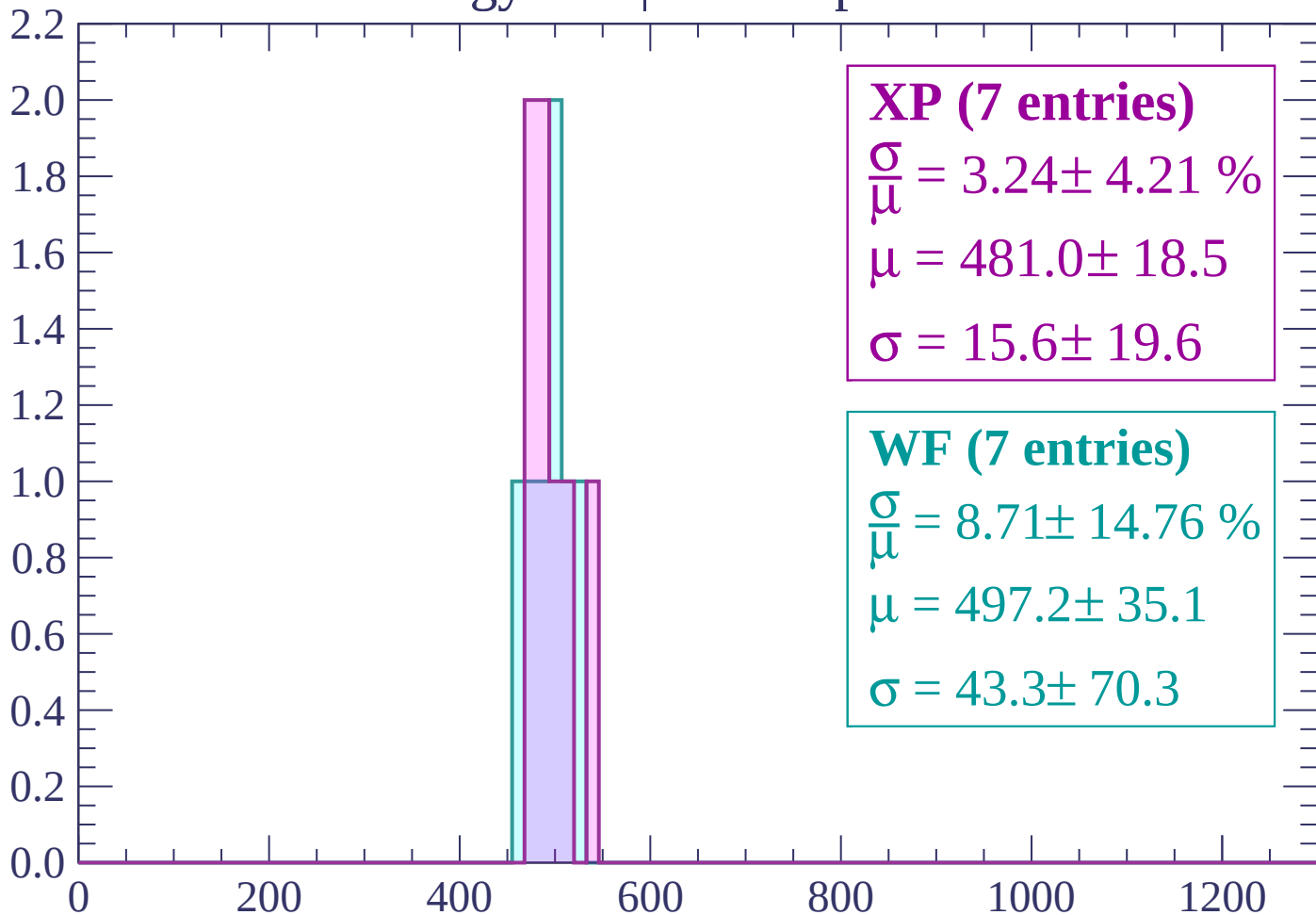
$\sigma = 0.3 \pm 2.3$

dE/dx (ADC counts/cm)



Energy loss | $2220 < p < 2280$

Count



XP (7 entries)

$\frac{\sigma}{\mu} = 3.24 \pm 4.21 \%$

$\mu = 481.0 \pm 18.5$

$\sigma = 15.6 \pm 19.6$

WF (7 entries)

$\frac{\sigma}{\mu} = 8.71 \pm 14.76 \%$

$\mu = 497.2 \pm 35.1$

$\sigma = 43.3 \pm 70.3$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (3 entries)

$\frac{\sigma}{\mu} = 0.05 \pm 0.30 \%$

$\mu = 487.2 \pm 2.4$

$\sigma = 0.3 \pm 1.5$

WF (3 entries)

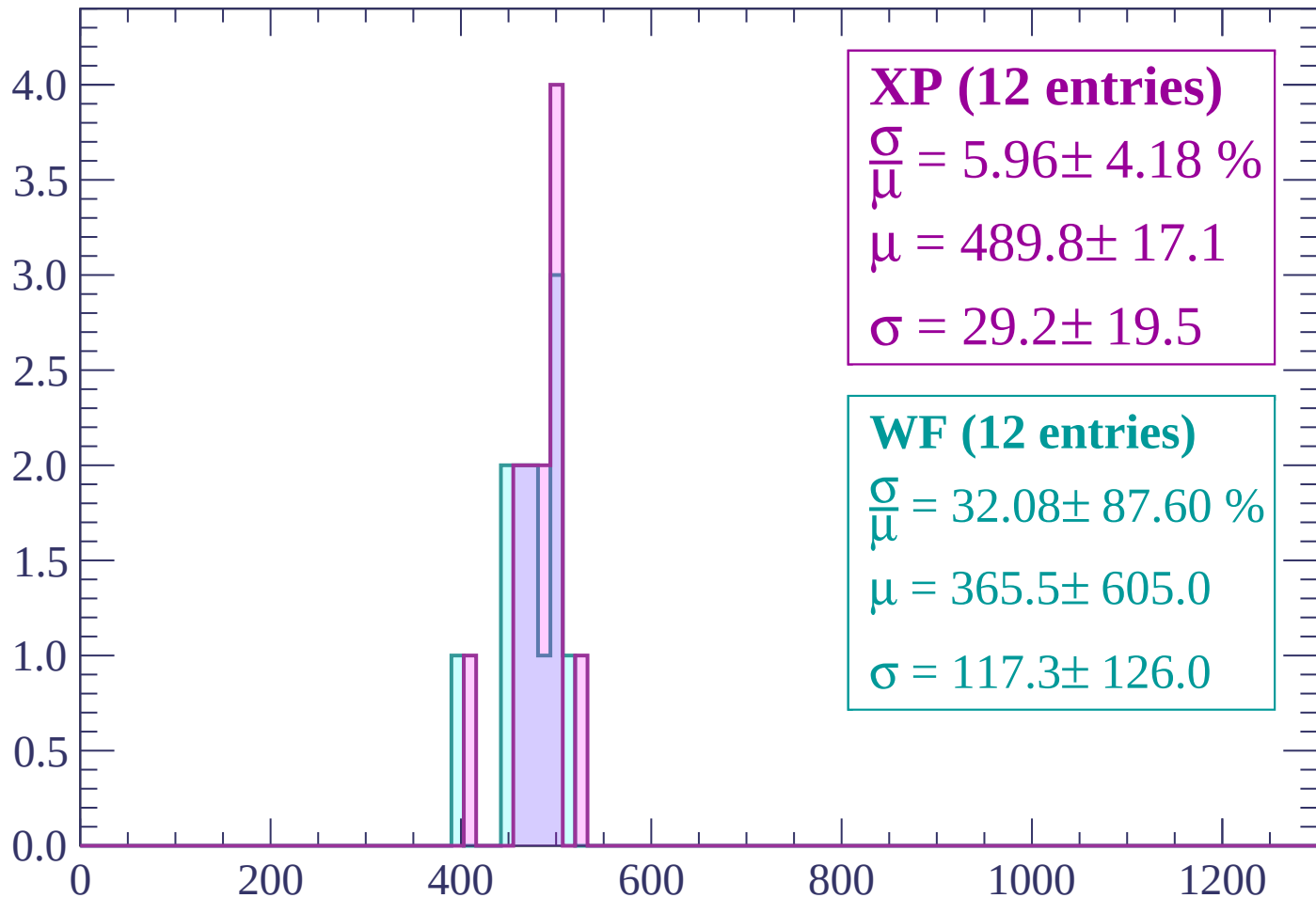
$\frac{\sigma}{\mu} = 5.19 \pm 38.65 \%$

$\mu = 500.5 \pm 81.9$

$\sigma = 26.0 \pm 189.2$

Energy loss | $2340 < p < 2400$

Count



XP (12 entries)

$\frac{\sigma}{\mu} = 5.96 \pm 4.18 \%$

$\mu = 489.8 \pm 17.1$

$\sigma = 29.2 \pm 19.5$

WF (12 entries)

$\frac{\sigma}{\mu} = 32.08 \pm 87.60 \%$

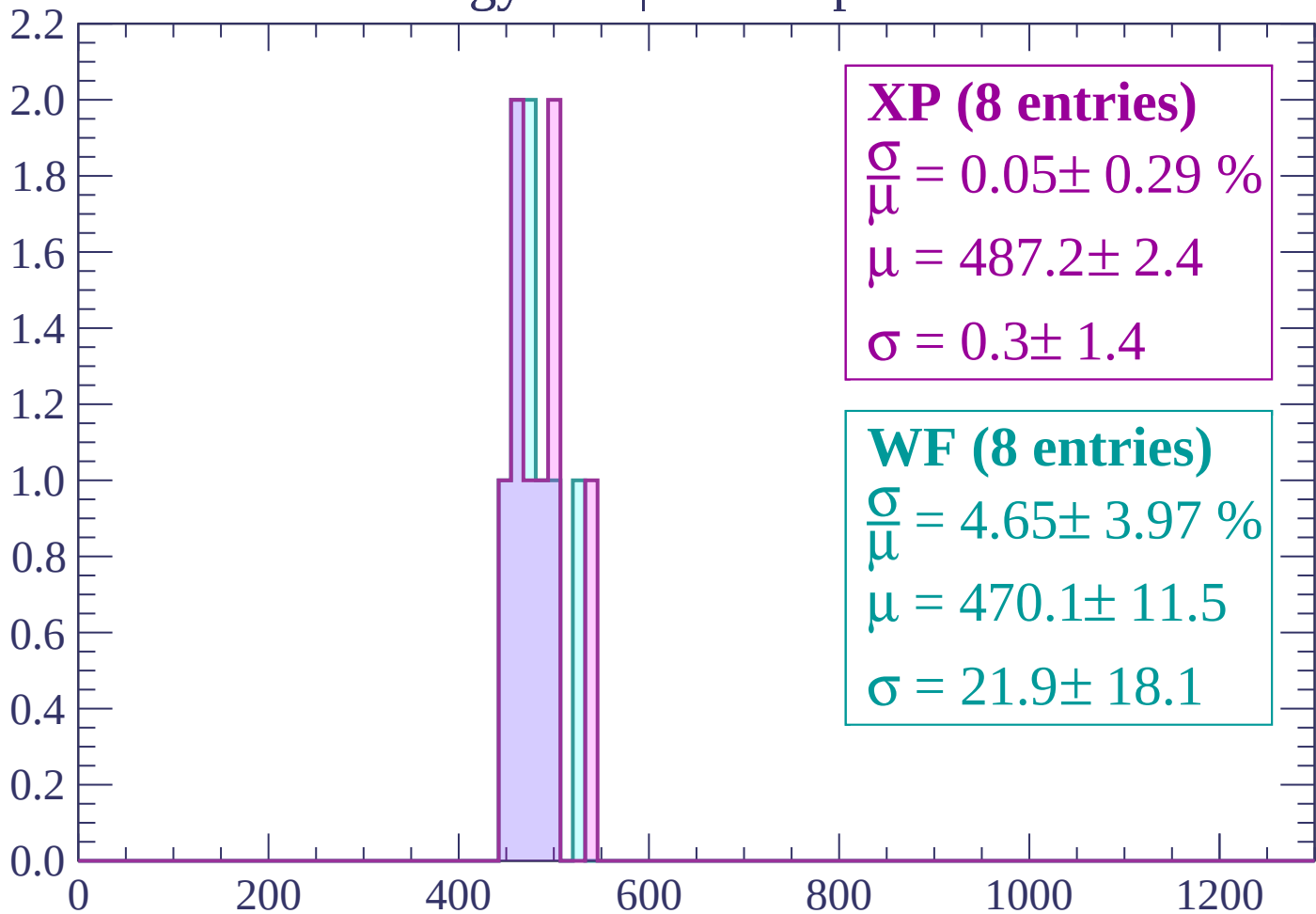
$\mu = 365.5 \pm 605.0$

$\sigma = 117.3 \pm 126.0$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



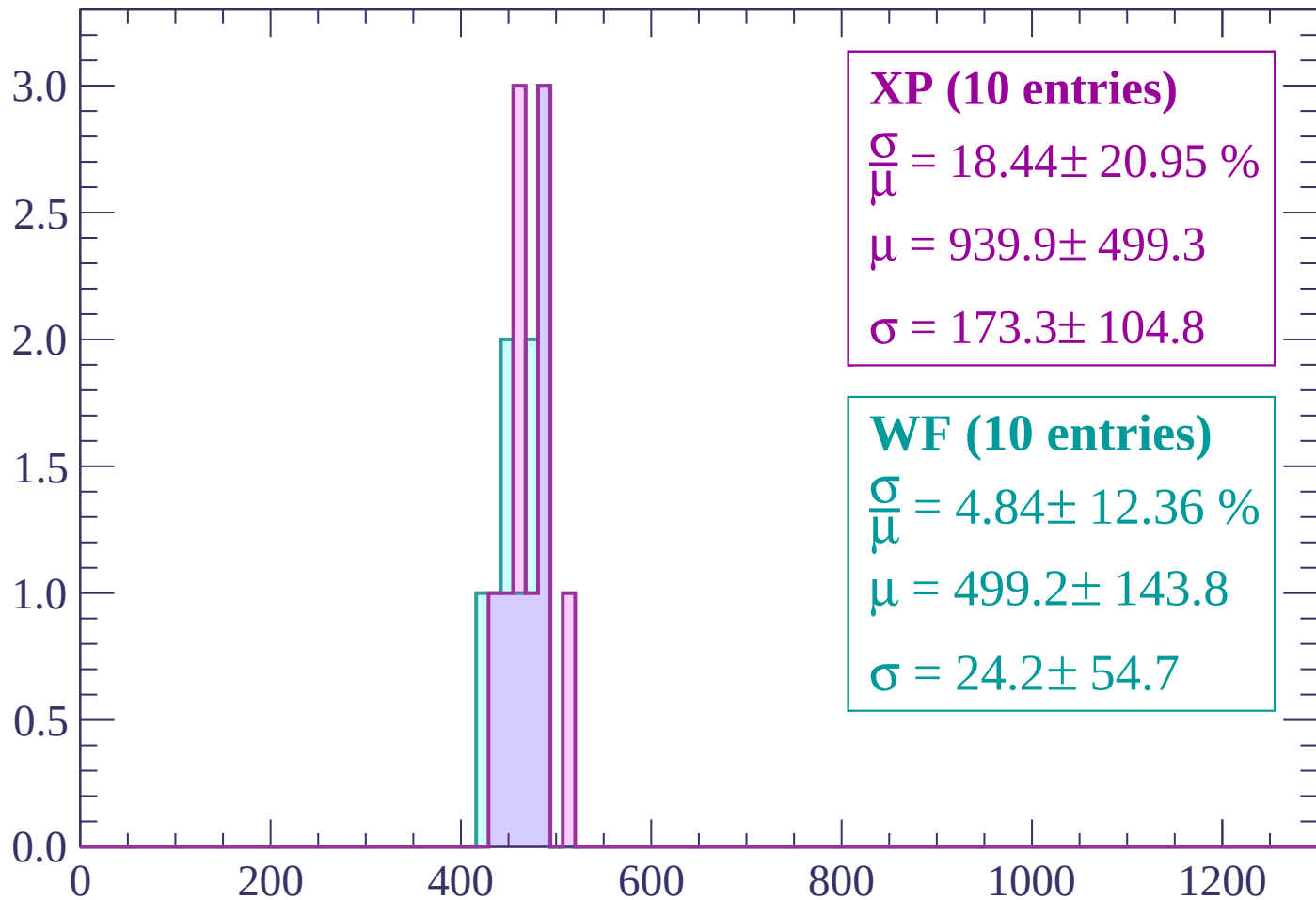
XP (8 entries)
 $\frac{\sigma}{\bar{\mu}} = 0.05 \pm 0.29 \%$
 $\mu = 487.2 \pm 2.4$
 $\sigma = 0.3 \pm 1.4$

WF (8 entries)
 $\frac{\sigma}{\bar{\mu}} = 4.65 \pm 3.97 \%$
 $\mu = 470.1 \pm 11.5$
 $\sigma = 21.9 \pm 18.1$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP (10 entries)

$$\frac{\sigma}{\mu} = 18.44 \pm 20.95 \%$$

$$\mu = 939.9 \pm 499.3$$

$$\sigma = 173.3 \pm 104.8$$

WF (10 entries)

$$\frac{\sigma}{\mu} = 4.84 \pm 12.36 \%$$

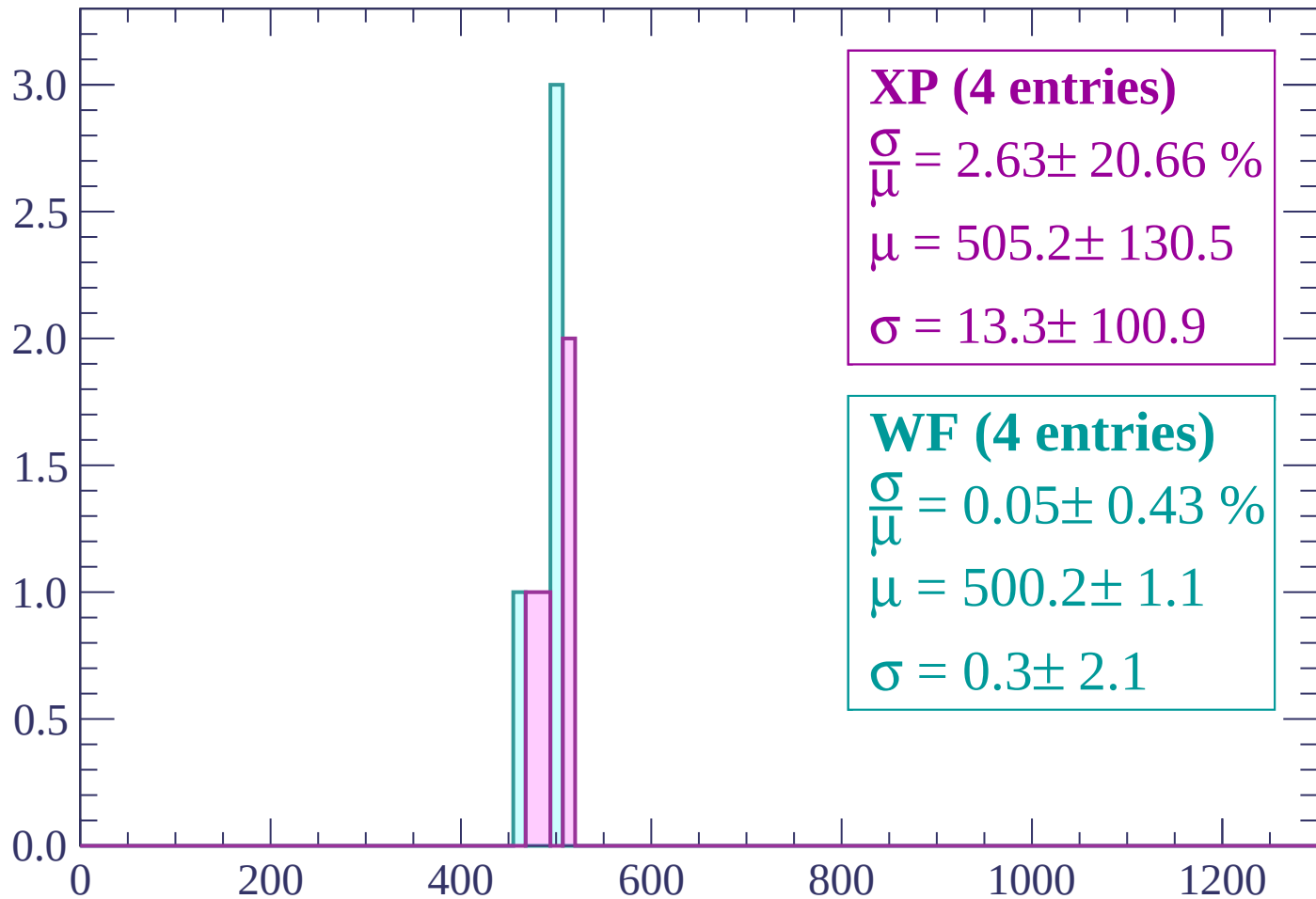
$$\mu = 499.2 \pm 143.8$$

$$\sigma = 24.2 \pm 54.7$$

dE/dx (ADC counts/cm)

Energy loss | $2520 < p < 2580$

Count



XP (4 entries)

$$\frac{\sigma}{\mu} = 2.63 \pm 20.66 \%$$

$$\mu = 505.2 \pm 130.5$$

$$\sigma = 13.3 \pm 100.9$$

WF (4 entries)

$$\frac{\sigma}{\mu} = 0.05 \pm 0.43 \%$$

$$\mu = 500.2 \pm 1.1$$

$$\sigma = 0.3 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (5 entries)

$\frac{\sigma}{\mu} = 8.70 \pm 51.03 \%$

$\mu = 448.5 \pm 118.4$

$\sigma = 39.0 \pm 218.6$

WF (5 entries)

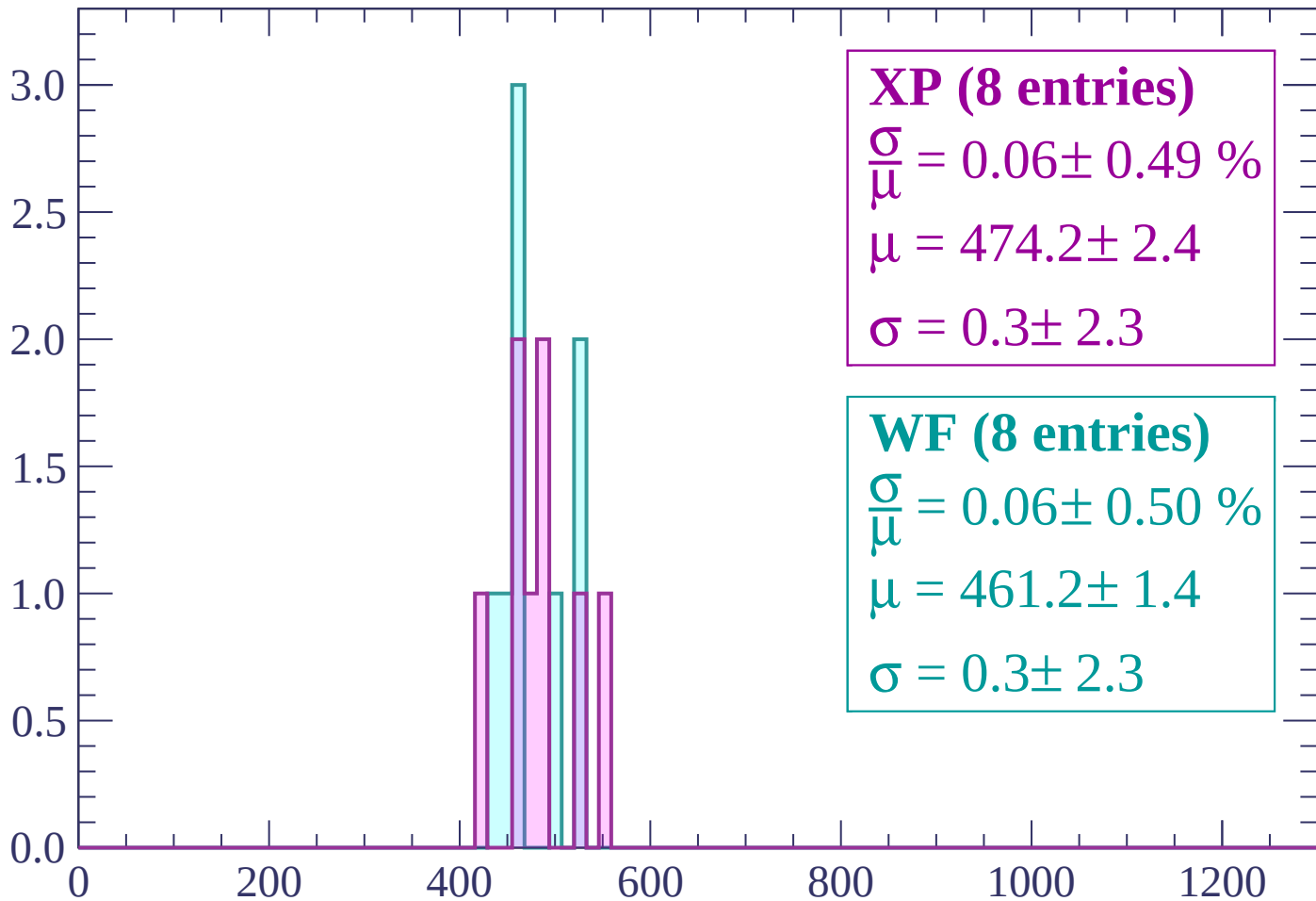
$\frac{\sigma}{\mu} = 10.29 \pm 61.13 \%$

$\mu = 442.0 \pm 139.7$

$\sigma = 45.5 \pm 255.8$

Energy loss | $2640 < p < 2700$

Count



XP (8 entries)

$\frac{\sigma}{\mu} = 0.06 \pm 0.49 \%$

$\mu = 474.2 \pm 2.4$

$\sigma = 0.3 \pm 2.3$

WF (8 entries)

$\frac{\sigma}{\mu} = 0.06 \pm 0.50 \%$

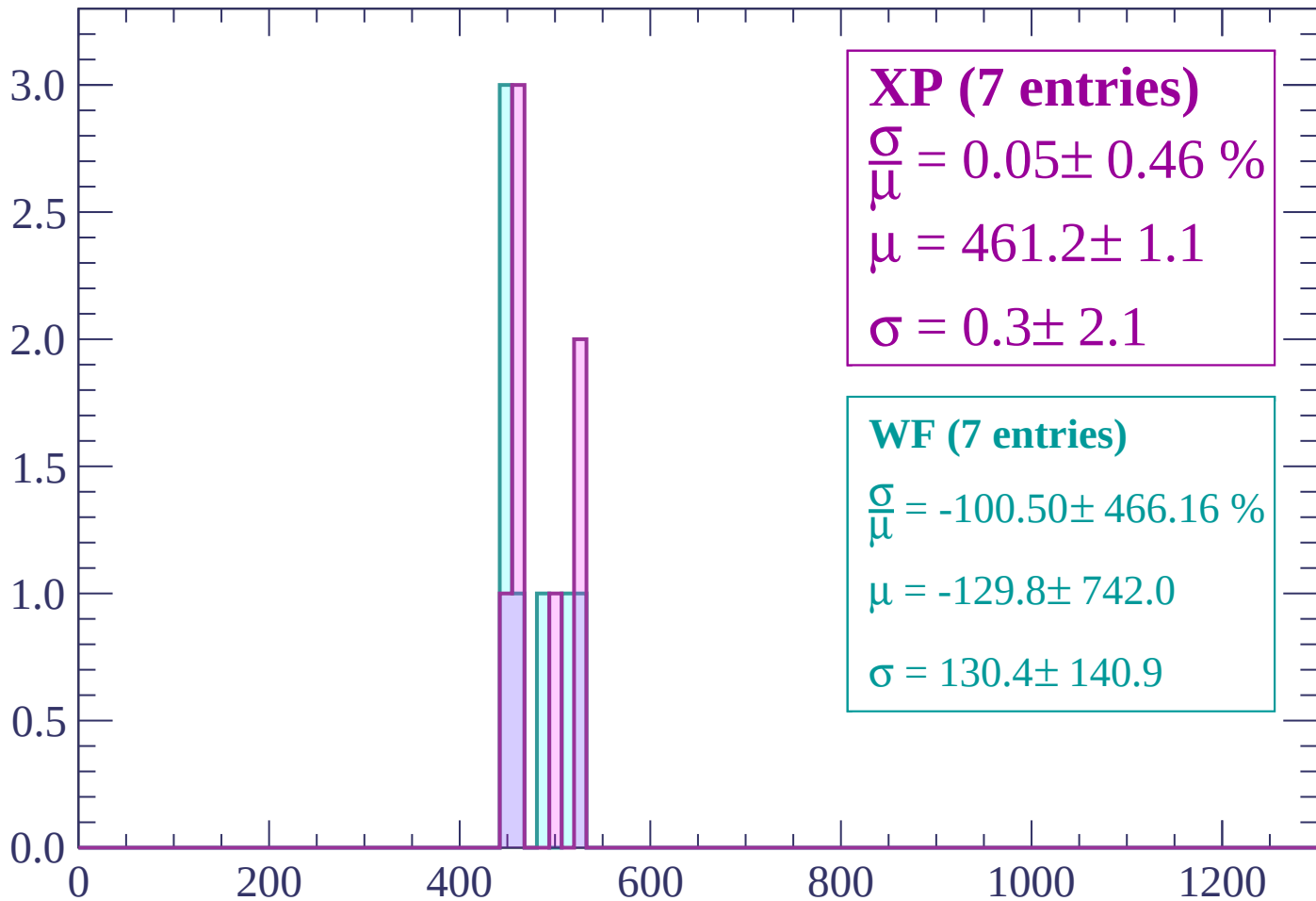
$\mu = 461.2 \pm 1.4$

$\sigma = 0.3 \pm 2.3$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



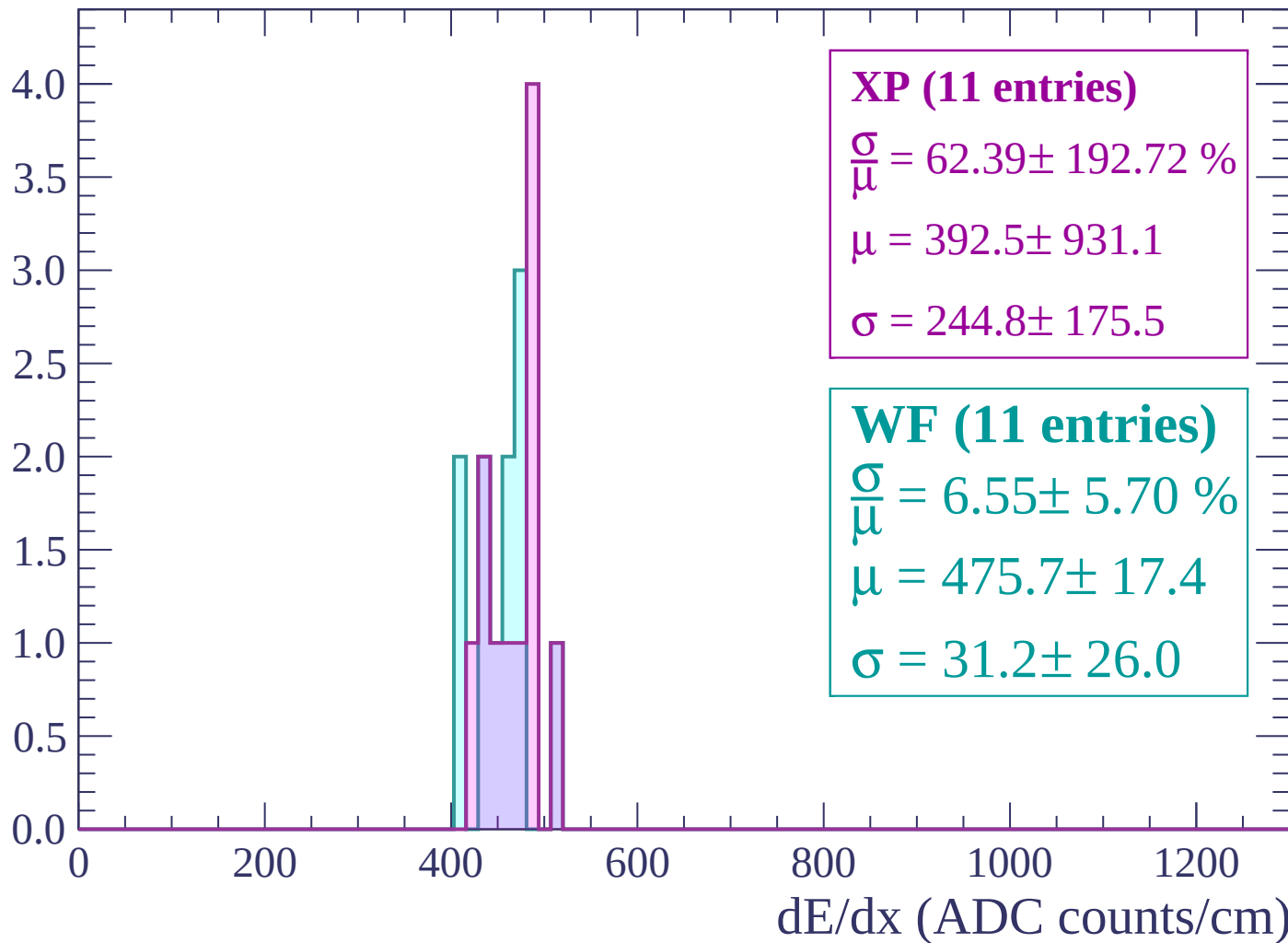
XP (7 entries)
 $\frac{\sigma}{\mu} = 0.05 \pm 0.46 \%$
 $\mu = 461.2 \pm 1.1$
 $\sigma = 0.3 \pm 2.1$

WF (7 entries)
 $\frac{\sigma}{\mu} = -100.50 \pm 466.16 \%$
 $\mu = -129.8 \pm 742.0$
 $\sigma = 130.4 \pm 140.9$

dE/dx (ADC counts/cm)

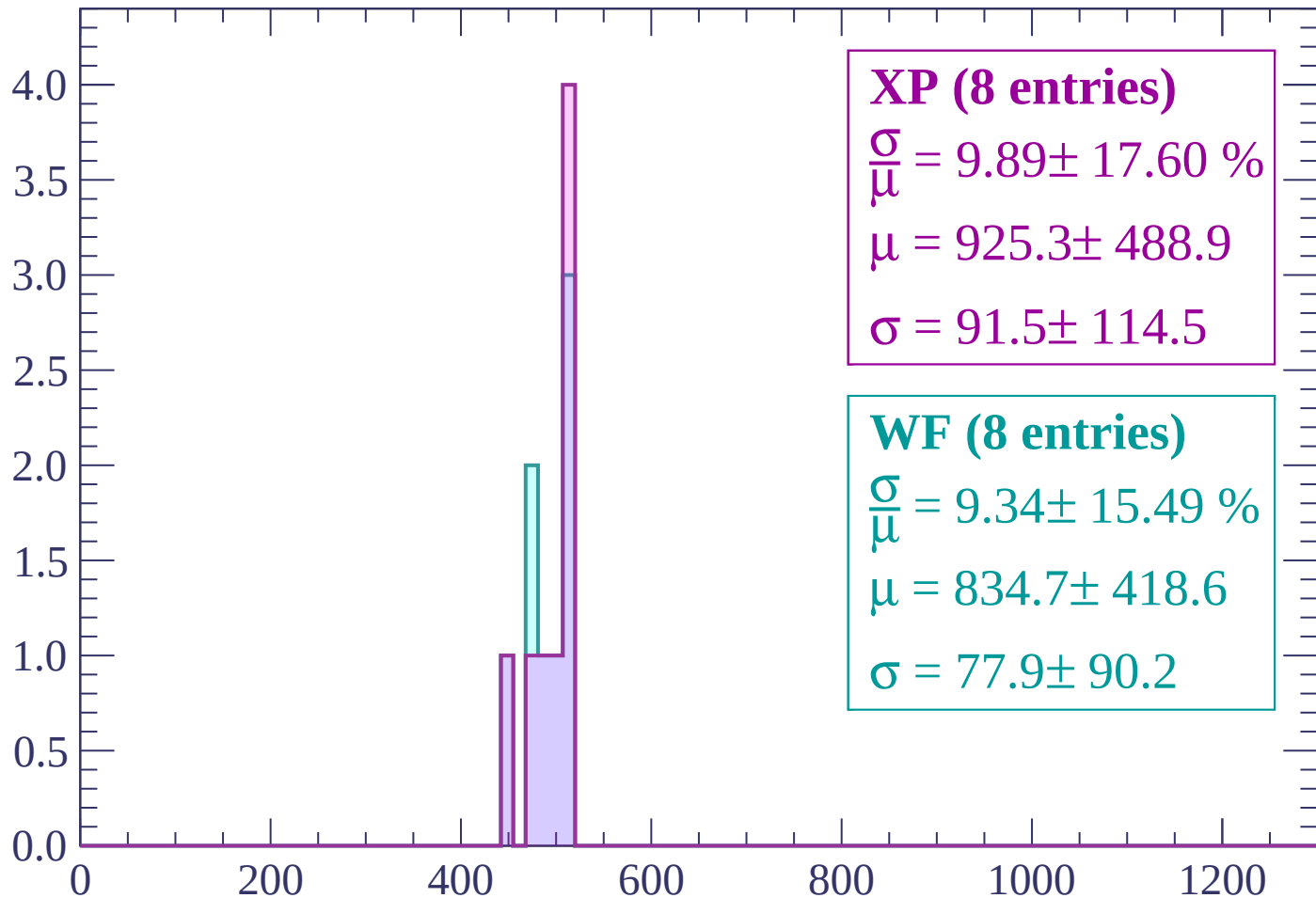
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (8 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 17.60 \%$$

$$\mu = 925.3 \pm 488.9$$

$$\sigma = 91.5 \pm 114.5$$

WF (8 entries)

$$\frac{\sigma}{\mu} = 9.34 \pm 15.49 \%$$

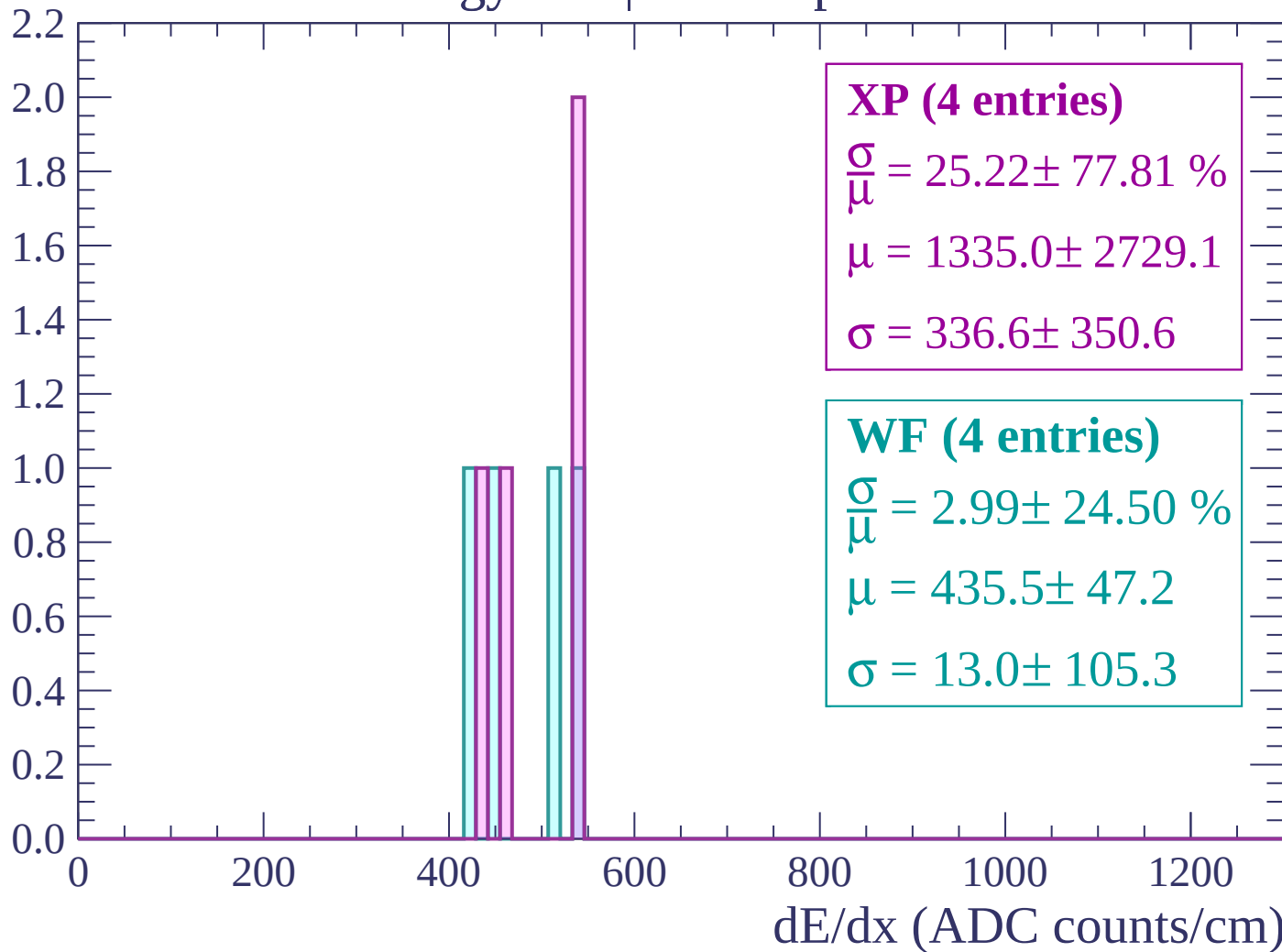
$$\mu = 834.7 \pm 418.6$$

$$\sigma = 77.9 \pm 90.2$$

dE/dx (ADC counts/cm)

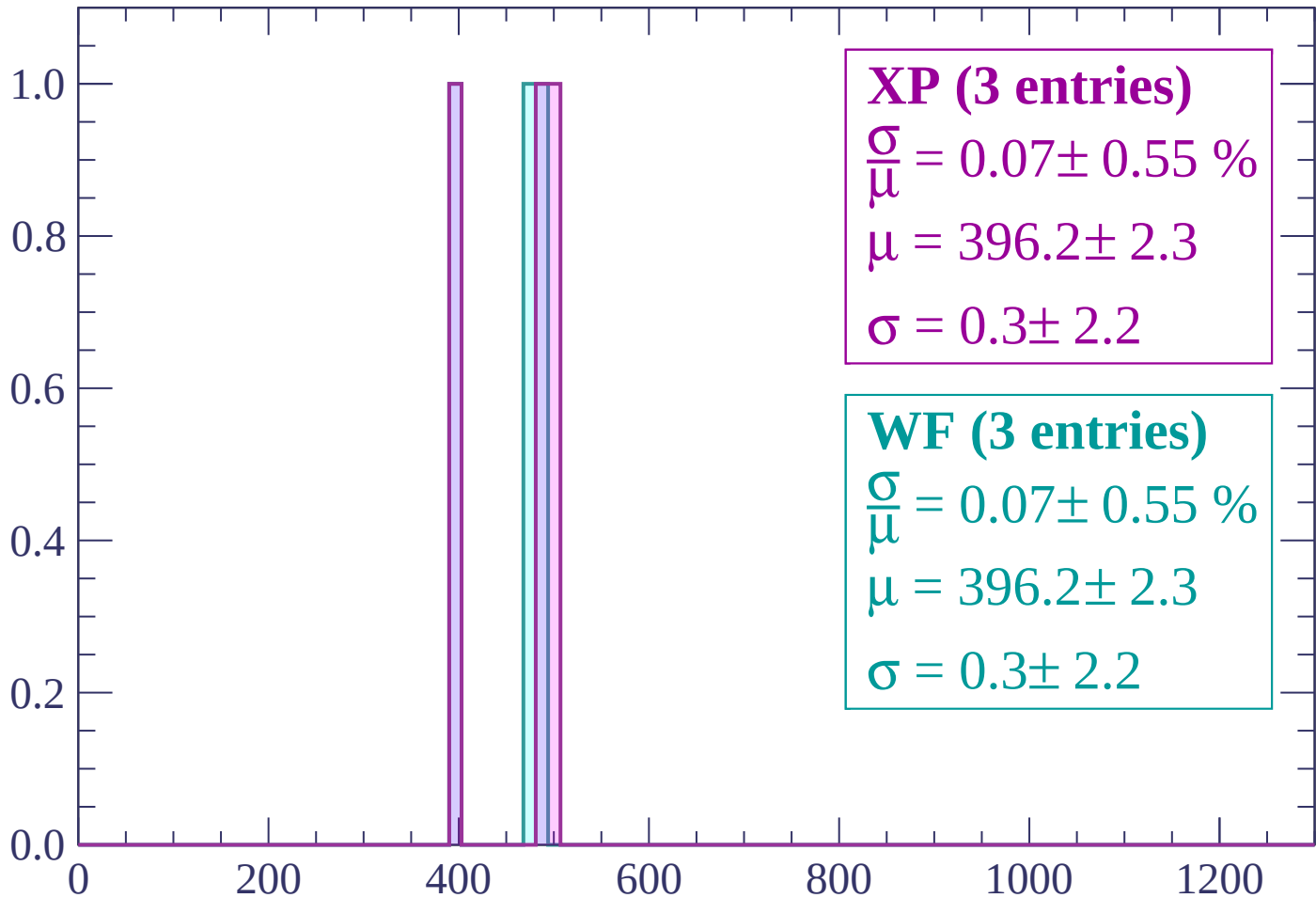
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP (3 entries)

$\frac{\sigma}{\mu} = 0.07 \pm 0.55 \%$

$\mu = 396.2 \pm 2.3$

$\sigma = 0.3 \pm 2.2$

WF (3 entries)

$\frac{\sigma}{\mu} = 0.07 \pm 0.55 \%$

$\mu = 396.2 \pm 2.3$

$\sigma = 0.3 \pm 2.2$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

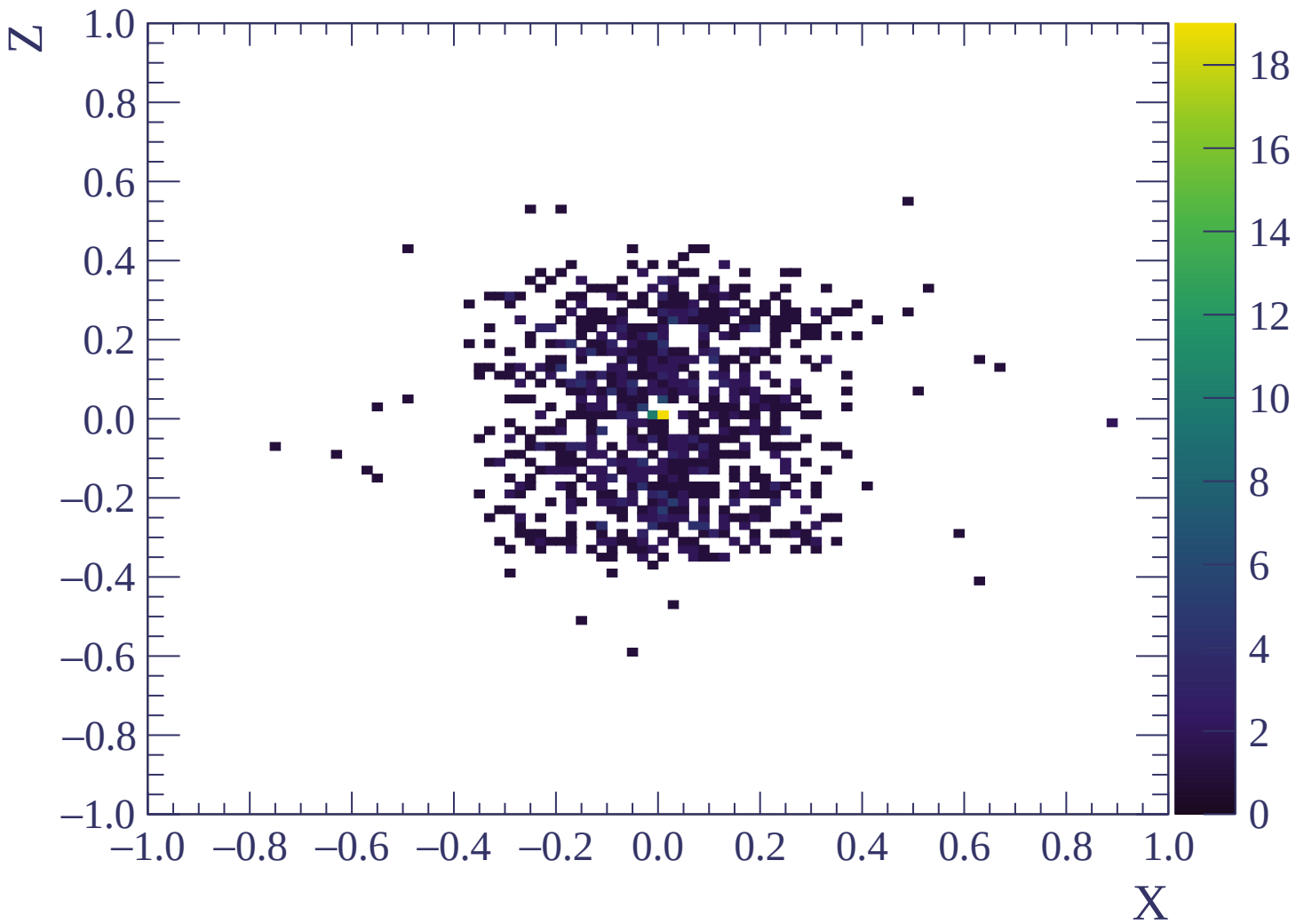
$\sigma = 0.0 \pm 0.0$

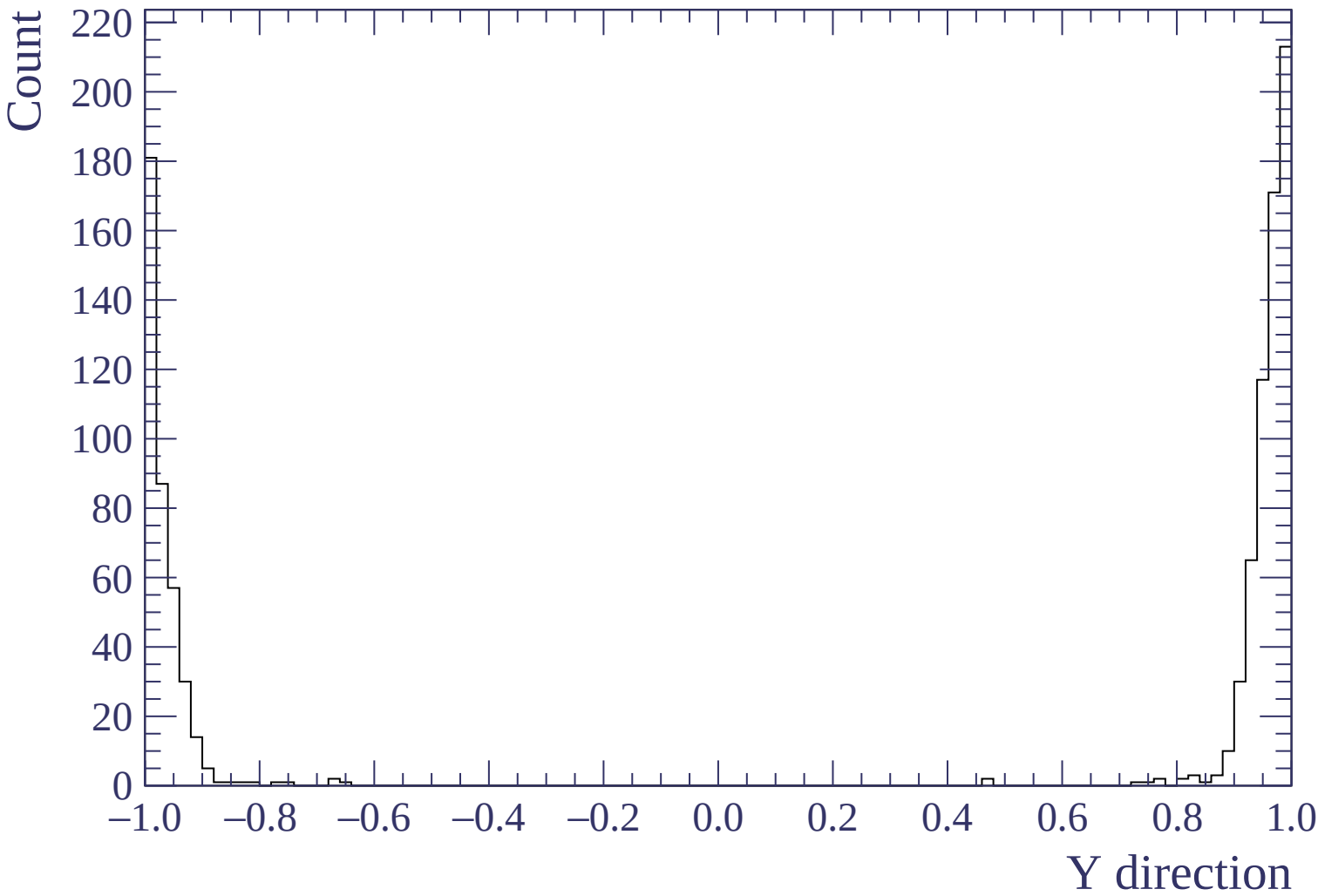
WF (0 entries)

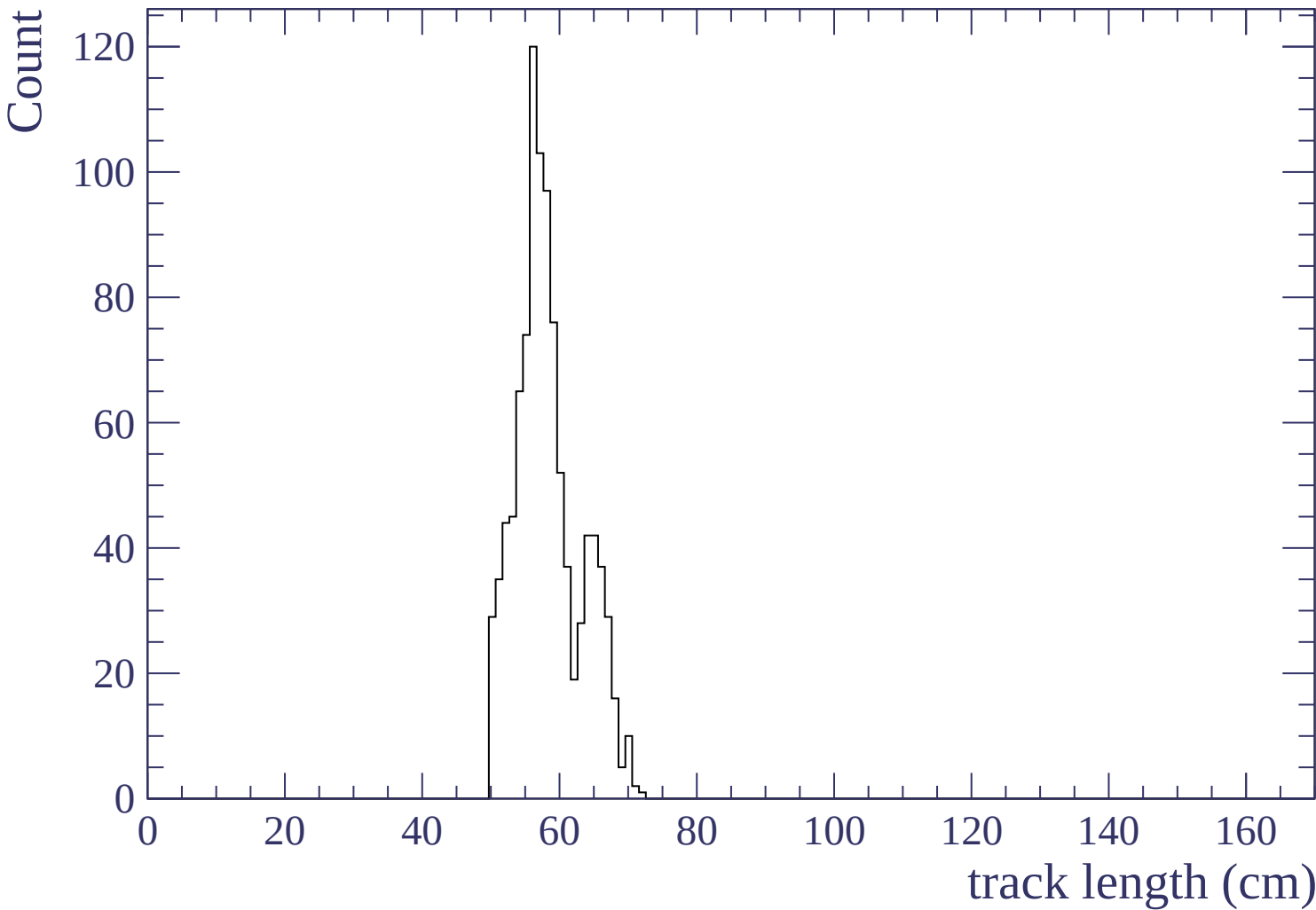
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

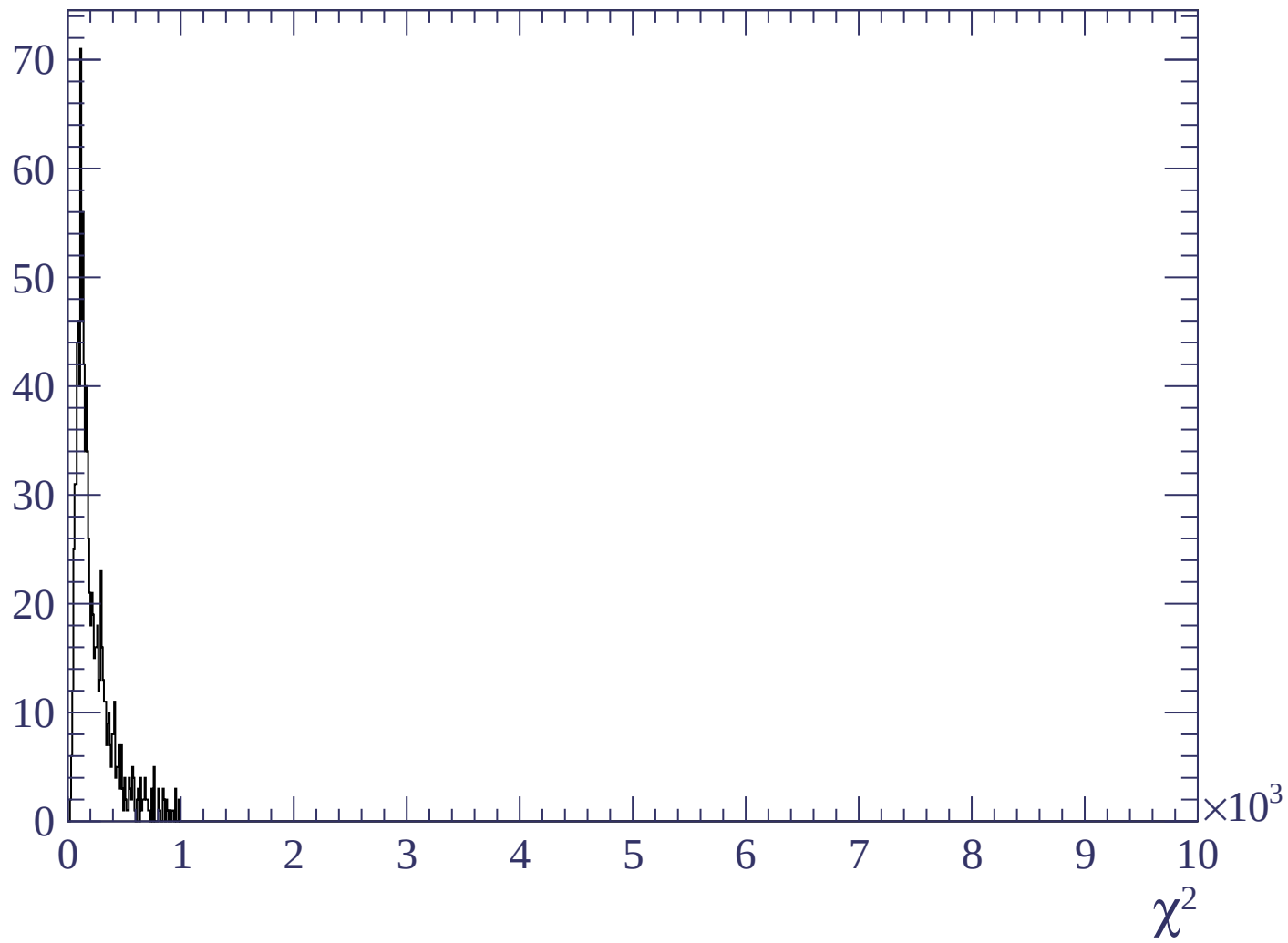
$\sigma = 0.0 \pm 0.0$

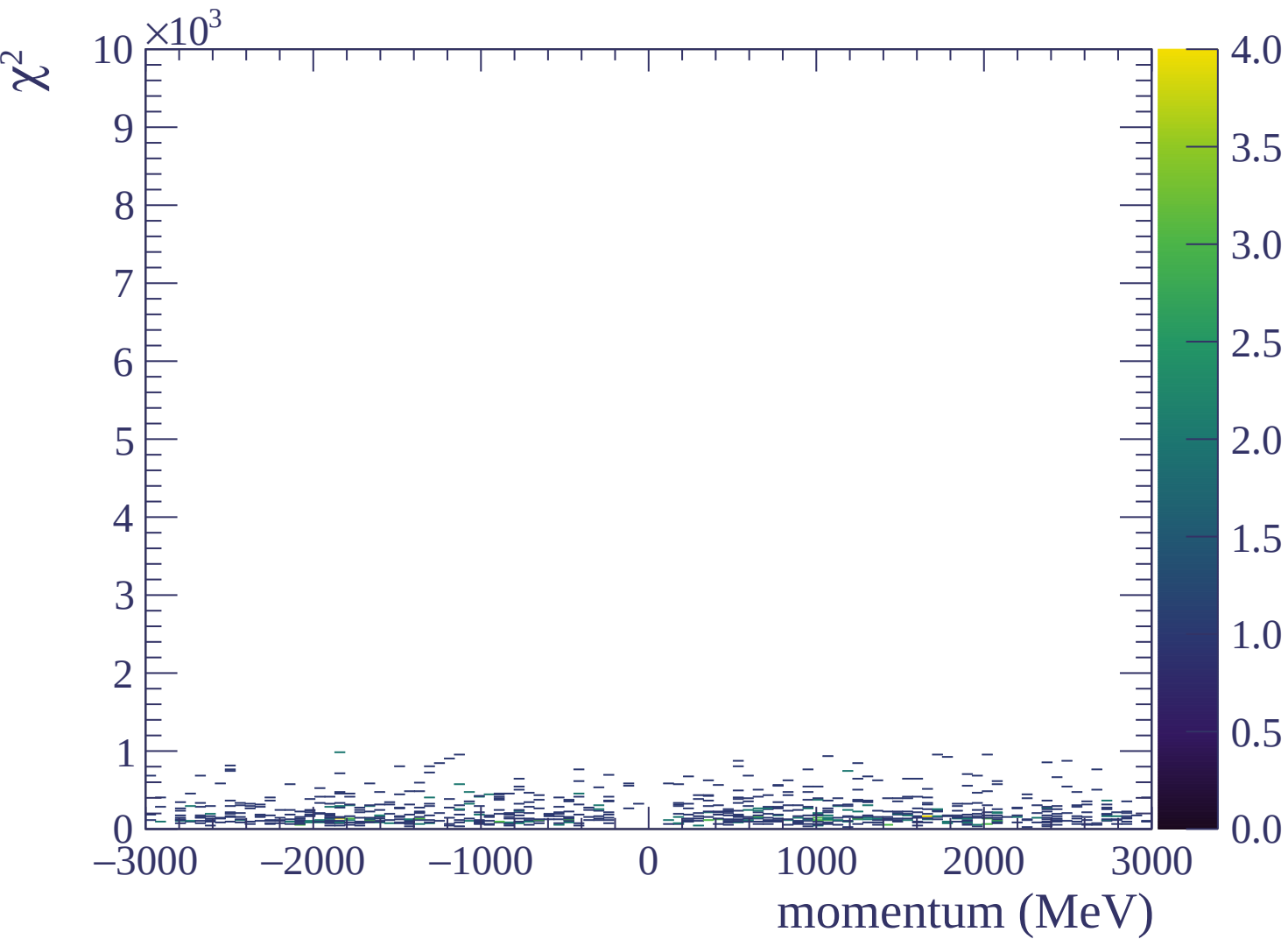




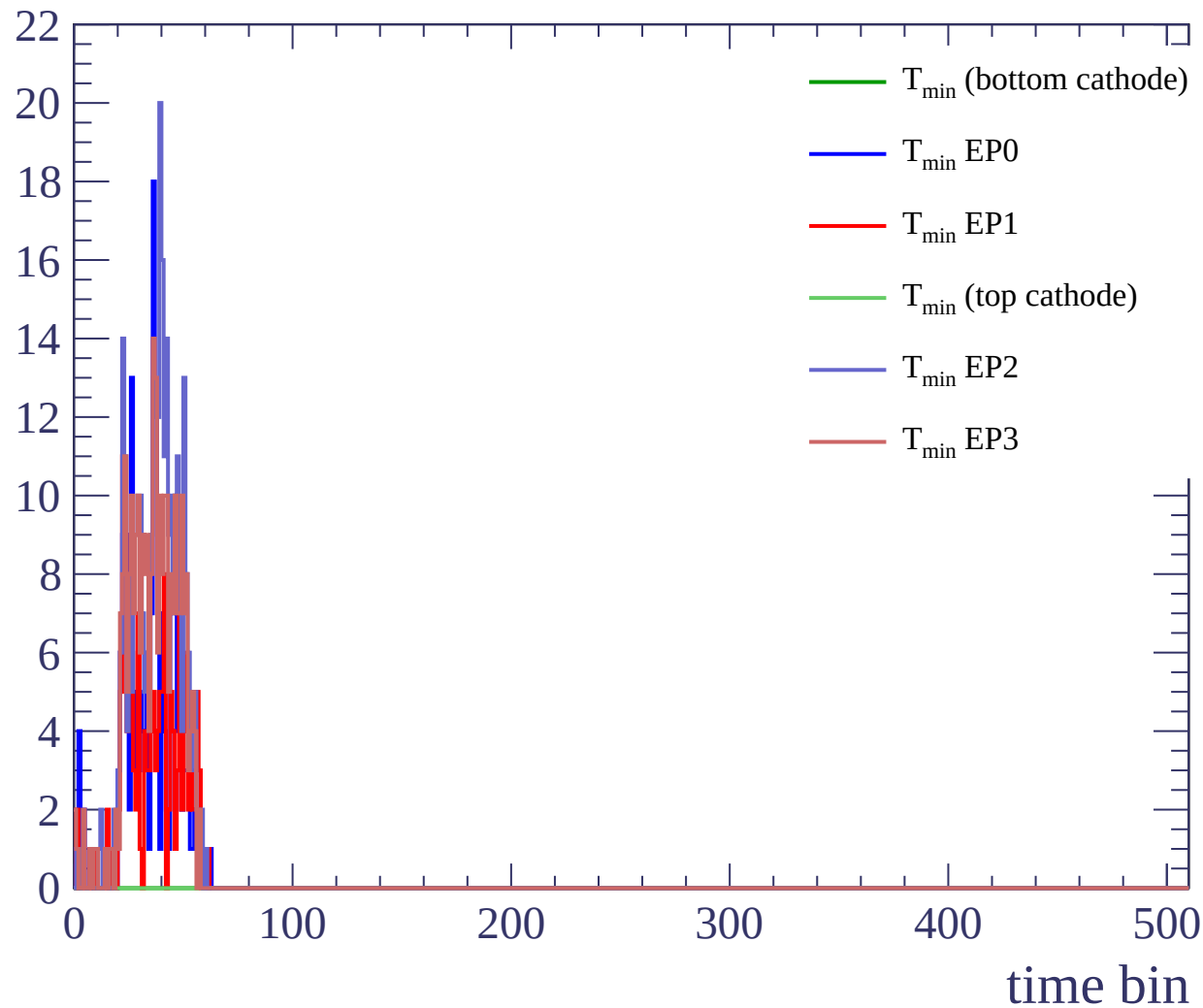


Count





Count



Count

